



UNITED NATIONS
INDUSTRIAL DEVELOPMENT ORGANIZATION



International Yearbook of
Industrial Statistics

Edition 2025

Progress by innovation with reliable industrial statistics

International Yearbook of Industrial Statistics

Edition 2025

UNIDO Statistics



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Abbreviations

AI artificial intelligence

CIP competitive industrial performance

CO₂ carbon dioxide

COVID-19 coronavirus disease

CPC Central Product Classification

EAEU Eurasian Economic Union

EIE emerging industrial economy

EU European Union

Eurostat Statistical Office of the European Union

FGP factoryless goods production (*see glossary*)

GCC Cooperation Council for the Arab States of the Gulf

GDP gross domestic product (*see glossary*)

GHG greenhouse gas

Gt gigatonnes

GVC global value chain

HDI Human Development Index (*see glossary*)

HHI Herfindahl-Hirschman Index

HS Harmonized Commodity Description and Coding System

IAEG-SDGs Inter-agency and Expert Group on SDG Indicators

ICLS International Conference of Labour Statisticians

IEA International Energy Agency

IIP index of industrial production (*see glossary*)

ILO International Labour Organization

IMF International Monetary Fund

ISIC International Standard Industrial Classification of All Economic Activities (*see glossary*)

ISIC Rev. 3 Revision 3 of ISIC

ISIC Rev. 4 Revision 4 of ISIC

ISID inclusive and sustainable industrial development

IVA industry value added (*see glossary*)

LCU large case unit (*see glossary*)

LDC least developed country (*see glossary*)

LLDC landlocked developing country

LoIP List of Industrial Products

MERCOSUR Southern Common Market

MHT medium-high and high technology

MMTD sub-annual manufacturing trade data (*see glossary*)

MNE multinational enterprise (*see glossary*)

MUVA mining and utilities value added (*see glossary*)

MVA manufacturing value added (*see glossary*)

NSO national statistical office

OECD Organisation for Economic Co-operation and Development

R&D research and development

RCEP Regional Comprehensive Economic Partnership

SBR statistical business register

SBS structural business statistics (*see glossary*)

SDC statistical disclosure control (*see glossary*)

SDG Sustainable Development Goal (*see glossary*)

SEEA system of environmental-economic accounting (*see glossary*)

SIDS small island developing State

SIEC Standard International Energy Product Classification

SITC Standard International Trade Classification

SME small and medium-sized enterprise

SNA system of national accounts (*see glossary*)

TCE tons of coal equivalent

UN United Nations

UNCTAD United Nations Trade and Development

UNDP United Nations Development Programme

UNIDO United Nations Industrial Development Organization

UNSD United Nations Statistics Division

USD United States dollar

USMCA United States-Mexico-Canada Agreement

WMC World Mining Congress

WMD World Mining Data

Foreword

The global industrial landscape continues to evolve and 2025 was a noteworthy year. Countries around the world had to steer their economies through a challenging environment marked by geopolitical tensions, the restructuring of supply chains, a rise in protectionist measures, and the need to transition towards more sustainability. Amid these uncertainties, industry remains a central driver of economic growth and resilience, as evidenced by the worldwide return of industrial policy to the centre of economic agendas.

This latest edition of the *International Yearbook of Industrial Statistics* offers an authoritative, data-based overview of these shifting dynamics and their implications for inclusive and sustainable industrial development.

Recent trends confirm that the manufacturing sector continues to play a decisive role in global economic performance, with multiplier effects that generates additional output, income, and jobs elsewhere in the economy. Manufacturing activity continues to expand at a moderate pace worldwide, reaching a growth rate of 2.9 per cent in 2024. Although this confirms the sector's resilience, it remains below the momentum observed before the pandemic. Middle-income economies again showed the strongest results, as many of them expand their participation in global value chains and invest in technological capabilities that enable a gradual shift toward higher-value and innovation-intensive industries.

The results here also show that while manufacturing value added has continued to grow, this has not consistently translated into broad gains in employment, especially for women and youth. Automation, digitalization, and the emergence of artificial intelligence are reshaping production, demanding new ways to provide people with the skills and opportunities needed for the future of work. Ensuring that technology supports decent jobs instead of displacing them is essential for maintaining trust in the current industrial transformation.

In terms of environmental sustainability, the evidence encouragingly shows that most economies have begun to decouple industrial growth from carbon emissions. Technological upgrading, greater energy efficiency, and an expanding market for low-carbon products demonstrate that greener industry is not only possible, but increasingly economically viable. However, the pace of change is uneven. Many low-income economies still face barriers to adopting cleaner production methods due to limited financing, technical capacity, or a lack of access to affordable technologies. Strengthening international cooperation,

technology transfer, and climate finance is absolutely critical in the years ahead.

The need for reliable and up-to-date industrial statistics has never been greater. Amid the rapid transformation and the volatile international environment, policymakers require accurate data to identify opportunities, measure progress, and ensure that industrialization leads to lasting impact and shared prosperity. UNIDO's statistical work, including this *Yearbook* and the open-access UNIDO Statistics Portal, is a core contribution to global evidence-based decision-making. The *Yearbook* offers a clear and objective picture of global industrial performance and long-term structural trends.

Looking ahead, we must ensure that industrialization allows opportunities for all, advances climate and environmental goals, and serves as a force for human well-being. The evidence presented in this *Yearbook* affirms that progress is possible and that there are important lessons to be learned from economies that have transitioned to innovation-based industrial sectors. Strategic investments, strong partnerships, and sound policies are all essential and should be prioritized.

I encourage policymakers, researchers, and all those driving industrial change to draw on the insights of this *Yearbook* to turn plans into action as we build a more inclusive, resilient, and sustainable future where no one is left behind.



Gerd Müller
Director General, UNIDO

PROGRESS BY INNOVATION WITH RELIABLE INDUSTRIAL STATISTICS





1 Introduction

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1.1 Preface

According to the Economic Policy Uncertainty Index and the World Uncertainty Index, two widely cited measures in the literature [1; 2], global economic uncertainty is at its highest level in at least three decades, since the beginning of the indices' historical records. The current landscape, characterized by persistent geopolitical tensions, prolonged disruptions in production chains, rising trade protectionism, environmental pressures, and rapid technological change, has resulted in economic volatility and uncertainty exceeding even the levels observed during the coronavirus disease (COVID-19) pandemic.

In this volatile context, the ability to rely on consistent, comparable data is more essential than ever. Industrial statistics play a vital role in understanding global industrial developments, identifying industries and groups that are advancing or lagging behind, and formulating effective strategies for inclusive and sustainable industrialization.

Industrial sectors are closely linked to productivity improvements, employment creation, innovation, and technological upgrading. Their evolution therefore has direct implications for broader development outcomes, including human development, food security, equity, and environmental sustainability. However, manufacturing has not been immune to the current uncertainty. As countries work towards achieving the industry-related Sustainable Development Goals (SDGs) targets in this challenging context, statistics remain indispensable for monitoring progress, diagnosing structural constraints and growth opportunities, guiding investment in infrastructure and innovation, and crafting effective industrial policy.

The *International Yearbook of Industrial Statistics* is UNIDO's flagship statistical publication. It provides a concise and accessible overview of recent global trends in industrial sectors, including the impact of ongoing uncertainty, based on authoritative indicators that allow for meaningful comparisons across countries, regions, and over time. This edition presents a snapshot of the latest data available as of October 2025.

This introductory chapter outlines the definition and significance of industry in sustainable development. Chapter 2 then provides a global overview of industrial performance, examining recent developments across all industrial sectors (including mining, manufacturing, and utilities). It also reviews progress towards SDG 9 industry targets and summarizes the latest results of the UNIDO competitive industrial performance (CIP) Index.

Thereafter, Chapter 3 focuses on the manufacturing sector, analysing recent global and regional production trends, structural patterns across industries, and key indicators related to international trade, employment, productivity, gender equality, and environmental impact. Chapter 4 turns to mining and utilities, discussing their role in regional industrial structures and their linkages with energy systems and resource use.

Finally, Chapter 5 is a thematic chapter that discusses the ongoing evolution and future transformation required for industrial statistics to remain responsive to policy needs in a world shaped by globalization, environmental pressures, technological change, and deepening inequalities.

All indicators featured in the *Yearbook* are based on official national and international statistics. The vast majority are drawn from UNIDO's own industrial statistics databases, developed in close collaboration with national statistical offices and international partners. Reflecting UNIDO's commitment to transparency and open data, all data featured in this publication are freely accessible through the [UNIDO Statistics Portal](#) [3], where users can explore, visualize, download, and reuse the information.

1.2 What are industrial statistics?

In statistics, having a clear and unambiguous definition is the essential first step before measuring or describing any concept. This ensures the production of consistent, harmonized, and internationally comparable data across statistical sources. According to current international standards, *industrial statistics* are defined in relation to a specific set of productive activities, as determined by the International Standard Industrial Classification of All Economic Activities (ISIC) [4]. These encompass four broad economic sectors, as outlined in the box below.

Definition of industrial statistics

Industrial statistics cover the characteristics and economic activities of all resident units in the reporting country that are primarily engaged in the following productive activities, as defined by the ISIC [4, p. 12; 5, pp. 79–172]:

- ▶ *Mining and quarrying* (ISIC Rev. 4 section B): The extraction of minerals that occur naturally as solids (e.g. coal and ores), liquids (e.g. petroleum), or gases (e.g. natural gas).
- ▶ *Manufacturing* (ISIC Rev. 4 section C): The physical or chemical transformation of materials, substances, or components into new products.
- ▶ *Electricity, gas, steam and air-conditioning supply* (ISIC Rev. 4 section D): The provision of electric power, natural gas, steam, hot water, and the like through a permanent infrastructure of lines, mains, and pipes.
- ▶ *Water supply; sewerage, waste management and remediation activities* (ISIC Rev. 4 section E): Activities related to the management (including collection, treatment, and disposal) of various forms of waste; water supply activities are also included in this section, as they are often carried out by the same establishments that provide sewage management.

Sections D and E are usually combined under the *utilities* sector. Given manufacturing's greater data availability and economic weight, separate statistical information for this sector is typically compiled, while data for the other three sectors are frequently grouped together under *mining and utilities*. UNIDO's statistical products, including this *Yearbook*, adhere closely to this definition and the associated terminology.

It is important to note that the everyday or specialized use of the term *industry* may deviate from this statistical definition. For instance, the term is often used narrowly to refer only to manufacturing activities, whereas in official statistics, manufacturing represents just one sector

Mining and quarrying



Manufacturing



Electricity, gas, steam and air-conditioning supply



Water supply; sewerage, waste management and remediation activities



within industry. This narrower interpretation is common in economics and in the literature on industrialization and structural transformation, including in UNIDO.

In the current version of the *Classification of Industrial Activities* [6], the term *industrial statistics* is not explicitly mentioned but falls under Domain 2 (economic statistics), Subdomain 2.3 (sectoral statistics). Within this framework, the four sectors listed in the box above partially overlap with Activities 20302 (energy), 20303 (mining, manufacturing, and construction), and 20399 (sectoral statistics not elsewhere classified).

Given these variations, some fluidity in the use of the term *industry* and related concepts should be expected. Finally, it is worth noting that construction (ISIC Rev. 4 section F) is not considered part of *industry* as defined in official statistics.

The definitions used here are based on the fourth revision of the ISIC, issued in 2008 [5]. A fifth revision, together with its implementation plan, has been endorsed by the United Nations (UN) Statistical Commission [7; 8]. While the new revision does not alter the scope of industrial statistics, it enhances the classification's relevance by better reflecting the structure of the contemporary global economy, including at a more detailed activity level. Relevant updates for industrial statistics will be reflected in future UNIDO statistical products, in line with their implementation by NSOs worldwide.

1.3 Why is industry important?

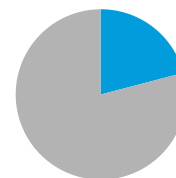
In 2024, industry accounted for 20.9 per cent of gross domestic product (GDP) globally, reflecting its central role in the world economy. Within industry, manufacturing remained the dominant sector, generating 78.7 per cent of total industrial value added, while the combined mining and utilities sector contributed the remaining share [9]. Manufacturing continues to stand out as one of the most dynamic components of economic activity, both because of its scale and its strong linkages with other sectors.

Historically, industrialization has been the engine of economic progress, driven by manufacturing's unique capacity to foster technological advancement, productivity growth, and innovation. Economies of scale are typically easier to achieve in manufacturing than in other sectors, allowing productivity gains to spread more broadly. As manufacturing expands, its cross-sectoral linkages stimulate activity in services, infrastructure, and agriculture, helping to lift the entire economy. Structural change towards higher value-added manufacturing has

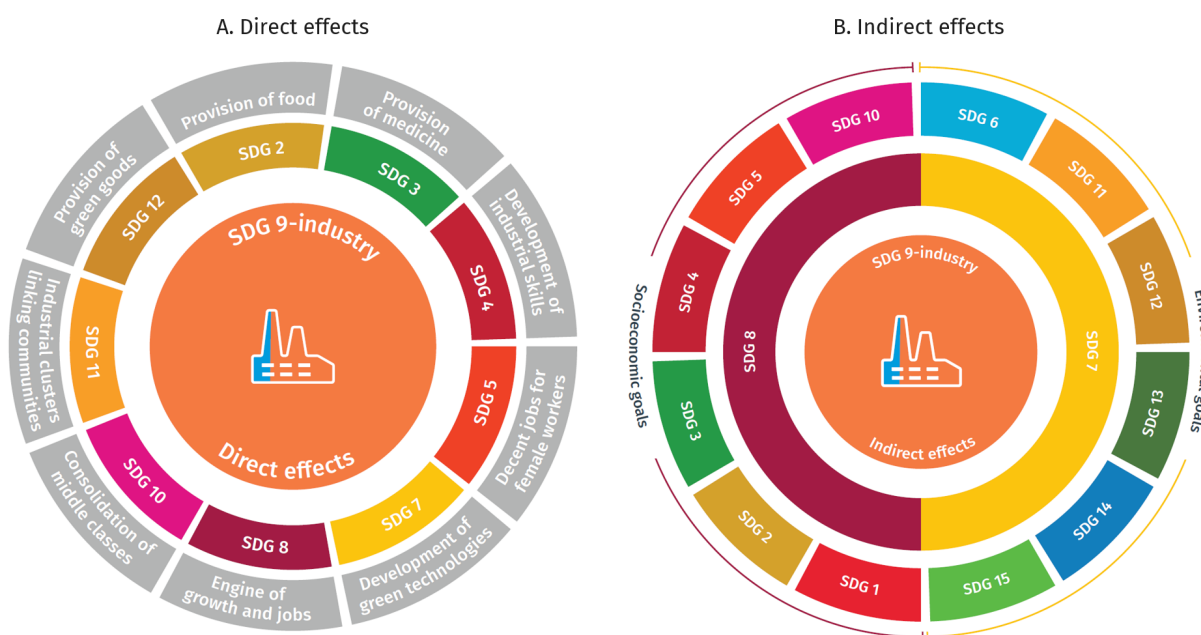
In official statistics,
manufacturing
is only one of four
industrial sectors



The **fifth revision**
of the **ISIC**
was globally endorsed in
2023–2024



Industrial sectors
account for
20.9%
of the global economy

Figure 1.1 | Industry and its direct and indirect links with other SDGs

Source: [10, pp. 37, 44]

long been recognized as the main pathway to higher income levels and sustainable livelihoods [10; 11].

The role of industry extends far beyond economic growth and productivity. Industrialization contributes directly and indirectly to the economic, social, and environmental pillars of the SDGs. By generating employment, improving living standards, and driving technological innovation, including advances in green production and circular economy practices, industry supports progress across multiple Goals (Figure 1.1). Research shows that manufacturing has accounted for roughly two-thirds of all country-level growth episodes over the past five decades, has led patenting activity related to frontier technologies and green innovation, and exhibits one of the strongest employment multiplier effects across sectors [10].

Given these broad spillover effects, the contribution of industry to sustainable development must be assessed beyond economic growth alone. Policies should capture industry's wider impacts on social inclusion, innovation, and environmental sustainability, to fully realize its potential contribution to these areas [12; 13].

Analysis based on UNIDO country groups highlights this multidimensional role. ⁱ As illustrated in Figure 1.2, countries with stronger industrial performance, measured by manufacturing value added (MVA) per capita, tend to perform better on the SDG Index, which tracks overall progress towards the 2030 Agenda. ⁱ The correlation is particularly strong among low- and middle-income countries, especially for MVA



ⁱ Annex C provides further information on UNIDO's country classification.

ⁱ The SDG Index is produced by the Sustainable Development Solutions Network. It represents a composite assessment of each country's overall performance on the 17 SDGs based on official and non-official data. The methodology and 2025 scores are presented in [14].

Figure 1.2 | Relationship between MVA per capita and the overall SDG Index in low- and middle-income economies

Source: [9; 14]

Note: The figure shows MVA per capita in 2024 and the SDG Index 2025. The horizontal axis is in logarithmic scale. The SDG Index is a composite assessment of each country's overall performance on the 17 SDGs based on official and non-official data. It ranges from 0 to 100, with higher values indicating a better performance; for more information, see [14]. High-income economies are excluded from this graph, as the relationship between manufacturing and development changes once countries reach advanced stages of development (see e.g. [15]).

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

per capita below 1,000 USD at constant 2015 prices. This indicates that industrialization, even from a modest starting point, can generate substantial gains towards sustainable development.

The connection between industrial growth and human development is equally strong. Figure 1.3 shows that, between 2013 and 2023, economies that achieved faster growth in MVA per capita generally recorded greater improvements in their Human Development Index (HDI).ⁱ Notably, all middle-income industrial economies improved their human development outcomes during this period.

Figure 1.4 provides stronger support, by first separating economies according to the UNIDO country classification and thus accounting for initial conditions. This additional evidence confirms that economies with stronger industrial growth also achieved the highest average annual increases in HDI. This pattern holds across income groups, with low-income economies experiencing robust industrial expansion showing the largest gains in human development, a clear reflection of the crucial role of structural transformation as a driver of inclusive progress.

ⁱ The HDI is a long-standing indicator measuring human development, defined as the process of expanding people's freedoms and opportunities and improving their well-being. The HDI is a composite index of three dimensions: 1) the ability to lead a long and healthy life; 2) the ability to acquire knowledge; and 3) the ability to achieve a decent standard of living. This indicator is published by the United Nations Development Programme (UNDP) [16].

Figure 1.3 | Relationship between the growth rate of MVA per capita and HDI in low- and middle-income economies, 2013–2023

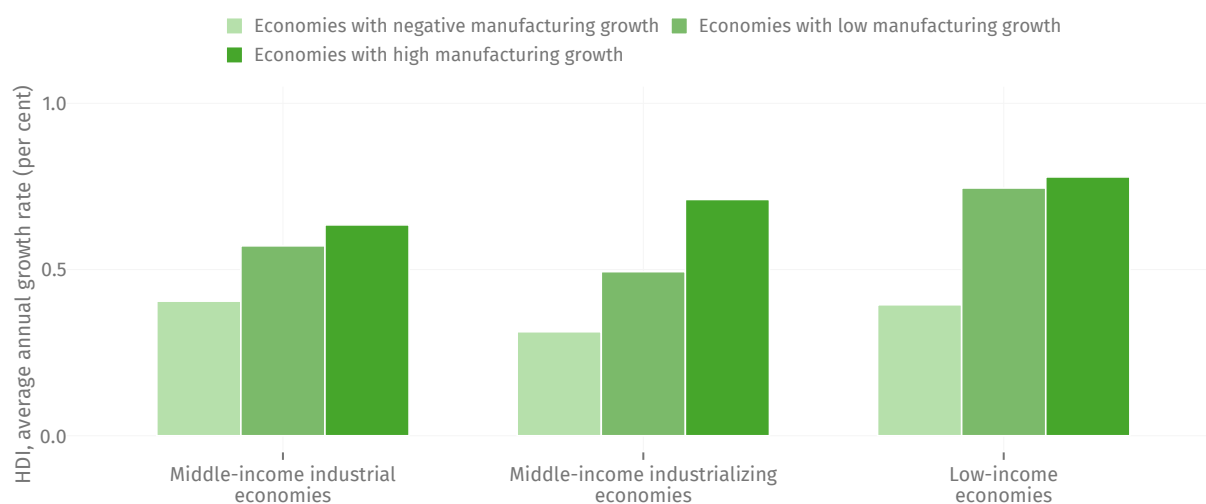


Source: [9; 16]

Note: Growth rates of MVA per capita are calculated over the values in constant 2015 USD. The HDI is a composite measure of human development. Higher values in the Index indicate a better performance; for more information, see [16]. High-income economies are excluded from this graph, as the relationship between manufacturing and development changes once countries reach advanced stages of development (see e.g. [15]).

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 1.4 | Average annual HDI growth rate in low- and middle-income economies by manufacturing performance, 2013–2023



Source: [9; 16]

Note: This graph shows the average annual HDI growth in 2013–2023 for low- and middle-income economies classified in three groups according to their average annual MVA per capita growth over the same period: economies with negative manufacturing growth (average annual MVA per capita growth smaller than 0 per cent), economies with low manufacturing growth (average annual MVA per capita growth larger than 0 per cent but smaller than 3 per cent), and economies with high manufacturing growth (average annual MVA per capita growth larger than 3 per cent). Economies are further partitioned according to the UNIDO country classification in order to reduce the impact of their pre-existing industrialization stage. High-income economies are excluded from this graph, as the relationship between manufacturing and development changes once countries reach advanced stages of development (see e.g. [15]). For more information on the HDI, see [16].

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

In sum, industry remains a cornerstone of sustainable development. Its impact extends well beyond economic growth, influencing social and environmental domains. The 2030 Agenda for Sustainable Development explicitly recognizes inclusive and sustainable industrial development (ISID) as a standalone goal (SDG 9), while also emphasizing its cross-cutting relevance to all other SDGs [17]. Empirical evidence further reinforces these interlinkages [18]. As economies industrialize, the positive ripple effects of industrial growth multiply, enhancing productivity, expanding opportunities, and strengthening the foundations for shared prosperity and advancing progress across the SDGs.



INDUSTRIAL STATISTICS: DATA AND VISUALIZATIONS





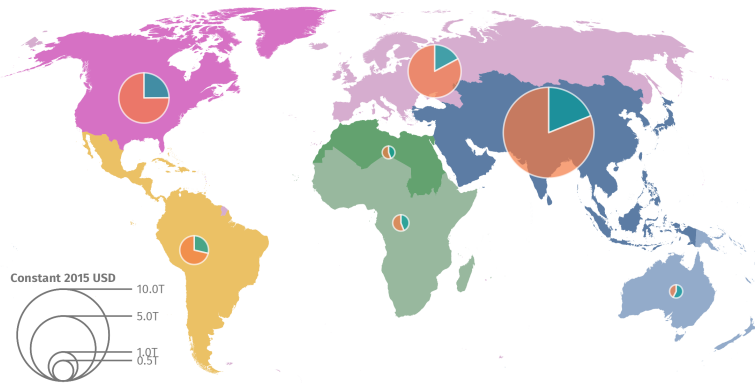
2 Global industrial statistics

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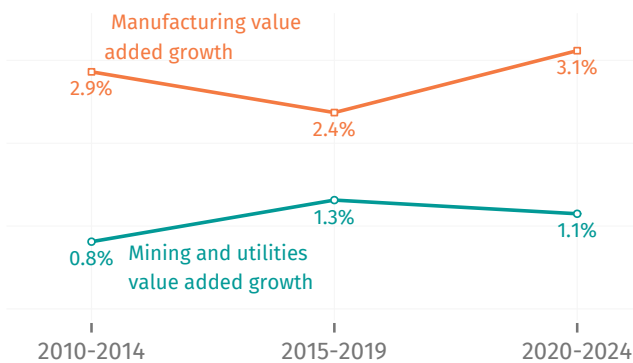
2 Global industrial statistics

Key figures

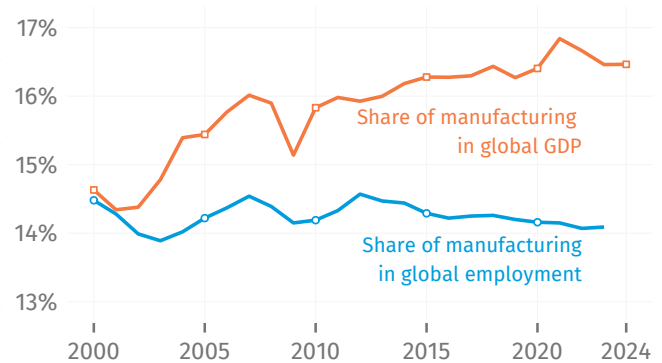
Industrial sector ● Manufacturing ● Mining and utilities



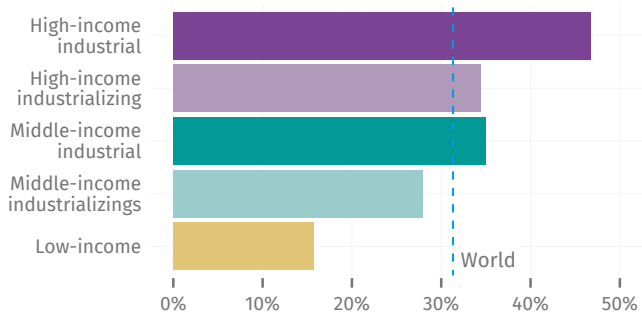
In 2024 **manufacturing** accounted for **78.7%** of total industry. **Oceania** was the only region where the **mining and utilities** sector dominated.



In terms of value added, global **manufacturing** has achieved stronger growth than global **mining and utilities** in recent years.

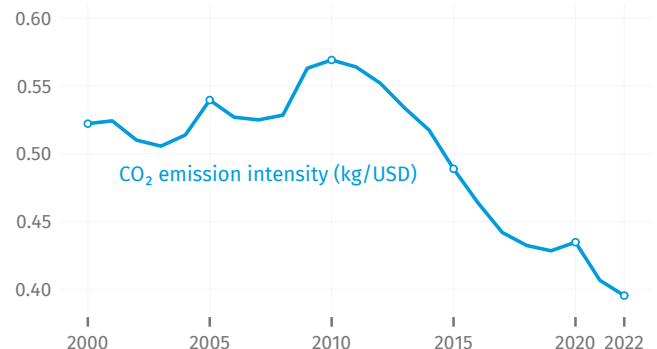


Manufacturing value added share in global GDP has increased steadily without a matching rise in **employment**.



Share of small manufacturing firms with access to finance

Globally, only **1 in 3** small manufacturing enterprises has a **loan** or line of credit.



Global **CO₂ emissions** per unit of manufacturing value added (in kg per constant 2015 USD) reached a **historical low** in 2022.

2.1 Recent trends in industrial sectors

This chapter examines the latest global developments shaping the industrial landscape. As outlined in Section 1.2, the term *industry* encompasses mining, manufacturing, and utilities activities. The following analysis offers a comprehensive overview of recent trends across these sectors from a comparative perspective and presents UNIDO's 2025 global growth forecasts. In addition, it reviews progress in key indicators related to SDG 9 and industrial competitiveness, highlighting how countries and regions are adapting to shifting economic and technological conditions. A more detailed analysis of manufacturing performance is provided in Chapter 3, while Chapter 4 explores the dynamics of the mining and utilities sectors.

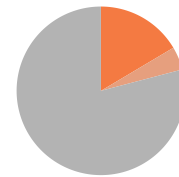
2.1.1 Annual industrial production

Industrial sectors accounted for 20.9 per cent of world GDP in 2024, equivalent to 20.1 trillion USD at constant 2015 prices. Manufacturing was the largest industrial sector, representing 78.7 per cent of total industry in 2024, while the combined mining and utilities sector was responsible for the remaining 21.3 per cent.

The predominance of manufacturing is evident across most regions, though with significant variation. As illustrated in Figure 2.1, in terms of value added, manufacturing was the main industrial sector in Europe (with 82.8 per cent of industry value added in 2024), Asia (81.1 per cent), Northern America (75.1 per cent), Latin America and the Caribbean (71.4 per cent), Sub-Saharan Africa (56.0 per cent), and Northern Africa (55.8 per cent). Oceania was the only region where mining and utilities prevailed, with manufacturing accounting for just 42.4 per cent of industry value added.

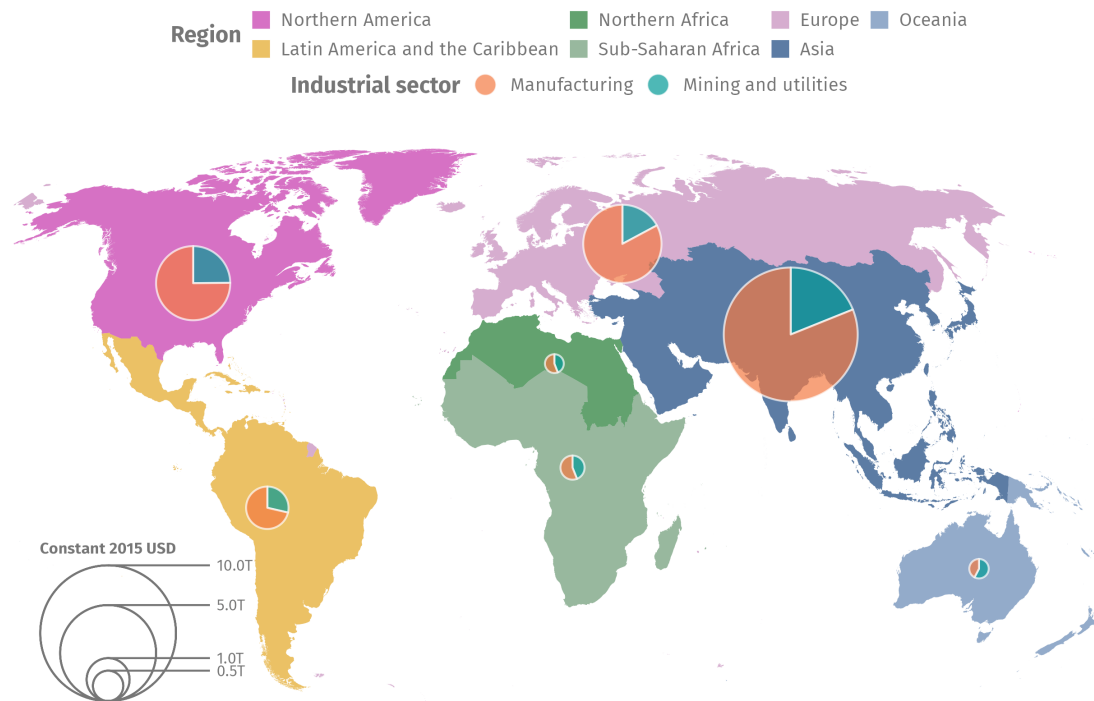
While industrial activity rebounded strongly following the COVID-19 crisis, growth has remained moderate. Figure 2.2 shows that industry has consistently grown at a slower pace than GDP since 2022. In 2024, global industry value added (IVA) rose by 2.5 per cent, below the 2.8 per cent increase in global GDP. This gap is expected to persist, though to narrow, in 2025.

Middle-income industrial economies were the most dynamic country group in 2024, recording an annual industrial growth rate approaching 4 per cent, followed by middle-income industrializing economies with growth of 3.5 per cent. In contrast, industrial growth in high-income economies remained close to 1 per cent in 2024 and is expected to continue at this subdued pace in 2025.



In 2024,
manufacturing and
mining and utilities
together accounted for more than
one fifth
of global economic activity

Globally,
industry value added
grew by
2.5%
in 2024

Figure 2.1 | Distribution of value added by industrial sector and region, 2024

Source: [9]

Note: The pie charts show the distribution of industry value added (IVA) between manufacturing (MVA) and mining and utilities (MUVA). The distribution is calculated using figures in constant 2015 USD.

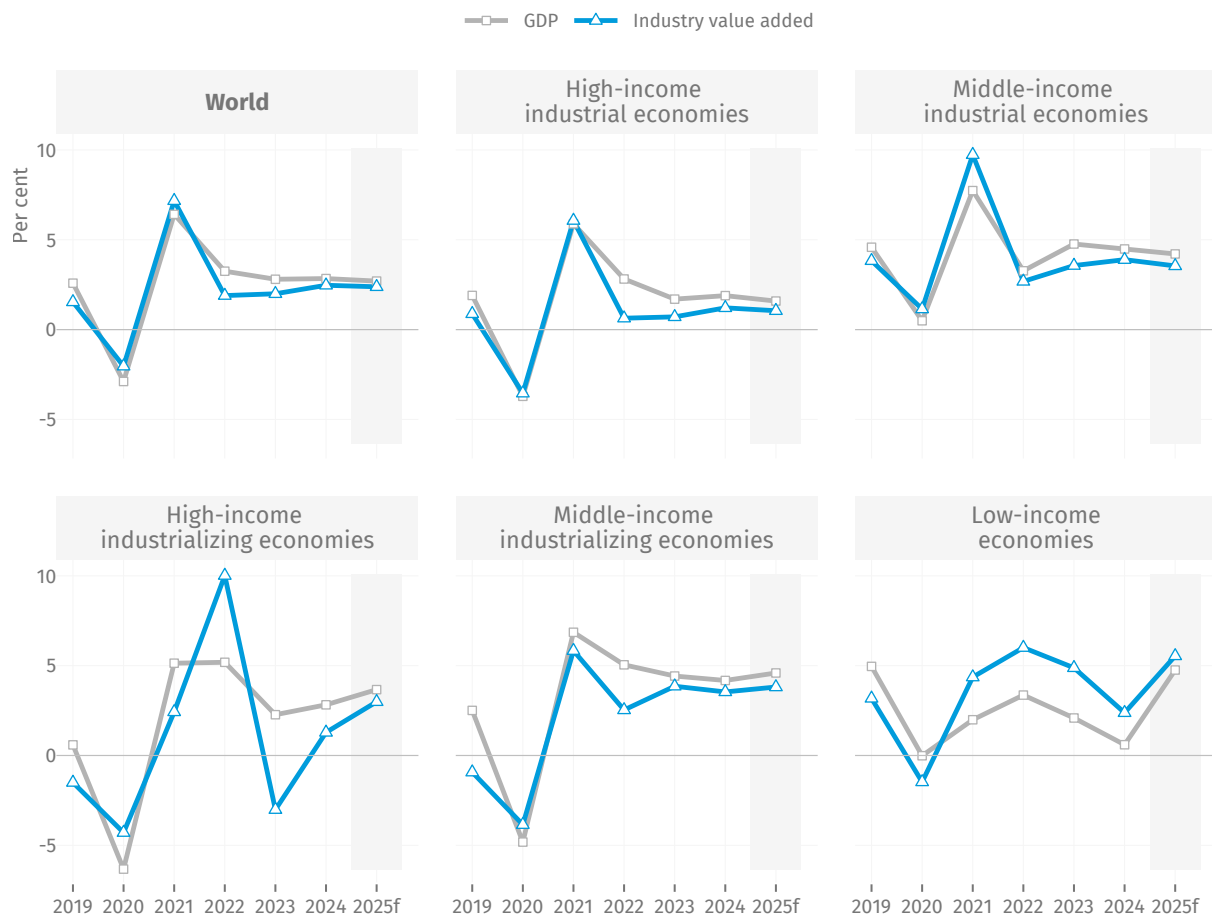
Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

It is worth noting that industrial activity in low-income economies has remained resilient in recent years. As presented in the corresponding panel of Figure 2.2, this is the only country group where industry has outpaced overall GDP growth. Although IVA growth in this group slowed to 2.5 per cent in 2024, it is projected to rebound in 2025, surpassing the 5 per cent mark. However, as will be discussed later, this strong performance has been driven mainly by the mining and utilities sectors rather than by manufacturing.

When comparing the performance of manufacturing with that of the combined mining and utilities sector, the former has consistently grown at a faster pace (see Figure 2.3). Although the gap narrowed during the pre-COVID-19 period, owing to a boom in mineral and mining commodities alongside a slowdown in manufacturing, it has widened again since then. Between 2020 and 2024, the manufacturing sector expanded at an average annual rate of 3.1 per cent, two percentage points higher than the growth rate of mining and utilities. This difference is expected to persist in 2025, with MVA projected to grow by 2.8 per cent, compared with only 0.80 per cent for mining and utilities.



Industrial production in
low-income economies
has shown
stronger momentum
than in any other country group

Figure 2.2 | Growth rate of GDP and industry value added by country group

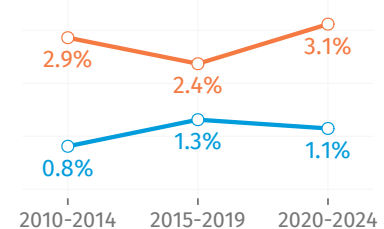
Source: [9]

Note: Growth rates are calculated over the values in constant 2015 USD. 2025f refers to UNIDO forecasts for the year 2025. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

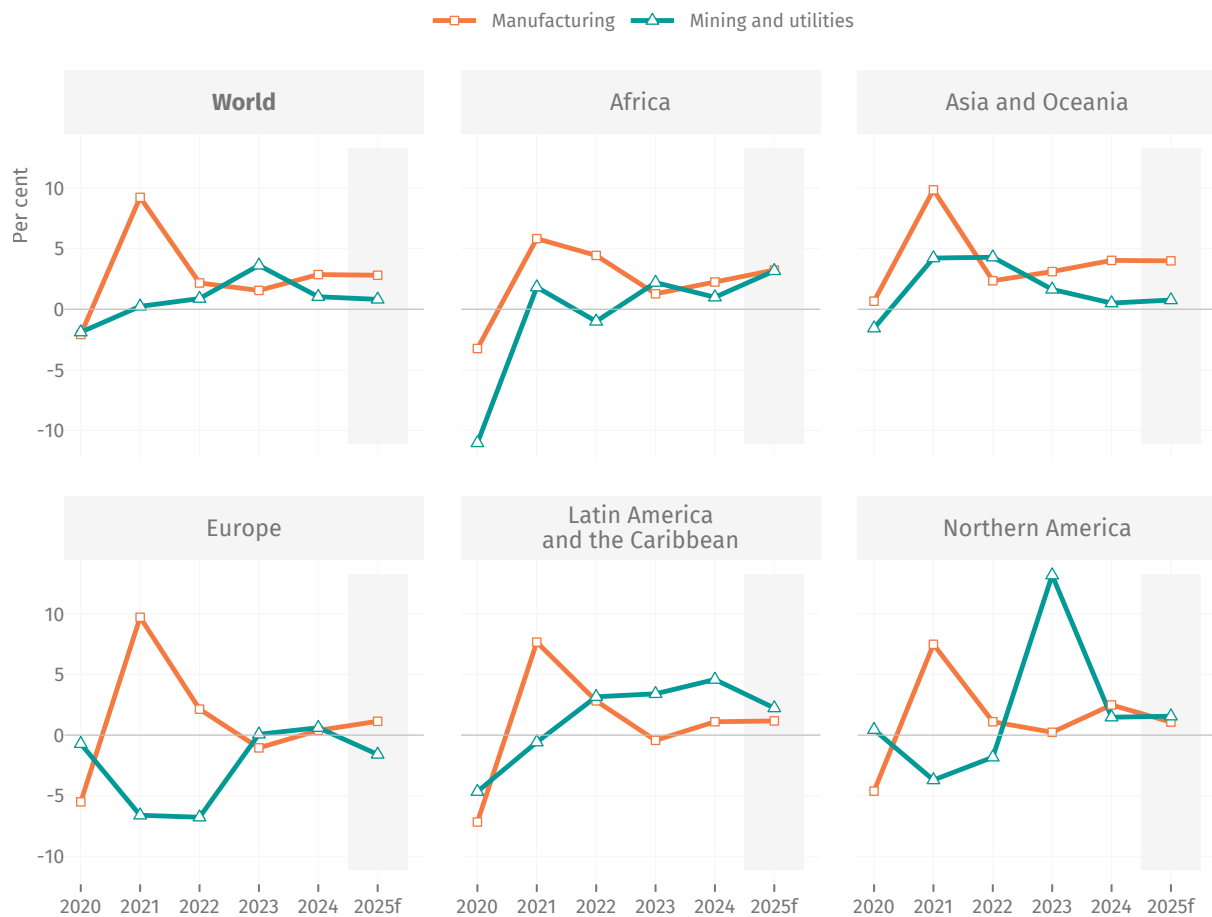
Figure 2.3 provides further insights into recent trends across industrial sectors. With the exception of the COVID-19 crisis in 2020, manufacturing growth has remained positive in all regions. The only exception was Europe, where manufacturing slowed and eventually turned negative in 2023. Latin America and the Caribbean is the only region where, in the post-pandemic period, manufacturing has been consistently outpaced by mining and utilities.

In 2025, manufacturing is expected to remain the main driver of industrial performance in Asia and Oceania as well as in Europe, while both broad sectors are projected to contribute equally to growth in Africa and Northern America. Finally, in Latin America and the Caribbean, the mining and utilities sectors are anticipated to maintain a higher growth rate than manufacturing, although the gap is expected to narrow compared with recent years.

With differing growth patterns across industrial sectors and varying



In terms of value added, **manufacturing** has achieved stronger growth than **mining and utilities** in recent years

Figure 2.3 | Growth rate of value added by sector and region

Source: [9]

Note: Growth rates are calculated over the values in constant 2015 USD. 2025f refers to UNIDO forecasts for the year 2025. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

levels of importance of each sector within overall industrial activity, the contribution of individual sectors to industrial growth varies significantly across country groups, as shown in Figure 2.4. The chart illustrates the respective contributions of manufacturing and of the mining and utilities sectors to the growth of IVA.

At the global level, industrial dynamics are predominantly driven by the manufacturing sector, owing to its larger relative weight and stronger performance in recent years. A similar pattern is observed among industrial economies. In both high- and middle-income industrializing economies, industrial growth reflects contributions from all sectors, although the trend is generally shifting towards greater reliance on manufacturing. It is worth highlighting the variability observed among high-income industrializing economies, reflecting their heterogeneity and diverse sources of growth, as many of these countries pursue a transition towards more manufacturing based activities.

In contrast with global trends, the industrial performance of low-


Globally,
industrial growth
is projected to remain relatively
stable in 2025

Figure 2.4 | Contribution of manufacturing and mining and utilities to industry value added growth by country group

Source: [9]

Note: A sector's contribution to growth depends on its growth rate, calculated over the values in constant 2015 USD, and its respective weight in total industry. 2025f refers to UNIDO forecasts for the year 2025.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

income economies is heavily dominated by the mining and utilities sectors. As noted earlier, the remarkable industrial growth recorded in these economies, well above their GDP growth rate, has largely originated outside the manufacturing sector. This raises concerns about the sustainability of such growth and its effectiveness in advancing structural transformation.

The outlook for industrial production in 2025 varies across country groups and industrial sectors. While global industrial production is projected to grow by 2.4 per cent, all country groups, except high-income industrial economies, are expected to achieve growth rates above 3 per cent. As shown in Figure 2.4, this projected dynamism is driven primarily by the manufacturing sector, with the sole exception of low-income economies, where mining and utilities sectors are expected to continue making an outsized contribution to overall industrial growth.



Mining and utilities
continue to play a key role
in **low-income economies**

2.1.2 Quarterly industrial production

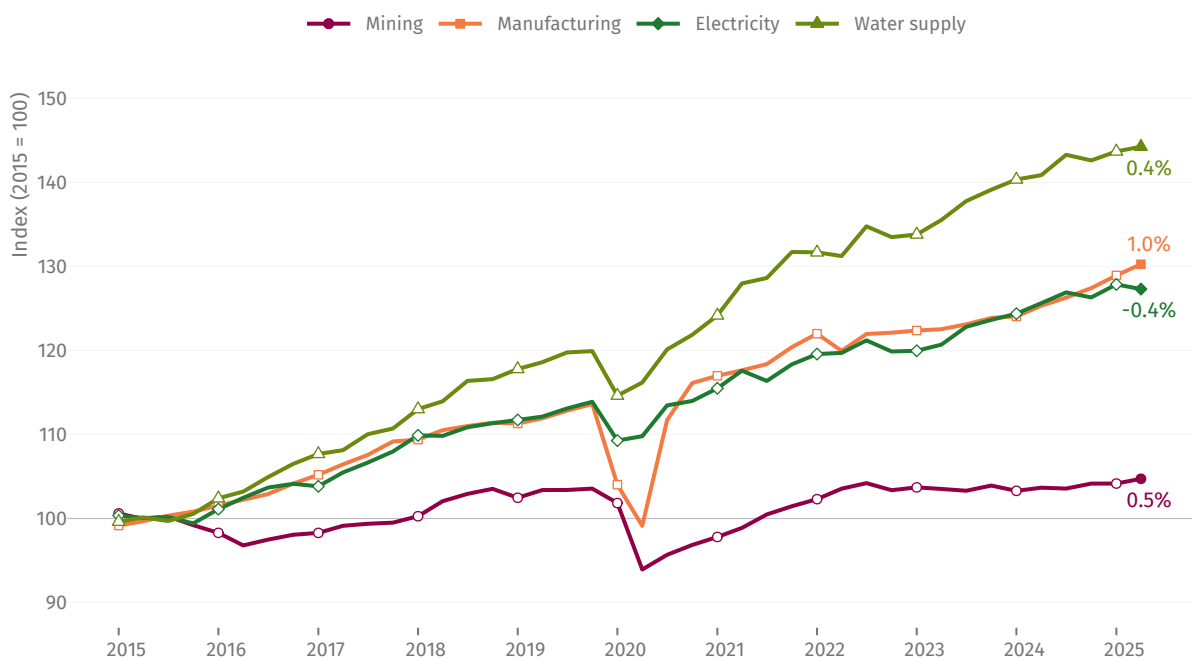
In addition to the information based on annual national accounts data described above, UNIDO also compiles sub-annual IIPs covering all industrial sectors: mining and quarrying; manufacturing; electricity, gas, steam and air-conditioning supply; and water supply; sewerage, waste management and remediation activities. Owing to their higher frequency, the IIPs enable a more detailed and timely analysis of sectoral trends and current performance. Annex C summarizes the main characteristics and methodological aspects of these indices.

Figure 2.5 provides a general overview of the industrial sectors and compares their trajectories since 2015. In-depth information on recent developments specific to the manufacturing sector and to the mining and utilities sectors can be found in Sections 3.1.2 and 4.1.2, respectively.

Index of industrial production

The sub-annual index of industrial production (IIP) tracks changes in industrial production in volume terms relative to a base year. It provides timely information on industrial activity over short- to medium-term periods.

Figure 2.5 | Global index of industrial production by industrial sector



Source: [19]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

As shown in the graph, all industrial sectors, except for mining activities, have demonstrated notable dynamism in recent years. The manufacturing sector recorded the highest growth in the second quarter of 2025, expanding by 1.0 per cent over the previous quarter. On the other hand, electricity production lost momentum and declined by 0.44 per cent. Although data for the water supply sector remain limited, available information indicates that this sector has shown the strongest long-term performance among industrial sectors over the past decade, with a further increase of 0.39 per cent in the latest



Global output in
manufacturing,
electricity, and
water supply
continued on a
dynamic trend

quarter. Meanwhile, economic activity in the mining and quarrying sector has remained sluggish since 2015, with production levels largely stagnant since mid-2022. The figure shows that manufacturing and electricity production followed comparable trends, with only minor deviations observed between the two sectors.

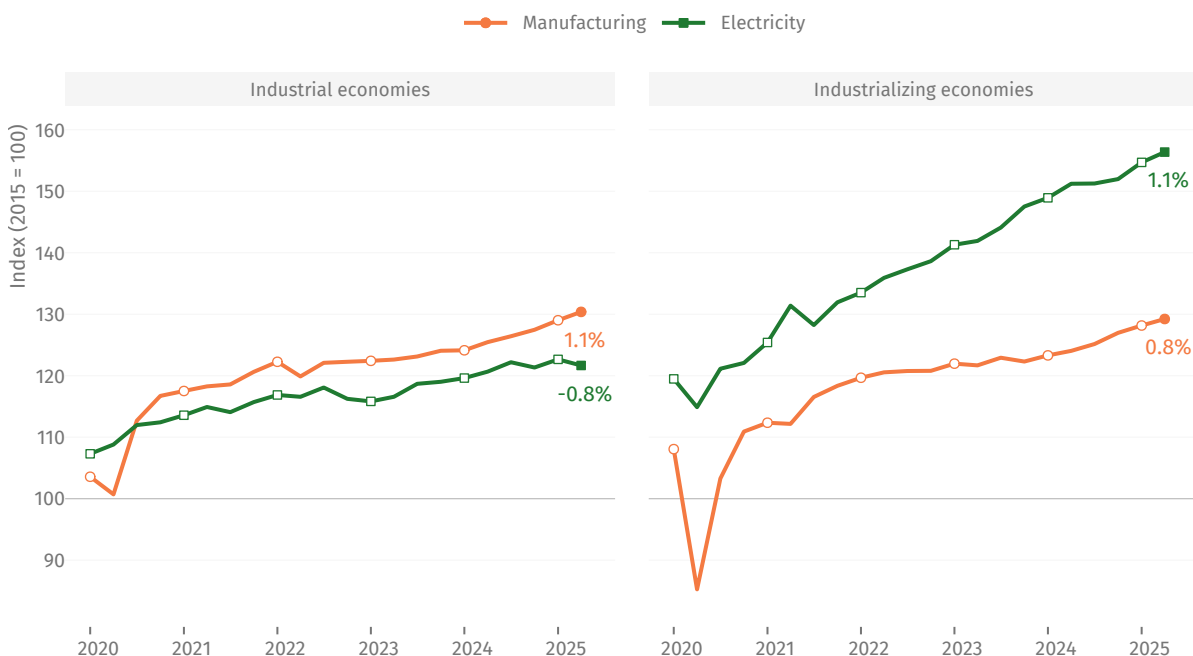
When focusing on UNIDO country groups (see Figure 2.6), it becomes apparent that industrial economies, which account for nearly 90 per cent of global MVA, have maintained a moderate growth trajectory in manufacturing output in recent years, while electricity production has displayed a more subdued and volatile pattern. In contrast, industrializing economies experienced a contraction in manufacturing activity in 2020 due to the COVID-19 crisis, followed by a more restrained recovery relative to the electricity sector. Nevertheless, in these economies both sectors recorded moderate quarterly growth in the second quarter of 2025.

A comparison between the first quarter of 2022, the first period unaffected by pandemic-related disruptions, and the second quarter of 2025 shows that manufacturing output grew at a similar cumulative rate in industrial (6.6 per cent) and industrializing (8.0 per cent) economies. Electricity production, however, exhibited a greater divergence, rising by 4.1 per cent in industrial economies but expanding by more than 17 per cent in industrializing economies.



Electricity production
in **industrializing**
economies
reached the **fastest**
expansion
among all industrial sectors
and country groups

Figure 2.6 | Global index of manufacturing and electricity production by country group



Source: [19]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

2.2 SDG 9

Industrialization is central to SDG 9, which seeks to build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation. Its role and outcomes, however, vary across development stages. In lower-income economies, industrialization marks a structural shift from traditional agriculture or low-value-added activities toward higher productivity in modern industry. In higher-income economies, it involves diversification, technological upgrading, and specialization in innovation-driven sectors. A unifying feature across all contexts is the transition toward environmentally sustainable manufacturing practices.

Industrialization drives productivity growth, job creation, and improved living standards, thereby contributing to poverty reduction and overall economic resilience [11]. Moreover, it supports broader SDG objectives by promoting innovation, technological progress, and green production and consumption. Evidence shows that, over the past five decades, the manufacturing sector has accounted for about two-thirds of global growth episodes, stimulated frontier and green technology development, and generated strong employment multipliers [10].

Given these wide-ranging impacts, assessing industrial performance requires going beyond traditional measures of economic growth to include broader dimensions of sustainable development when designing effective industrial policies. The industry-related targets of SDG 9 aim to capture these different dimensions and measure progress through internationally-agreed indicators.

2.2.1 SDG 9.2

Global manufacturing has maintained a steady growth in recent decades, only interrupted by the global financial crisis in 2008–2009 and the COVID-19 pandemic in 2020. The left-side panel of Figure 2.7 shows that global MVA per capita reached an all-time high in 2024, at 1,937 USD in constant 2015 prices. This represents an increase of 17.4 per cent with respect to 2015.

However, as shown in the right-hand panel of Figure 2.7, the contribution of manufacturing to total economic activity has slowed in recent years. After declining to 15.1 per cent in 2009, the share of MVA in GDP gradually recovered, peaking at 16.8 per cent in 2021. However, it has slightly declined since then, reaching 16.5 per cent in 2023, where it is projected to remain in 2024 according to UNIDO estimates. Despite some gains, the dynamism seen before the global financial crisis has not been regained, and the current trajectory reflects a period of relative stagnation.



Sustainable Development Goal
with three focus areas:

Industry
Innovation
Infrastructure

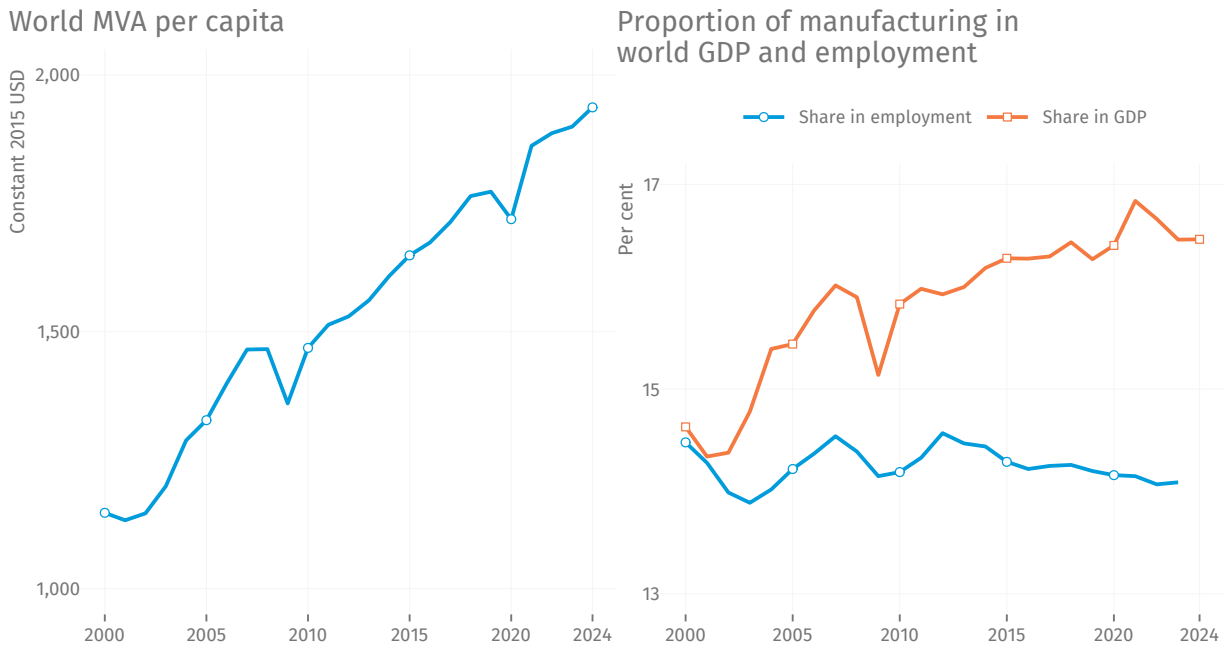
SDG target 9.2



Promote inclusive and sustainable industrialization and, by 2030, significantly raise industry's share of employment and gross domestic product, in line with national circumstances, and double its share in least developed countries.

- **9.2.1** Manufacturing value added as a proportion of GDP and per capita
- **9.2.2** Manufacturing employment as a proportion of total employment

Figure 2.7 | Global trends in SDG target 9.2: World MVA per capita (left) and proportion of manufacturing in world GDP and total employment (right)



Source: [20]

Note: Figures for 2024 are UNIDO estimates.

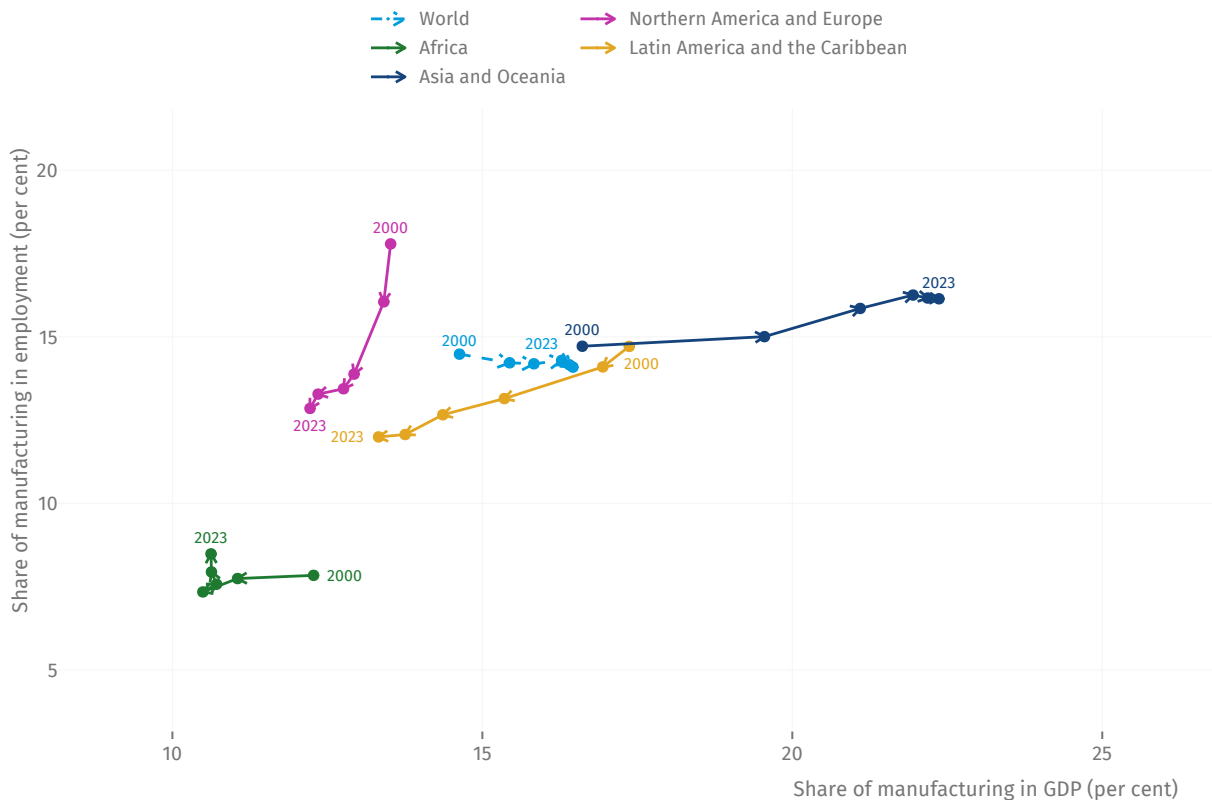
Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

In addition to value added, SDG target 9.2 also emphasizes employment. The second panel of Figure 2.7 presents the share of manufacturing in total global employment. After peaking at 14.6 per cent in 2012, manufacturing employment has followed a steadily declining trend, falling to 14.1 per cent in 2023, the latest year with complete data. This pattern of output growth without commensurate job creation raises concerns regarding the sustained capacity of manufacturing to generate employment and support livelihoods.

Regionally, trends in the proportion of manufacturing in GDP (9.2.1a) and in total employment (9.2.2) since the start of the century have varied considerably, as shown in Figure 2.8. Among all regions, only Asia and Oceania shows simultaneous growth in both indicators, suggesting a sustained path of industrialization. By 2023, this region achieved the world's highest shares in both value added and employment, positioning it as a leader in industrial development. In contrast, Latin America and the Caribbean as well as Northern America and Europe are experiencing deindustrialization, reflected in declining shares of manufacturing in both GDP and employment. Africa presents a more nuanced picture: while the manufacturing share in GDP is falling, the share in employment is rising, indicating shifts in production and labour dynamics in this region.

In 2024, global
MVA per capita
climbed to
\$1,937
(at constant 2015 prices),
an **all-time high**

The increase in the contribution
of manufacturing
value added to GDP
was not matched by
manufacturing
employment

Figure 2.8 | SDG 9.2.1a and 9.2.2: Proportion of manufacturing in GDP and in employment by region

Source: [20]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

2.2.2 SDG 9.3

SDG target 9.3 underscores the importance of small-scale enterprises, particularly those operating in the manufacturing sector, and their crucial role in fostering inclusive economic development. These enterprises typically operate with limited capital, rely heavily on local resources, and are a major source of employment in low- and middle-income economies. They also play a pivotal role in promoting entrepreneurship and creating job opportunities for women, youth, and vulnerable groups, helping to reduce socioeconomic disparities.

SDG target 9.3 includes two indicators: one measuring the share of MVA generated by small-scale enterprises (9.3.1), and another tracking their access to financial services (9.3.2). Due to a lack of consistent official data for indicator 9.3.2, results from the World Bank Enterprise Surveys are currently used as proxy values. For indicator 9.3.1, UNIDO compiles official data on MVA by firm size. A key challenge in producing internationally comparable data lies in the wide variation of national size classifications. Countries differ not only in the thresholds they apply, such as number of employees, turnover, or assets, but also in how these categories are defined and integrated into national legal and policy frameworks. These classifications often determine business

SDG target 9.3

TARGET 9-3
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

INCREASE ACCESS TO FINANCIAL SERVICES AND MARKETS

Increase the access of small-scale industrial and other enterprises, in particular in developing countries, to financial services, including affordable credit, and their integration into value chains and markets.

- ▶ **9.3.1** Proportion of small-scale industries in total industry value added
- ▶ **9.3.2** Proportion of small-scale industries with a loan or line of credit

registration procedures, taxation rules, and eligibility for support schemes targeting small-scale enterprises.

Since few economies report data following the international definition of small-scale enterprises, fewer than 20 employees, as per international guidelines [4], the indicator remains geographically and temporally limited. To address this, starting in 2025, UNIDO began collecting two separate series for indicator 9.3.1, as described in the box below.

Dual reporting of SDG 9.3.1

With the aim of increasing data availability, UNIDO started collecting two separate data series for indicator 9.3.1 in 2025.

- **9.3.1a:** Indicator based on UNIDO's standardized international classification of small-scale enterprises, ensuring cross-country comparability and maintaining continuity with previously reported data.
- **9.3.1b:** Indicator based on national definitions of small-scale enterprises to broaden country coverage, but with limited comparability across countries.

This dual-reporting approach was endorsed by the Inter-agency and Expert Group on SDG Indicators (IAEG-SDGs) and the UN Statistical Commission as part of the 2025 Comprehensive Review.

Despite the option to use national classifications, only seven additional countries were included under the expanded reporting framework (i.e. 9.3.1b) in 2025. However, as this is the first year of implementation, it is expected that adoption will gain momentum over time.

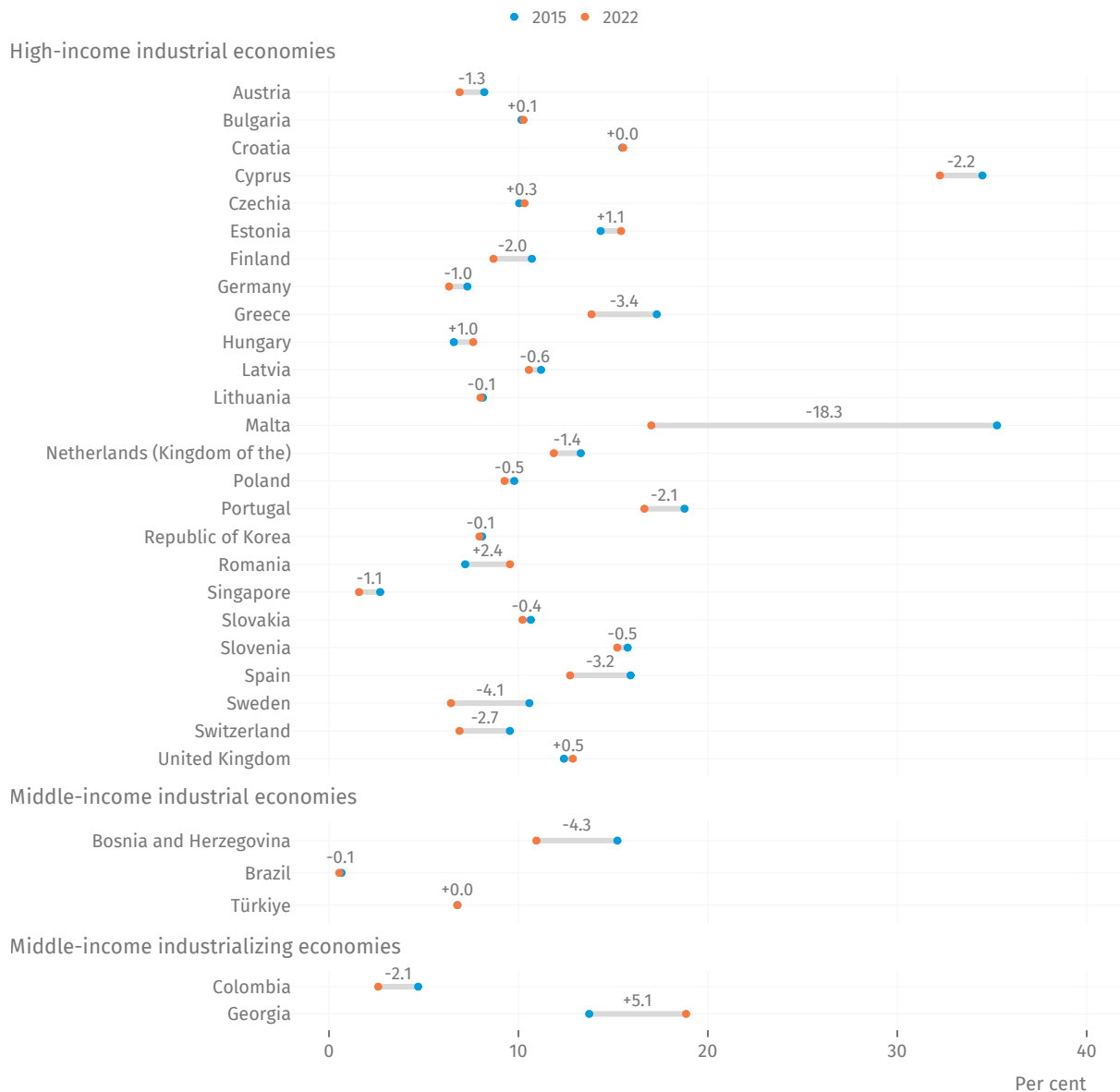
Figure 2.9 presents recent trends for indicator 9.3.1a among economies with available data for both 2015 and 2022. It conveys a rather sobering picture: from 2015 to 2022, the share of small-scale industries in total MVA has either stagnated or declined in most economies. Malta recorded the most significant drop, with a decrease of 18.3 percentage points. Only Georgia and Romania registered increases above 2 percentage points in the same period, rising by 5.1 and 2.4 percentage points, respectively. Overall, the share of small-scale industries in manufacturing remains under 20 per cent across all countries, with the sole exception of Cyprus, where it reached 32.2 per cent in 2022, despite declining by 2.2 percentage points since 2015. High-income industrializing economies and low-income economies are absent from the chart due to the complete lack of official data. This is particularly concerning in the case of low-income economies, where small-scale enterprises play a vital role in employment generation, poverty reduction, and broader economic development.



Small firms play a
pivotal role
in creating **job**
opportunities
for woman, youth, and
vulnerable groups

World Bank
Enterprise Survey
is the main **data source**
for SDG indicator **9.3.2**

Starting in 2025, UNIDO
implements a
dual reporting for
indicator **9.3.1:**
9.3.1a (standardized, comparable)
9.3.1b (national, broader
coverage)

Figure 2.9 | SDG 9.3.1: Share of small-scale industries in total industry value added

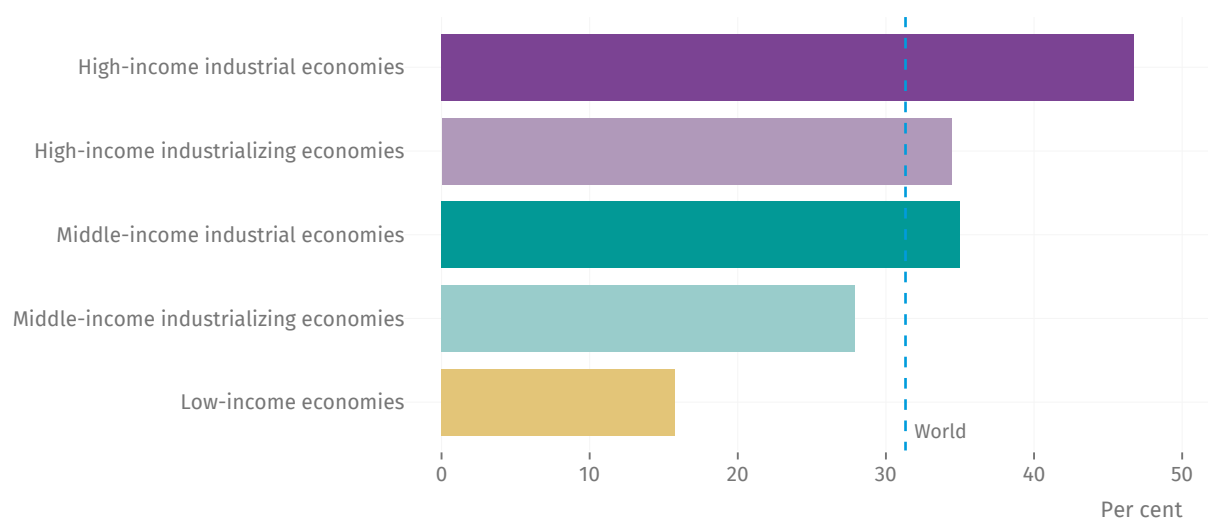
Source: [20]

Note: The graph only includes economies with available data in both 2015 and 2022. The numbers above the bars indicate the change in the proportion (in percentage points) between 2015 and 2022.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Similarly, access to credit remains uneven across economies. Figure 2.10 shows that there is a clear positive correlation between national income and access to financial services by small industrial firms. In particular, financial credit is scarcest among low-income economies. Globally, almost one in three small manufacturing enterprises has a loan or line of credit. To meet SDG target 9.3, inclusive industrial policies are vital to promote the growth, competitiveness, and resilience of small enterprises. Additionally, strengthening social safety nets will help support employment opportunities within small firms, particularly for women, youth, low-skilled, and informal workers.

Globally,
one in three small
manufacturing enterprises
have access to a
loan or line of credit

Figure 2.10 | SDG 9.3.2: Proportion of small-scale industries with a loan or line of credit, by country group

Source: [20]

Note: Group averages are computed as a simple average of available country-level point estimates. For each economy, only the latest available year of survey data, which can be between 2006 and 2023, is used in this computation. Only surveys adhering to the World Bank Enterprise Surveys methodology [21] are considered.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

2.2.3 SDG 9.4

After maintaining CO₂ emissions from fuel combustion below 6 gigatonnes (Gt) for five consecutive years, the manufacturing sector surpassed this threshold in 2021, largely due to the post-pandemic rebound. However, emissions declined again in 2022, falling just slightly below the 6 Gt mark to 5.98 Gt. Thereby, manufacturing accounted for 17.5 per cent of total CO₂ emissions from fuel combustion in 2022, a decrease of 0.35 percentage points from the previous year.

The CO₂ emission intensity of global manufacturing declined from 0.49 kg per USD in 2015 to 0.39 kg per USD in 2022, marking the first time it has fallen below the 0.40 threshold. The statistical evidence therefore corroborates that global progress towards SDG target 9.4 has taken place.

Figure 2.11 disaggregates the trends in indicator 9.4.1, the CO₂ emission intensity of manufacturing, since the adoption of the SDGs in 2015. It shows that all country groups have achieved reductions, albeit to varying degrees.

China recorded the largest decline, with a 36.6 per cent drop from 0.90 kg per USD in 2015 to 0.57 kg per USD in 2022. This substantial reduction is largely attributable to a shift toward renewable energy sources, particularly wind and solar photovoltaics, as well as energy efficiency improvements across its manufacturing sector. Furthermore, China's major investments in clean energy technologies have contributed to lowering global costs, thereby accelerating the global transition to

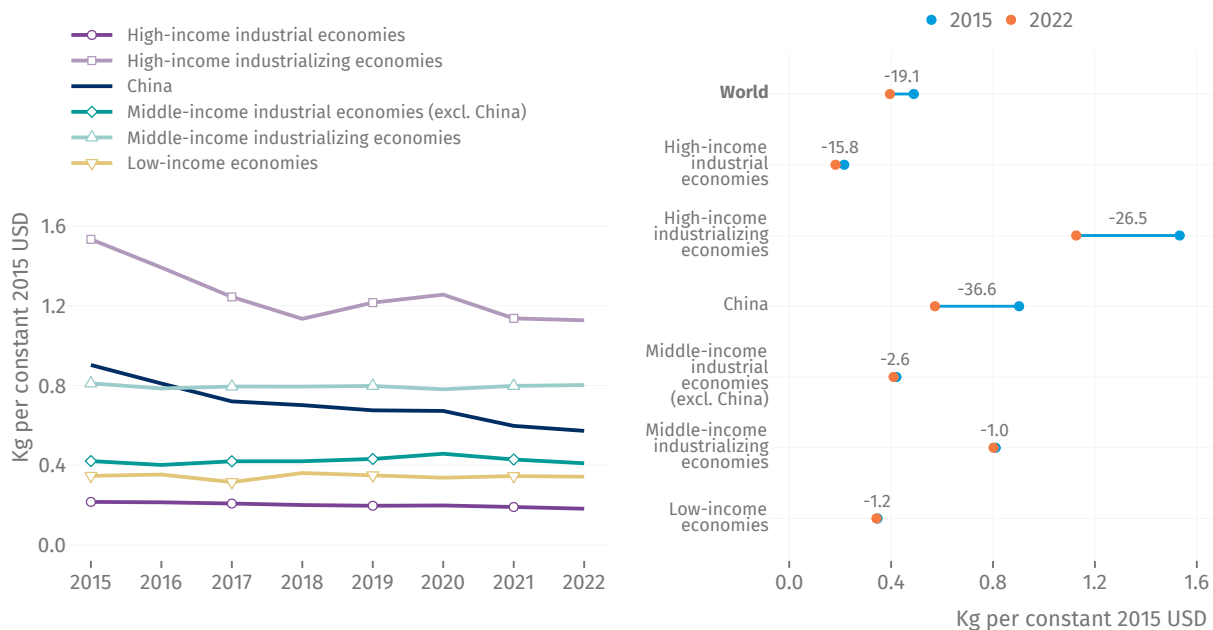
SDG target 9.4



By 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities.

► **9.4.1** Carbon dioxide (CO₂) emission per unit of value added

Figure 2.11 | SDG 9.4.1: CO₂ emissions from manufacturing per unit of MVA (left) and its comparison between 2015 and 2022 (right) by country group



Source: [20]

Note: In the right-side chart, the numbers above the lines or points indicate the change in the CO₂ emissions intensity of manufacturing (in percentage growth rate) between 2015 and 2022 for each country group.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

renewable energy [22]. Nevertheless, despite this progress, China's emissions intensity remains almost three times higher than that of high-income industrial economies, the group with the lowest CO₂ intensity.

High-income industrializing economies achieved a significant 26.5 per cent reduction in CO₂ intensity between 2015 and 2022. Despite this progress, they still recorded the highest levels of CO₂ intensity. Nonetheless, their values are gradually converging toward those of middle-income industrializing economies, which hold the second-highest intensity. This group registered only a modest 1.0 per cent decline over the same period, the smallest improvement among all groups included in the graph.

Due to China's dominant role in global manufacturing, its data is presented separately to provide a clearer picture of global trends without the outsized influence of a single economy. While middle-income industrial economies excluding China and low-income economies saw declines of 2.6 per cent and 1.2 per cent, respectively, between 2015 and 2022, low-income economies recorded the second-lowest CO₂ emissions intensity at 0.34 kg per USD, reflecting both lower industrial activity and a less carbon-intensive production profile.

CO₂ emissions
intensity

reached an
all-time low
in 2022

High-income
industrial economies

recorded the lowest
CO₂ emissions intensity in
manufacturing

2.2.4 SDG 9.b

Structural change entails not only a redistribution of economic activity from agriculture to industry, but also a shift towards higher-value added, higher-productivity sectors. In manufacturing, this is generally associated with an increasing share of innovation-focused industries that rely on advanced technologies and that operate with sophisticated digital production processes.

A transition towards these sectors would lead to a more diversified, resilient, and stable economy [23], a continuous support for innovation and knowledge creation, an upgrade in skills and jobs, and a progression to green growth, as these activities are generally less energy and emissions intensive [24]. Given manufacturing's potential for knowledge transmission and economic spillover, the reliance on technology and innovation in manufacturing could have significant benefits throughout the economy [25].

This is the subject of SDG target 9.b, which highlights the role of technology development, research, and innovation, and their relevance for promoting industrial diversification and boosting value addition. The target is monitored through indicator 9.b.1, which measures the contribution of medium-high and high technology (MHT) industries to total MVA. Manufacturing industries are grouped by their level of technological intensity according to a taxonomy developed by the Organisation for Economic Co-operation and Development (OECD), which is based on average research and development (R&D) expenditure by sector [26; 27]. The classification is presented in Annex C.

Globally, the share of MHT industries in total manufacturing has remained relatively stable in recent years, fluctuating within a narrow range. As shown in Figure 2.12, the share declined steadily between 2000 and 2009, reaching a low of 43.9 per cent.¹ This trend reversed after 2009, with the indicator rising to 45.1 per cent by 2018. It then dipped slightly to 44.7 per cent in 2019 and remained at that level during the COVID-19 pandemic, highlighting the sector's resilience. In 2021, however, the share dropped to 43.8 per cent, its second-lowest level since the start of the century, before quickly recovering to 44.5 per cent in 2022. Nonetheless, this still represents a decline of 0.49 percentage points compared to 2015.

Figure 2.13 shows that only high-income industrial economies registered an increase in the share of MHT industries in total manufacturing, albeit a modest rise of 0.54 percentage points. This group also maintains the highest overall share, fluctuating around 50 per cent, significantly ahead of other country groups.

They are followed by middle-income industrial economies, which have largely stagnated since 2015. In contrast, high-income industrializing economies experienced the most substantial decline, with a drop

SDG target 9.b



TARGET 9-B

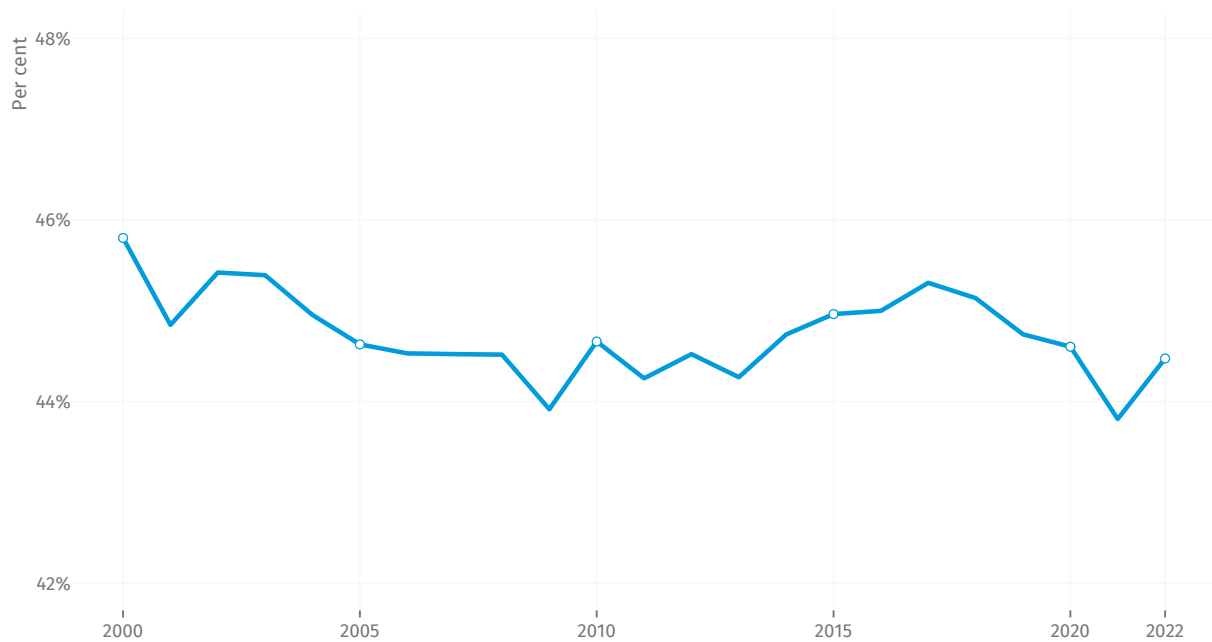
SUPPORT DOMESTIC TECHNOLOGY DEVELOPMENT AND INDUSTRIAL DIVERSIFICATION

Support domestic technology development, research and innovation in developing countries, including by ensuring a conducive policy environment for, inter alia, industrial diversification and value addition to commodities.

► **9.b.1** Proportion of medium and high-tech industry value added in total value added

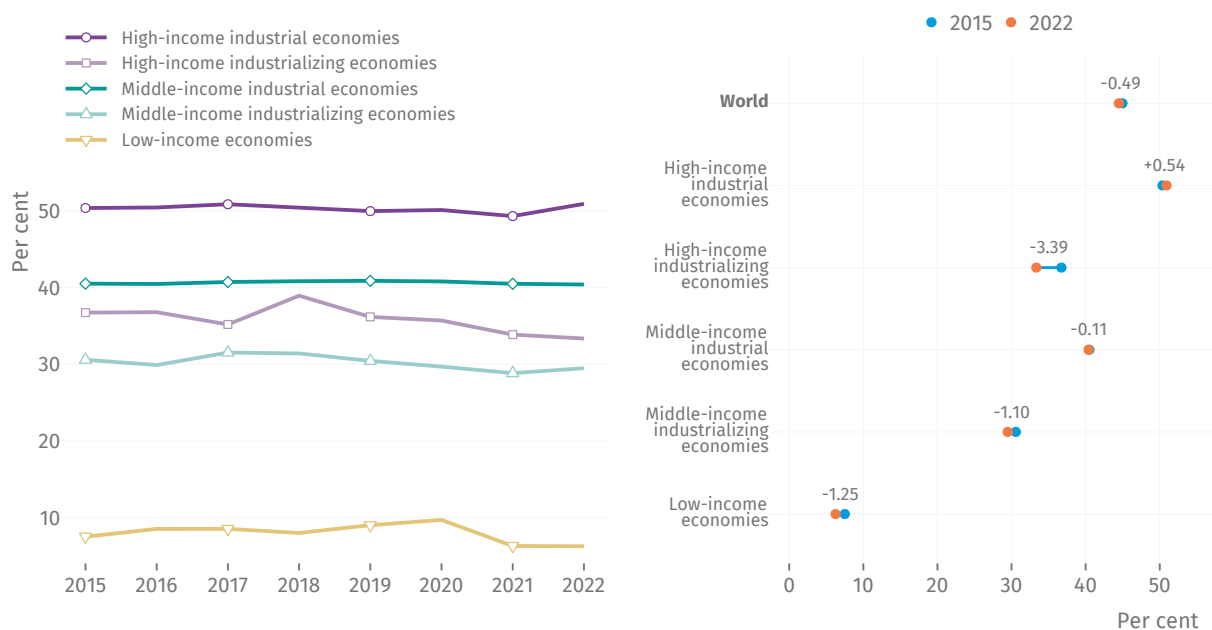
ⁱ For SDG 9.b.1, UNIDO relies exclusively on officially reported national data. Consequently, the resulting figures may differ from the MHT estimates presented in the *Yearbook*, which also incorporate UNIDO's own statistical estimates.

Only
high-income
industrial economies
registered an
increase
in the proportion of MHT
industries

Figure 2.12 | Global trends in SDG target 9.b: Proportion of medium- and high-tech manufacturing value added in total MVA

Source: [20]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 2.13 | SDG 9.b.1: Proportion of medium and high-tech manufacturing value added in total MVA (left) and its comparison between 2015 and 2022 (right) by country group

Source: [20]

Note: In the right-side chart, the numbers above the lines or points indicate the change in the proportion (in percentage points) between 2015 and 2022 for each country group.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

of 3.4 percentage points between 2015 and 2022. Middle-income industrializing and low-income economies saw similar levels of decline over the same period, though the latter continued to lag significantly behind all other groups in this indicator.

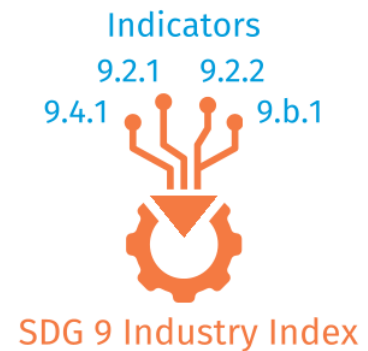
2.2.5 Overall performance towards SDG 9 industry targets

The UNIDO SDG 9 Industry Index is a composite measure of the industry-related indicators considered in SDG 9. It was introduced as a tool for supporting policymakers and other users in measuring a country's overall progress towards industrialization and evaluating performance along the different dimensions of industrial development, as defined in the 2030 Agenda.

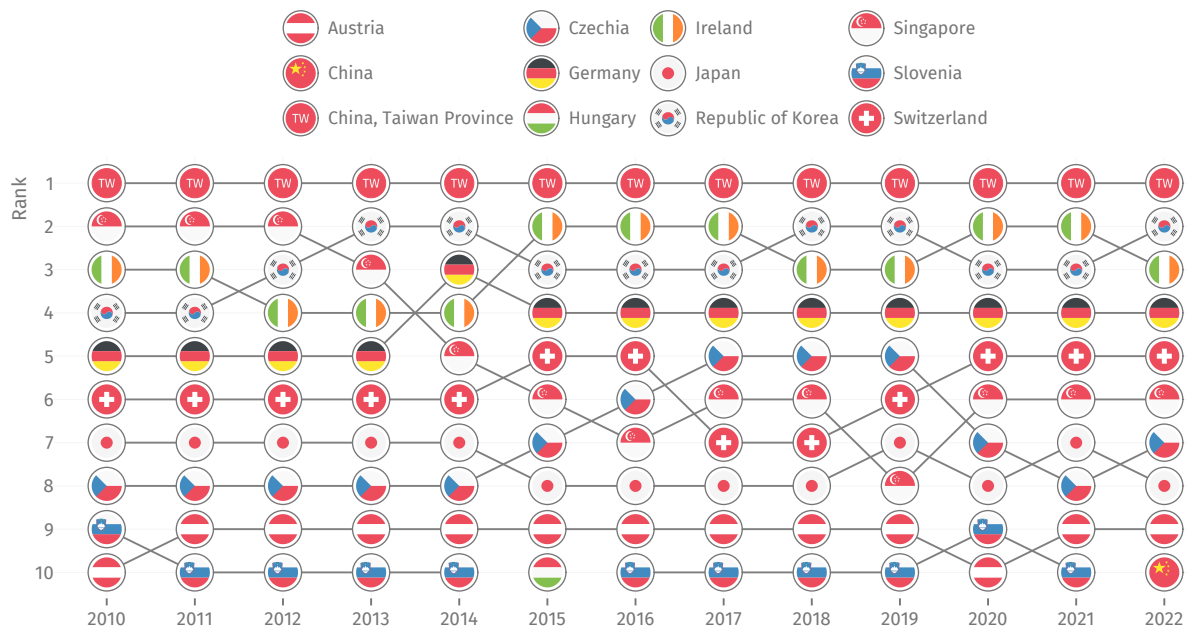
The Index encompasses the SDG 9 targets directly related to industrialization. These targets are monitored through eight indicators, referring to all three dimensions of inclusive and sustainable industrial development: economic (9.2.1a, 9.2.1b, 9.3.1a, 9.3.1b, 9.3.2, and 9.b.1), social (9.2.2), and environmental (9.4.1). However, due to insufficient data coverage across countries and time, SDG indicators 9.3.1 and 9.3.2 (*Proportion of small-scale industries in total industry value added* and *Proportion of small-scale industries with a loan or line of credit*) are excluded. The Index is therefore constructed from the other five indicators. Future editions may include these omitted indicators if data availability improves, offering a more comprehensive assessment of inclusive industrial development. The Index ranges from zero to one, with higher values reflecting better performance. A brief explanation of the Index and its methodological background can be found in Annex C, while a comprehensive coverage is provided in [28; 29].

The current edition of the SDG 9 Industry Index benchmarks 138 economies based on 2022 data. Although some indicators are available for more recent years, the Index requires data for all five indicators, which is only possible up to 2022. Figure 2.14 presents the ranking of the top ten economies based on the SDG 9 Industry Index. Compared to 2021, the ranking in 2022 is marked by a position exchange between Ireland and the Republic of Korea, as well as between Japan and Czechia. The Republic of Korea climbed from the third position to the second, while Ireland fell from the second to the third position. Japan dropped from the seventh to the eighth position, while Czechia moved from the eighth to the seventh. China entered the top ten for the first time, climbing from 11th to 10th place.

Since the launch of the SDGs in 2015, the top four economies have remained consistent, comprising Taiwan Province of China, the Republic of Korea, Ireland, and Germany. While Taiwan Province of China and



For the first time, a
middle-income economy
 entered the
top 10
 of the SDG 9 Industry Index
 ranking

Figure 2.14 | Top ten economies in the SDG 9 Industry Index

Source: [20]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Germany have consistently held the first and fourth positions, respectively, Ireland and the Republic of Korea have alternated between the second and third positions over the years.

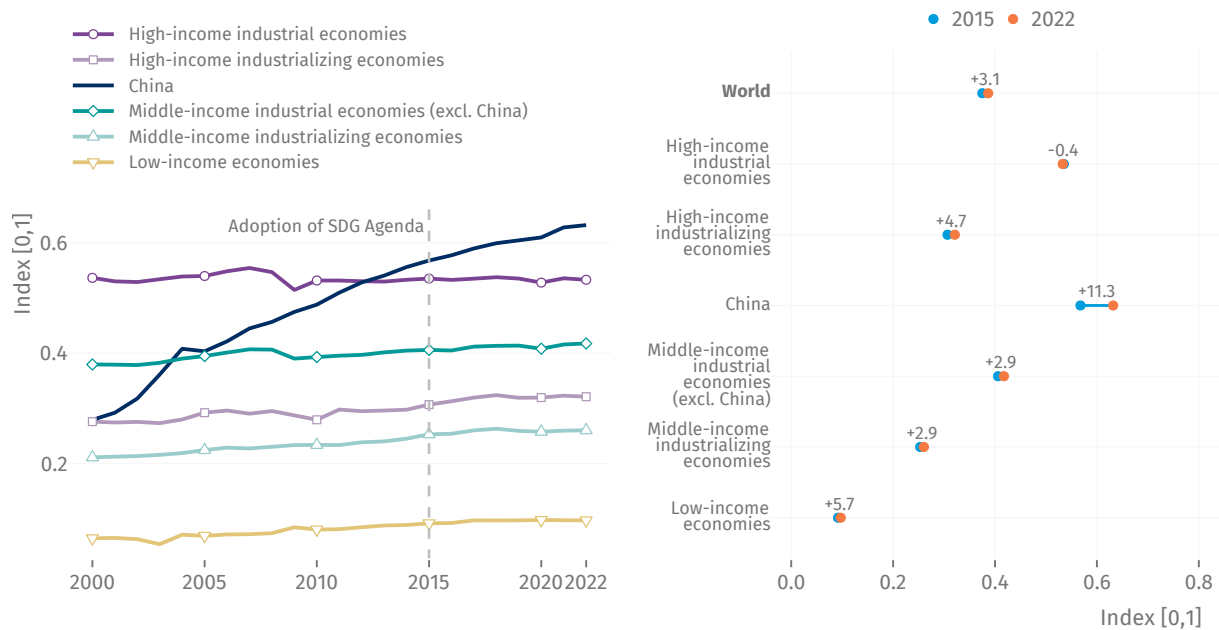
Until 2021, the top ten ranked economies were all high-income industrial economies. This changed in 2022 when China, a middle-income industrial economy, entered the list. Nonetheless, only three of the top 20 economies in 2022 are classified as middle-income industrial economies, as opposed to high-income industrial economies: China ranked 10th, Malaysia ranked 17th, and Mexico ranked 20th. The highest rank achieved by a high-income industrializing economy was the United Arab Emirates, at rank 40th. Hence, there is a strong correlation between industrialization levels, as classified by UNIDO country groups, and performance in the Index.

Figure 2.15 presents the average score in the SDG 9 Industry Index by UNIDO country group, providing an overview of their progress since the start of the century, while also highlighting the ranking of these groups over time. Due to China's dominant role in the manufacturing industry, its score is presented separately to mitigate the outsized influence of a single economy. Since the beginning of the century, China has made the most significant progress, surpassing first the other middle-income industrial economies in 2004, and then the high-income industrial economies in 2013. Indeed, since the adoption of the SDG Agenda in 2015, China has shown the most significant progress, recording an 11.3 per cent increase between 2015 and 2022. In contrast, high-income industrial economies have reported a gradual deindustrialization, and

1st
position
consistently held by
China, Taiwan Province



Since 2015,
China
has shown the
strongest progress
in the **SDG 9 Index**

Figure 2.15 | Average score in the SDG 9 Industry Index by country group

Source: [20]

Note: Group aggregates are calculated as weighted averages of economies with available information, using population as weights. In the right-side chart, the numbers above the lines or points indicate the change (in percentage growth rate) in the Index between 2015 and 2022 for each country group.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

are the only group experiencing a decline, dropping by 0.40 per cent between 2015 and 2022.

All other country groups have made varying degrees of progress since the adoption of the SDGs in 2015: low-income economies improved by 5.7 per cent, high-income industrializing economies saw a 4.7 per cent increase, middle-income industrial economies (excluding China) and middle-income industrializing economies both grew by 2.9 per cent. Despite low-income economies recording the second highest growth rate, the gap between them and the rest remains substantial.

The SDG 9 Industry Index is constructed exclusively from the official indicators of the SDG global indicator framework and their available data. Several potential extensions could be considered for a more complete version of the Index, such as including sex-disaggregated indicators to evaluate gender inclusiveness in manufacturing and the impact of policies on gender outcomes. Although indicators on SDG target 9.3 are currently excluded due to low data availability, once the data become available they would allow a more complete overview of the performance of small-scale industrial enterprises. Furthermore, other SDG 9 indicators beyond industry could also be included in the Index, with the objective of studying the interlinkages between the three focus areas of this Goal: industry, innovation, and infrastructure.



High-income industrial economies
are undergoing a
gradual process of
deindustrialization

2.3 Competitive industrial performance

UNIDO has tracked industrial competitiveness through the CIP Index for more than two decades. This Index assesses how effectively an economy's manufacturing sector produces and sells goods in domestic and international markets while advancing technologically. A core feature of the CIP Index is its comparative nature: each economy's performance is evaluated relative to all other economies, thereby embedding a true measure of global industrial competitiveness. ⁱ

The Index enables cross-country comparisons of industrial competitiveness and highlights key industrial development challenges. It is important to note, however, that the CIP Index focuses exclusively on economic competitiveness.

The CIP Index is constructed using statistical indicators across three dimensions:

1. Capacity to produce and export manufactured goods;
2. Technological deepening and upgrading;
3. World impact.

In its latest update, the Index covers 153 economies from 1990 to 2023. The overall CIP Index score ranges from zero to one, with higher scores indicating stronger industrial competitiveness. In practice, the observed range is narrower, as no country excels simultaneously across all dimensions. This section presents key findings from the most recent update of the CIP Index. The latest scores and global rankings for all economies are presented in Annex A.

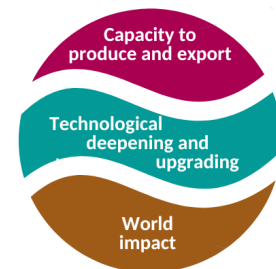
2.3.1 Global results

Figure 2.16 presents the economies with the most competitive manufacturing sector according to the CIP Index. Germany, China, and Ireland currently hold the top three positions. These economies have ranked among the three global leaders since 1990, 2014, and 2020, respectively. While the composition of the top 10 remained unchanged between 2022 and 2023, slight shifts occurred in the relative rankings. China, Taiwan Province fell one position to sixth place, exchanging ranks with the United States of America, which moved up to fifth place. Japan likewise dropped one position to eighth, continuing its gradual decline since it ranked as the world's most competitive industrial economy in 1994. Switzerland, by contrast, improved its position, rising from eighth to seventh rank.

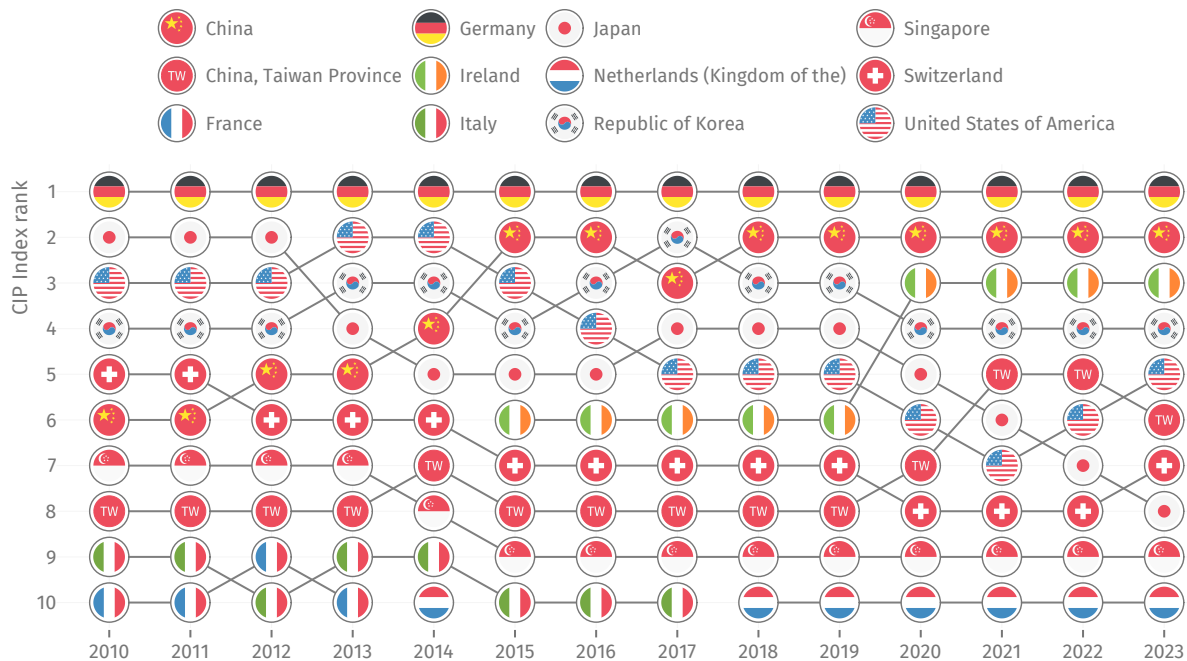
The full results reveal a strong correlation between income level and industrial competitiveness. The top ten economies in the CIP Index ranking are almost exclusively high-income economies. China, a

ⁱ See [30; 31; 32; 33] for a detailed description of the CIP Index's methodology and historical results. A brief overview of the Index's dimensions and the calculation of the composite index is presented in Annex C. The complete dataset is available in [34].

Three dimensions of the CIP Index



Germany, China, and Ireland
have been the three
most competitive industrial economies
for four years in a row

Figure 2.16 | Top ten economies in the CIP Index

Source: [34]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

middle-income industrial economy, is the sole exception. This pattern extends throughout the top quintile of the Index, where 24 out of the 31 economies are classified as high-income industrial economies. On the other hand, low-income economies are predominantly concentrated in the bottom quintile.

Industrial competitiveness generally evolves slowly, with structural transformation and changes in performance typically occurring incrementally over time. This is reflected in CIP Index scores and ranks, which tend to remain stable from one year to the next. A longer-time perspective is therefore required to identify improvements and declines. The left panel of Figure 2.17 shows the economies with the largest gains in ranking between 2010 and 2023, led by Armenia (up 28 positions), Nicaragua (27), Viet Nam (26), Cambodia (24), and Panama (23). The right panel highlights the economies with the largest declines over the same period, including the Bolivarian Republic of Venezuela (recording a loss of 62 positions), the Syrian Arab Republic (29), and the Bahamas (25).

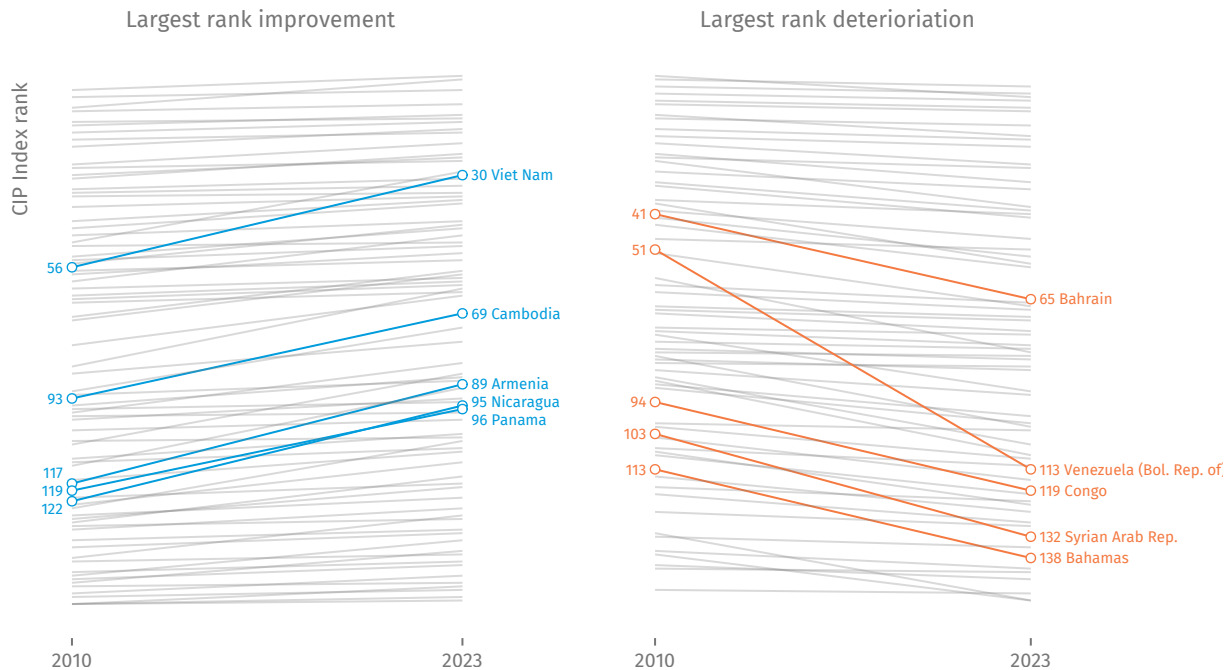
Changes to
industrial competitiveness
are gradual processes requiring
a long-term **statistical perspective**

2.3.2 CIP results by UNIDO country group

As noted above, there is a clear positive correlation between national income level and industrial competitiveness. However, Figure 2.18 shows that this association is not perfect and appears to have weakened in recent years. High-income industrial economies remain the

Economies with the best improvement in the global CIP rankings between 2010 and 2023

1. Armenia
2. Nicaragua
2. Viet Nam
4. Cambodia
5. Panama

Figure 2.17 | Economies with the largest increases and decreases in the CIP ranking between 2010 and 2023

Source: [34]

Note: The graphs show the economies that recorded an improvement (left-side chart) or a deterioration (right-side chart) in the CIP ranking between 2010 and 2023. Each line represents one economy, with the five largest rank improvements and deteriorations highlighted. In case of tied results, more than five economies may be highlighted.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

most competitive group, but as their economic structures become less manufacturing intensive, the gap with other country groups has narrowed. Their average rank in the CIP Index fell from the eighth place in 2010 to the tenth place in 2023. High-income industrial economies and low-income economies were the only country groups in the graph to record a decline in their average global ranking. ⁱ

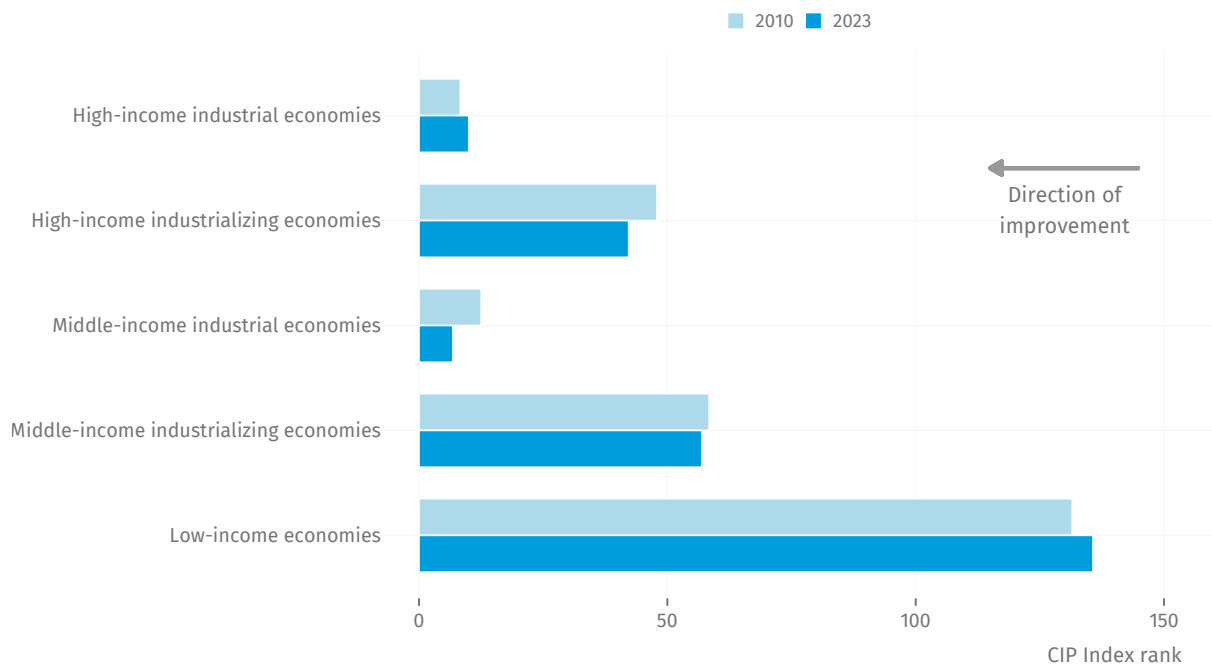
On the other hand, middle-income industrial economies improved markedly, moving on average from the 12th to the seventh position between 2010 and 2023. This was one of the largest improvements among all country groups and it reflects, to a big extent, the strong performance of China. High- and middle-income industrializing economies followed as the third and fourth-ranked groups, with average positions of 42nd and 57th in 2023, respectively. As illustrated in the figure, their industrial performance has gradually strengthened in recent years. Low-income economies remain at the bottom of the ranking, placed on average at the 136th position, with no indication of a reversal in the current trend.

Figure 2.19 presents the average rank of each group disaggregated by CIP dimension, allowing for the identification of relative strengths and weaknesses. High-income industrial economies perform strongly across all three dimensions. However, as their economic structures shift towards activities with a lower manufacturing share, they are

ⁱ The average scores and ranks for country groups and geographical regions presented in this section are weighted averages, using MVA as the weighting variable.

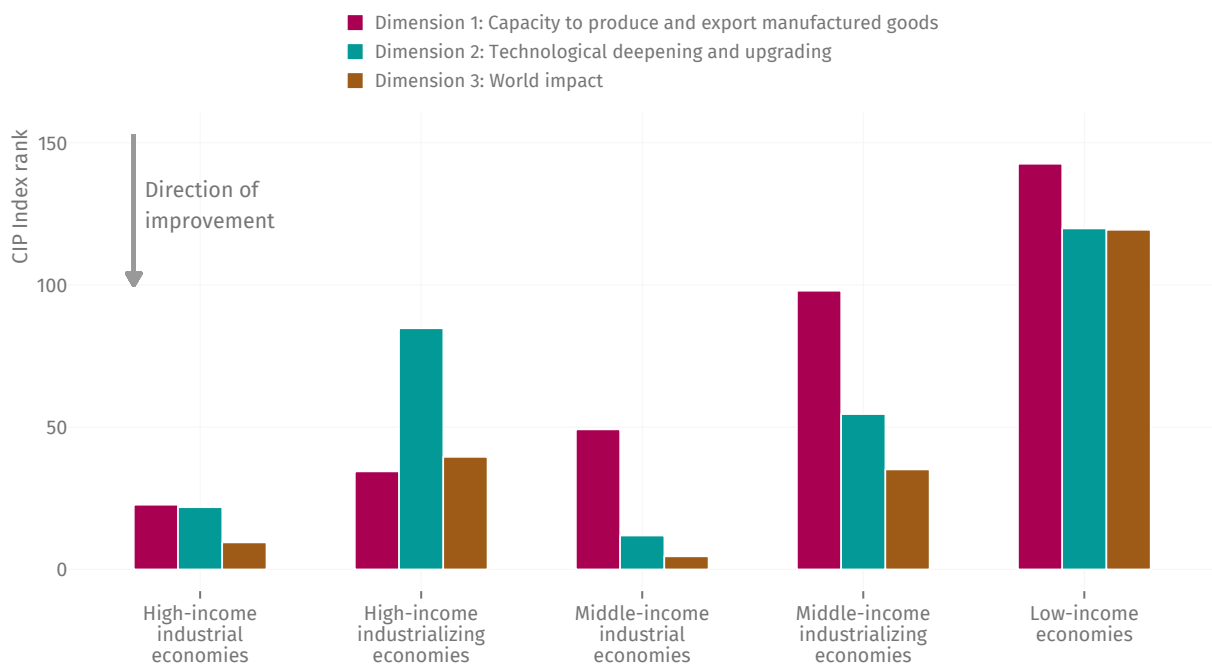
Strong
but weakening
correlation

between national income levels
and industrial competitiveness

Figure 2.18 | Average rank in the CIP Index by country group

Source: [34]

Note: The bars show weighted average ranks of economies with available information, using their MVA as weights. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 2.19 | Average rank in the CIP Index by dimension and country group, 2023

Source: [34]

Note: The bars show weighted average ranks of economies with available information, using their MVA as weights. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

expected to continue declining in the global ranking, particularly in their share of world manufacturing activity (third dimension: *World impact*) and in their relative capacity to produce and export manufactured goods (first dimension). Nonetheless, they are likely to retain strong technological and innovation capabilities (second dimension: *Technological upgrading and deepening*). High-income industrializing economies perform best in the first dimension, although they remain considerably behind industrial economies at the same income level.

Middle-income industrial economies perform strongest in the dimension related to *World impact*. This reflects the presence of several major global manufacturers in this group, including Brazil, China, Mexico, Thailand, and Türkiye. However, this group has also improved significantly in the second dimension, now outperforming even high-income industrial economies. Middle-income industrializing economies perform best in the third dimension but appear less competitive in the first. As shown in the figure, low-income economies face challenges across all competitiveness dimensions.



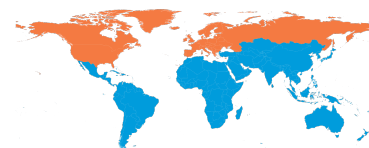
High-income industrial economies
remain at the forefront of industrial competitiveness, but **middle-income industrial economies** are gradually closing the gap

2.3.3 Regional competitiveness

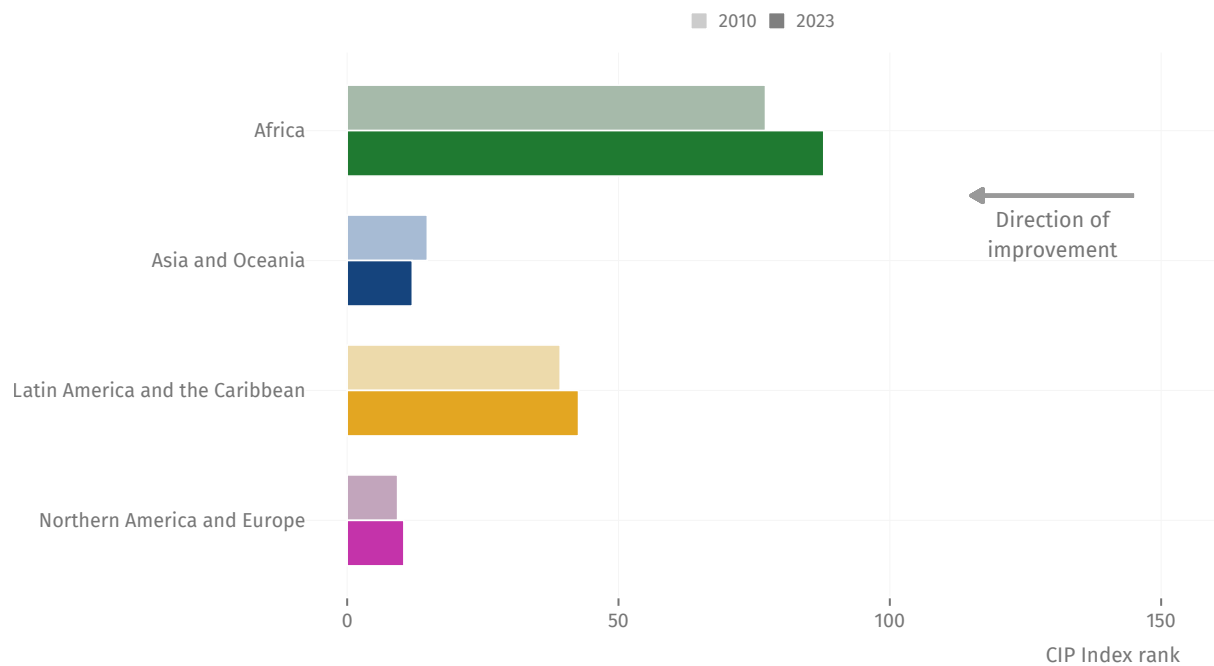
Industrial competitiveness also varies considerably across regions. Figure 2.20 presents the average CIP Index ranking by geographical region. The joint region of Northern America and Europe records the highest level of competitiveness, with an average rank of tenth in 2023. Notably, 14 economies from this region appear among the top 20 of the global CIP ranking.

Close behind, Asia and Oceania achieves an average position of 12th, led by China, Japan, the Republic of Korea, and Taiwan Province of China. Since 2010, this region has climbed three positions on average and now places five economies among the world's top 20. By contrast, Latin America and the Caribbean recorded an average rank of 43rd in 2023, four positions lower than in 2010. Africa shows the steepest decline, dropping nine positions over the same period and currently averaging 88th place.

The results by CIP dimensions are summarized in Figure 2.21. Asia and Oceania leads in the second and third pillars of the Index, closely followed by Northern America and Europe. However, the latter is the world's best performing region in the first dimension. Latin America and the Caribbean is relatively well placed in the *World impact* dimension, driven by the presence of major manufacturers, such as Brazil and Mexico, but remains at a considerable distance from the two leading regions in the other two dimensions. Africa faces the greatest challenges, ranking last among world regions across all three dimensions of industrial competitiveness.

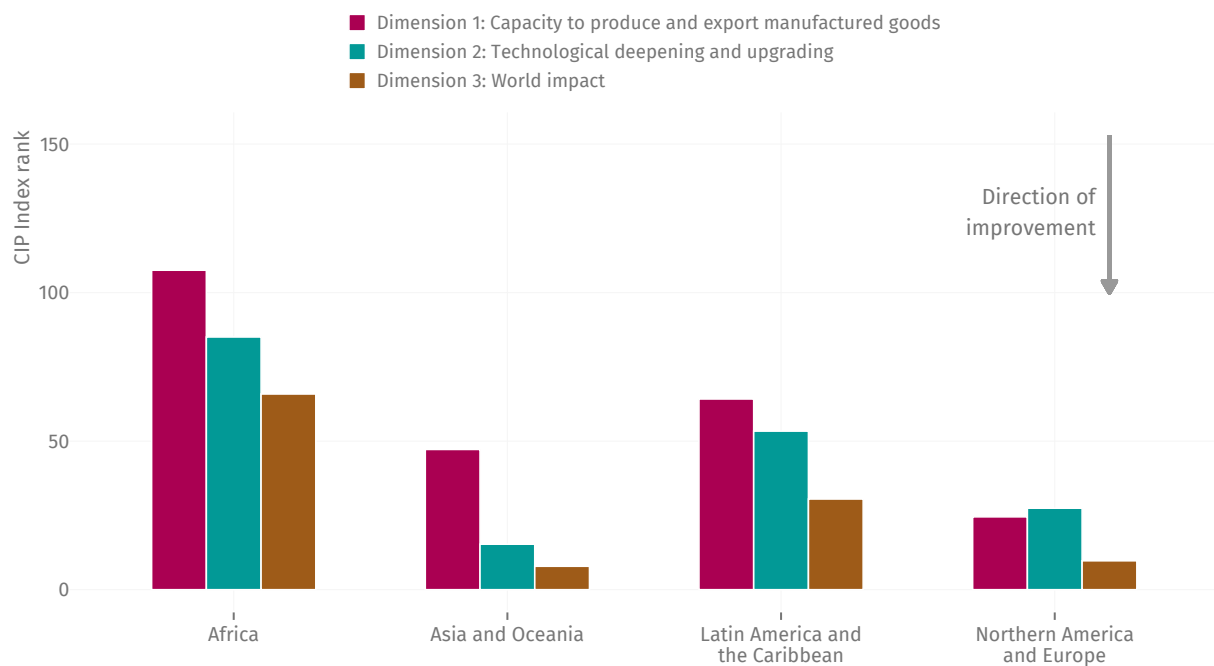


The region of **Northern America and Europe** accounts for **14** of the **top 20** economies in the CIP ranking

Figure 2.20 | Average rank in the CIP Index by region

Source: [34]

Note: The bars show weighted average rank of economies with available information, using their MVA as weights. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 2.21 | Average rank in the CIP Index by dimension and region, 2023

Source: [34]

Note: The bars show weighted average rank of economies with available information, using their MVA as weights. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 2.22 provides a more granular overview of industrial competitiveness, enabling a detailed assessment of current performance and long-term trends across subregions. Within Africa, Southern and Northern Africa are the strongest performers, although both have slipped in the global ranking in recent years, particularly the former subregion. Central Africa is the only African subregion that registered an improvement between 2010 and 2023, though the gain was marginal and from a low starting point.

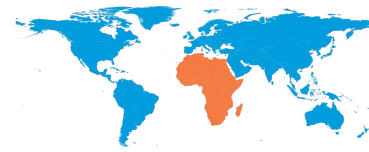
In Asia and Oceania, Eastern Asia is not only the highest-performing subregion within the region but also the strongest worldwide. It currently holds an average rank of third, driven by the robust competitiveness of China, Japan, the Republic of Korea, and Taiwan Province of China. South-Eastern Asia follows, reaching an average of 32nd in 2023 and benefiting from solid results in Malaysia, Singapore, Thailand, Viet Nam, and others. With the exception of Oceania, all subregions in Asia and Oceania improved their ranking between 2010 and 2023.

In Latin America and the Caribbean, two of the three subregions experienced declines in the CIP ranking. South America fell by 10 positions between 2010 and 2023, reaching a rank of 54th, while the Caribbean dropped from 88th to 100th over the same period. Central America is the best-performing subregion, although its progress has stalled; it gained only one position between 2010 and 2023, reaching a rank of 25th. Taken together, these trends suggest that Latin America and the Caribbean may be undergoing a deindustrialization process.

Finally, Northern America and Europe includes several of the world's most advanced manufacturers, a pattern reflected in its subregional outcomes. Western Europe, Northern America, and Northern Europe are among the best-ranked subregions in the world. However, their performance has been gradually declining. Among the five subregions in this group, only Eastern Europe recorded a rank improvement between 2010 and 2023.

Towards a next-generation CIP Index

The CIP Index has served as a global benchmark for measuring countries' industrial competitiveness for over two decades. However, evolving industrial paradigms and the growing need to integrate inclusiveness and sustainability have prompted a comprehensive revision of the Index. UNIDO is currently developing a revamped CIP Index grounded in an enhanced conceptual framework, stronger statistical robustness, and greater policy relevance. The updated Index, scheduled for launch in mid-2026, will provide policymakers with a broader and more actionable tool for evaluating and advancing sustainable industrial competitiveness.



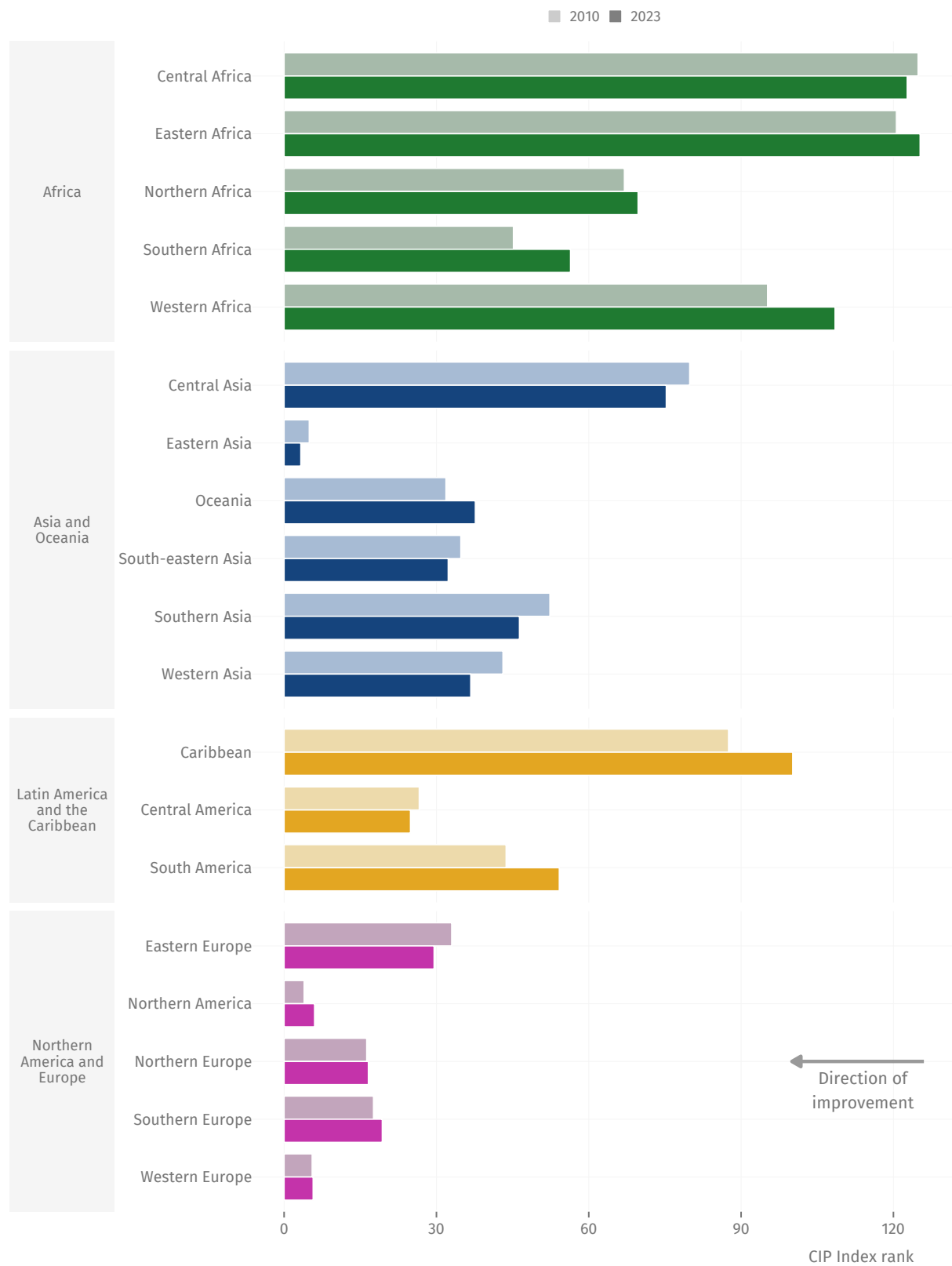
African economies
face challenges across all
competitiveness
dimensions

Top 5 subregions in the global CIP Index 2023

1. Eastern Asia
2. Western Europe
3. Northern America
4. Northern Europe
5. Southern Europe



The region of
**Latin America and
the Caribbean**
is following a
declining trend
in CIP rankings

Figure 2.22 | Average rank in the CIP Index subregion

Source: [34]

Note: The bars show weighted average ranks of economies with available information, using their MVA as weights. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

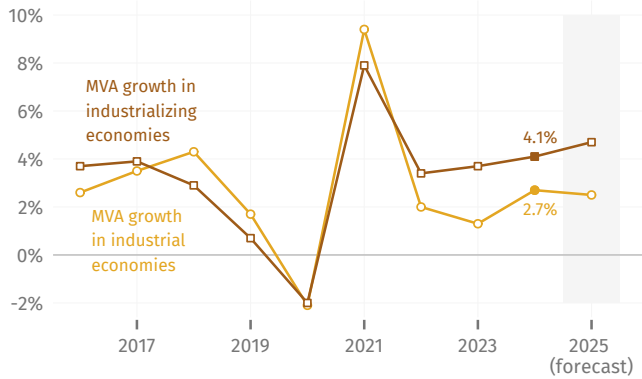


3 Spotlight on manufacturing

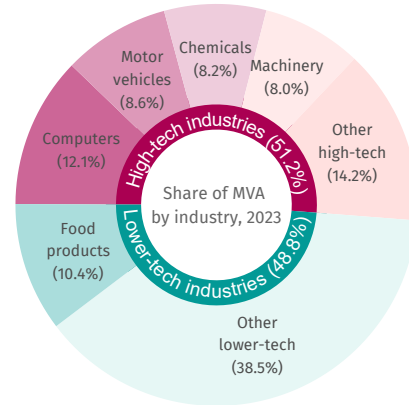
3.1 Current trends and distribution of world manufacturing	42
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3 Spotlight on manufacturing

Key figures

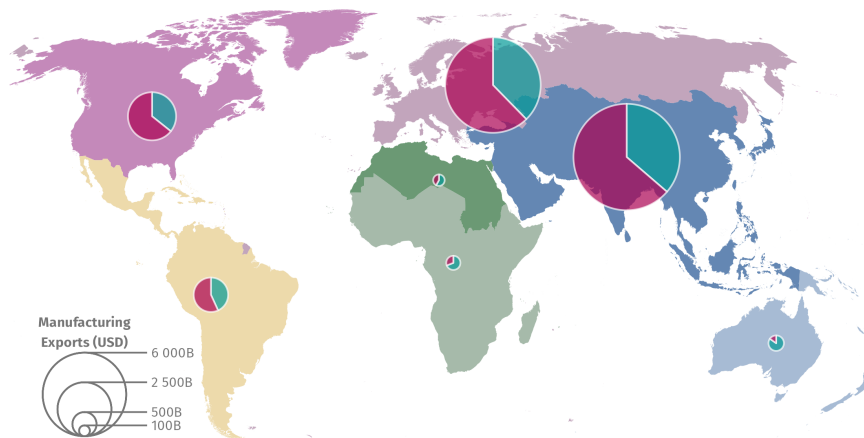


Industrializing economies accounted for only 12.3% of global MVA, yet recorded robust growth of 4.1% in 2024, compared with a slower pace of 2.7% among industrial economies.



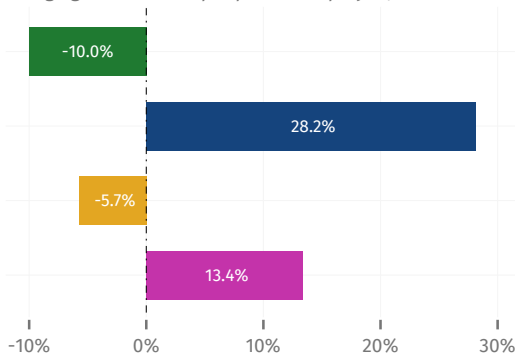
Higher-tech industries accounted for more than half of global MVA, with four MHT sectors among the top five largest industries, led by the production of computers and electronics.

Manufacturing exports by technological intensity ● Higher-tech goods ● Other manufactured goods



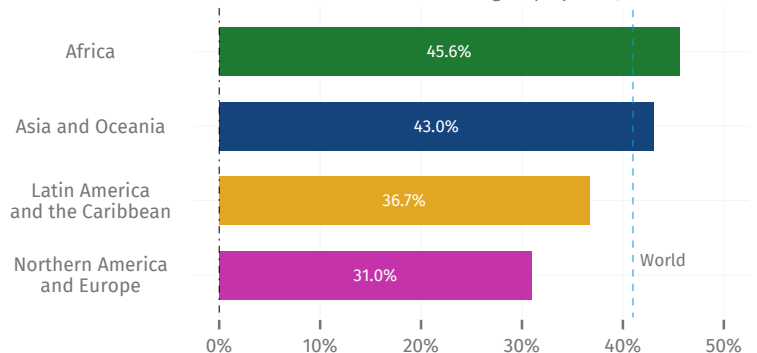
Manufacturing exports are concentrated primarily in Asia, Northern America, and Europe. Together, these regions accounted for nearly 95% of global higher-tech exports in 2024.

Average growth of MVA per person employed, 2015-2023



From 2015 to 2023, MVA per person employed increased by nearly 30% in Asia and Oceania, followed by more moderate growth in Northern America and Europe.

Share of women in manufacturing employment, 2023



Women represented 41.0% of the global manufacturing workforce. Their share reached 45.6% in Africa, but remained below one-third in Northern America and Europe.

3.1 Current trends and distribution of world manufacturing

This chapter examines both the short- and long-term developments shaping the manufacturing sector, with particular attention to their potential influence on the global distribution and evolution of manufacturing activity. It explores how recent shifts in production patterns, technology adoption, and market integration may redefine the geography of industrial output worldwide. A more granular breakdown of production across individual industries is presented in Section 3.2, providing deeper insight into sectoral dynamics. The subsequent four sections expand the analysis to encompass a range of complementary indicators, such as international trade performance, employment trends, productivity levels, gender equality dimensions and environmental implications, offering a broader view of the sector's overall transformation.



3.1.1 Annual manufacturing production

Globally, the manufacturing sector grew by 2.9 per cent in terms of value added in 2024. Although this marks a gradual acceleration compared with the 1.6 per cent recorded in 2023, it continues the pattern of subdued expansion in the sector, remaining below pre-pandemic growth levels. This outcome reflects the impact of moderating global economic growth and persistent headwinds, including armed conflicts, geopolitical instability, and elevated energy and commodity prices. The effects of recent trade restrictions and tariff disputes are not yet captured in these annual figures but will be discussed later in this section based on quarterly IIP data.

Table 3.1 presents the latest MVA growth rates for selected country groups and regions, illustrating that manufacturing performance varied considerably across the world. Among the UNIDO groups by industrial development, middle-income industrial economies recorded the strongest performance in 2024. This was partly driven by China's robust growth rate of 4.9 per cent, while the remaining economies in this group expanded at a combined rate closer to the world average, around 3.2 per cent. This solid performance is expected to continue in 2025.

By contrast, manufacturing in low-income economies has grown at a more modest pace in recent years, reaching only 2.0 per cent in 2024, though it is projected to accelerate significantly in 2025. High-income industrial economies, meanwhile, recovered from the decline in MVA observed in 2023 but registered only 1.2 per cent growth in 2024, the



Table 3.1 | Growth rate of MVA for selected country groups and regions

	2022	2023	2024	2025f
	Per cent			
World	2.2	1.6	2.9	2.8
UNIDO groups by stage of industrial development				
High-income industrial economies	1.4	-0.5	1.2	1.2
High-income industrializing economies	6.5	-1.8	3.9	5.8
Middle-income industrial economies	2.8	3.6	4.6	4.3
Middle-income industrial economies (excl. China)	3.6	0.8	3.2	2.6
China	2.6	4.2	4.9	4.6
Middle-income industrializing economies	3.1	4.4	4.2	4.6
Low-income economies	1.7	1.8	2.0	4.4
Main regions				
Africa	4.4	1.3	2.2	3.2
Asia and Oceania	2.4	3.1	4.0	4.0
Europe	2.1	-1.1	0.4	1.1
Latin America and the Caribbean	2.8	-0.4	1.1	1.2
Northern America	1.1	0.2	2.5	1.1
Other groups				
Emerging industrial economies	2.6	4.7	5.1	4.9
Least developed countries (LDCs)	7.1	5.4	3.6	3.5
Landlocked developing countries (LLDCs)	5.4	4.0	3.9	4.2
Small island developing States (SIDS)	1.8	-2.8	3.2	0.4

Source: [9]

Note: 2025f refers to UNIDO forecasts for the year 2025.

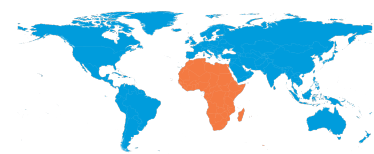
Data used to create this table can be downloaded at the [UNIDO Statistics Portal](#).

lowest among the UNIDO groups, and are expected to maintain this subdued pace into 2025.

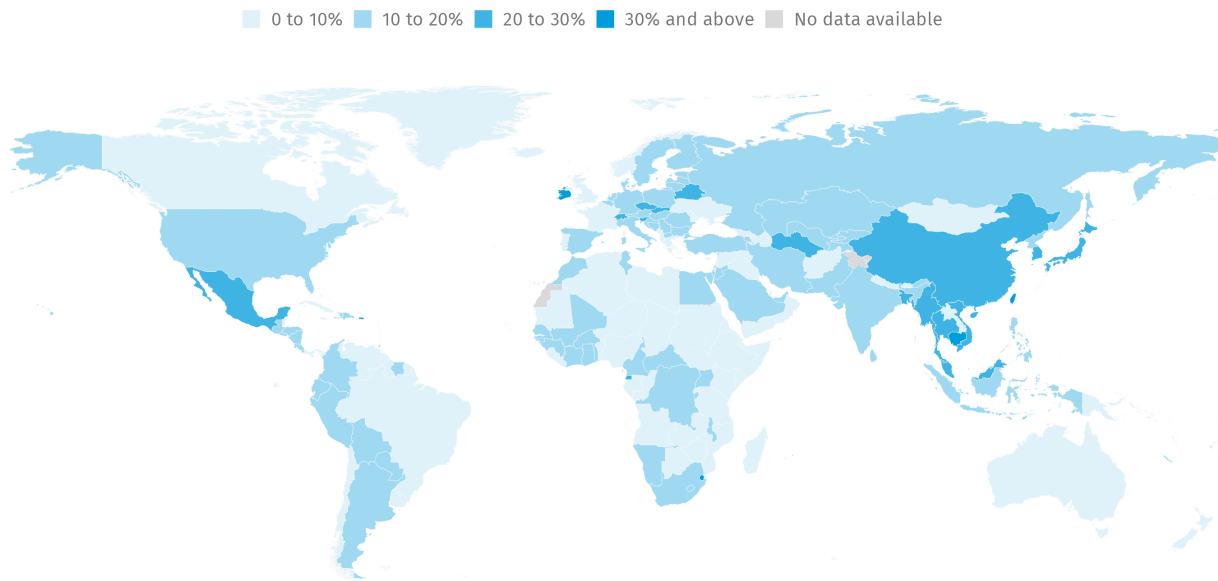
Looking at the major regions, Asia and Oceania recorded the highest MVA growth rate in 2024, at 4.0 per cent, a pace that is expected to be sustained in 2025. This dynamism reflects not only China's strong performance, but also the solid growth of Cambodia, Georgia, India, Indonesia, Kyrgyzstan, Malaysia, Taiwan Province of China, Viet Nam, and other economies in the region. Northern America expanded by 2.5 per cent in 2024, supported by strong growth in the United States of America. Africa grew by 2.2 per cent in the same year, driven by robust performance in Côte d'Ivoire, Ethiopia, Rwanda, the United Republic of Tanzania, Togo, and Uganda, among others. Although this represents a slight deceleration compared with recent years, Africa's manufacturing growth is expected to regain momentum in 2025.

In Latin America and the Caribbean, MVA growth remained subdued in 2024 and is projected to expand at a similar rate in 2025 due to mixed performance among the region's economies. Europe recorded marginal growth of 0.39 per cent, following a contraction in the previous year. While some European countries, such as Poland and Slovakia, achieved solid gains, many others stagnated or experienced declines in manufacturing output.

A positive development, also shown in Table 3.1, is the robust manu-



Africa's MVA
grew moderately in 2024
but is expected to
accelerate
in 2025

Figure 3.1 | MVA as a proportion of GDP by economy, 2024

Source: [9]

Note: The share is based on MVA and GDP in constant 2015 USD.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

facturing growth achieved by the group of least developed countries (LDCs), which is expected to continue into 2025. However, as shown in Section 2.2, this progress may still fall short of meeting the SDG 9 targets by 2030. Manufacturing in landlocked developing countries (LLDCs) has also expanded strongly, while growth among SIDS remains lower and more volatile.

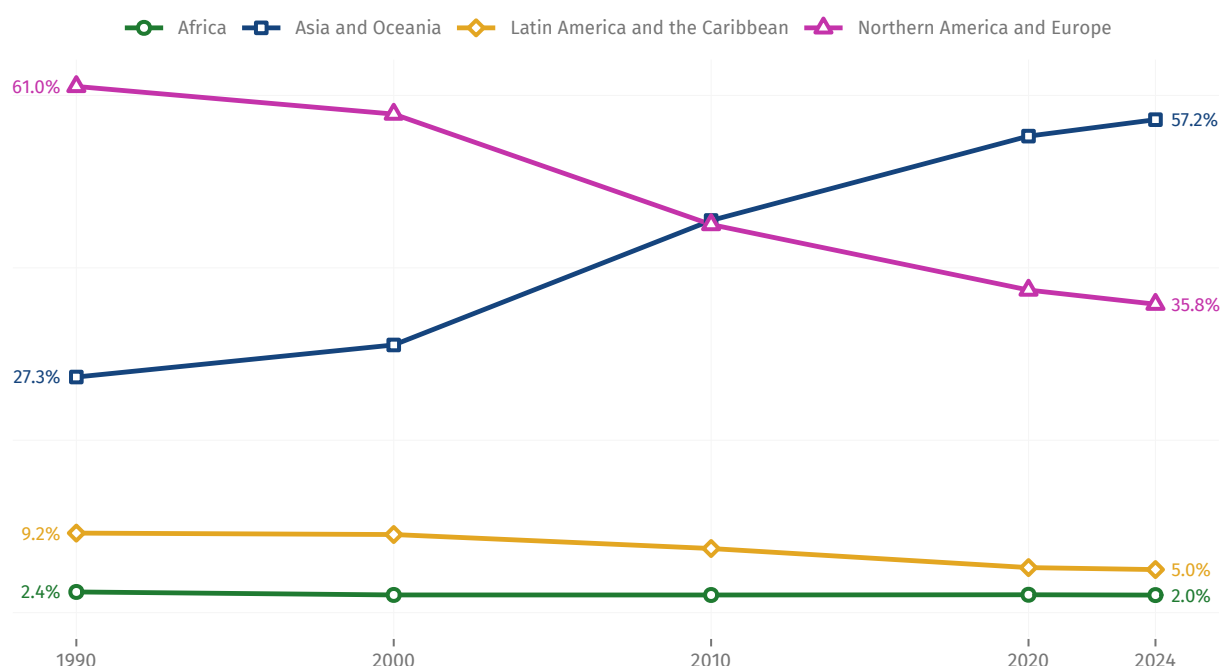
The map in Figure 3.1 illustrates the share of MVA in GDP across the world.ⁱ Most economies with high MVA-to-GDP ratios are located in South-East Asia and East Asia, although several are also found in Europe as well as Latin America and the Caribbean. In contrast, the economic structure of many of African countries remains more concentrated in other sectors, indicating that their structural transformation is still under way.

The global distribution of manufacturing activity is undergoing a significant transformation, as traditional industrial centers in the Global North increasingly give way to complex production networks distributed across the world, most prominently in Asia. This global rebalancing is illustrated in Figure 3.2, which shows the distribution of world MVA across four broad regions at five points in time: 1990, 2000, 2010, 2020, and 2024.

In 1990, Northern America and Europe accounted for 61.0 per cent of global MVA, while Asia and Oceania contributed 27.3 per cent. Three and a half decades later, the positions of these two regions have

Least developed countries
have recorded
robust growth
in recent years

ⁱ This indicator, corresponding to SDG indicator 9.2.1, measures the contribution of manufacturing to a country's economy. It serves as a key marker of industrial development and enables cross-country comparisons.

Figure 3.2 | MVA as a proportion of world MVA by region

Source: [9]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

effectively reversed. By 2024, the share of Northern America and Europe had declined to 35.8 per cent, while Asia and Oceania's share had risen to 57.2 per cent. However, this shift appears to be stabilizing, with the regional distribution approaching a new equilibrium.

This transformation is driven by the remarkable rise of China as the world's leading manufacturing powerhouse. China's share of global manufacturing production increased from 2.8 per cent in 1990 to 32.0 per cent in 2024. This exceeds the combined share of the next four largest manufacturing economies: United States of America (15.0 per cent), Japan (6.3 per cent), Germany (4.6 per cent), and the Republic of Korea (3.3 per cent). According to UNIDO projections, China's share will expand further in 2025, surpassing the combined share of the next five manufacturing economies.

By contrast, the weight of Latin America and the Caribbean in global MVA declined from 9.2 per cent in 1990 to 5.0 per cent in 2024, reflecting a continued trend of deindustrialization in the region [10]. Africa accounted for the smallest share of global manufacturing among the main regions, at only 2.0 per cent, a level that has remained virtually unchanged over the past three decades.

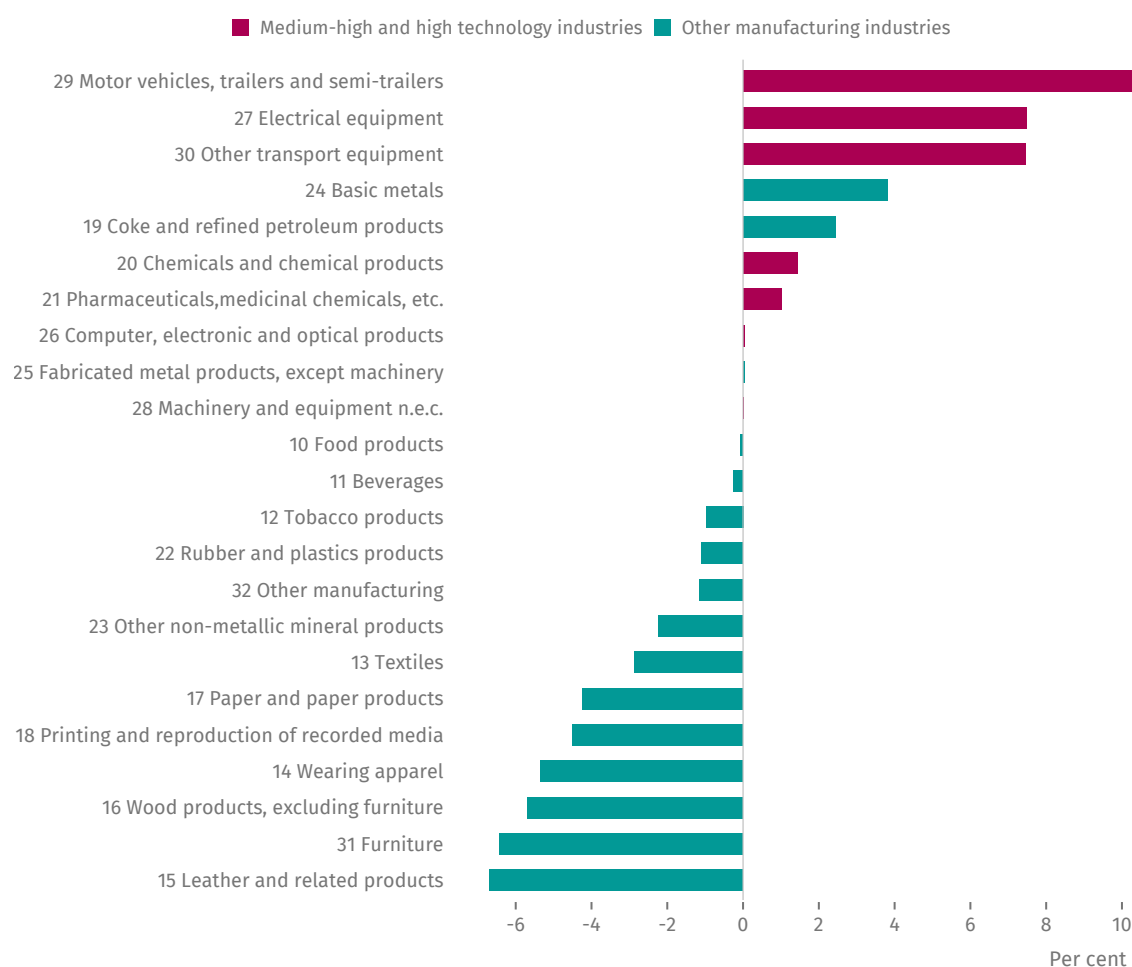
Disaggregated data by manufacturing industry at the ISIC Rev. 4 division level are presented in Figure 3.3, allowing for a more detailed analysis of recent trends. In 2023, the latest year with relatively comprehensive data coverage, most of the industries that performed best were classified as MHT sectors. The three industries with the highest

Economies with the largest gains in global MVA share between 1990 and 2024

1. China +29.2%
2. India +2.1%
3. Republic of Korea +1.8%
4. China, Taiwan Province +1.0%
5. Ireland +0.8%

Economies with the largest declines in global MVA share between 1990 and 2024

1. Japan -7.2%
2. United States -7.1%
3. Germany -5.0%
4. Italy -3.0%
5. Russian Federation -2.1%

Figure 3.3 | Growth rate of global value added by manufacturing industry, 2023

Source: [35; 36]

Note: Manufacturing industries correspond to ISIC Rev. 4 divisions.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

growth rates in value added were motor vehicles, electrical equipment, and other transport equipment. Notably, none of the MHT industries recorded negative growth.

On the other hand, with the exception of basic metals and petroleum-related manufacturing, nearly all industries relying on lower technologies either stagnated or experienced declines in 2023. This shift towards higher-technology industries represents another megatrend in global manufacturing [37, pp. 105–106], underscoring the growing importance of transitioning to sectors with greater innovation and technological content, which have consistently demonstrated stronger performance. Achieving this requires supportive policies to foster technological adoption, including financing instruments and mechanisms for technology transfer [10].



Technological progress

is a pathway towards
stronger and more resilient
manufacturing growth

3.1.2 Quarterly manufacturing production

In addition to the yearly statistics presented above, the remainder of this section draws on seasonally-adjusted IIPs on a quarterly basis to provide a more up-to-date view of trends in the manufacturing sector. The IIP is a key indicator for monitoring economic activity in a timely and consistent manner, given its broad availability across countries. The main features of this index are summarized in Annex C.

Figure 3.4 traces the long-term evolution of global manufacturing production. Beyond major setbacks during the 2008–2009 financial crisis and the COVID-19 pandemic in 2020–2021, the sector experienced only temporary slowdowns followed by strong recoveries, underscoring its overall resilience. The latest data indicate a solid quarter-on-quarter increase of 1.0 per cent in the second quarter of 2025, maintaining the positive quarterly growth trend since mid-2022. This steady improvement, particularly in recent periods, can be partly explained by increased defence spending and military equipment production across major economies. Overall, global manufacturing has demonstrated remarkable momentum despite tariff escalation, inflation, supply chain disruptions, and geopolitical instability.

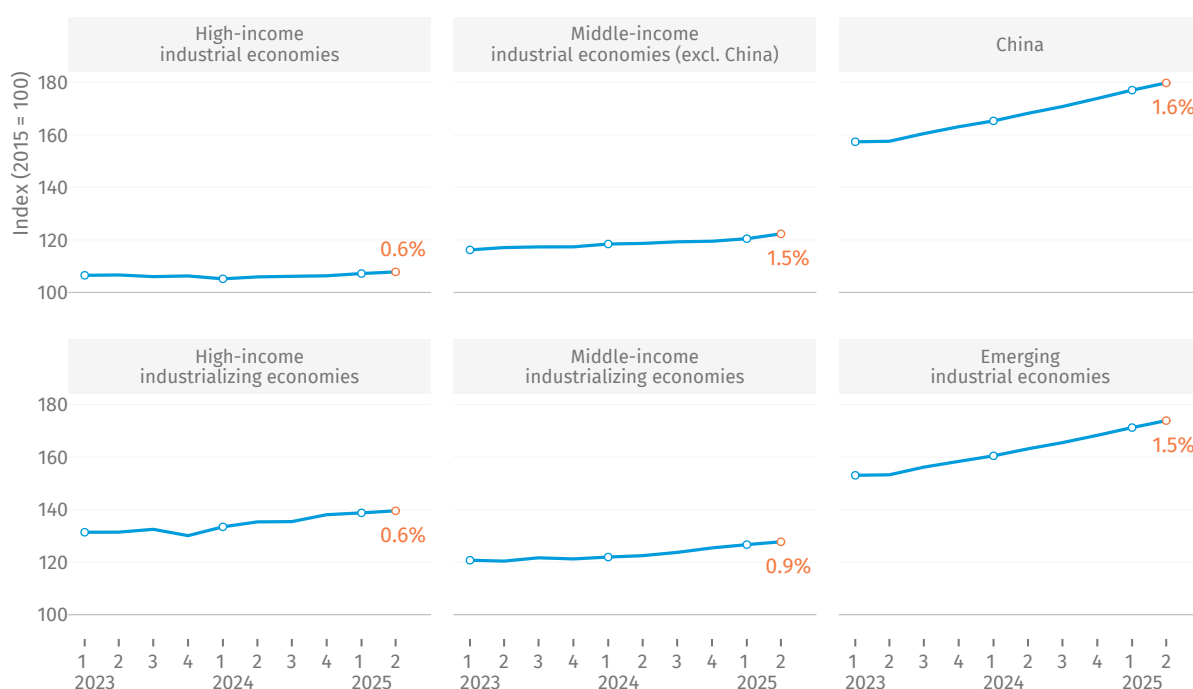


Figure 3.4 | Index of global manufacturing (2015 = 100)



Source: [19]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.5 | Index of manufacturing production by country group

Source: [19]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Only country groups with sufficient data coverage are included.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Trends disaggregated by country group and region

The most recent trends disaggregated by UNIDO country group are presented in Figure 3.5. Most groups have registered output increases since early 2023, though with varying magnitudes. High-income industrial economies recorded the lowest growth among the groups presented, both compared to the first quarter of 2023 and also when comparing to the previous quarter. Similarly, high-income industrializing economies achieved quarterly growth of 0.59 per cent, but moderately increased its production since 2023. In contrast, China, the world's largest manufacturer, has maintained steady quarterly growth rates close to or above 1.5 per cent since mid-2023, with only minor fluctuations.

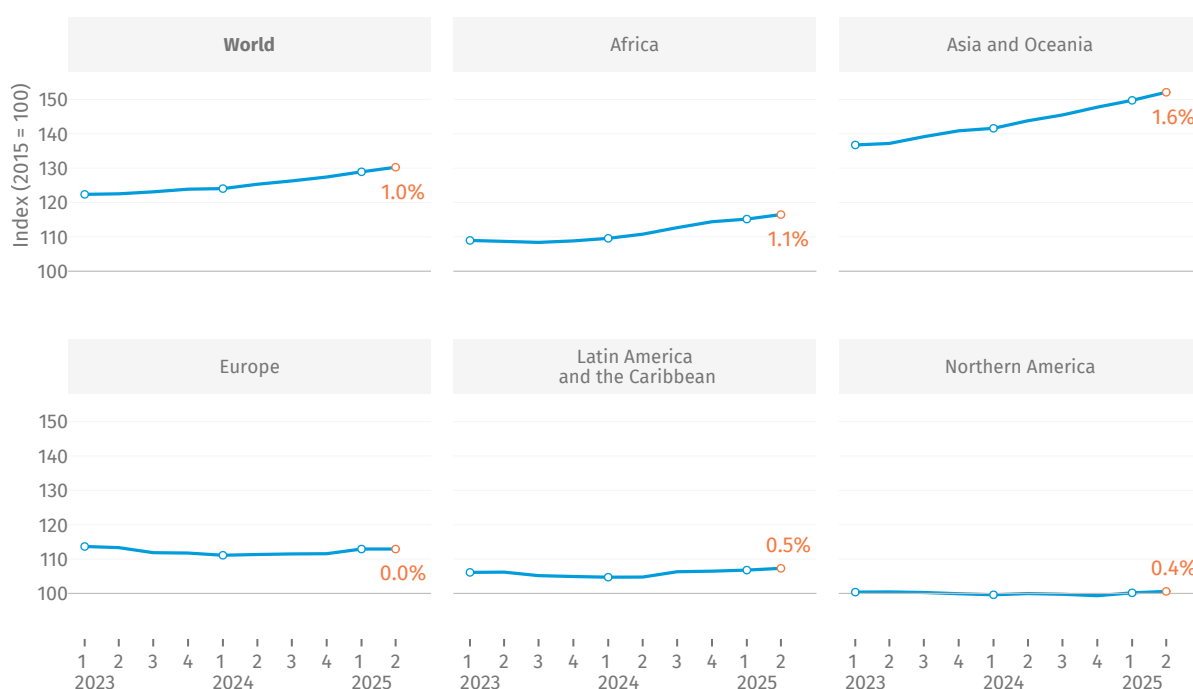
The figure also shows that both subgroups of middle-income economies have recorded growth of similar magnitude since early 2023. Middle-income industrial economies (excl. China) registered a notable quarterly growth rate of 1.5 per cent in the second quarter of 2025, while middle-income industrializing economies, the largest UNIDO group by number of countries, achieved slightly lower but still solid growth of 0.86 per cent.

The graph also includes the group of emerging industrial economies (EIEs)ⁱ, which encompass some established industrial economies as

Top 10 manufacturers in 2024 and their quarter-over-quarter growth in Q2 2025

1. China (1.6%)
2. United States (0.6%)
3. Japan (0.0%)
4. Germany (-1.3%)
5. Republic of Korea (-0.2%)
6. India (0.8%)
7. Italy (0.4%)
8. United Kingdom (0.3%)
9. Mexico (1.1%)
10. France (0.7%)

ⁱ Emerging industrial economies (EIEs) is defined as the group of low- and middle-income economies with the most dynamic manufacturing sector. For more details on the composition of this group, see Annex E.5.

Figure 3.6 | Index of manufacturing production by region

Source: [19]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

well as other industrializing economies that have recently experienced strong, resilient manufacturing progress. The manufacturing output of this group has demonstrated solid performance, with a quarterly growth of 1.5 per cent, significantly exceeding the world average of 1.0 per cent. It is worth highlighting that the number of low-income economies consistently reporting manufacturing production data has declined in recent periods. Due to inadequate data coverage, this group cannot be included in the analysis.

A regional perspective is shown in Figure 3.6. The latest results in global manufacturing are distributed in a relatively uneven manner. Asia and Oceania, responsible for more than half of global manufacturing output, achieved modest growth of 1.6 per cent, followed by Africa with quarterly growth of 1.1 per cent. Considering the trajectories since 2023, these two regions maintained continuous growth, while the manufacturing sectors of other regions mainly stagnated. Latin America and the Caribbean, Northern America as well as Europe also recorded subdued growth or stagnation in the past quarter, at 0.48 per cent, 0.42 per cent and 0.00 per cent respectively.

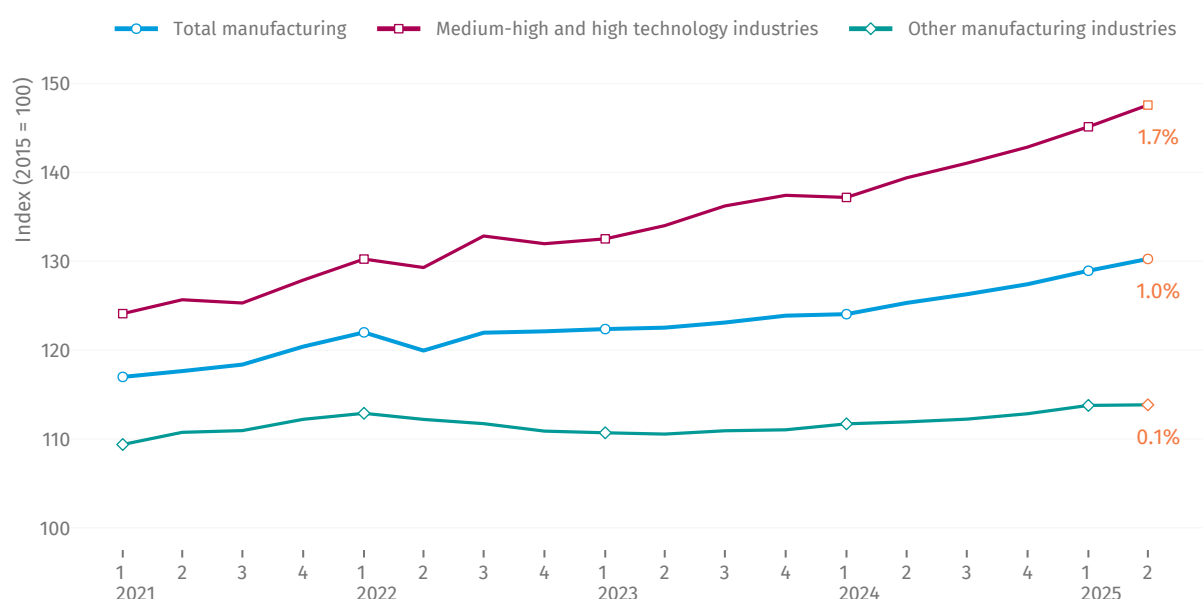


Asia and Oceania
maintained the most
dynamic path
among world regions

Developments by manufacturing sector

The current trajectory of production according to industries aggregated by technological intensity is shown in Figure 3.7. ⁱ A robust 1.7 per cent increase in higher-technology industries during the second quarter of 2025 outweighed the slower growth observed in other

ⁱ The list of medium-high and high technology industries is included in Annex E.2.

Figure 3.7 | Index of manufacturing production by technological intensity

Source: [19]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

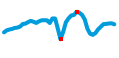
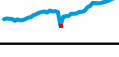
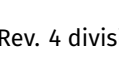
manufacturing sectors. This was driven by the strong performance of computers and electronics, electrical equipment, motor vehicles, and other transport equipment, which achieved quarter-over-quarter growth rates above 1 per cent. Low-technology industries experienced a stalled production level in the latest quarter, and have barely expanded since early 2021.

Table 3.2 presents recent developments by industry, illustrating trends since 2015 and quarterly growth rates for the four most recent quarters. The data reveal substantial variation across industries. The manufacturing of computers and electronics has sustained robust growth, achieving the highest quarter-over-quarter growth rate in the second quarter of 2025 and nearly doubling output since early 2015. Quarterly performance among other higher-technology industries varied considerably. Production of chemicals, electrical equipment, motor vehicles, and other transport equipment recorded significant increases, while pharmaceuticals and machinery sectors exhibited more subdued growth. Overall, most MHT industries demonstrated solid growth with only minor setbacks, underscoring the significance of innovation and technological development. Notably, all seven higher-technology industries ranked within the top ten for cumulative growth since 2015. Conversely, lower-technology industries, including furniture, non-metallic minerals, textiles, wearing apparel, and wood products, experienced predominantly stagnant or declining output levels in recent years, in addition to significant volatility.

Top 10 manufacturing industries by cumulative growth since Q1 2015

1. Computers and electronics (95.5%)
2. Electrical equipment (74.4%)
3. Pharmaceuticals (54.8%)
4. Motor vehicles (28.8%)
5. Machinery (27.3%)
6. Basic metals (27.1%)
7. Chemicals (27.1%)
8. Other transport equipment (23.6%)
9. Food products (22.9%)
10. Beverages (20.3%)

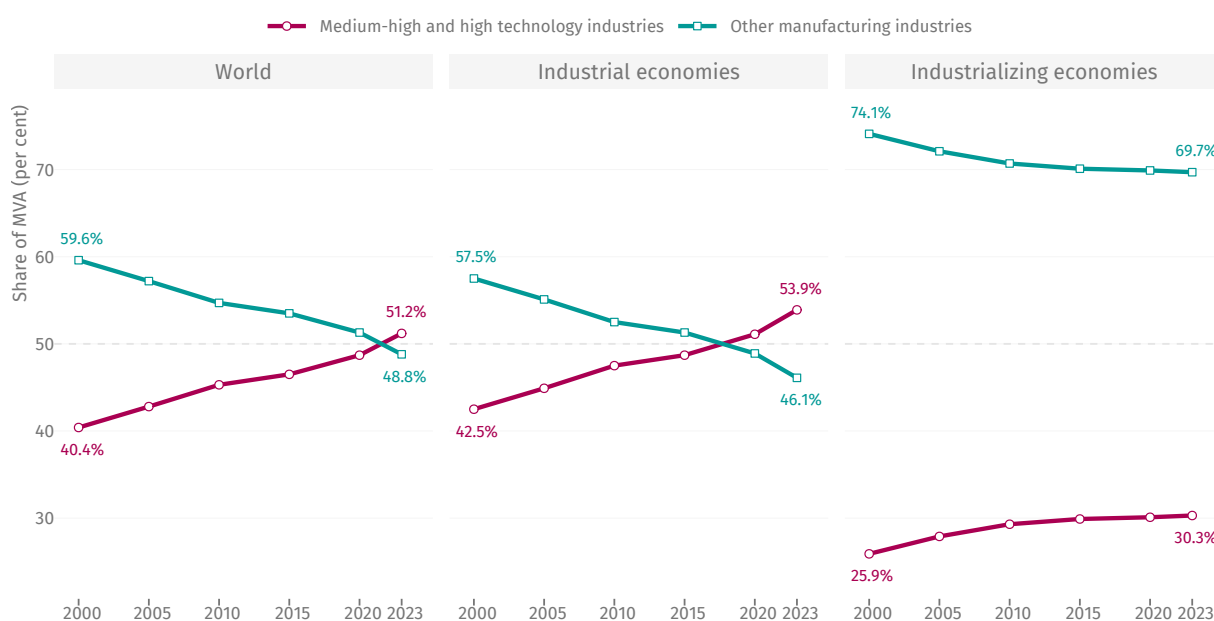
Table 3.2 | Quarter-over-quarter growth rates of global production by manufacturing industry

Manufacturing industry		Q3 2024	Q4 2024	Q1 2025	Q2 2025
		Per cent			
10 Food products		0.7	0.9	1.2	0.1
11 Beverages		-0.2	-0.3	1.2	-0.7
12 Tobacco products		1.2	-0.9	0.1	1.0
13 Textiles		0.6	0.5	0.7	-0.6
14 Wearing apparel		0.8	0.3	-0.3	-1.2
15 Leather and related products		0.4	-0.5	0.4	-0.4
16 Wood products, excluding furniture		0.3	0.6	1.3	-1.0
17 Paper and paper products		0.0	0.1	0.0	-0.3
18 Printing and reproduction of recorded media		-0.6	-1.3	1.9	-0.8
19 Coke and refined petroleum products		0.1	0.8	-0.2	1.2
20 Chemicals and chemical products		0.6	1.1	1.1	0.7
21 Pharmaceuticals, medicinal chemicals, etc.		2.2	-1.3	1.3	-0.2
22 Rubber and plastics products		0.1	0.2	1.1	0.2
23 Other non-metallic mineral products		-0.8	0.9	0.6	-0.7
24 Basic metals		0.4	1.1	1.1	1.3
25 Fabricated metal products, except machinery		0.6	0.5	0.6	0.2
26 Computer, electronic and optical products		2.1	2.5	2.3	3.5
27 Electrical equipment		1.6	2.5	2.3	1.1
28 Machinery and equipment n.e.c.		0.2	1.3	0.3	0.7
29 Motor vehicles, trailers and semi-trailers		0.6	0.9	0.3	1.7
30 Other transport equipment		0.2	-0.5	6.1	3.3
31 Furniture		-0.5	-0.6	0.6	-1.5
32 Other manufacturing		0.0	0.5	0.8	-0.3
33 Repair and installation of machinery		0.7	0.9	1.7	0.6

Source: [19]

Note: Manufacturing industries correspond to ISIC Rev. 4 divisions. Line charts show indices (2015=100) by manufacturing industry from Q1 2015 to Q2 2025, with the red points indicating minimum/maximum values.

Data used to create this table can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.8 | MVA composition by industries according to technological intensity

Source: [9; 19; 35; 36]

Note: The classification of ISIC Rev. 4 industries by categories of technological intensity are described in Annex E.2.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

3.2 Structure and transformation of manufacturing industries

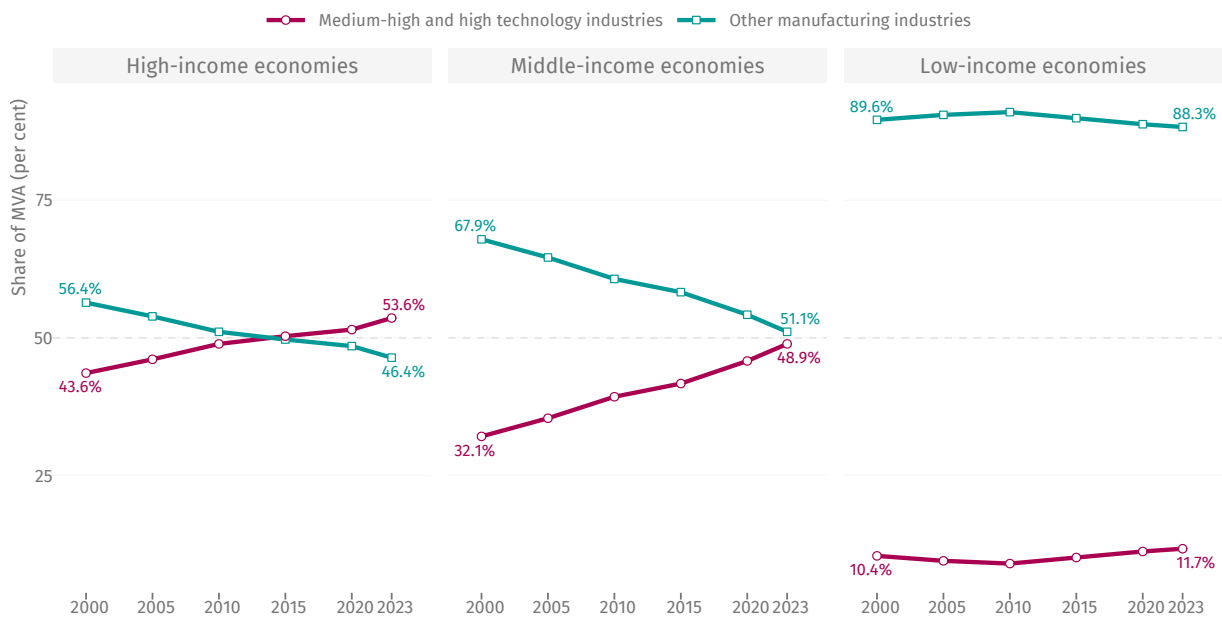
This section focuses on the contribution of different industrial activities to total manufacturing. It presents industry-level information according to technological intensity and more detailed ISIC Rev. 4 levels. It also highlights evolving trends among leading manufacturers within selected country groups, alongside the transformations observed over the past two decades.

The growing importance of MHT industries,ⁱ already highlighted in previous editions of this *Yearbook*, persisted in 2023, according to the most recent UNIDO estimations at constant prices. The trend observed in industrial economies has played a pivotal role in reinforcing the global dominance of MHT sectors. In these economies, MHT industries have consistently outpaced other manufacturing sectors in terms of their share in MVA, as illustrated in Figure 3.8. Their share rose from 42.5 per cent in 2000 to 53.9 per cent in 2023, corresponding to a compound average annual growth rate of approximately 1.0 percentage point over the full period. Notably, the pace of growth has intensified in recent years: between 2020 and 2023, the share increased from 51.1 per cent to 53.9 per cent, implying an average annual growth rate of 1.8 percentage points, nearly double the long-term average. This acceleration highlights the faster recovery of MHT industries following the COVID-19 pandemic, benefitting from resilient global demand in health-related products, strong investment in innovation, and the

Methodological explanation

The figures presented throughout Section 3.2 are based on value added, compiled using base weights for the reference year 2015. These are mainly calculated using data from UNIDO's Industrial Statistics Databases (INDSTAT). However, value-added figures are only available in current prices. The annual IIP is used to present trends over time, free from currency and price variations.

ⁱ The composition of these categories in the sense of the ISIC classification of economic activities is presented in the Annex E.2.

Figure 3.9 | MVA composition by industries according to technological intensity by income groups

Source: [9; 19; 35; 36]

Note: The classification of ISIC Rev. 4 industries by categories of technological intensity are described in Annex E.2. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

deepening structural shift towards technology-intensive production, driven by digitalization and the expanding role of automation. Industries such as pharmaceuticals, electronics and machinery have increasingly displaced traditional manufacturing activities and reinforced their upward trajectory within industrial countries.

In comparison, the picture for industrializing economies remains markedly different. Despite some growth in MHT industries reflecting an increase in the proportion from 25.9 per cent in 2000 to 29.3 per cent in 2010, their share has remained relatively modest and stagnant from 2011 onward. In 2023, other manufacturing industries still dominate, accounting for nearly 70 per cent of MVA. This reflects earlier stages of structural transformation, with continued reliance on labour-intensive and resource-based manufacturing. Factors such as limited access to advanced technologies, lower R&D capacity, and dependence on global value chains for intermediate goods may be constraining the expansion of MHT sectors. This divergence is not unexpected, given that UNIDO's classification of countries as 'industrial' or 'industrializing' is partly based on the share of MHT industries in their manufacturing structure [38]. In essence, economies are considered 'industrial' because they have successfully transitioned towards more technology-intensive production. When analysed by income group, the pattern remains consistent and reinforces the structural narrative. High-income economies had already achieved MHT predominance around 2015, with MHT industries accounting for 53.6 per cent of MVA in 2023. Middle-income economies are converging rapidly,

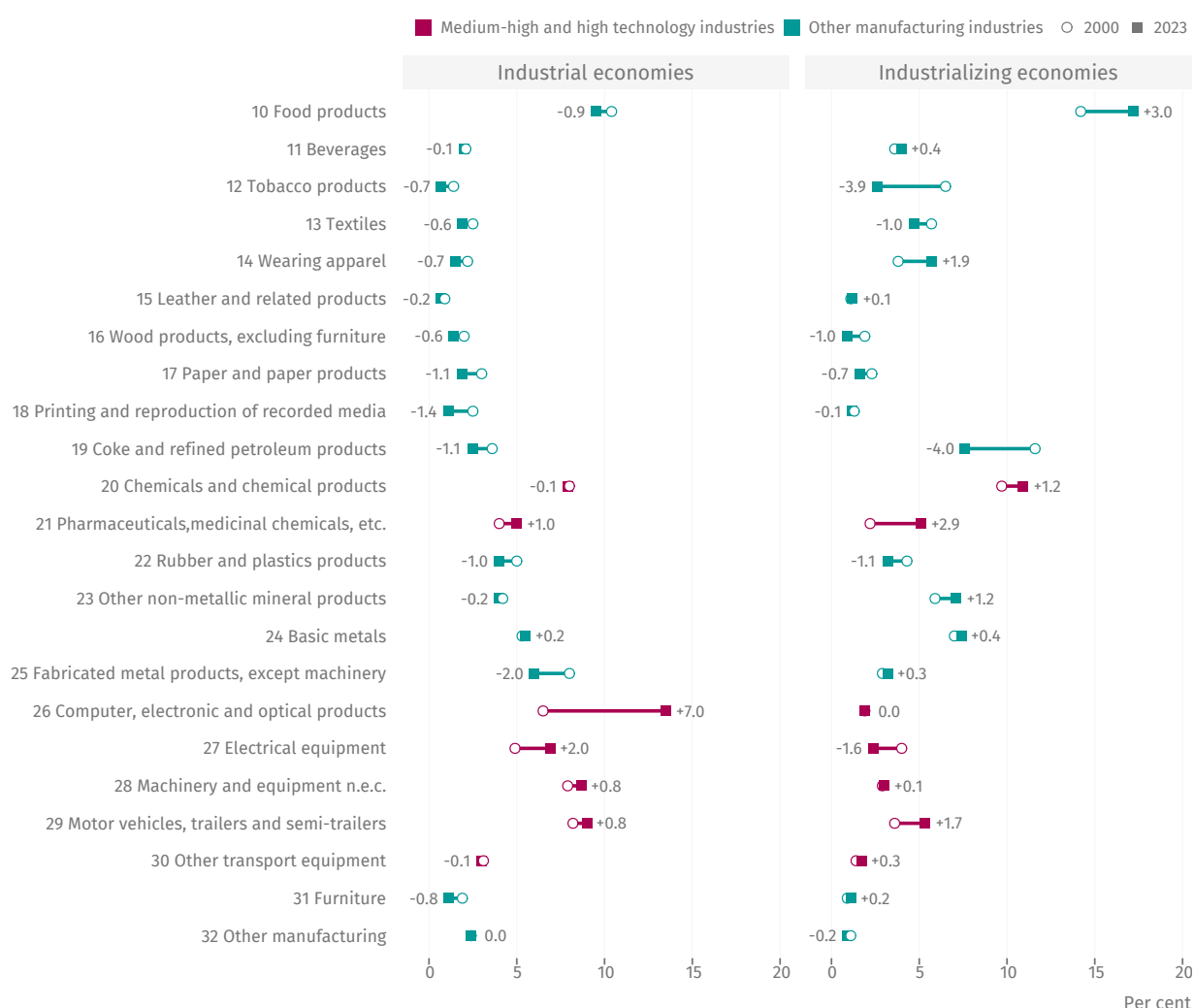


In 2023, more than **half** of global MVA originated in **higher-technology** industries

High-income economies
already achieved predominance of higher-tech industries

The share in **middle-income economies** is converging rapidly to 50%

Low-income economies
remain significantly behind

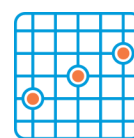
Figure 3.10 | Share of manufacturing industries in MVA by country groups

Source: [9; 19; 35; 36]

Note: The chart indicates the evolution of each industry's share in MVA between 2000 and 2023, for economies grouped by their stage of industrial development. The labels next to the points show the change in the share over the same period. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

with the share rising from 32.1 per cent in 2000 to 48.9 per cent in 2023. However, low-income economies remain significantly behind, with MHT industries contributing only 11.7 per cent in 2023, while other manufacturing industries still comprise nearly 90 per cent of their manufacturing base.

In development economics, the link between industrial structure and income level is well-established and has recently been re-affirmed with empirical evidence in the *SDG 9 Progress Report 2025* [39], which shows a strong correlation between income level and industrial performance, with high-income countries consistently exhibiting higher shares of MHT industries and stronger innovation capacity. The report also highlights that technological upgrading remains a challenge for many economies, reinforcing the need for targeted industrial policies to foster diversification and economic advancement. Building upon



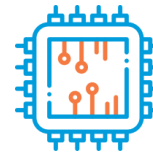
The **strong correlation** between **income level** and **industrial performance** has been reaffirmed by the latest *SDG 9 Progress Report*

the earlier assessment, the refined ISIC Rev. 4 division-level data reveal more pronounced structural contrasts between industrial and industrializing economies.

In industrial economies, the predominance of MHT industries is largely driven by sustained growth in computer, electronic, and optical products (ISIC Rev. 4 division 26), which increased its share by 7.0 percentage points, from 6.5 per cent in 2000 to 13.5 per cent in 2023. Similarly, electrical equipment (ISIC Rev. 4 division 27) and machinery and equipment (ISIC Rev. 4 division 28) recorded notable expansions. Electrical equipment, in particular, grew by 2.0 percentage points and is expected to keep expanding as a key industry for the green transition and electrified mobility. Pharmaceuticals (ISIC Rev. 4 division 21) and motor vehicles (ISIC Rev. 4 division 29) also posted significant gains, reinforcing the shift toward innovation-led manufacturing. By contrast, sectors such as textiles and wearing apparel (ISIC Rev. 4 divisions 13 and 14) as well as fabricated metal products (ISIC Rev. 4 division 25) experienced declining shares, signalling a continued transition away from low-technology, labour-intensive production.

While in both industrial and industrializing economies, the manufacture of food products (ISIC Rev. 4 division 10) remains at prominent levels ensuring food security, the structural composition of the latter group of countries is characterized by a relatively limited share of MHT in total MVA. Although there has been modest growth in some MHT divisions—pharmaceuticals (ISIC Rev. 4 division 21) rose from 2.5 per cent to 5.1 per cent, motor vehicles (ISIC Rev. 4 division 29) from 4.1 per cent to 5.3 per cent and chemicals (ISIC Rev. 4 division 20) from 9.7 to 10.9 per cent—the majority of value added is still derived from other manufacturing industries. Besides food products (ISIC Rev. 4 division 10), which increased from 14.8 per cent to 17.2 per cent, coke and refined petroleum products (ISIC Rev. 4 division 19), remains a significant contributor, despite a slightly declining share. Labour-intensive industries such as wearing apparel (ISIC Rev. 4 division 14) and textiles (ISIC Rev. 4 division 13) also continue to play a prominent role, alongside basic metals (ISIC Rev. 4 division 24) and non-metallic mineral products (ISIC Rev. 4 division 23).

To further illustrate the evolving landscape of manufacturing, it is essential to examine the regional share estimates of MHT industries from 2000 to 2023 in more detail. During this period, there have been notable shifts in the distribution of the MHT share in MVA across different regions. In Northern America and Europe, the region's share expansion from 41.2 per cent in 2000 to 50.8 per cent in 2023 was due to a broad-based development across many countries. This is shown in Figure 3.11 by the simultaneous upward shift in the median value of countries within this region. By contrast, in Asia and Oceania, although the regional share increased, the median share remained relatively stable. This indicates that the regional growth was



Electronics nearly **doubled** its share in MVA since 2000, reflecting global demand for digital technologies and automation.

Top 5 industries and their share in MVA in 2023

World

- ▶ Computers and electronics (12.1%)
- ▶ Food products (10.4%)
- ▶ Motor vehicles (8.6%)
- ▶ Chemicals (8.2%)
- ▶ Machinery (8.0%)

Industrial economies

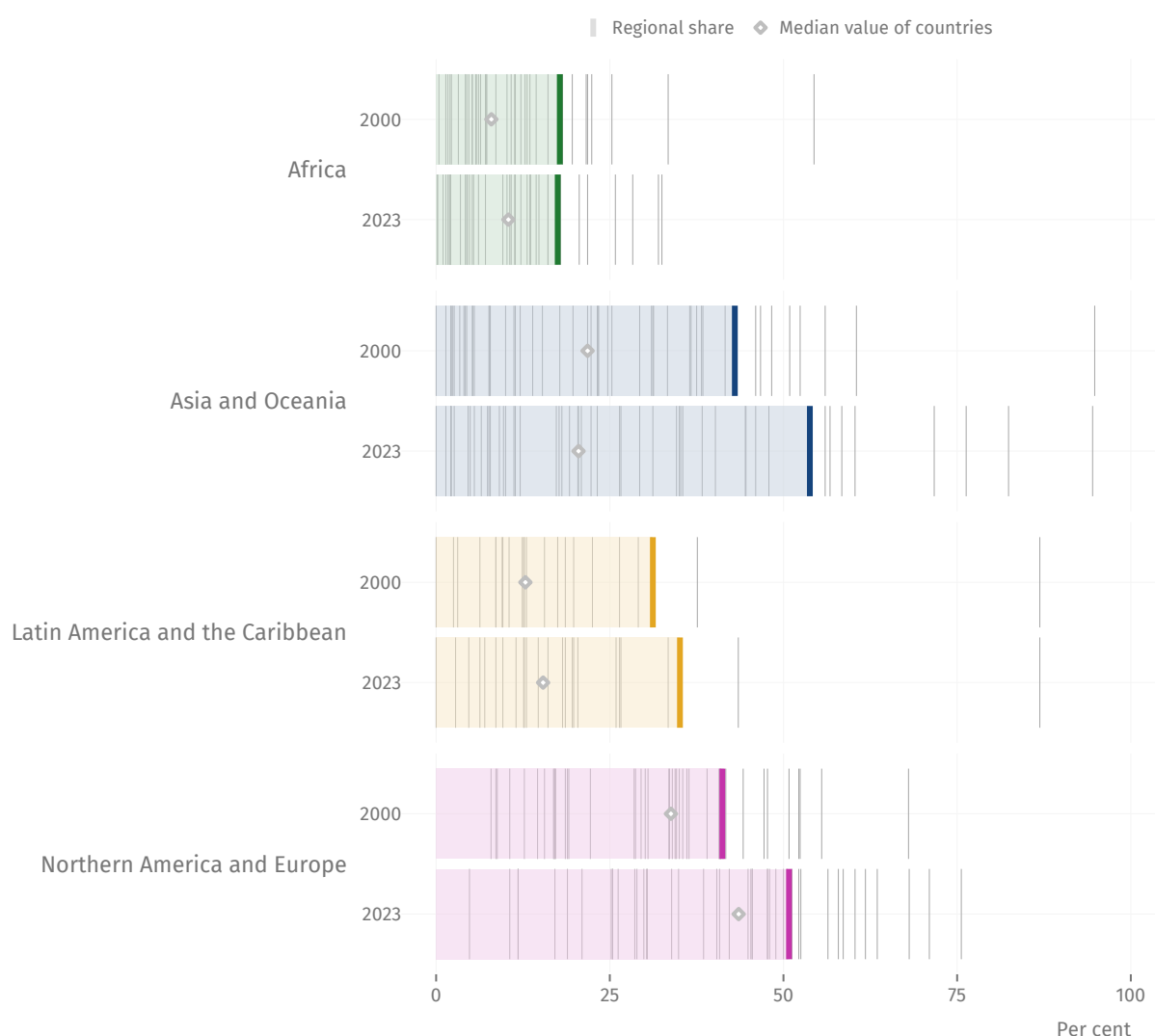
- ▶ Computers and electronics (13.5%)
- ▶ Food products (9.5%)
- ▶ Motor vehicles (9.0%)
- ▶ Machinery (8.7%)
- ▶ Chemicals (7.9%)

Industrializing economies

- ▶ Food products (17.2%)
- ▶ Chemicals (10.9%)
- ▶ Coke and refined petroleum products (7.6%)
- ▶ Basic metals (7.4%)
- ▶ Other non-metallic mineral products (7.1%)



Labour-intensive industries (textiles, apparel) persist in **industrializing** economies, while **industrial** economies pivot toward **higher-tech** manufacturing

Figure 3.11 | Proportion of medium and high-tech manufacturing value added in total MVA by main regions

Source: [9; 19; 35; 36]

Note: The grey lines show MHT shares of individual economies within each group. The colored bars shows the regional shares, while the points represents the median values for the regions' economies.

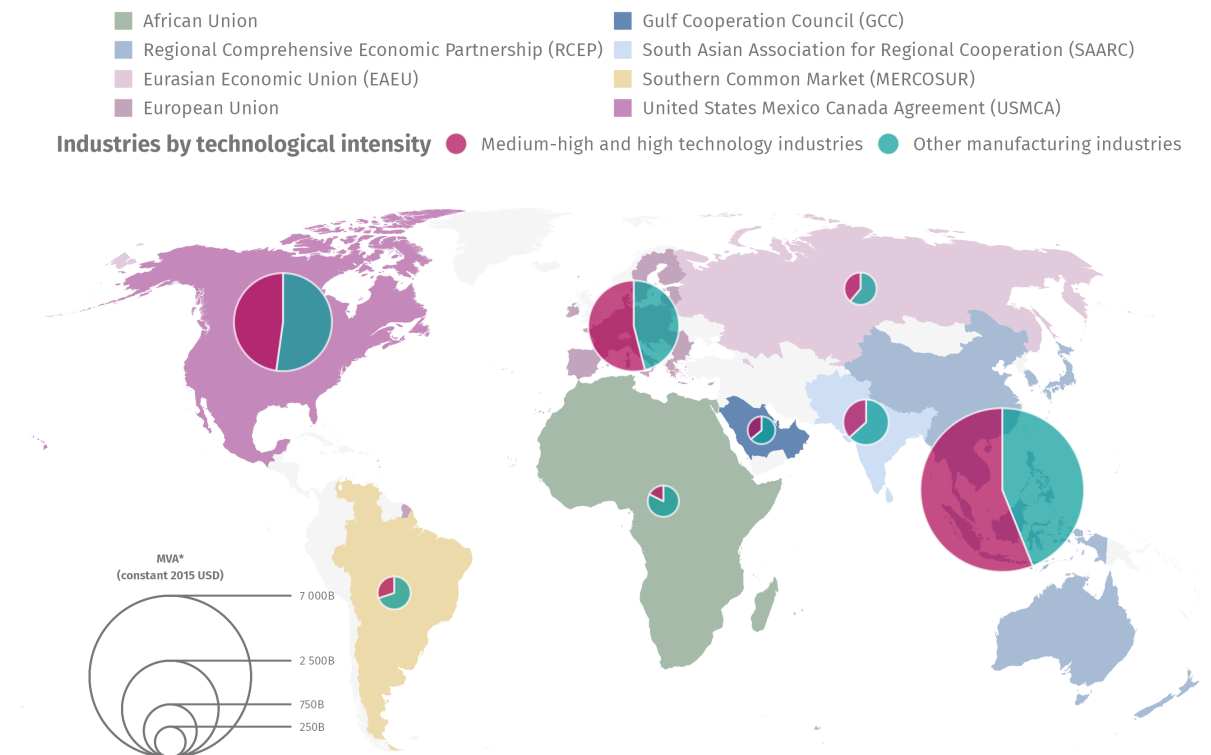
Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

driven primarily by a small number of high-performing economies, while many others, particularly SIDS, have yet to experience significant structural upgrading in their manufacturing sectors. In Latin America and the Caribbean, the expansion of MHT industries was more evenly distributed across countries, pointing to a more inclusive, albeit moderate, transformation. In Africa, despite some dynamism in individual economies, reflected in a rising median value, the overall regional share remained almost unchanged. These insights deepen our understanding of regional dynamics and highlight the need for tailored strategies to foster MHT growth in diverse regional contexts.

Over recent decades, economic and political alliances have played an increasingly pivotal role in shaping the trajectory of manufacturing development. Economic groupings such as BRICS, analysed in the pre-



Regional differences
in MHT trends call for
customized
industrial strategies

Figure 3.12 | MVA composition by industries according to technological intensity for selected economic blocs, 2023

Source: [9; 19; 35; 36]

Note: Member countries of the selected economic blocs are listed in Annex E.5. The pie charts embedded in the regional map illustrate the MVA contribution of industries based on their technological intensity as defined in Annex E.2. Due to the applied methodology and the exclusion of ISIC Rev. 4 division 33 from the calculations, MVA figures can deviate from the figures published in [9].

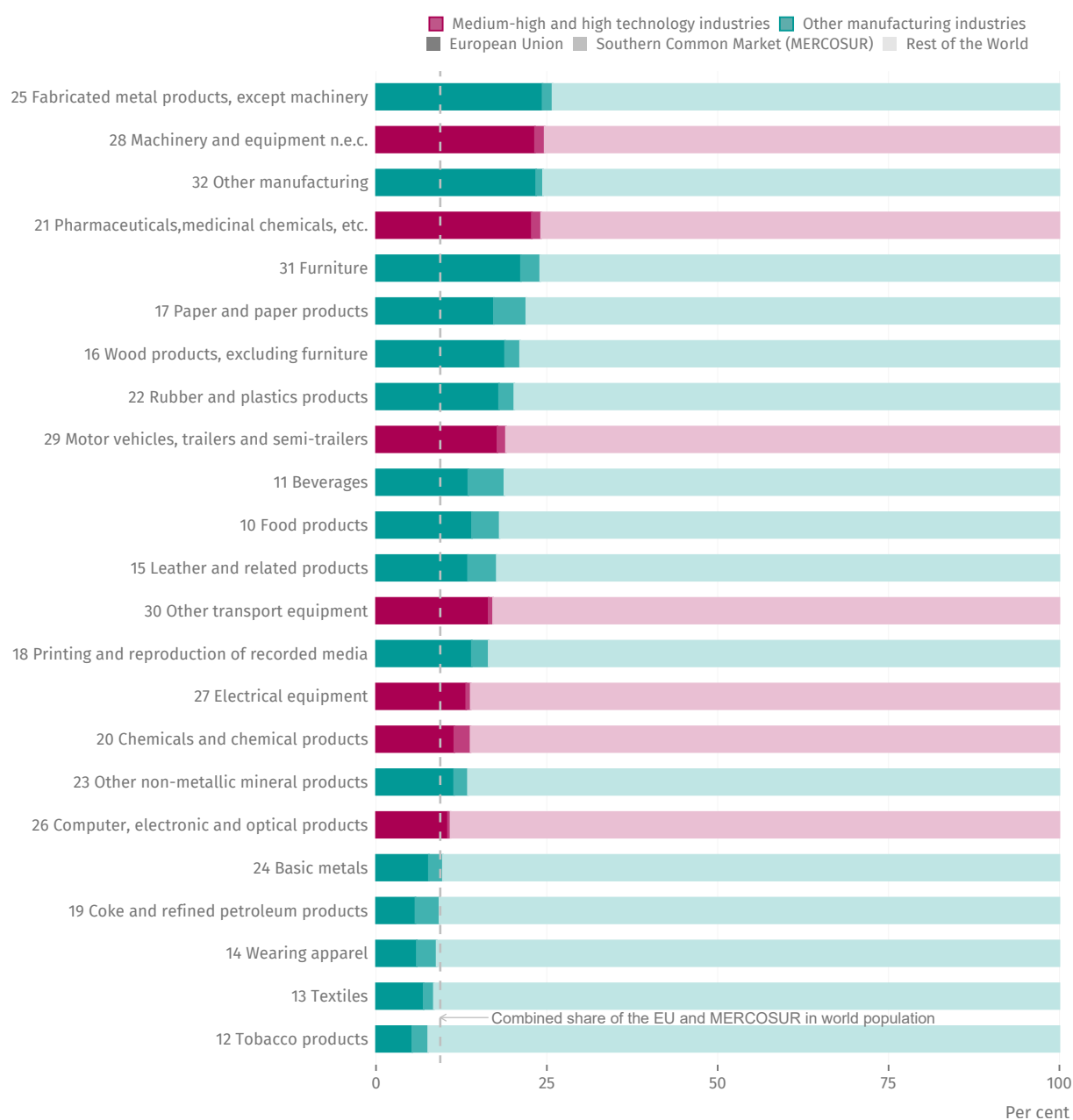
Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

vious edition of the *Yearbook* and whose global influence has further expanded following Indonesia's accession in 2025, alongside the European Union (EU), the Regional Comprehensive Economic Partnership (RCEP), the United States-Mexico-Canada Agreement (USMCA), and other groups, have emerged as key platforms for industrial coordination and technological advancement. The contribution of selected economic blocs to global manufacturing output, as well as the composition of this output by industries according to technological intensity, is illustrated in Figure 3.12.

According to estimates for 2023, RCEP, the largest trade bloc in history formally signed in November 2020 and entering into force in January 2022 for the first ratifying countries, leads this list of selected economic blocs with a manufacturing output exceeding 7.2 trillion USD, of which 56.1 per cent is attributed to MHT industries. This underscores RCEP's position as a global manufacturing powerhouse, integrating advanced industrial capabilities. USMCA follows with 2.6 trillion USD in MVA, with 47.8 per cent attributed to MHT industries, which reflects the region's strong technological base and its continued relevance in



The **Regional Comprehensive Economic Partnership (RCEP)**, the largest trade bloc in history, has become the **global manufacturing powerhouse**

Figure 3.13 | Contribution of EU and MERCOSUR to global manufacturing activities, 2023

Source: [9; 19; 35; 36]

Note: Member countries of the EU and MERCOSUR are listed in Annex Tables E.5.7 and E.5.8. The bars show the share of global value added within each manufacturing industry in per cent. The vertical dashed line indicates the combined share of the EU and MERCOSUR in world population.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

global industrial production. In contrast, blocs such as the African Union, the Eurasian Economic Union (EAEU), and the Cooperation Council for the Arab States of the Gulf (GCC) continue to exhibit more traditional industrial structures, with lower shares of MHT industries and a stronger reliance on resource-based manufacturing.

The Southern Common Market (MERCOSUR) and the EU also illustrate two contrasting industrial realities. The latter stands out as one of



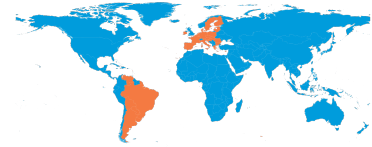
the global manufacturing leaders, with a structure deeply rooted in technology-intensive sectors indicated by a MHT share in MVA of 54.0 per cent. Industries such as machinery (ISIC Rev. 4 division 28), pharmaceuticals (ISIC Rev. 4 division 21), and motor vehicles (ISIC Rev. 4 division 29) dominate its profile, as shown in Figure 3.13, each commanding significant shares of global output and reflecting decades of investment in innovation and advanced production systems. The EU's technological depth, combined with a large and affluent internal market of approximately 450 million people, positions it at the forefront of global value chains.

MERCOSUR, in comparison, retains a more traditional industrial base. Food products (ISIC Rev. 4 division 10), beverages (ISIC Rev. 4 division 11), leather goods (ISIC Rev. 4 division 15), and paper products (ISIC Rev. 4 division 17) remain central pillars, while MHT industries occupy only a modest share of its manufacturing structure. Although chemicals (ISIC Rev. 4 division 20) and motor vehicles (ISIC Rev. 4 division 29) show some presence, their global footprint is limited, underscoring the scope for structural upgrading. With a population of around 300 million, MERCOSUR offers scale and resource advantages (see also Section 4.2), yet its per-capita industrial output lags far behind that of the EU, particularly in technology-driven sectors.

The proposed EU–MERCOSUR partnership agreement may serve as a catalyst for a deeper industrial transformation. If ratified, the agreement could reduce tariffs, harmonize standards, and facilitate technology transfer, thereby creating opportunities for MERCOSUR to diversify beyond resource-based production and integrate more deeply into global supply chains.

Brazil, as the largest economy in MERCOSUR and a key member of the BRICS grouping, appears uniquely positioned to leverage its dual connectivity by drawing on both its regional leadership and global partnerships to attract investment and promote industrial upgrading. In a global context increasingly shaped by shifting trade alliances and rising protectionism, enhanced cooperation between MERCOSUR and the EU holds the potential to play a transformative role in reshaping industrial dynamics across the Atlantic. Such collaboration could accelerate the transition towards more inclusive, innovation-led, and technology-intensive models of growth.

This strategic alignment is particularly pertinent in light of the prevailing dominance of China and the United States of America in global manufacturing. As illustrated in Table 3.3, these two economies lead production across all 23 major industrial sectors, with China ranking first in 19 industries and the United States of America in four, underscoring their significant influence on the future trajectory of industrial development.



The **EU** and **MERCOSUR**
together represent
18.6%
of the world's population,
and account for
17.4%
of global MVA in 2023



China and the
United States
of **America**
dominate global manufacturing,
jointly leading **all major**
industries

Table 3.3 | Leading manufacturers for selected ISIC divisions (World)

	2000	2023		2000	2023		2000	2023
10 Food products			18 Printing and reproduction of recorded media			26 Computer, electronic and optical products		
China	5.6	22.5	China	3.4	23.4	China	14.5	43.6
United States	23.8	17.2	United States	30.8	21.0	United States	10.5	11.0
Japan	10.5	5.6	Japan	13.8	11.0	Korea, Rep. of	6.3	9.7
Indonesia	1.2	3.2	United Kingdom	3.5	3.9	China, Taiwan Prov.	3.9	6.7
United Kingdom	4.1	2.7	Germany	5.6	3.3	Ireland	3.9	4.7
11 Beverages			19 Coke and refined petroleum products			27 Electrical equipment		
China	3.6	18.4	United States	20.6	19.5	China	9.6	58.7
United States	18.8	17.3	China	6.1	17.3	United States	24.9	8.6
Mexico	5.2	5.5	India	7.9	11.7	Japan	13.0	5.6
Japan	10.1	5.3	Russian Federation	4.0	5.8	Germany	9.6	5.3
United Kingdom	4.9	3.2	Mexico	10.1	5.5	Korea, Rep. of	3.6	2.5
12 Tobacco products			20 Chemicals and chemical products			28 Machinery and equipment n.e.c.		
China	4.5	22.7	China	7.1	33.9	China	8.1	38.0
United States	30.7	17.2	United States	29.8	17.7	United States	20.9	10.1
Indonesia	4.6	11.9	Japan	8.6	4.8	Japan	17.1	9.4
Myanmar	0.1	4.1	India	4.1	3.5	Germany	13.3	8.9
Japan	2.9	3.4	Germany	7.7	3.3	Italy	7.2	3.9
13 Textiles			21 Pharmaceuticals, medicinal chemicals, etc.			29 Motor vehicles, trailers and semi-trailers		
China	17.7	48.6	China	4.0	20.9	China	5.3	32.2
Türkiye	4.2	5.0	United States	38.6	19.5	United States	19.5	13.0
India	6.9	4.8	Switzerland	3.0	6.9	Japan	22.2	11.4
United States	16.2	4.6	India	3.0	6.1	Germany	14.0	8.3
Turkmenistan	0.8	2.4	Japan	11.5	4.9	Mexico	4.1	5.2
14 Wearing apparel			22 Rubber and plastics products			30 Other transport equipment		
China	15.4	42.0	China	6.0	27.1	United States	40.0	29.7
Bangladesh	3.2	13.9	United States	25.7	17.8	China	5.0	21.3
Türkiye	2.4	5.6	Japan	14.6	8.7	United Kingdom	7.5	5.2
India	2.2	2.7	Germany	6.6	5.2	Germany	3.3	4.9
Viet Nam	1.0	2.3	Korea, Rep. of	3.9	3.2	France	4.7	4.4
15 Leather and related products			23 Other non-metallic mineral products			31 Furniture		
China	19.4	51.1	China	8.4	40.9	China	4.7	24.2
Italy	16.2	6.5	United States	18.6	9.9	United States	31.4	17.1
Viet Nam	0.9	6.0	Japan	9.4	3.7	Italy	6.8	4.4
Indonesia	4.1	3.9	India	2.3	3.2	Germany	6.8	4.3
France	4.8	2.9	Germany	4.9	2.7	Japan	5.7	3.2
16 Wood products, excluding furniture			24 Basic metals			32 Other manufacturing		
China	3.9	30.0	China	14.6	48.8	United States	39.5	26.6
United States	25.4	18.2	United States	16.3	7.7	China	8.7	25.0
Japan	8.4	4.2	India	2.1	5.4	Germany	6.3	6.3
Germany	4.5	3.9	Japan	11.9	4.8	Ireland	2.3	5.9
Canada	4.8	3.7	Russian Federation	4.2	3.6	Japan	7.5	4.7
17 Paper and paper products			25 Fabricated metal products, except machinery					
United States	36.2	23.1	China	3.7	25.1			
China	4.8	21.8	United States	28.7	19.1			
Japan	9.2	6.3	Germany	7.5	7.4			
Germany	5.0	3.9	Japan	12.7	6.0			
Brazil	2.2	3.1	Italy	7.2	3.7			

Source: [9; 19; 35; 36]. Note: Shares in per cent of value added of the respective group; red points show minimum/maximum values. Data used to create this table can be downloaded at the [UNIDO Statistics Portal](#).

3.3 The manufacturing sector trade

This section examines various aspects of manufacturing trade. While both exports and imports will be analysed, the focus will remain on exports to maintain consistency with earlier discussions on manufacturing production, as domestic manufacturing output is closely linked to export performance. The first part of this section presents the latest developments on global manufacturing trade, drawing from both annual and quarterly data. Thereafter, an analysis of the structure of global manufacturing trade and the trade balance across country groups and regions will be presented. Annex C summarizes the key features of both annual and sub-annual manufacturing trade data.

Manufacturing trade data

Annual trade data with full country coverage can be found in the Annual Manufacturing Trade Database [40], while sub-annual trade data with higher frequency, but lower data coverage, can be found in the Monthly Manufacturing Trade Database [41], both accessible from the [UNIDO Statistics Portal](#) [3].

3.3.1 Global, regional, and country-level performance

The latest annual growth rates of exports of manufactured and non-manufactured goods are presented in Figure 3.14. ⁱ Overall, non-manufactured goods have exhibited greater volatility compared to manufactured goods. In 2024, both categories posted modest growth, following declines of varying magnitude in the previous year that offset the exceptional gains seen in previous years. Although the downturn and subsequent recovery of exports during the COVID-19 pandemic were smoother than, for instance, the trends observed in manufacturing production (see Section 3.1.2), it is worth noting that global exports had already begun to decline since 2019.

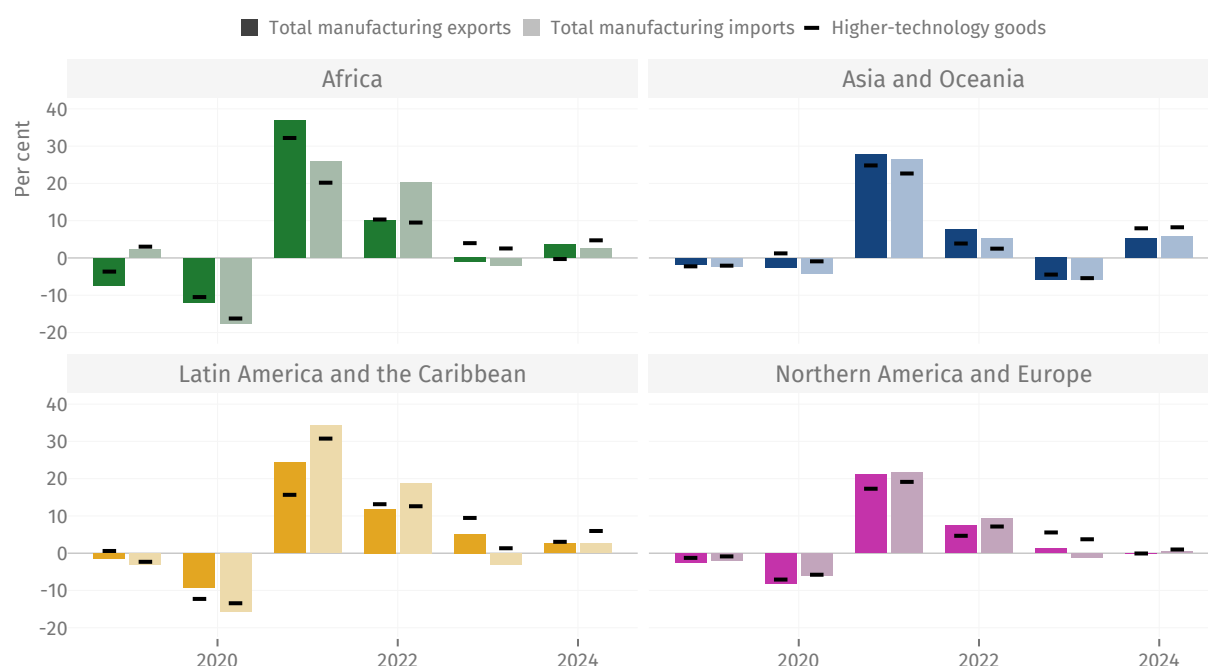
ⁱ UNIDO Statistics applies the Lall classification [42], under which traded manufacturing goods include all commodities excluding primary agricultural and mineral commodities, non-monetary gold, electric current, and unallocated commodities.

Figure 3.14 | Annual growth rate of global exports



Source: [40]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.15 | Annual growth rate of manufacturing exports and imports, by region

Source: [40]

Note: The black marks indicate the growth rate of the corresponding higher-technology manufacturing exports and imports. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Regional disparities in manufacturing trade are shown in Figure 3.15. In 2023, all regions recorded subdued or negative annual growth in both manufacturing exports and imports, with only a few exceptions, most notably Latin America and the Caribbean. By 2024, however, most regions returned to moderate growth, with the exception of Northern America and Europe, where trade remained largely stagnant. More specifically, Europe registered a slight decline in exports, while Northern America experienced a modest increase.

Over the past two years, higher-technology manufacturing exportsⁱ have outperformed exports of other manufactured goods. This trend was observed in most, but not all, regions.

Seasonally adjusted sub-annual manufacturing trade data provide a more up-to-date and granular view of trade dynamics but entail a trade-off in terms of country and time coverage. Multiple factors have contributed to the growing uncertainty in global trade since early 2025. The data available at the time of writing this *Yearbook* extend up to June 2025 and already capture some of the initial effects of this uncertainty on manufacturing trade.

In April 2025, the United States of America announced a series of protectionist measures targeting most countries, prompting retaliatory actions, renewed negotiations, delays in investment decisions, measures to promote diversification in products and trade partners, and shifts in ongoing trade dynamics.

Total trade
includes both
re-exports and
re-imports

ⁱ The detailed list of traded commodities classified as MHT products, based on the Standard International Trade Classification (SITC) Revision 3, is provided in Annex E.2.

Top 10 exporters of manufactured goods and their export growth in 2024

1. China (4.6%)
2. Germany (-3.7%)
3. United States (2.1%)
4. Netherlands (-1.9%)
5. Republic of Korea (8.4%)
6. Italy (-0.4%)
7. Japan (-3.2%)
8. France (4.2%)
9. China, Hong Kong SAR (9.9%)
10. Mexico (3.0%)

Figure 3.16 | Quarter-over-quarter growth rate of manufacturing exports

Source: [41]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

At the same time, a depreciation of the United States dollar against other major currencies altered relative export competitiveness, strengthening that of the United States of America while weakening that of many other economies. These developments have compounded pre-existing challenges to global trade, including rising geopolitical tensions, vulnerabilities in global shipping, rapid technological change, and environmental pressures.

Figure 3.16 illustrates the quarterly trajectory of world manufacturing exports since early 2022. In the second quarter of 2025, exports expanded robustly by 4.1 per cent, following a 1.3 per cent increase in the previous quarter. The strong expansion observed in the first half of the year may partly reflect trade diversion and front-loading of shipments in anticipation of expected trade measures.

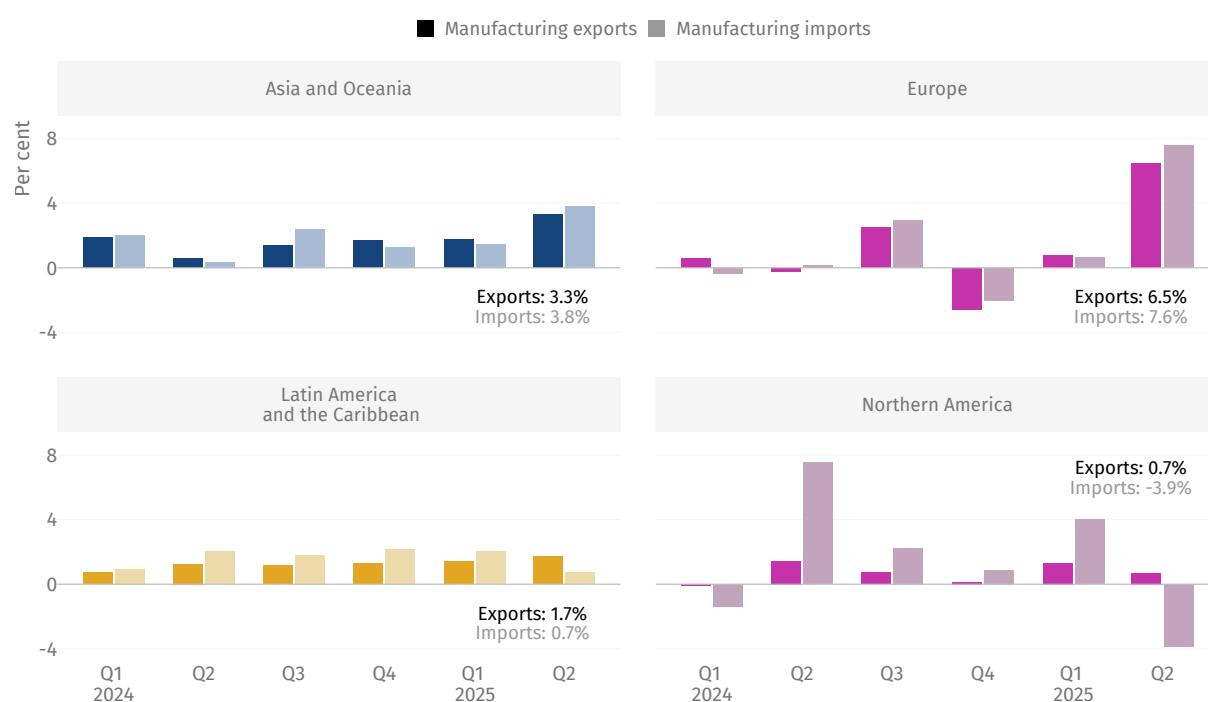
In the second quarter of 2025, higher-technology exports rose by 4.4 per cent, while other manufactured goods grew by 3.0 per cent relative to the previous quarter. Since mid-2022, higher-technology exports have outperformed lower-technology goods, which saw subdued activity until early 2025. The sharp contraction at the end of 2022 mainly reflected China's export downturn resulting from COVID-19 measures, an effect also visible in the production data (see Figure 3.4).

Figure 3.17 provides further insights into the most recent quarterly developments, highlighting the effects of the newly introduced or announced tariff measures. Europe recorded the strongest export growth, rising by 6.5 per cent compared with the first quarter of 2025. Most European economies performed well, particularly the major exporters:

In 2025,
trade positions
were strongly shaped by
tariff disputes,
geopolitical tensions,
and heightened
uncertainty



Global **manufacturing exports** grew by **4.1%** in Q2 2025, following a year of **stagnation** in 2024

Figure 3.17 | Quarter-over-quarter growth rate of manufacturing exports and imports, by region

Source: [41]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Only country groups with sufficient data coverage are included.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

France, Germany, Italy, and the Netherlands. Asia and Oceania showed similar momentum, with both exports and imports expanding by more than 3 per cent. Quarterly growth in the second quarter of 2025 was broadly positive across the region, led by Hong Kong SAR of Chinaⁱ, Singapore, Taiwan Province of China, and Viet Nam. Latin America and the Caribbean continued its upward trend, with exports increasing by 1.7 per cent and imports by 0.73 per cent over the same period. In Northern America, imports fell sharply by nearly 4 per cent, as both the United States of America and Canada registered substantial declines. By contrast, the region's exports rose by 0.70 per cent, driven by a moderate 3.1 per cent rise in exports from the United States of America, while Canada's exports dropped by 10.2 per cent, partly reflecting the impact of higher tariffs imposed by the United States of America on aluminum and steel imports.

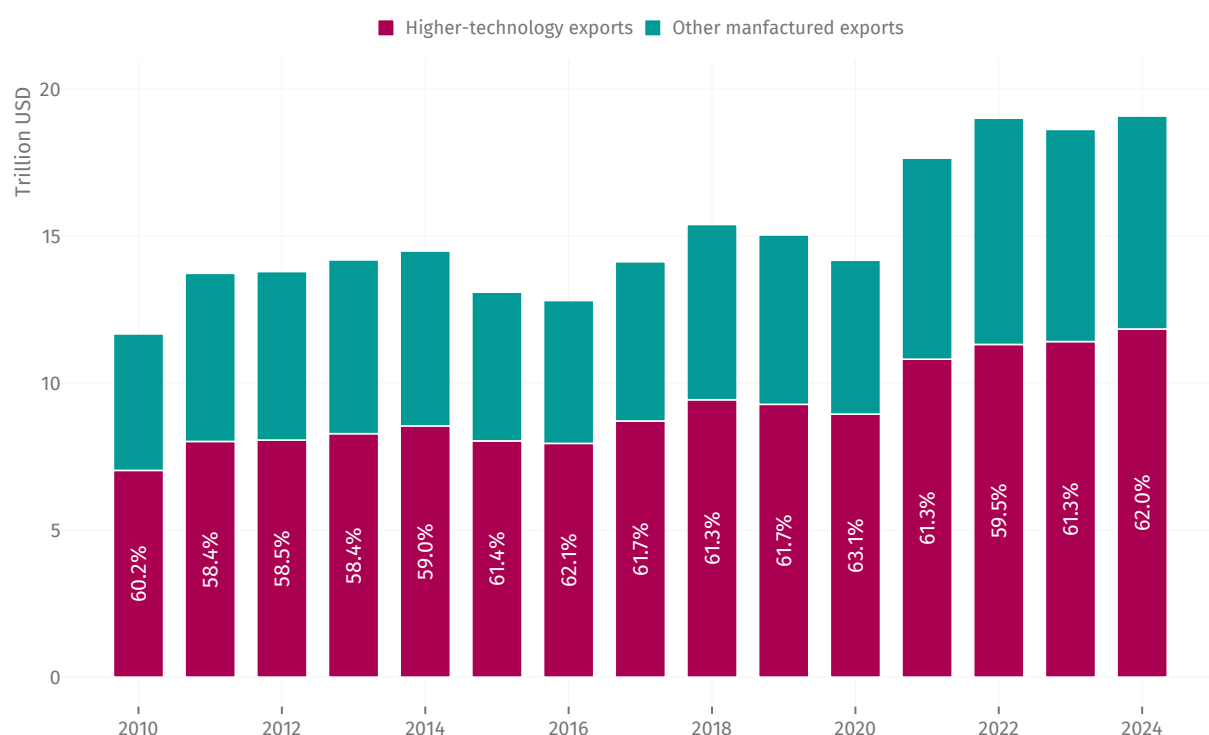
ⁱ China, Hong Kong SAR's exports consist mainly of re-exports, which account for nearly 99 per cent of total exports. The same pattern is observed for imports [43].

Top 10 importers of manufactured goods and their import growth in Q2 2025

1. United States (-4.3%)
2. China (2.8%)
3. Germany (9.0%)
4. Netherlands (7.8%)
5. France (8.0%)
6. China, Hong Kong SAR (3.6%)
7. United Kingdom (6.6%)
8. Mexico (0.29%)
9. Japan (3.1%)
10. Italy (7.1%)

3.3.2 Structure of global and regional manufacturing trade

Manufactured goods represented 78.8 per cent of total exports in 2024. Within total manufacturing exports, higher-technology products accounted for 62.0 per cent. When measured relative to global GDP,

Figure 3.18 | Distribution of global manufacturing exports in absolute values

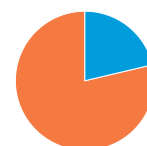
Source: [40]

Note: The labels indicate the percentage of higher-technology manufacturing exports on total manufacturing exports. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

manufacturing exports represented 17.3 per cent of GDP in 2023, the latest year for which GDP figures in current USD are available.

The trajectory of the distribution of manufacturing exports is shown in Figure 3.18. In recent years, MHT exports accounted for more than 60 per cent of manufacturing exports, with only minor fluctuations, resulting in a 62.0 per cent share for 2024, while lower-tech manufactured goods represented the remaining 38.0 per cent. Since 2021, following the recovery from the COVID-19 pandemic, global exports have grown significantly, whereas in preceding years export values followed a rather subdued trajectory.

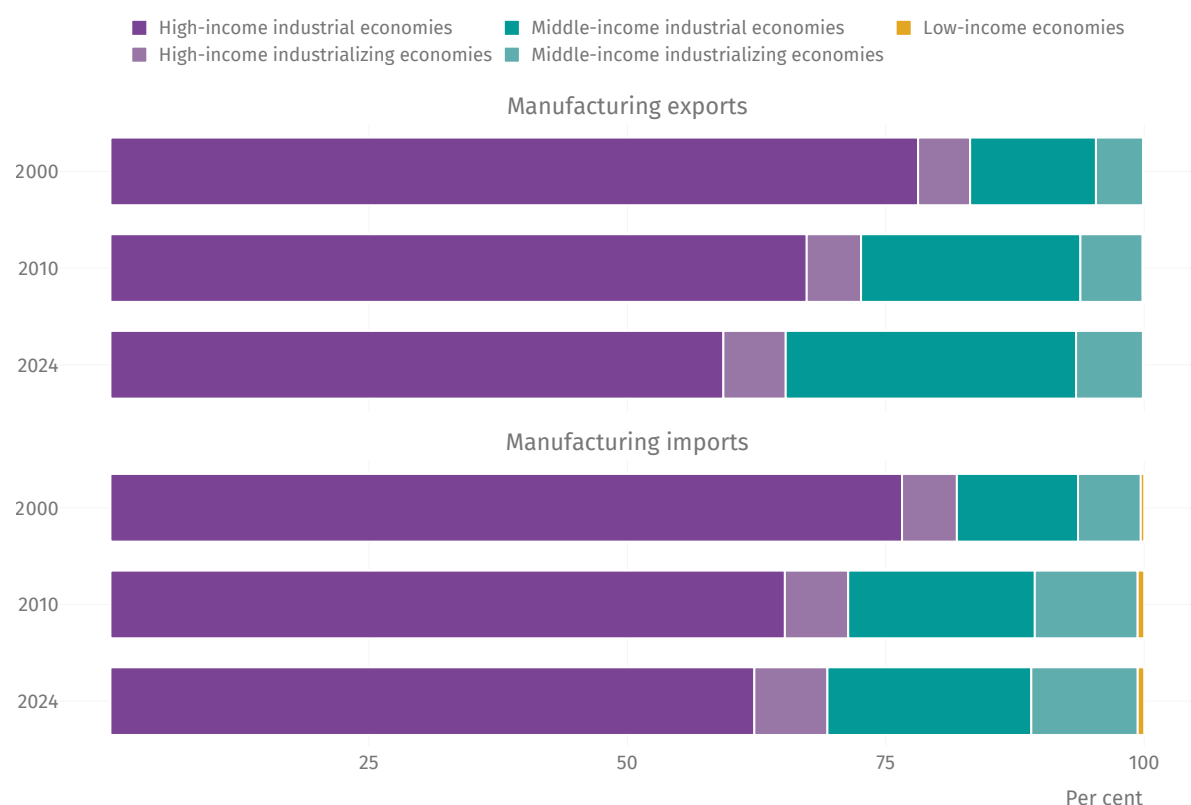
A more detailed distribution of manufacturing exports and imports by UNIDO country groups is presented in Figure 3.19. Since 2000, the share of high-income industrial economies, the largest group in absolute trade values, has gradually declined, though they still accounted for more than 60 per cent of global manufacturing exports and imports in 2024. In contrast, the share of middle-income industrial economies, which include the world's largest exporter, China, has expanded steadily over the past two decades. This group's share of global manufacturing exports increased from 12.2 per cent in 2000 to 28.1 per cent in 2024. The shares of other country groups have remained broadly stable over this period, although low-income economies



Manufacturing exports
accounted for
78.8% of total exports



In 2024,
MHT exports
accounted for
62.0% of manufacturing exports

Figure 3.19 | Distribution of global manufacturing exports and imports, by country group

Source: [40]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

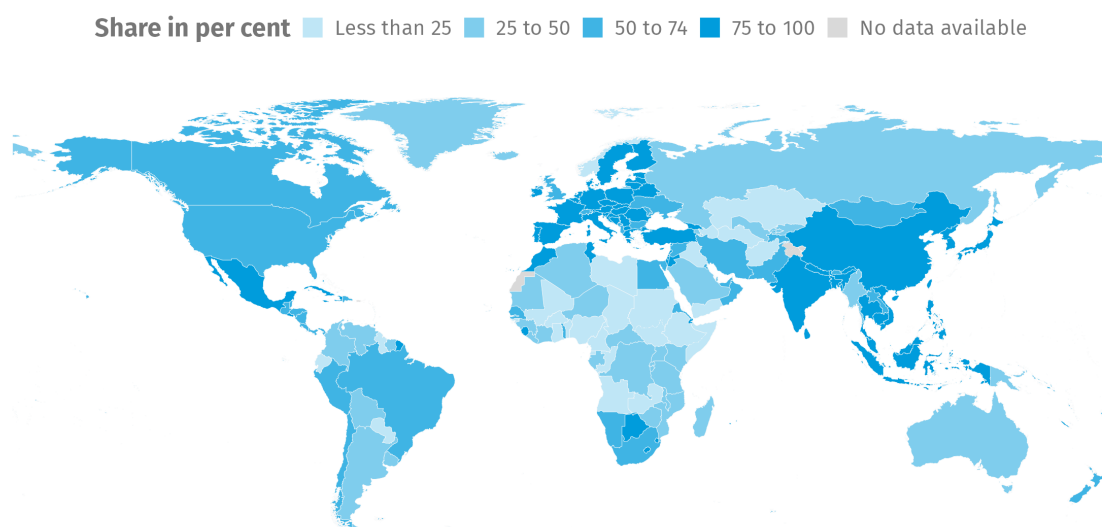
nearly doubled their share of imports, from 0.34 per cent in 2000 to 0.63 per cent in 2024, albeit from a very low base.

The share of manufactured goods in total exports across economies is illustrated in Figure 3.20. Northern America and Europe are the most homogeneous regions, with exports in most countries dominated by manufactured goods. Only a few economies deviate from this pattern: Greenland, Iceland, Norway, and the Russian Federation. Eastern and South-Eastern Asia also feature predominantly manufacturing-based exporters, including economies such as the Republic of Korea, Singapore, and Viet Nam. In contrast, the rest of Asia and Oceania, as well as Latin America and the Caribbean, exhibit greater heterogeneity. Economies such as Afghanistan, Argentina, Azerbaijan, Colombia, and Iraq rely mainly on natural resources and other primary commodities for their exports. Africa displays the widest contrasts. Some economies, including Botswana, Eswatini, Morocco, and Tunisia, report a high share of manufactured exports, while others, such as Angola, Chad, and Somalia, show very low levels, reflecting a continued dependence on primary commodities.

Exports of MHT goods as a share of total manufacturing exports serve as a key indicator of a country's progress on technological upgrading. While this metric does not replace direct data on the production of

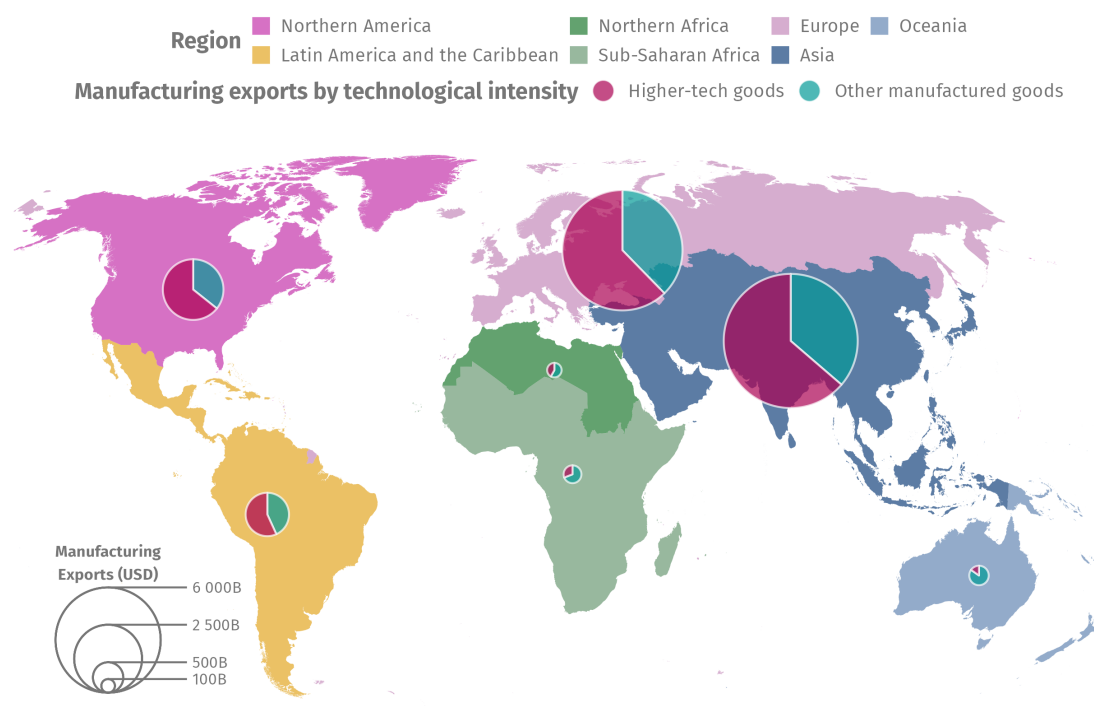


In **Europe**
and **Northern America**
exports in most
economies are
predominantly composed of
manufactured goods

Figure 3.20 | Share of manufacturing exports in total exports, 2024

Source: [40]

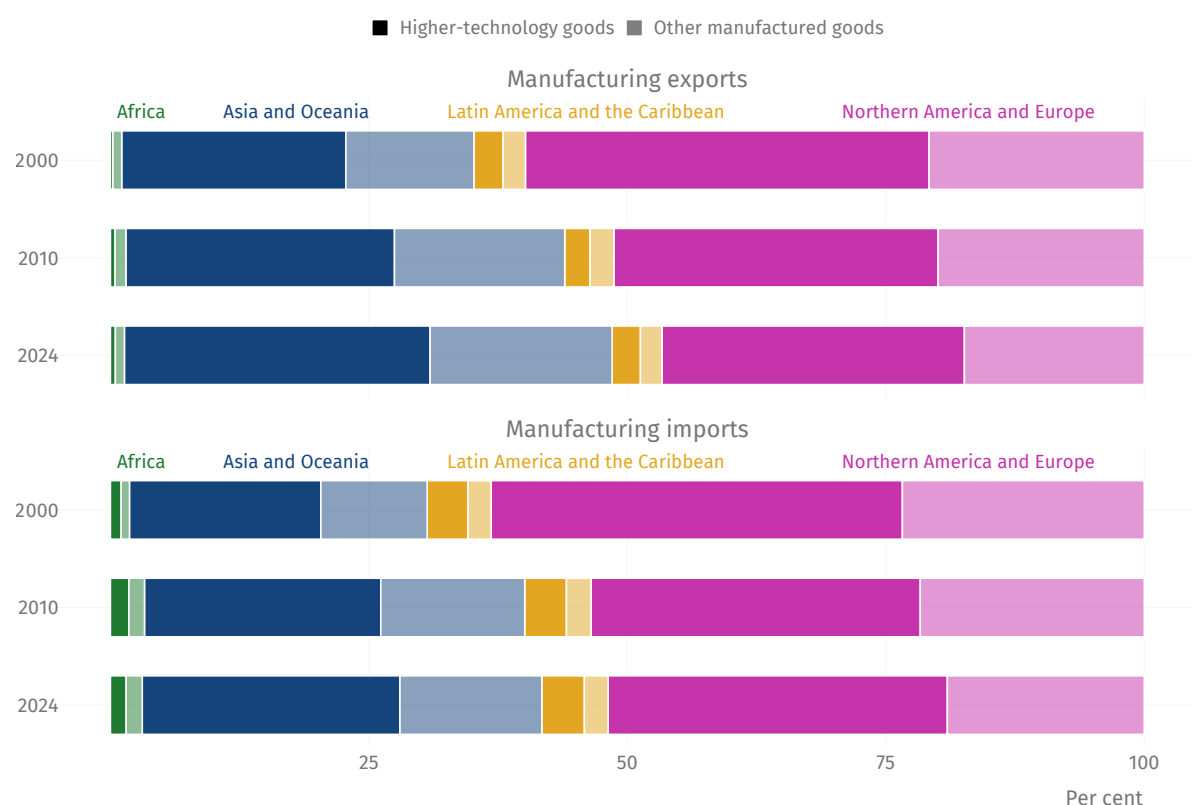
Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.21 | Distribution of manufacturing exports by products according to technological intensity and region, 2024

Source: [40]

Note: The pie charts embedded in the regional map show the exports of manufacturing goods according to their technological intensity, as defined in Annex Table E.2.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.22 | Distribution of global manufacturing exports and imports by type of goods and region

Source: [40]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

higher-technology manufactured goods, it provides a timely and widely available proxy. Figure 3.21 presents the distribution of manufacturing exports in 2024 by technology intensity, highlighting major regional differences. Europe, Eastern Asia, Northern America, and Latin America and the Caribbean had more than half of their manufacturing exports classified as MHT goods. In contrast, Oceania and Africa, especially Sub-Saharan Africa, recorded the lowest share, reflecting their limited participation in higher-technology manufacturing trade. Globally, MHT exports remain highly concentrated: Northern America, Europe, and Asia and Oceania together accounted for 94.8 per cent of world MHT exports in 2024, only slightly below the 96.6 per cent registered in 1995. These patterns closely mirror the production structures by technological intensity discussed in Section 3.2.

Figure 3.22 illustrates how the regional distribution of manufacturing exports and imports by technology intensity has evolved over recent decades. The share of Northern America and Europe in global manufacturing trade has gradually declined, offset by a steady rise in Asia and Oceania. Across these regions, the composition of traded goods has remained broadly similar, with higher-tech products representing the majority of both exports and imports. By contrast, the shares of Latin America and the Caribbean as well as Africa have remained

Top five higher-tech exporters and their share in global higher-tech exports in 2024

- ▶ China (18.0%)
- ▶ Germany (9.3%)
- ▶ United States (8.3%)
- ▶ Republic of Korea (4.2%)
- ▶ China, Hong Kong SAR (4.2%)

Top five higher-tech importers and their share in global higher-tech imports in 2024

- ▶ United States (15.8%)
- ▶ China (9.5%)
- ▶ Germany (6.0%)
- ▶ China, Hong Kong SAR (4.1%)
- ▶ Netherlands (3.1%)

relatively stable. A notable distinction emerges in Africa: exports remain concentrated in lower-technology manufactured goods, which accounted for 64.7 per cent of the region's manufacturing exports in 2024, while MHT exports represented only 35.3 per cent. On the import side, however, higher-technology accounted for around half of Africa's total manufacturing imports, underscoring the region's reliance on imported sources of technology-intensive products, rather than on domestic production.

3.3.3 Manufacturing trade balance

The trajectories of the manufacturing trade balance in absolute terms across UNIDO country groups are shown in Figure 3.23. High-income and middle-income industrial economies (excl. China) recorded steadily rising manufacturing exports and imports, resulting in a largely balanced trade position over time, with high-income industrial economies gradually shifting towards a deficit in recent years. China's manufacturing exports, by contrast, have grown far more rapidly than its imports, leading to a substantial and expanding manufacturing trade surplus. High-income industrializing economies exhibited trade deficits of varying magnitude. Middle-income industrializing and low-income economies showed the largest manufacturing trade deficits relative to their overall trade values, with an increasing share attributable to higher-technology imports. Notably, the deficit of the low-income economies has widened significantly in recent years.

Because of the large differences in the scale across country groups, Figure 3.23 applies different axis limits. To enable a better cross-regional comparison, Figure 3.24 presents the normalized manufacturing trade balance, revealing pronounced disparities.ⁱ China's manufacturing trade surplus exceeded 25 per cent, indicating that its manufacturing exports surpassed its imports by one-quarter of its total trade value. In contrast, Northern America's deficit has widened significantly in recent years, reaching about 25 per cent of total trade. Latin America and the Caribbean also registered a sizable but relatively stable deficit of around 15 per cent, while Africa posted the highest normalized deficit among all regions, accompanied by considerable volatility over time.

For higher-technology goods, the right-side panel of Figure 3.24 shows more balanced trade patterns across regions. Northern America and China continued to hold notable deficit and surplus positions, respectively, whereas other regions maintained relatively stable, near-balanced outcomes in recent years. Asia and Oceania (excl. China) and Europe reported comparable normalized trade balances across both total manufactured goods and higher-technology products.

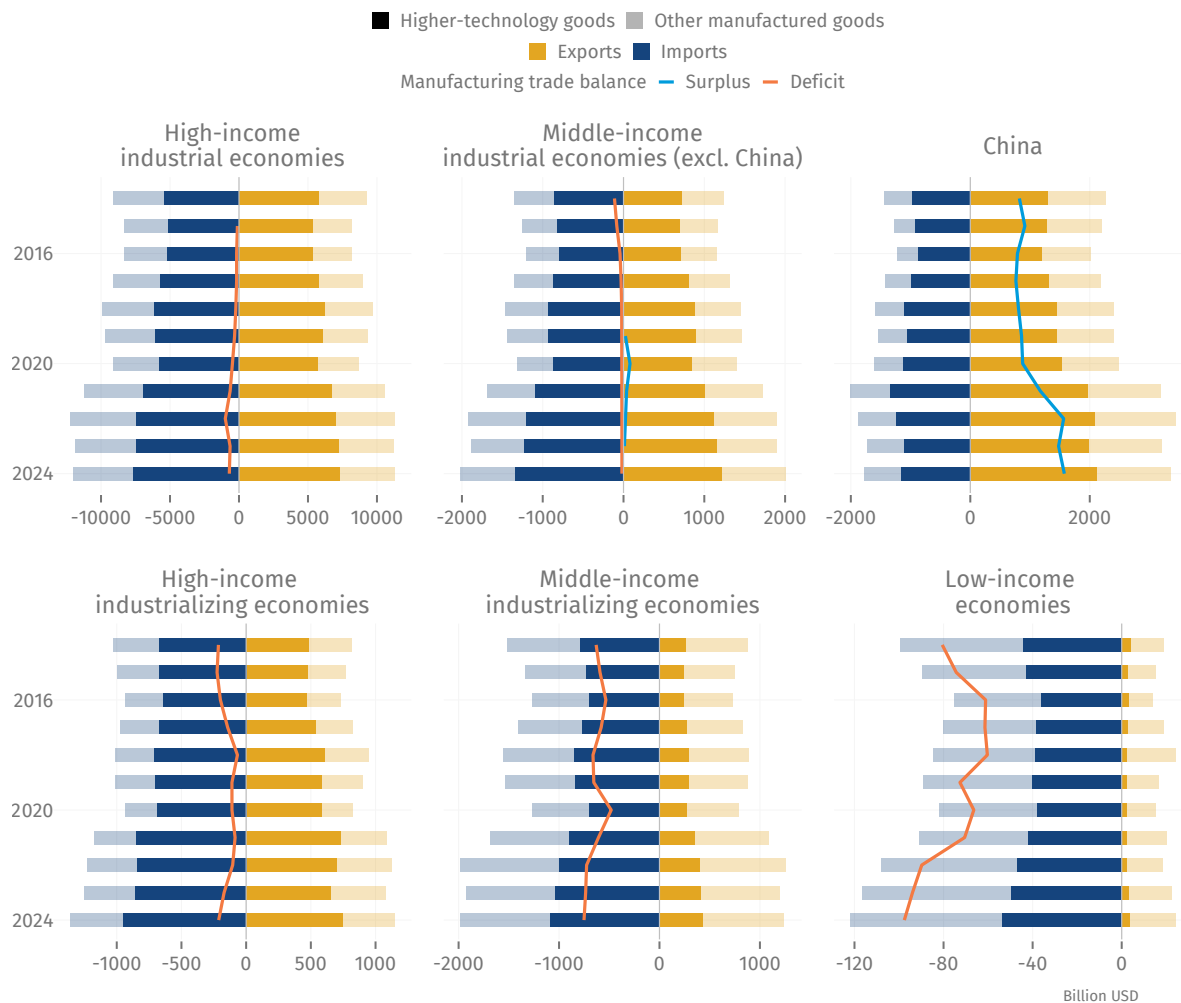


Asia, Northern America, and Europe
accounted for almost
95% of world
higher-tech exports in 2024



Middle-income industrializing and low-income economies
are facing persistent
trade deficits

ⁱ The normalized manufacturing trade balance expresses a region's manufacturing trade surplus or deficit as a share of its total trade (exports plus imports), allowing for meaningful comparisons of manufacturing trade performance across regions.

Figure 3.23 | Manufacturing exports, imports, and trade balance in absolute values, by UNIDO country group

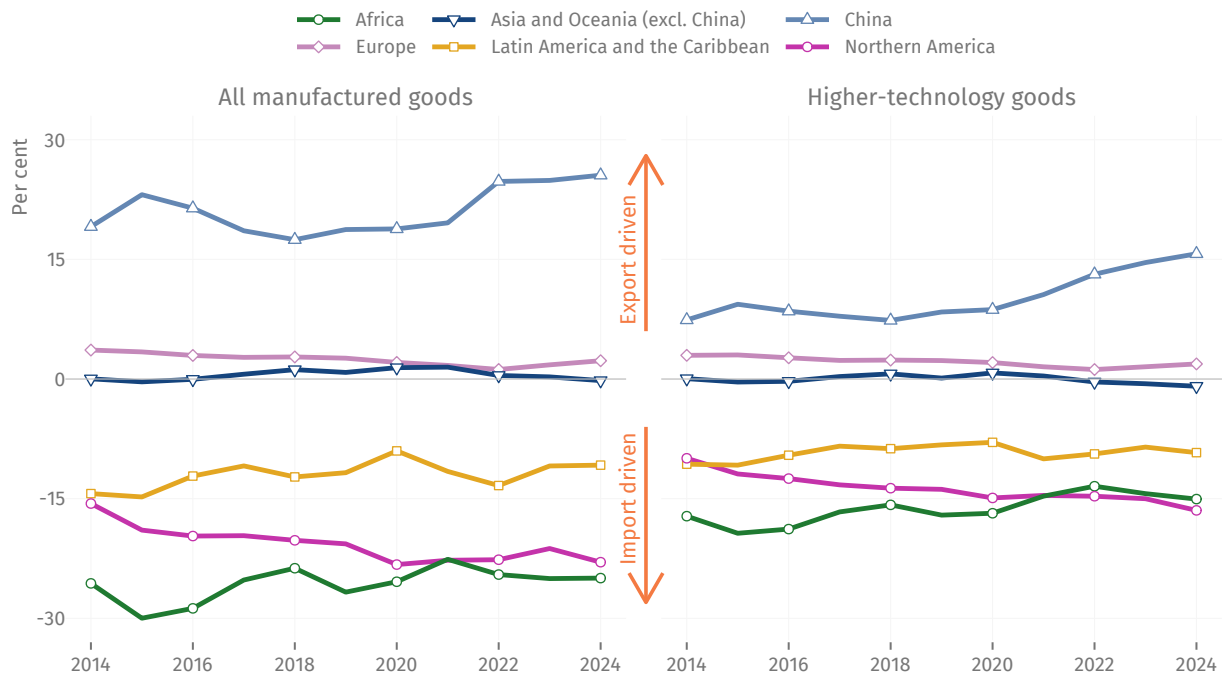
Source: [40]

Note: Due to the varying trade volumes across groups, different scales are applied. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Furthermore, Figure 3.25 illustrates recent developments in the trade balance of manufactured goods on a quarterly basis, allowing an examination of changes linked to the various challenges that have emerged since early 2025. In the second quarter of 2025, Northern America was the only region to display a marked adjustment, with its deficit in manufactured goods narrowing moderately, likely reflecting the recent tariff measures and the depreciation of the United States dollar. However, this change did not carry over to higher-technology goods, where the deficit remained broadly unchanged. By contrast, other regions maintained relatively stable and balanced positions in both manufactured and higher-technology goods, showing little variation over the past two years.



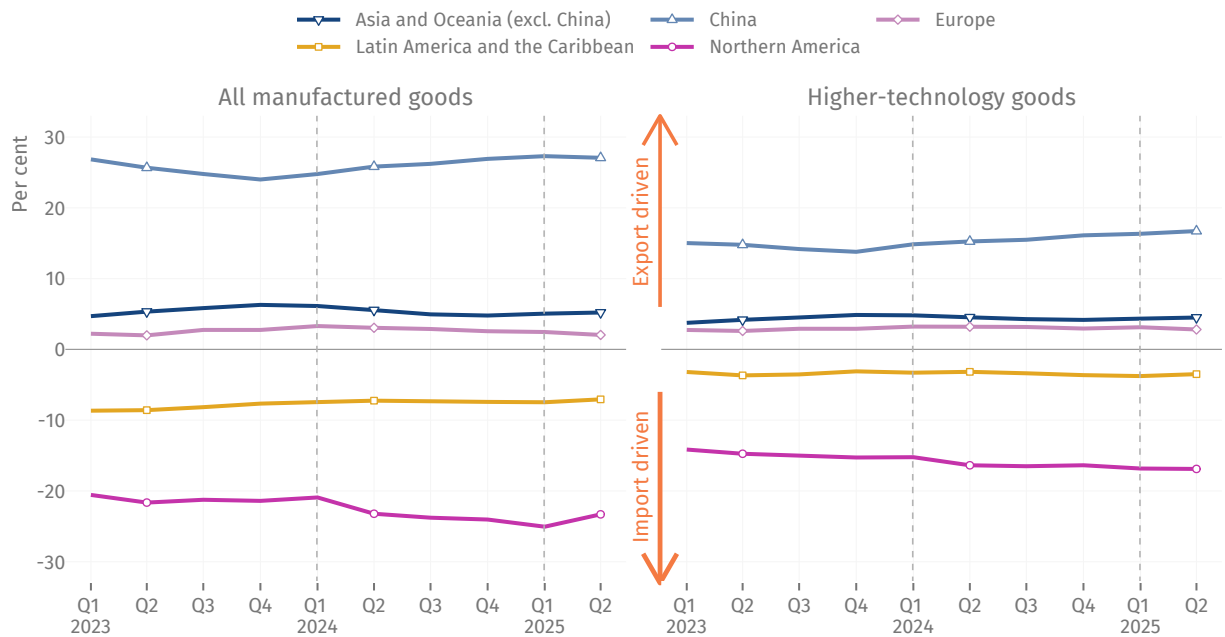
In Q2 2025, **Northern America** recorded a somewhat **narrowed** manufacturing **trade deficit**

Figure 3.24 | Normalized trade balance of total and higher-technology manufactured goods, by region

Source: [40]

Note: The normalized trade balance shows the trade surplus or deficit as a proportion of total trade (the sum of exports and imports) to facilitate cross-regional comparison.

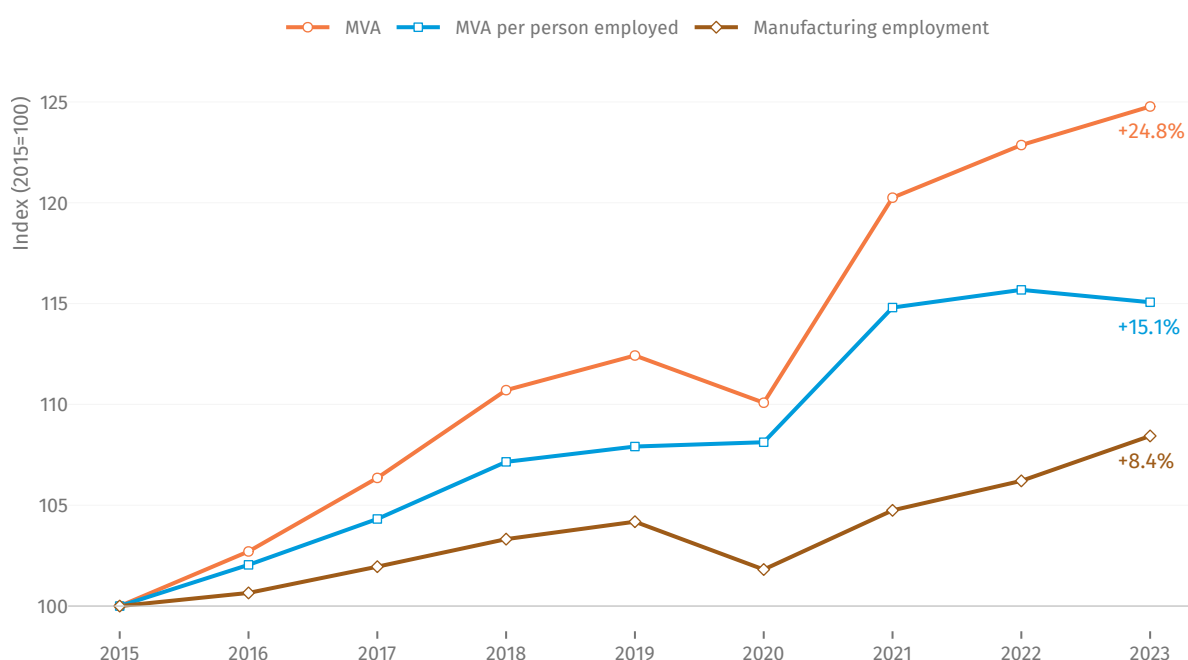
Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.25 | Quarterly normalized trade balance of total and higher-technology manufactured goods, by region

Source: [40]

Note: The normalized trade balance shows the trade surplus or deficit as a proportion of total trade (the sum of exports and imports) to facilitate cross-regional comparison. Only country groups with sufficient coverage are included.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.26 | Global MVA, manufacturing employment, and labour productivity

Source: [9; 44]

Note: Due to their different measurement units, all series have been converted to an index with base year 2015.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

3.4 Manufacturing employment and labour productivity

In recent years, the global labour market outlook has weakened amid persistent geopolitical tensions, trade disruptions, rising protectionism, and sluggish growth. Although unemployment remains low in many economies, declining job vacancies and business confidence point to softening labour demand. Over the past decade, global economic growth has been accompanied by persistently high informal employment, declining labour income shares, and a shift in employment structures toward higher-skill occupations, particularly in high-income countries. These dynamics underscore the fragility of recent progress and the need for policies that strengthen resilience, promote formalization, and foster skill development to ensure that growth translates into decent work for all [45].

Between 2015 and 2023, global labour productivity growth outpaced manufacturing employment growth (Figure 3.26). During this period, manufacturing employment expanded by 8.4 per cent, rising from 454 million to 492 million persons, while labour productivity, measured as MVA per person employed, increased by 15.1 per cent.

Regional data reveal substantial variation in manufacturing employment and productivity trends (Figure 3.27). Africa experienced robust employment growth of 34.8 per cent between 2015 and 2023, but

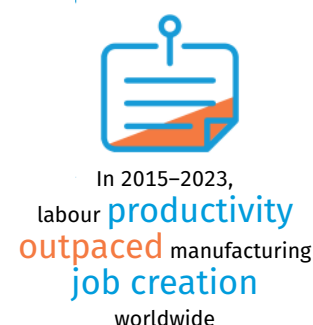
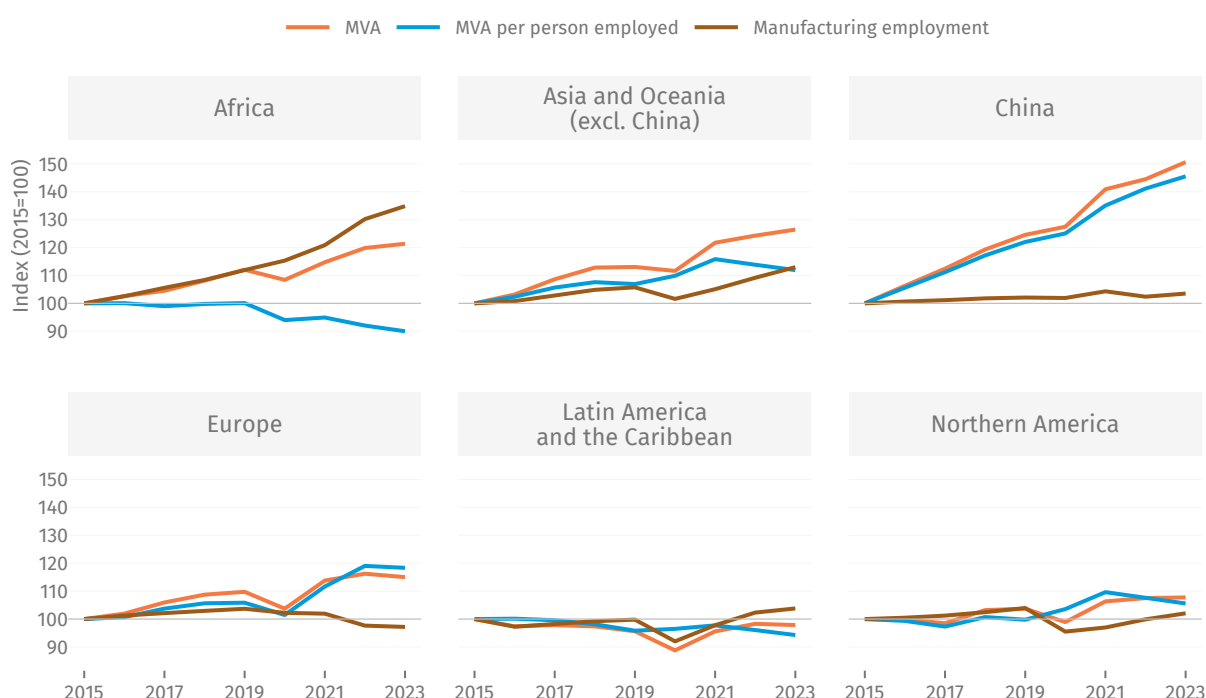


Figure 3.27 | MVA, manufacturing employment, and labour productivity by region

Source: [9; 44]

Note: Due to their different measurement units, all series have been converted to an index with base year 2015.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

only modest gains in MVA, resulting in a 10.0 per cent decline in productivity relative to 2015. In contrast, China's manufacturing growth since 2015 has been driven primarily by productivity improvement rather than job creation. Asia and Oceania (excluding China) present a mixed picture, combining economies with a rapidly expanding manufacturing base and advanced economies with mature, high-technology industries, reflecting both employment growth and productivity gains across the region. In Europe, manufacturing employment fell more sharply than output in the most recent years, making productivity appear to rise, though this largely reflects job contraction rather than sustained, innovation-led growth. In Latin America and the Caribbean, employment continued to grow despite weak output, leading to a 5.7 per cent decline in productivity compared to 2015. Northern America experienced a temporary dip in manufacturing employment during the pandemic, while productivity remained relatively resilient, outpacing MVA growth.

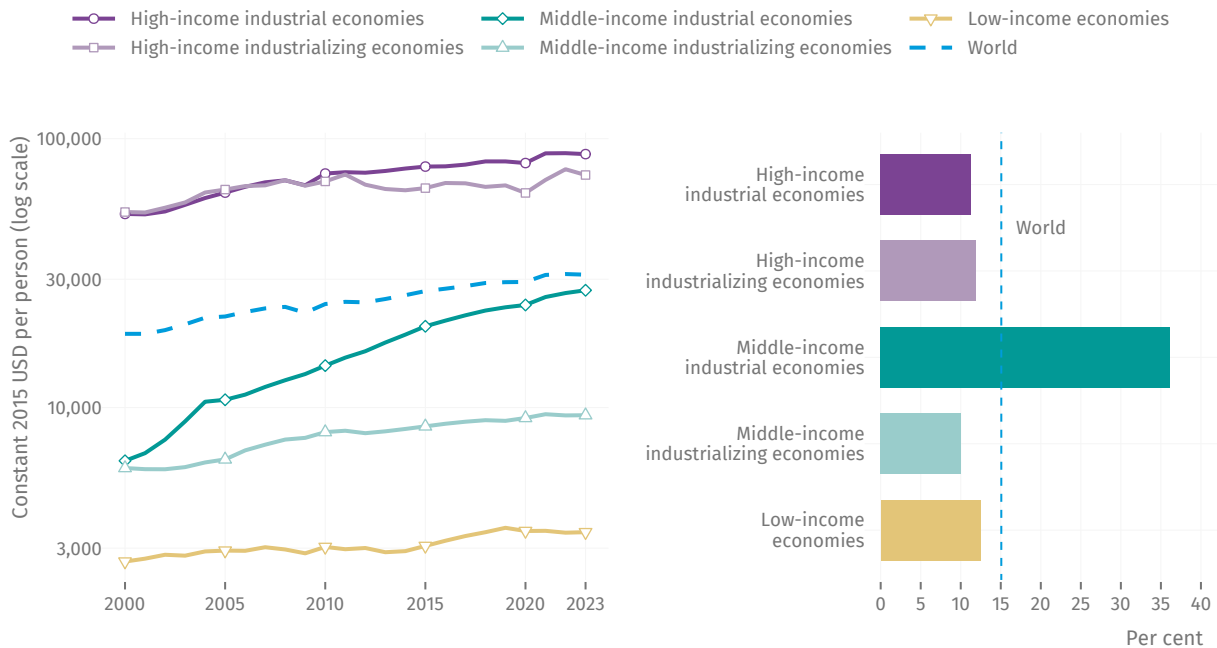
While Figure 3.27 uses indices to compare trends since 2015, absolute levels reveal substantial differences across regions. Northern America shows the highest labour productivity at 131,656 USD per person employed at constant 2015 prices, followed by Europe (63,915) and China (29,742). In contrast, Africa exhibits a relatively low productivity level of 6,982 USD per person employed.

Regional trends
in manufacturing employment
and labour productivity
diverged sharply

**Manufacturing labour
productivity by region, 2023
(constant 2015 USD)**

1. Northern America ... 131,656
2. Europe 63,915
3. China 29,742
4. Latin America
and the Caribbean ... 21,836
5. Asia and Oceania
(excl. China) 21,224
6. Africa 6,982

Figure 3.28 | MVA per person employed by country group (left-side chart) and the corresponding growth rate between 2015 and 2023 (right-side chart)



Source: [9; 44]

Note: The vertical axis of the left-side chart has been log transformed.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.28 illustrates the evolution of labour productivity across different income groups. As shown on the left-side of the figure, middle-income economies reached a record high in 2023. Although not all income groups peaked in that year, each recorded more than 10 per cent growth in labour productivity between 2015 and 2023. Middle-income industrial economies stood out with a significant increase of 36.1 per cent, steadily narrowing the gap with high-income economies. The second-highest growth was recorded by low-income economies, with a 12.5 per cent increase compared to 2015, followed by high-income industrializing economies (11.9 per cent), high-income industrial economies (11.3 per cent), and middle-income industrializing economies (10.0 per cent).

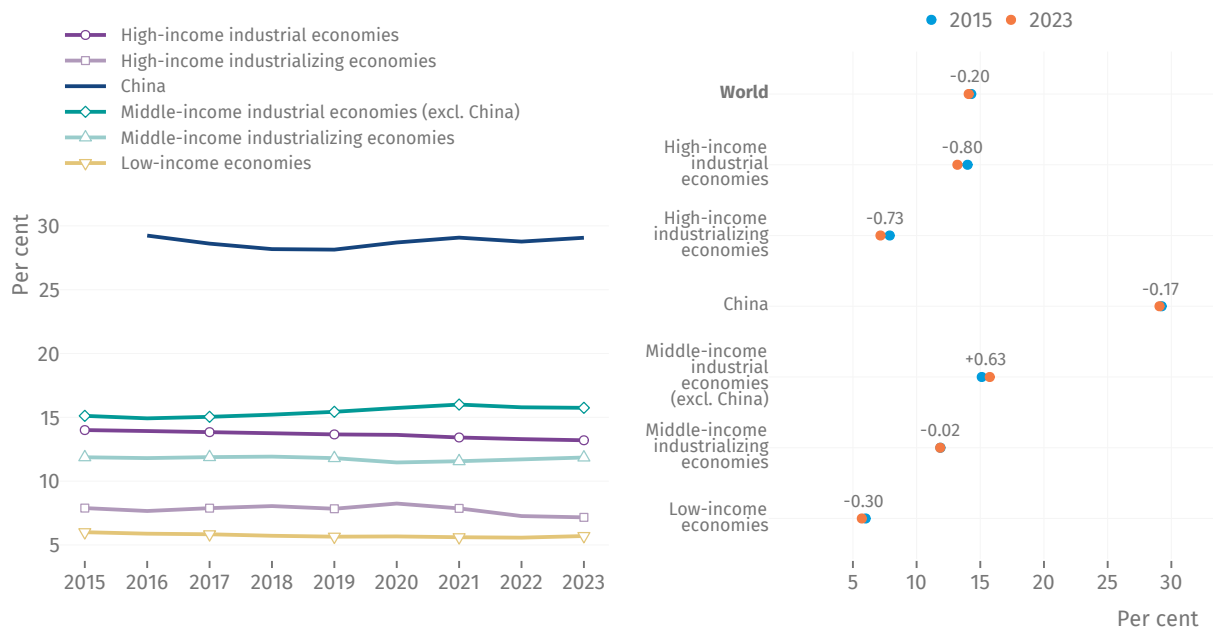
Overall, these patterns suggest that productivity trends are shaped by a complex set of factors. Understanding these drivers is essential for formulating policies that simultaneously promote job creation and productivity growth. Expanding manufacturing employment remains vital for supporting livelihoods and social stability. However, sustained improvements in living standards depend not only on job creation and productivity gains, but also on how the benefits of growth are distributed. Over the past decade, in fact, the labour income share has declined by 0.6 percentage points globally, contributing to widening inequality [45].



All income groups recorded
more than 10%
productivity growth
between 2015 and 2023

Over the past decade,
labour income share
declined by 0.6%
globally

Figure 3.29 | Manufacturing employment as proportion of total employment (left) and its comparison between 2015 and 2023 (right) by country group



Source: [20]

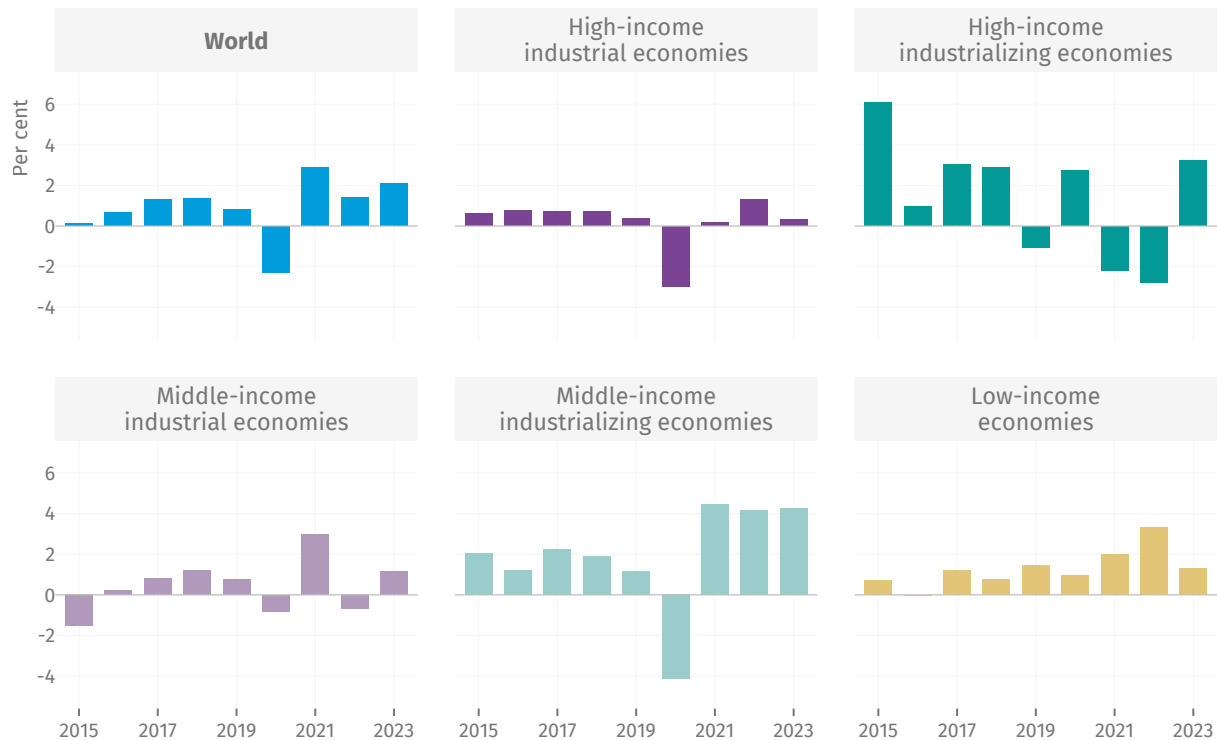
Note: In the right-side chart, the numbers above the lines or points indicate the change in the proportion (in percentage points) between 2015 and 2023 for each country group. For China, the graph shows data for 2016 and 2023 due to data availability. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Over the same period, the share of manufacturing in total employment declined slightly, from 14.3 to 14.1 per cent. This suggests that, despite its substantial job-creation potential, the manufacturing sector has been notably affected by recent crises.

A more detailed breakdown in Figure 3.29 reveals a downward trend in the share of manufacturing employment across most country groups. The only exception is middle-income industrial economies (excluding China), where the share modestly increased by 0.63 percentage points between 2015 and 2023. While China saw a slight decline of 0.17 percentage points over the same period, it still maintained the highest share of manufacturing employment at 29.1 per cent in 2023, well above all income groups, which remained below the 20 per cent mark. The lowest share was recorded in low-income economies, where manufacturing employment gradually declined from 6.0 per cent in 2015 to 5.7 per cent in 2023. This underscores the persistent disparity between low-income and industrial economies. Furthermore, given the limited growth in manufacturing output in these economies, the prospects for sustained employment growth in the sector remain challenging [39].

Figure 3.30 presents the year-over-year growth rates of manufacturing employment by UNIDO country classification. Although global manufacturing employment growth remained positive for three con-



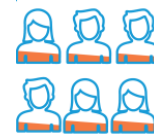
Figure 3.30 | Year-over-year growth rates of manufacturing employment by country group

Source: [44]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

secutive years, it was only in 2023 that all UNIDO country groups recorded simultaneous growth. This broad-based expansion built on the previous year's 1.4 per cent increase, accelerating to 2.1 per cent in 2023. Manufacturing employment in middle-income industrializing economies approached a 4.5 per cent expansion over the past three years, indicating a degree of resilience. Meanwhile, employment in middle-income industrial economies contracted by 0.69 per cent in 2022 but rebounded with a 1.2 per cent increase in 2023. High-income industrializing economies also experienced a recovery, with a 3.2 per cent rise following declines of 2.2 and 2.8 per cent in the preceding years. In contrast, employment in high-income industrial economies recorded only a modest 0.34 per cent increase, remaining nearly flat. Low-income economies have seen accelerating manufacturing employment growth in recent years, with the exception of 2016, when growth temporarily stalled.

Looking ahead, technological change, including the rapid emergence of artificial intelligence (AI), will increasingly shape the future of manufacturing employment and productivity. While many workers are in roles with little or no exposure to automation risk from AI, nearly one in four face some degree of risk, particularly in medium- and high-skill roles [45]. Rather than widespread job losses, the main impact is likely to be a reconfiguration of tasks, requiring policies that support adaptation and reskilling.



In 2023,
**manufacturing
employment**
grew across all
country groups



AI is emerging as a
transformative force
in the world of work

3.5 Gender and manufacturing

Women's participation in manufacturing is a critical factor in advancing inclusive and sustainable industrial development. Despite progress in education and growing commitment to gender equality, structural barriers, unequal access to opportunities, and persistent gender norms continue to constrain women's role in industry.

Four out of ten people employed in the manufacturing sector worldwide are women. A share that has largely remained stable in recent years, underscoring persistent structural imbalances and limited progress towards gender inclusivity in the sector.

Despite this relative stability at the global level, growth trends in female and male manufacturing employment have not always converged. Figure 3.31 highlights that during the 2010s, female employment growth generally lagged behind that of men. Both groups were equally affected by the COVID-19 pandemic in 2020, but female employment rebounded more strongly in 2021, rising by 3.5 per cent compared to 2.5 per cent for men. This momentum continued in 2022, before moderating in 2023, when male employment grew by 2.2 per cent, slightly outpacing the 2.0 per cent recorded for women.

In addition, a more detailed analysis reveals heterogeneous trends across country groups. Figure 3.32 shows that the proportion of women in manufacturing employment has declined in both high-income and middle-income industrial economies. Notably, middle-income industrial economies remains the only country group where women form the majority of manufacturing employment, although their share fell from 53.4 per cent in 2000 to 50.3 in 2023. By contrast, women's participation has increased in both middle-income industrializing and low-income economies, with the latter now exceeding the world average. Finally, in high-income industrializing economies, manufacturing employment is dominated by men, with women accounting for only one in ten positions.

While recent years have seen a modest rebound in women's manufacturing employment growth, this has not translated into improvements in the types of industries where they work. Women remain disproportionately concentrated in lower-productivity sectors, limiting the potential impact of any increase in participation. Figure 3.33 illustrates employment distribution across MHT and other manufacturing industries by sex, showing that women's share in MHT industries lags behind that of men. In 2023, 23.1 per cent of men employed in manufacturing worked in an MHT industry, compared to only 15.6 per cent of women.

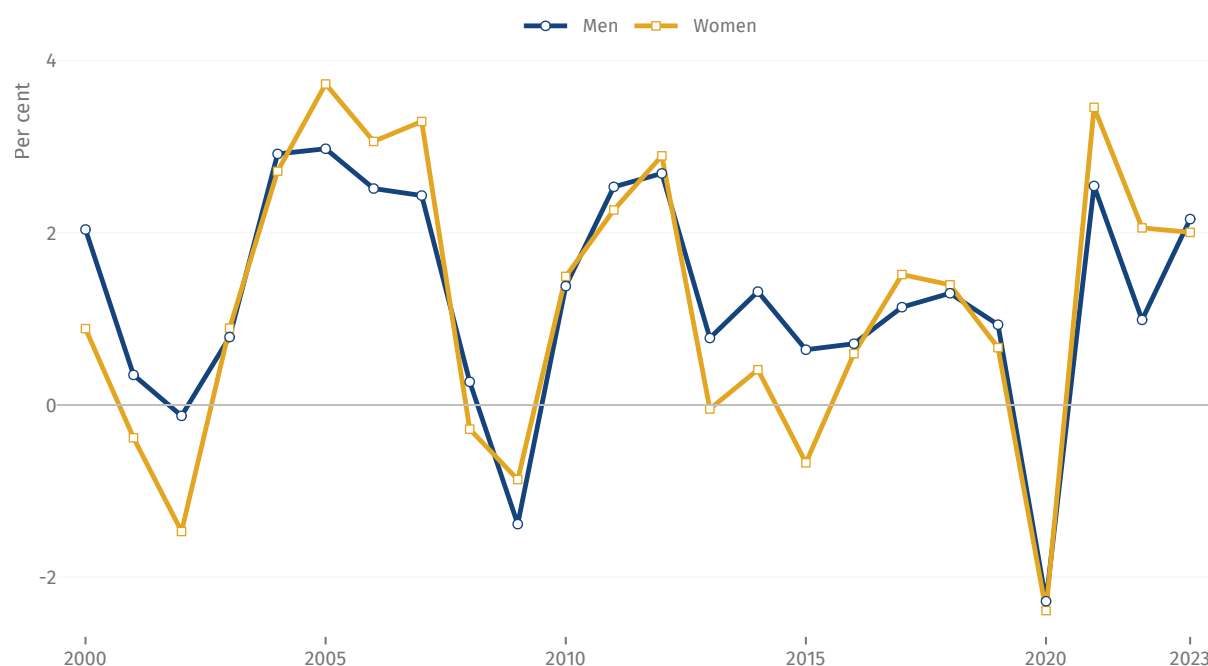
Since the onset of the COVID-19 crisis, women's participation in MHT industries has also declined more sharply than men's. Between 2019



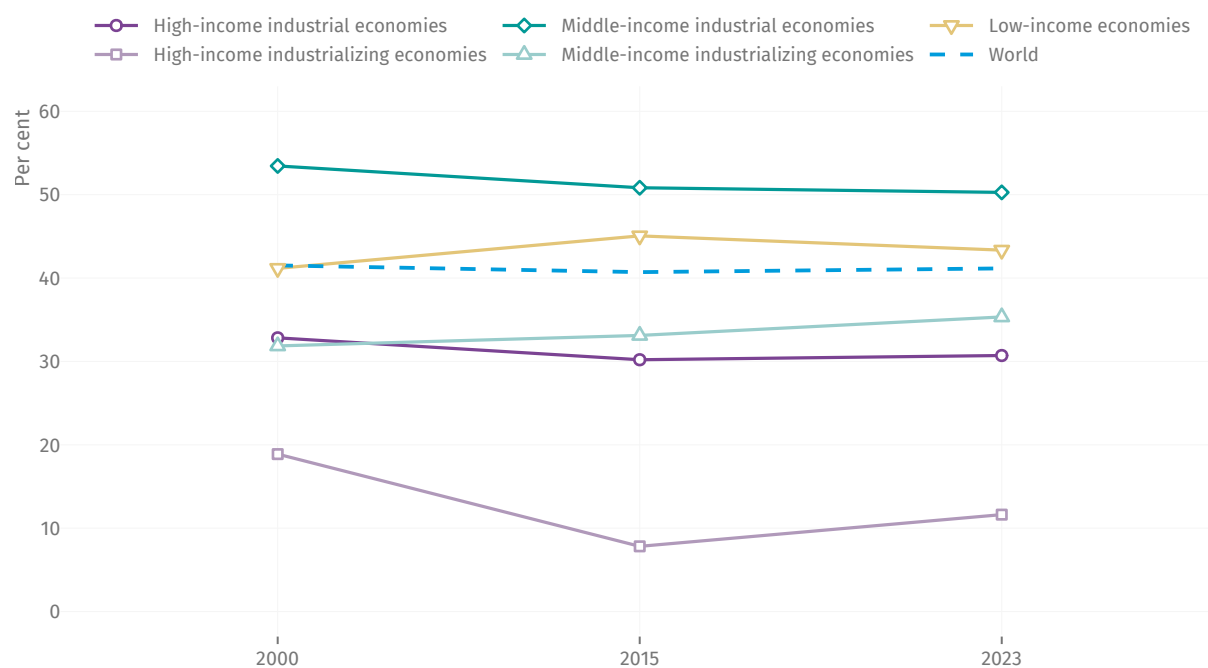
Women
remain
underrepresented
in the
manufacturing
workforce

Globally,
4 in 10 people
employed in the
manufacturing sector
are
women

The majority
of persons employed in
manufacturing in
middle-income
industrial economies
are women

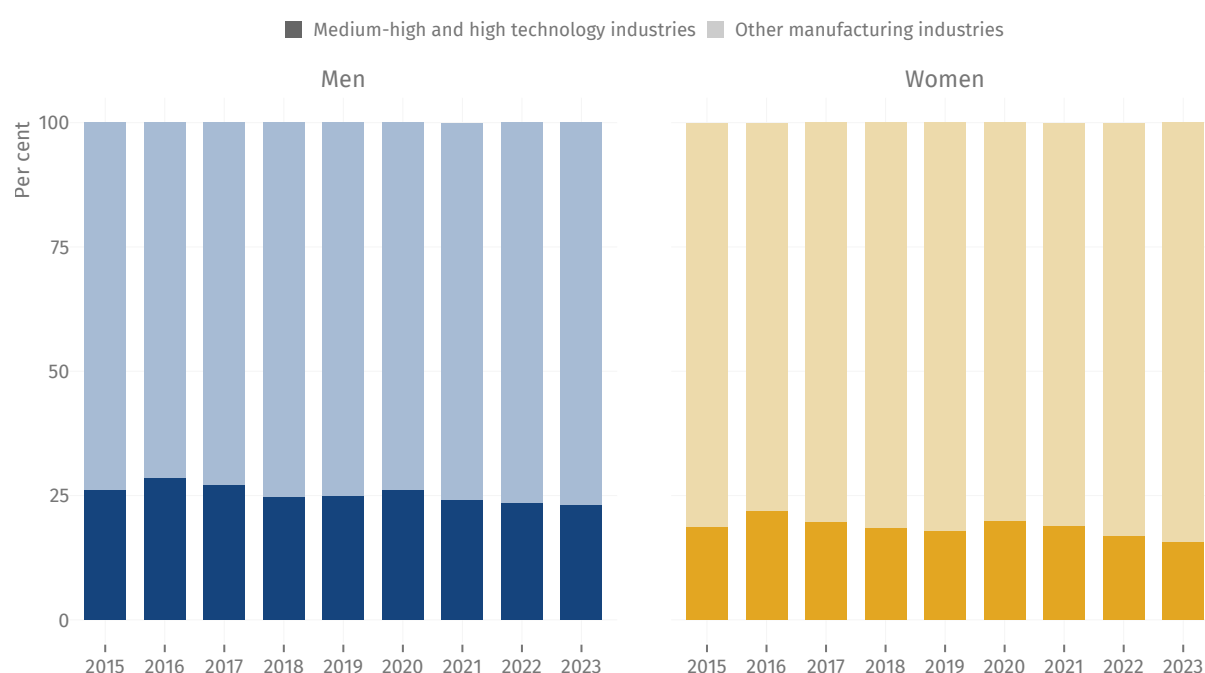
Figure 3.31 | Year-over-year growth rate of manufacturing employment by sex

Source: [44]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).**Figure 3.32** | Share of women in manufacturing employment by country group

Source: [44]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 3.33 | Distribution of global manufacturing employment by sex and industry group (technological intensity)

Source: [46]

Note: The chart is based on divisions of ISIC Rev. 4. The MHT category includes divisions 20, 21, 26, 27, 28, 29, and 30 (see Annex E.2). Division 33 has been omitted from the calculations due to data unavailability.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

and 2023, the share of men in MHT industries fell by 1.8 percentage points, while for women it fell by 2.3 percentage points. This sectoral pattern underscores a persistent imbalance: women remain underrepresented in higher-technology and higher-productivity industries, perpetuating gender disparities in manufacturing even as their overall employment growth has outpaced that of men in certain groups.

A closer look at manufacturing employment by educational attainment and occupational level reveals important nuances behind these gender disparities. While the imbalance remains more pronounced across occupations, particularly in higher-skilled roles, substantial progress has been made in narrowing gender gaps in education.

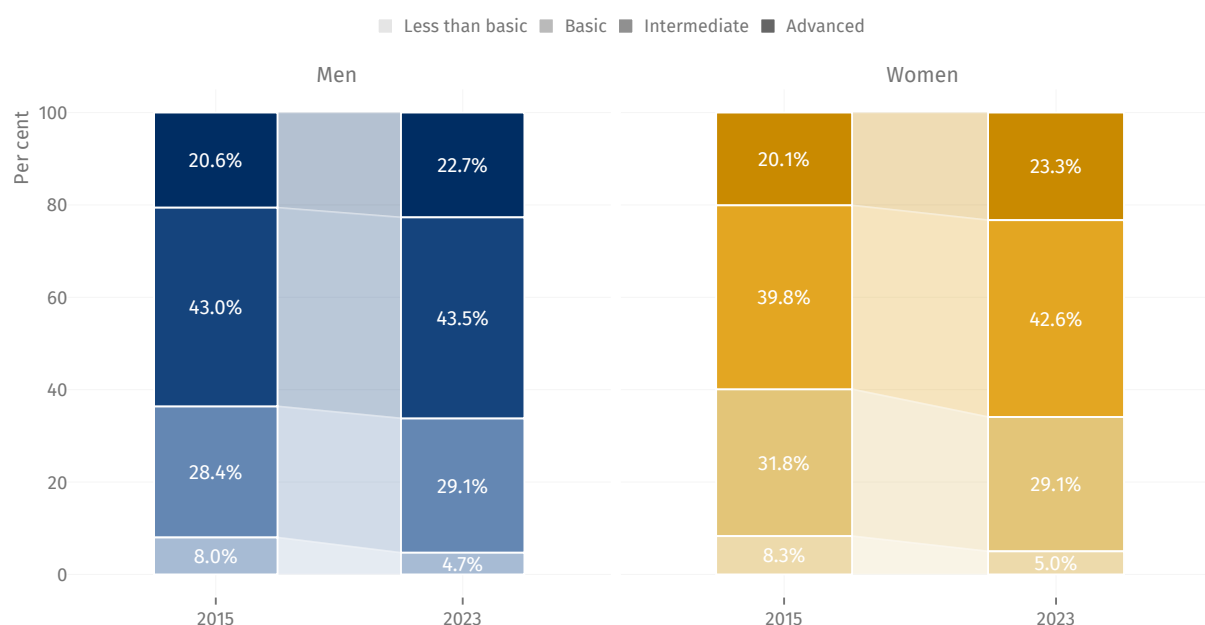
Figure 3.34 presents the global distribution of male and female employment in manufacturing according to education level. This figure confirms that, while women with a higher level of education have historically been underrepresented in the manufacturing workforce compared to men, this gap has narrowed considerably in recent years.

In 2015, 40.1 per cent of women employed in manufacturing had a basic or lower level of education, compared with 36.4 per cent of men, a difference of 3.7 percentage points. By 2023, this gap had almost disappeared, with a difference of only 0.30 percentage points. Overall, there is a clear trend towards higher educational attainment among the manufacturing workforce, with an increasing share of workers holding



Women's share in
MHT industries
lags behind
that of men

Manufacturing employment by
educational attainment
and
occupational level
reveal important nuances
behind existing
gender disparities

Figure 3.34 | Distribution of global manufacturing employment by sex and education level

Source: [47]

Note: The graph indicates the share of manufacturing employment by education level calculated relative to the total employed population in manufacturing by sex.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

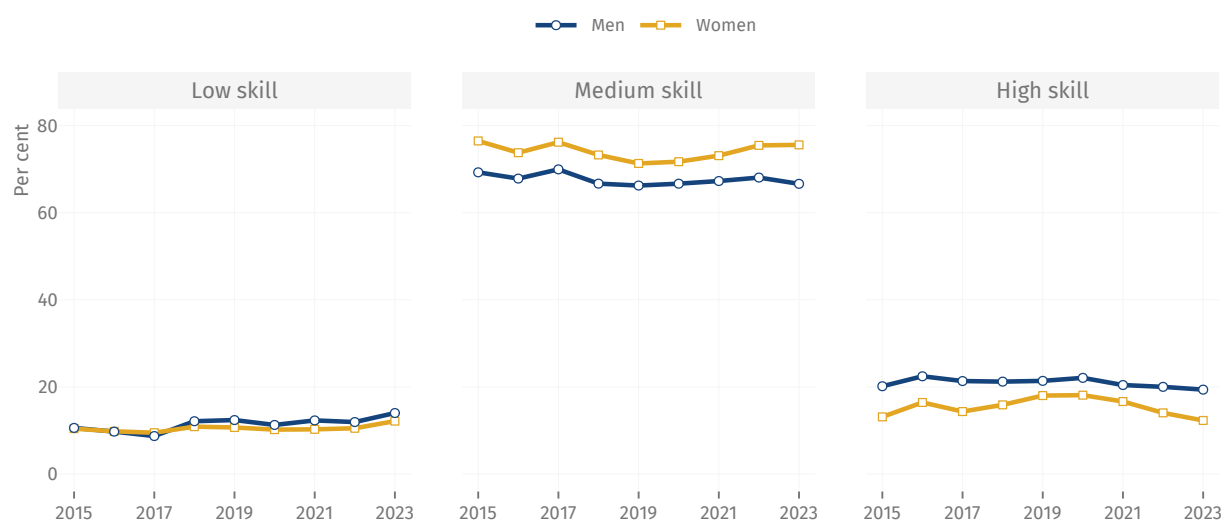
intermediate or advanced qualifications, and a corresponding decline in those with only basic education or less. This shift has benefited both women and men. For example, in 2023, 23.3 per cent of women employed in manufacturing had an advanced education, slightly higher than the 22.7 per cent recorded for men. These developments suggest that women are increasingly achieving higher qualifications, which may help reduce gender disparities over time.

Despite progress in education, substantial gender disparities remain across occupational levels, as illustrated in Figure 3.35, which depicts the global distribution of manufacturing employment by sex and occupation. To provide a clearer perspective, occupations are classified into three groups according to their skill level, as defined by the International Labour Organization (ILO) [48]: low skill, medium skill, and high skill.

The data confirm that women's employment in manufacturing is predominantly concentrated in medium-skilled occupations, at even higher rates than men. In 2023, 75.6 per cent of women employed in the sector were classified at this skill level, compared with 66.6 per cent of men. By contrast, women were less represented in high-skilled occupations, accounting for 12.3 per cent of total female employment, compared with 19.4 per cent for men. The gender gap in high-skilled occupations had narrowed in recent years, reaching just 3.4 percentage points in 2019. However, it subsequently widened, more than

Women employed in manufacturing
increasingly
have higher
educational
attainment

Women's employment
is predominantly
concentrated
in medium-skilled
occupations

Figure 3.35 | Distribution of global manufacturing employment by sex and occupational level

Source: [49]

Note: The graph indicates the share of manufacturing employment by skill level calculated relative to the total employed population in manufacturing by sex (i.e., for any given year and sex, the sum of the three skill levels equals 100 per cent). Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

doubling to 7.1 percentage points in 2023. In low-skilled occupations, the gender difference is the smallest. Overall, while high-skilled positions offer greater opportunities for job advancement, women remain underrepresented in these roles.

In general, these patterns point to a persistent gender inequality in manufacturing employment, encompassing not only participation rates but also education and skill levels. Moreover, although educational attainment among both male and female manufacturing employees has steadily improved in recent years, this progress has not translated into a proportional increase in medium or high-skilled employment, particularly for women.

Moving forward, national statistical systems and international agencies must prioritize the collection of sex-disaggregated industrial statistics, going beyond headline indicators to capture the qualitative dimensions of women's work. Thereby, the systematic integration of gender perspectives into industrial policy, supported by robust data, is essential to ensure that women are not only present in manufacturing but are also empowered to advance, innovate, compete, and lead. A more detailed analysis of gender dynamics in manufacturing can be found in the thematic chapter on women in manufacturing in UNIDO's *SDG 9 Progress Report 2025* [39].



Closing gender gaps
in the sector requires bold
evidence-informed
policies

Detailed
analysis of gender
dynamics
can be found in the
SDG 9 Progress
Report 2025

3.6 Environmental impact of manufacturing

The manufacturing sector plays a major role in shaping the environmental footprint of economic activity. This is a critical consideration: although manufacturing is one of the key engines of sustained growth and a central pillar of many development strategies, it can also generate significant environmental pressures if adequate policies and safeguards are not in place. These impacts include direct and indirect greenhouse gas (GHG) emissions from energy use; air and water pollution; intensive use of materials, particularly minerals, metals, and other natural resources; waste generation; and land-use pressures.

This section provides a brief, non-exhaustive overview of some of these environmental effects. Previous editions of the *Yearbook* have examined additional dimensions. As will be discussed later, however, the ability to undertake a comprehensive assessment remains limited by persistent data gaps and insufficiently harmonized environmental reporting.

Manufacturing is a significant contributor to global GHG emissions. This section presents recent data on CO₂ emissions, the largest component of GHGs and explicitly referenced under SDG 9. As noted in Section 2.2, manufacturing accounted for about 17.5 per cent of global CO₂ emissions from fuel combustion, about 6.0 Gt, in 2022, the latest year with data available. Although absolute emissions have remained relatively constant in recent years, they remain at or near historical peak levels.

Three industries together were responsible for 67.3 per cent of manufacturing's CO₂ emissions in 2022: iron and steel, non-metallic minerals, and chemicals and petrochemicals. Their combined contribution to the sector's total has increased over time, rising from 57.9 per cent in 2000. Figure 3.36 shows both the long-term trajectory of manufacturing CO₂ emissions and the evolving distribution of emissions across these sectors.

The iron and steel industry (ISIC Rev. 4 group 241 and class 2431), critical for infrastructure, construction, and automotive production, is the largest CO₂ emitter in manufacturing. It accounted for 35.9 per cent of the sector's total emissions in 2022, generating 2.1 Gt of CO₂, a 1.0 per cent increase compared with 2021. Emissions in this industry are still increasing, and while recent data suggest a possible deceleration, it remains too early to determine whether a plateau has been reached.

The non-metallic minerals sector (ISIC Rev. 4 division 23), which includes cement, ceramics, and glass production, ranked second in global manufacturing emissions. It released 1.1 Gt of CO₂ in 2022, 18.8 per cent of the sector's total. This industry reached its emissions



Manufacturing
plays a critical role
in mitigating the
environmental
impact

of overall economic activity

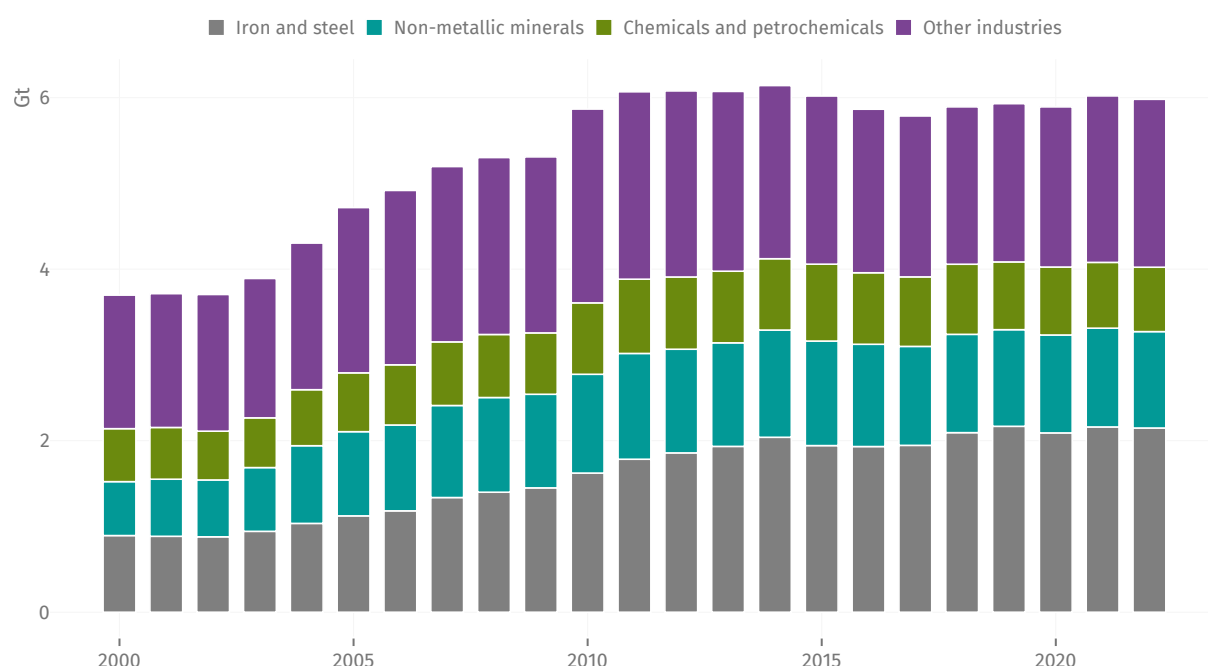


Iron & steel,
non-metallic
minerals, and
chemical sectors

collectively account for

67.3%

of CO₂ emissions from
manufacturing

Figure 3.36 | CO₂ emissions from fuel combustion by manufacturing industry

Source: [20; 50]

Note: Iron and steel corresponds to ISIC Rev. 4 group 241 and class 2431; non-metallic minerals to ISIC Rev. 4 division 23; and chemicals and petrochemicals to ISIC Rev. 4 divisions 20 and 21.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

peak in 2014, and by 2022 emissions were 10.1 per cent below that historical high.

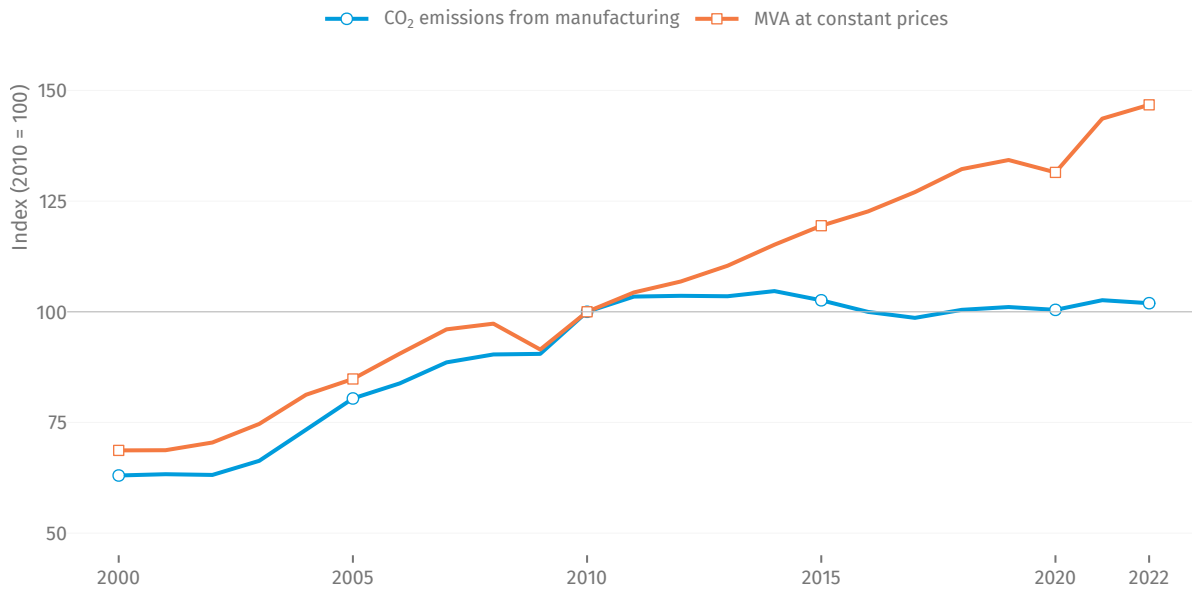
The chemical and petrochemical industries (ISIC Rev. 4 divisions 20 and 21), producing fertilizers, plastics, and a wide range of chemicals products, emitted 0.75 Gt of CO₂ in 2022, equivalent to 12.6 per cent of manufacturing's emissions. These industries are highly energy-intensive, with emissions generated both through fossil fuel combustion and chemical reactions intrinsic to production. Notably, emissions appear to have peaked in 2015, and by 2022 stood 16.2 per cent below that level.

The remaining manufacturing industries collectively emitted 1.96 Gt of CO₂ in 2022. While their individual contributions are smaller, they still present important environmental challenges, including high resource consumption and substantial waste generation.

Several industries have significantly reduced their CO₂ emissions over time. In addition to the non-metallic minerals and chemical sectors, substantial reductions have been achieved in food products (ISIC Rev. 4 divisions 10 to 12), machinery (ISIC Rev. 4 divisions 25 to 28), and transport equipment (ISIC Rev. 4 divisions 29 and 30). A particularly successful example is the textile industry (ISIC Rev. 4 divisions 13 to 15), which cut its emissions by 58.3 per cent between its peak in 2005 and 2022, despite continued growth in production. This



The **chemical sector** is among the industries whose CO₂ emissions have peaked and are now in a **declining trend**

Figure 3.37 | Global MVA and CO₂ emissions from manufacturing

Source: [9; 50]

Note: Due to their different measurement units, both series have been converted to an index with base year 2010. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

decline reflects cleaner and more efficient manufacturing processes, a shift towards green energy, and increased adoption of sustainable practices. For instance, since 2000, the EU Innovation Strategy for Textile Sustainability has increasingly focused on circularity, promoting durable, repairable, and recyclable products [51]. In Asia, major global producers have similarly reduced emissions through a greater reliance on renewable energy sources, enhanced energy efficiency measures, and waste-repurposing initiatives, among other factors [52].

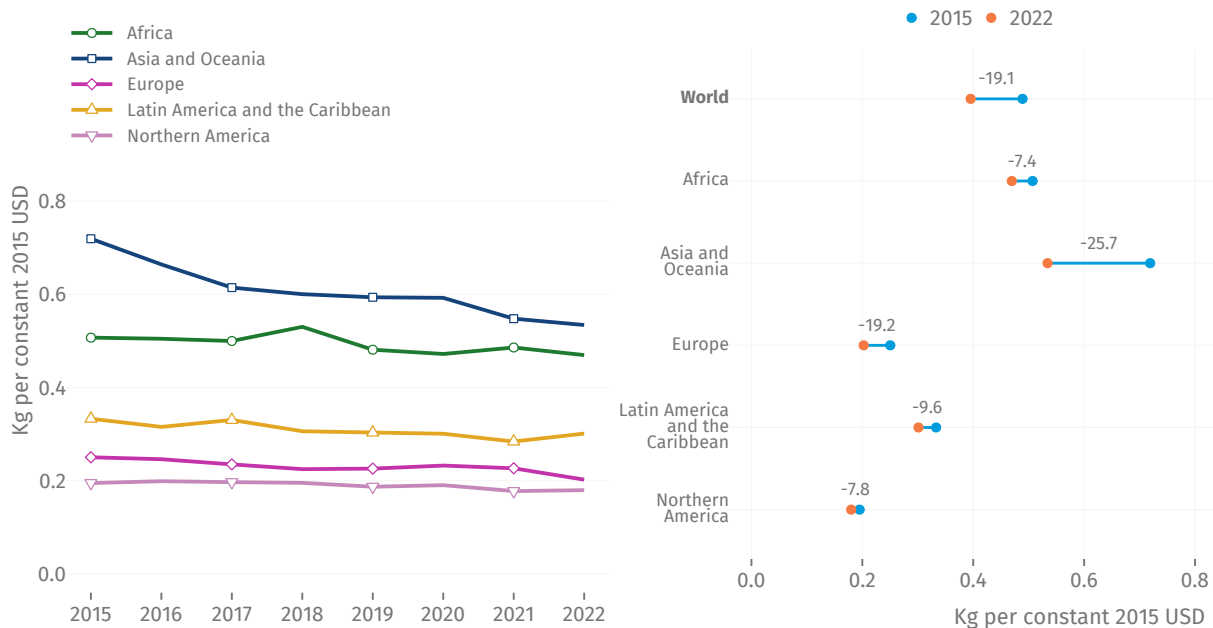
These developments point to a noticeable decoupling of CO₂ emissions from global manufacturing activity. As shown in Figure 3.37, despite the continuous rise in MVA, manufacturing-related CO₂ emissions have remained broadly flat since 2011. Mitigation efforts, together with the growing relative importance of higher-technology industries, have helped reduce both energy and carbon intensity, leading to lower emissions. The increasing affordability of low-emission technologies has further expanded the range of available options across manufacturing activities.

As a result, the global CO₂ emission intensity of the manufacturing (that is, the units of CO₂ emitted per unit of MVA) has steadily declined. By 2022, intensity had fallen 24.3 per cent relative to 2000, and 19.1 per cent relative to its 2015 level, at the start of the 2030 Agenda. Figure 3.38 shows that Asia registered the largest improvement, reducing emission intensity by 25.7 per cent since 2015, although it remains the region with the highest overall level. Europe achieved the second-largest reduction (19.2 per cent), narrowing the gap with Northern



A decoupling between
CO₂ emissions
and
manufacturing activity
has been evident
since 2011

Figure 3.38 | CO₂ emissions from manufacturing per unit of MVA (left) and its comparison between 2015 and 2022 (right) by region



Source: [20]

Note: In the right-side chart, the numbers above the lines or points indicate the change in the CO₂ emissions intensity of manufacturing (in percentage growth rate) between 2015 and 2022 for each country group.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

America, the best performing region globally. Africa and Latin America and the Caribbean also recorded improvements of 7.4 per cent and 9.6 per cent, respectively.

While this section focuses primarily on CO₂ emissions due to data availability, it is important to acknowledge the substantial role of other GHG, such as methane and nitrous oxide, both of which have far higher global warming potential. Methane, for instance, is 27-30 times more potent than CO₂ over a 100-year timescale, while nitrous oxide is 273 times more potent [53]. These gases are often released during industrial processes and the production of chemicals, meaning that exclusive attention to CO₂ provides only a partial view of manufacturing's climate impact.

Moreover, the environmental effects of manufacturing extend beyond GHG emissions. They include resource-intensive practices, such as excessive water use, pollution from industrial wastewater, large-scale extraction of materials, and significant land use. For example, the textile industry is a major consumer of water and chemicals, whereas the paper industry (ISIC Rev. 4 divisions 17 and 18) contributes to deforestation and water pollution. Transporting manufactured goods also contributes to the sector's broader footprint, adding to GHG emissions and resource depletion.

A comprehensive assessment of manufacturing's impact is currently

Other **greenhouse gases** can have a vast higher **global warming potential** than CO₂

Environmental considerations of manufacturing should also account for **water pollution, material extraction, land use,** and other factors

constrained by persistent data gaps across many environmental dimensions beyond CO₂ emissions. This lack of detailed and harmonized information remains a major obstacle to designing effective mitigation strategies, such as expanding circular economy practices, improving waste management, shifting to green energy, and adopting cleaner production techniques. Strengthening data collection systems and integrating diverse environmental indicators into manufacturing policy frameworks are essential for supporting the development of robust strategies that reduce emissions while also addressing issues like water pollution, waste generation, and resource overuse.



**Circular economy
principles**

can help reduce manufacturing's
**environmental
impacts**

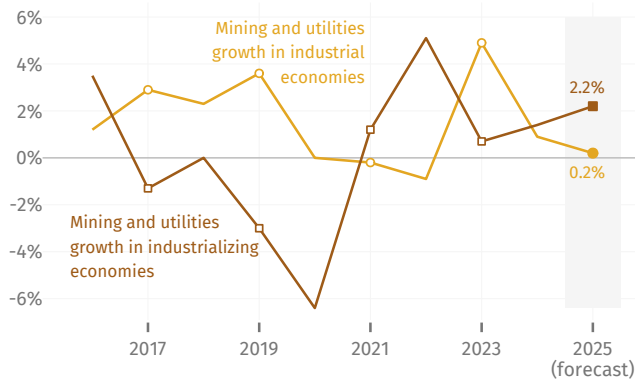


4 Mining and utilities in the spotlight

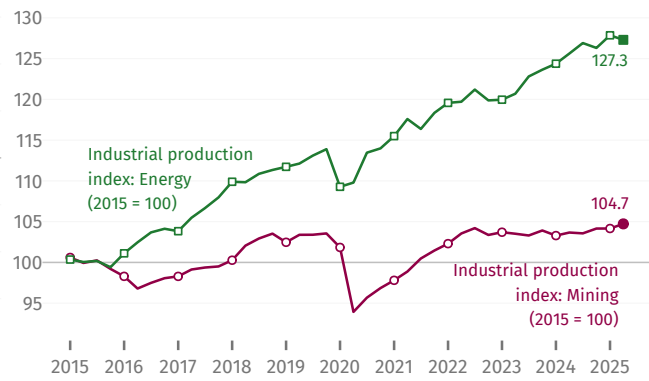
4.1 Current trends and distribution of world mining and utilities . .	89
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4 Mining and utilities in the spotlight

Key figures

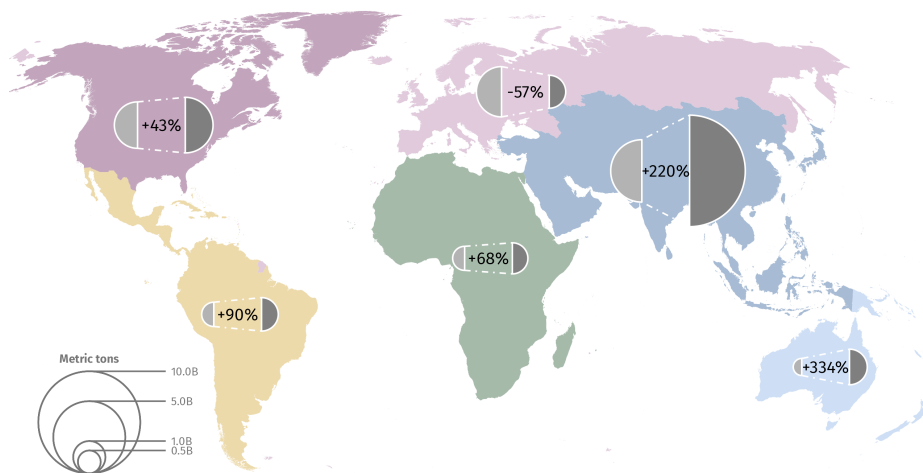


Growth rates of mining and utilities value added in **industrializing economies** follow a more volatile path than in **industrial economies**.



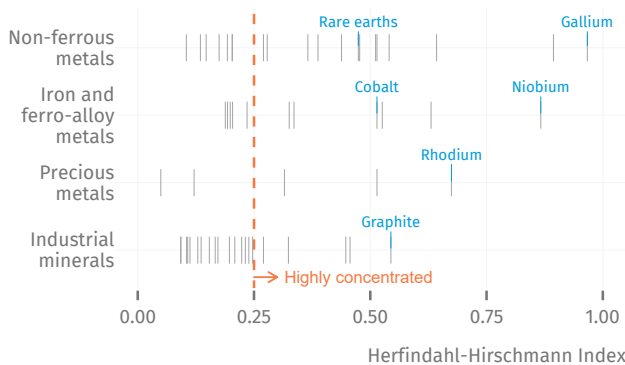
Energy production expanded by 27% from 2015 to the second quarter of 2025, while **mining** production has experienced slower growth.

Regional development of mineral raw materials production



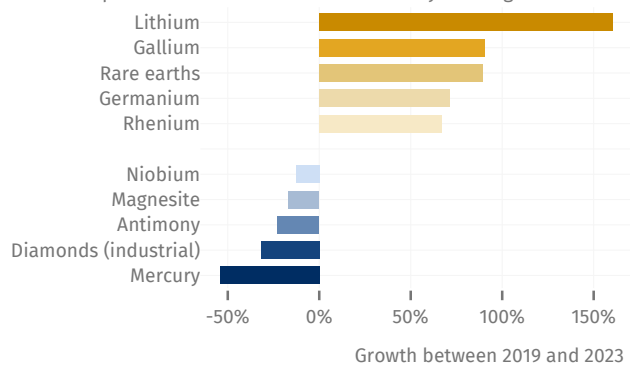
Between 1985 and 2023, **Asia** emerged as the leading producer of mineral raw materials. Concurrently, **Northern America**, **Latin America and the Caribbean** and **Oceania** overtook **Europe**, marking a notable reconfiguration of the global mining landscape.

Producer concentration of mineral commodities in 2023



In 2023, **production** of many **mineral raw materials**, like gallium, niobium, rhodium, graphite, and rare earths was highly **concentrated** in a few countries, revealing strategic **supply chain dependencies**.

Top and bottom mineral commodities by recent growth



Non-ferrous metals dominated recent growth trends, taking all **top five** spots. **Lithium** led with a **160%** increase between 2019 and 2023, followed by gallium and rare earths. Mercury saw the steepest decline.

4.1 Current trends and distribution of world mining and utilities

Although the mining and utilities sectors together account for only 4.5 per cent of global GDP, they exert an outsized influence on many areas of the economy that are strategically important or essential for daily life. For instance, these sectors encompass:

- ▶ The extraction of minerals and metals, including critical minerals, that form the foundation of many manufacturing products;
- ▶ The extraction of oil, gas, and other energy-related resources;
- ▶ The production and distribution of electricity and water, as well as the treatment of wastewater, relied upon daily by households and businesses;
- ▶ Recycling and material recovery, which play a key role in mitigating the environmental impacts of production and consumption.

Despite their strategic significance, statistical information on these sectors remains relatively limited. This is partly due to the fact that activity in each sector is typically concentrated among a small number of firms, which restricts data dissemination under statistical disclosure control (SDC) rules. As a result, substantial data gaps persist, and information is often available only in aggregated form rather than by individual sector.

This chapter therefore draws on available information from national accounts, IIPs, and novel data sources to present an overview of the combined mining and utilities sectors. The first section examines the short- and long-term trends in global production, followed by a detailed analysis of the global output of mining commodities in Section 4.2. The final section discusses the main environmental challenges associated with these sectors.

4.1.1 Annual production in mining and utilities

Figure 2.3 already compared the trajectories of broad industrial sectors, highlighting the generally weaker performance of the mining and utilities sectors relative to manufacturing in most, though not all, regions. Overall, the combined mining and utilities sector grew moderately by 1.0 per cent in 2024, marking a significant slowdown compared with 2023 and a return to the subdued trend observed in previous years. According to UNIDO estimates, output in this sector is expected to decelerate further to 0.82 per cent in 2025.

Table 4.1 presents an overview of recent trends in mining and utilities value added (MUVA) by region and country group. The slowdown observed in 2024 was widespread across most country groups by

Mining and utilities

are strategic sectors for **economic growth**, **energy security**, and **social well-being**



Statistical disclosure control

may pose challenges to the dissemination of **disaggregated** mining and utilities data



Globally,
mining and utilities
value added
grew by
1.0%
in 2024

Table 4.1 | Growth rate of MUVA for selected country groups and regions

	2022	2023	2024	2025f
	Per cent			
World	0.9	3.6	1.0	0.8
UNIDO groups by stage of industrial development				
High-income industrial economies	-2.8	5.9	1.1	0.5
High-income industrializing economies	11.7	-3.6	0.0	1.6
Middle-income industrial economies	2.3	3.4	0.5	-0.3
Middle-income industrial economies (excl. China)	-0.1	1.0	1.3	1.1
China	3.1	4.2	0.2	-0.7
Middle-income industrializing economies	1.4	2.7	2.1	2.2
Low-income economies	10.4	7.8	2.7	6.6
Main regions				
Africa	-1.0	2.2	1.0	3.2
Asia and Oceania	4.3	1.6	0.5	0.8
Europe	-6.8	0.1	0.6	-1.6
Latin America and the Caribbean	3.2	3.4	4.6	2.2
Northern America	-1.8	13.2	1.5	1.6

Source: [9]

Note: 2025f refers to UNIDO forecasts for the year 2025.

Data used to create this table can be downloaded at the [UNIDO Statistics Portal](#).

stage of industrial development, with the exception of high-income industrializing economies, which improved from a 3.6 per cent decline in 2023 to a near-zero growth rate in 2024. The best performance was recorded in low-income economies and middle-income industrializing economies, which grew by 2.7 per cent and 2.1 per cent, respectively. A similar pattern is expected in 2025, when low-income economies are projected to rebound to the high growth rates seen in the immediate post-COVID-19 period.

The lower part of the table presents results by region. In 2024, Latin America and the Caribbean recorded the strongest performance, accelerating relative to previous years and achieving 4.6 per cent growth. Northern America followed with 1.5 per cent, representing a strong correction after the double-digit increase observed in 2023. All other regions grew at subdued rates of 1 per cent or less. For 2025, the mining and utilities sectors in Africa are expected to gain momentum, whereas Latin America and the Caribbean is projected to slow slightly, and Europe to record a modest decline.

Similar to the manufacturing sector, China remained the world's largest producer in the mining and utilities sectors in 2024, accounting for 20.5 per cent of the sectors' global value added. However, the country's output growth slowed markedly during the year, with MUVA expanding by only 0.21 per cent. The United States of America remained the second-largest producer, contributing 16.9 per cent of global MUVA. The other economies in the top five (the Russian Federation, Saudi Arabia, and Japan) each represented less than 5 per cent of the global total. Compared to manufacturing, activity in the mining and utilities

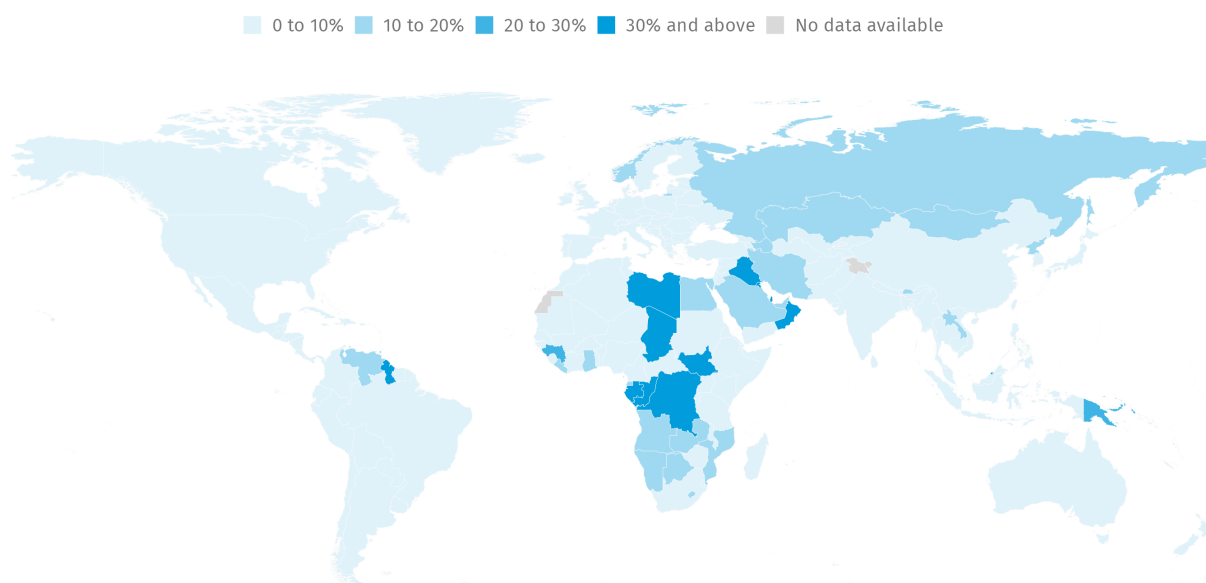


Latin America and the Caribbean

recorded the highest
growth rate
in mining and utilities value
added in 2024

Top 5 countries with the largest mining and utilities sectors and their global MUVA share in 2024

1. China (20.5%)
2. United States (16.9%)
3. Russian Federation (3.8%)
4. Saudi Arabia (3.7%)
5. Japan (3.4%)

Figure 4.1 | MUVA as a proportion of GDP by economy, 2024

Source: [9]

Note: The share is based on MUVA and GDP in constant 2015 USD.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

sectors is more evenly distributed across a broader range of countries.

A key difference from manufacturing lies in the relative weight of mining and utilities in national economies, as illustrated in Figure 4.1. Many African economies depend heavily on these sectors, with several registering MUVA-to-GDP ratios exceeding 30 per cent: Chad, Congo, the Democratic Republic of the Congo, Gabon, Libya, and South Sudan. Similarly, four economies in Western Asia (Iraq, Kuwait, Oman, and Qatar) also exhibited MUVA shares above this threshold.

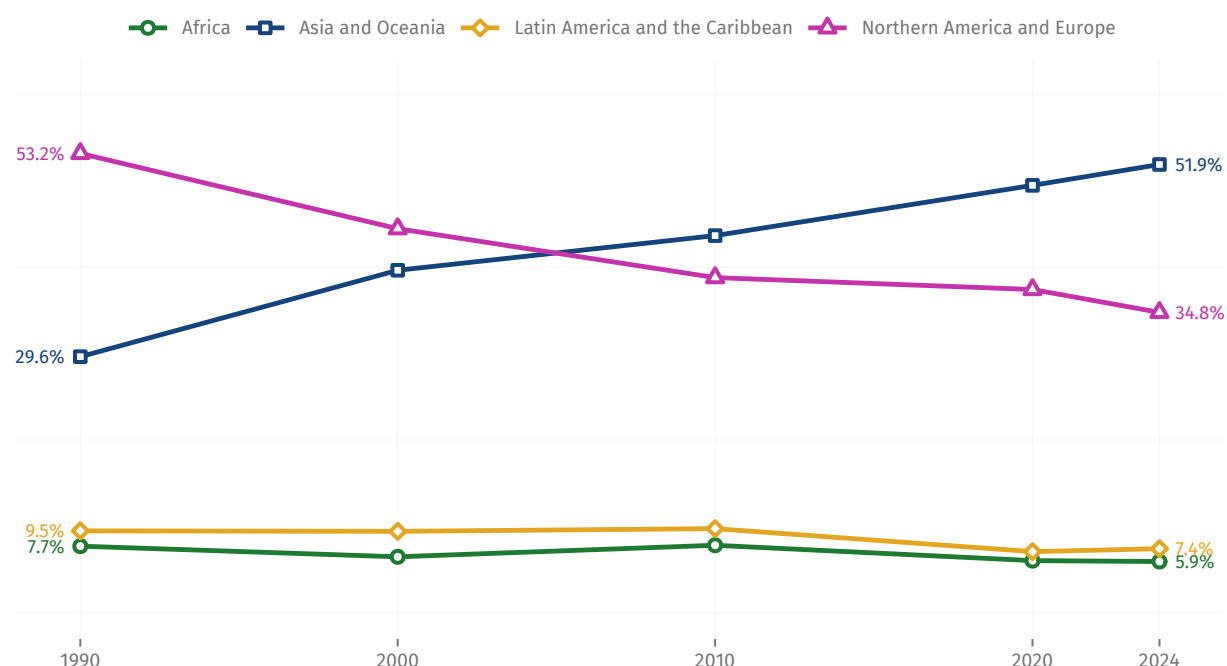
In other regions, the contribution of mining and utilities to GDP remains modest, with a few notable exceptions, such as Azerbaijan, Brunei Darussalam, the Lao People's Democratic Republic, Norway, and Papua New Guinea. The economy with the highest share of MUVA in GDP in the world was Guyana, where the extraction of recently discovered hydrocarbon resources pushed the sectors' contribution to over 70 per cent of GDP in 2024.

Industrial sectors also vary in the stability of their regional distribution. As discussed in Chapter 3, the geography of manufacturing has changed profoundly over the past decades. The mining and utilities sectors have experienced a similar reallocation of activity, though at a more gradual pace. Nonetheless, as shown in Figure 4.2, a clear shift from Northern America and Europe towards Asia and Oceania has taken place, albeit less dramatically than in the case of manufacturing.

Most economies with the highest contribution of mining and utilities to GDP are located in **Africa** and **Western Asia**



Guyana was the country with the highest MUVA-to-GDP ratio in 2024

Figure 4.2 | MUVA as a proportion of world MUVA by region

Source: [9]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

In 2024, Asia and Oceania accounted for an estimated 51.9 per cent of global MUVA, followed by Northern America and Europe with a combined share of 34.8 per cent. The shares of Africa (5.9 per cent) and Latin America and the Caribbean (7.4 per cent) have remained relatively stable, though both showed a gradual downward trend over time.

As in manufacturing, these regional shifts largely reflect the increasing concentration of global mining and utilities activity in China, whose share of world MUVA rose from 3.1 per cent in 1990 to 20.5 per cent in 2024. Other economies, such as Australia, India, and Qatar, also expanded their global shares in these sectors, although to a lesser extent. These country-level dynamics explain much of the overall increase in the weight of Asia and Oceania in global totals. Conversely, several economies in Europe and the Americas, but also in Asia, recorded declines in their global shares.

This transformation in the distribution of MUVA across economies and regions reflect not only the role of natural resource endowments, but also policy choices that prioritize sectoral development. Given the strategic importance of mining and utilities for economic growth, energy security, and social well-being, enhanced and more disaggregated statistics are essential for guiding evidence-informed policies and identifying sustainable growth opportunities in these sectors.

Economies with the largest gains in global MUVA share between 1990 and 2024

1. China +17.4%
2. India +1.8%
3. Australia +1.7%
4. Qatar +1.1%
5. Dem. Rep. of the Congo +0.7%

Economies with the largest declines in global MUVA share between 1990 and 2024

1. Russian Federation -4.2%
2. Venezuela (Bol. Rep.) ... -2.8%
3. United Kingdom -2.7%
4. Japan -2.1%
5. Germany -1.9%

4.1.2 Quarterly production in mining and utilities

As noted earlier, UNIDO also compiles and publishes sub-annual IIP data, which enable a more detailed and timely analyses than those based on annual national accounts. These data encompass four industrial sectors: mining and quarrying (ISIC Rev. 4 section B), manufacturing (section C), electricity and related industries (section D), and water supply and related sectors (section E).

Mining and electricity production indices are available for more than 100 economies, representing about 80 per cent of global mining and utilities value added, while data on water supply and other sectors included in ISIC Rev. 4 section E cover around 60 economies. Due to this limited availability of quarterly data as well as insufficient coverage for calculating country group and regional aggregates, this section is excluded from the analysis. Annex C summarizes the main characteristics and methodological aspects of these indices.

Recent developments in the mining and electricity sectors are illustrated in Figure 4.3. Global mining and quarrying activities have followed a volatile trajectory, registering only a modest production increase of 2.4 per cent in cumulative terms since the first quarter of 2022, the first year free from most pandemic-related disruptions. Production levels in this sector have remained relatively flat across both industrial and industrializing economies. Recent quarterly data indicate moderate growth of 0.73 per cent in industrial economies, while output in industrializing economies recorded a slight short-term decline. Given the dominant share of industrial economies in global mining output, the global growth rate largely reflects their recent performance and longer-term growth trends.

Global initiatives aimed at reducing the environmental impact of mining activities, as discussed in Section 4.3, will likely contribute to increased instability in the sector's prospects. At the same time, advances in technologies related to energy, batteries, machinery, and transportation may stimulate demand for minerals and metals, potentially leading to higher production levels in the sector.

In contrast, the production of electricity and other utilities classified under ISIC Rev. 4 section D exhibited stronger global momentum, increasing by 6.5 per cent in cumulative terms since early 2022. When disaggregated by country group, industrializing economies recorded a steady and sustained expansion, with electricity rising by more than 17 per cent over the past ten quarters. Conversely, industrial economies experienced a pronounced slowdown in recent years. In the most recent quarter, electricity output in industrializing economies continued to expand robustly by 1.1 per cent, while it declined by 0.81 per cent in industrial economies.



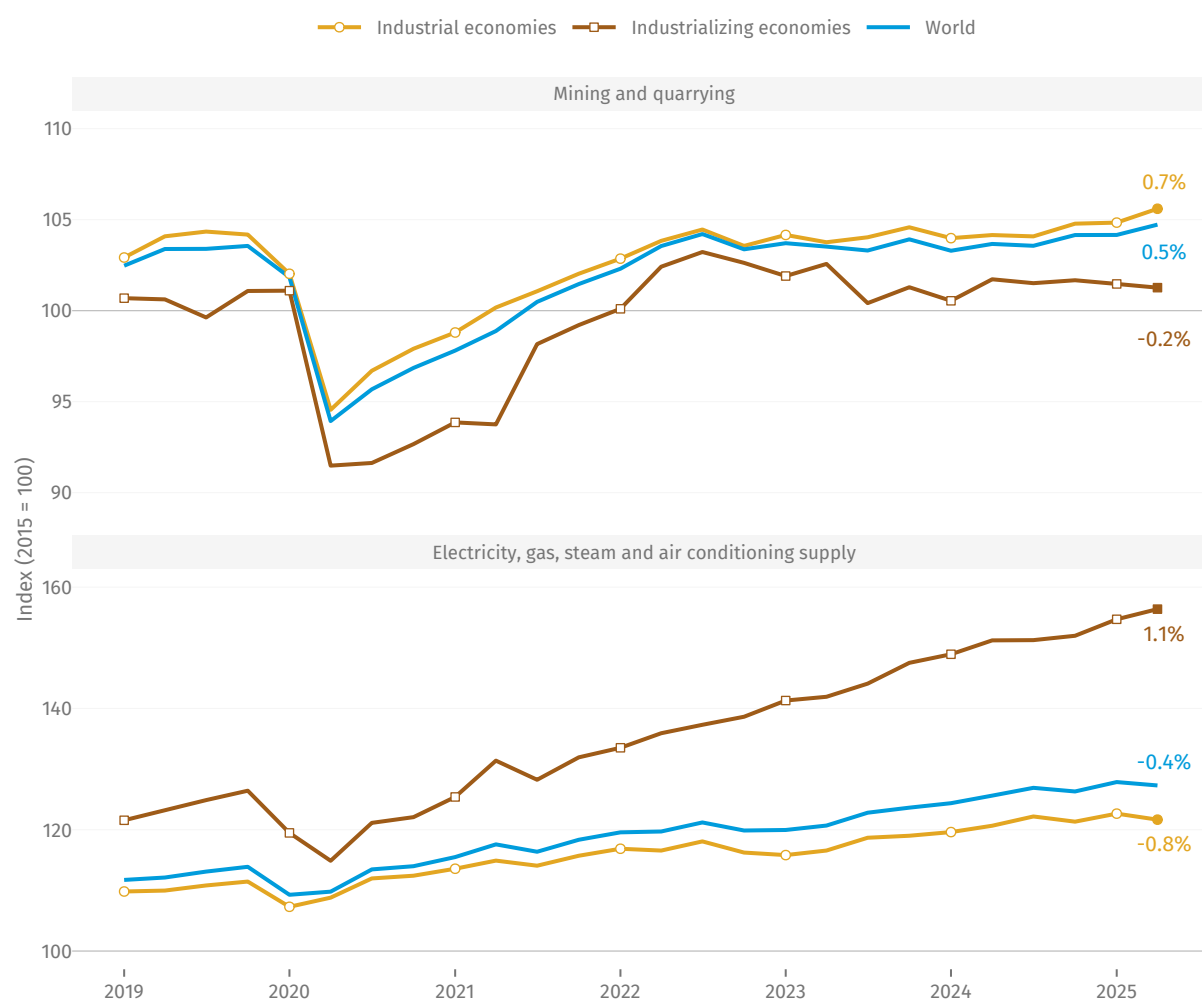
Quarterly
mining and utilities
production index
in UNIDO databases covers
around **100 economies**
representing **80%** of
global value added



Global **mining**
has followed a
volatile path,
currently growing
by only **0.5%**



Efforts towards
green industry
are likely **reinforcing**
volatility
in the mining sector

Figure 4.3 | Indices of production in the mining and electricity sectors by country group

Source: [19]

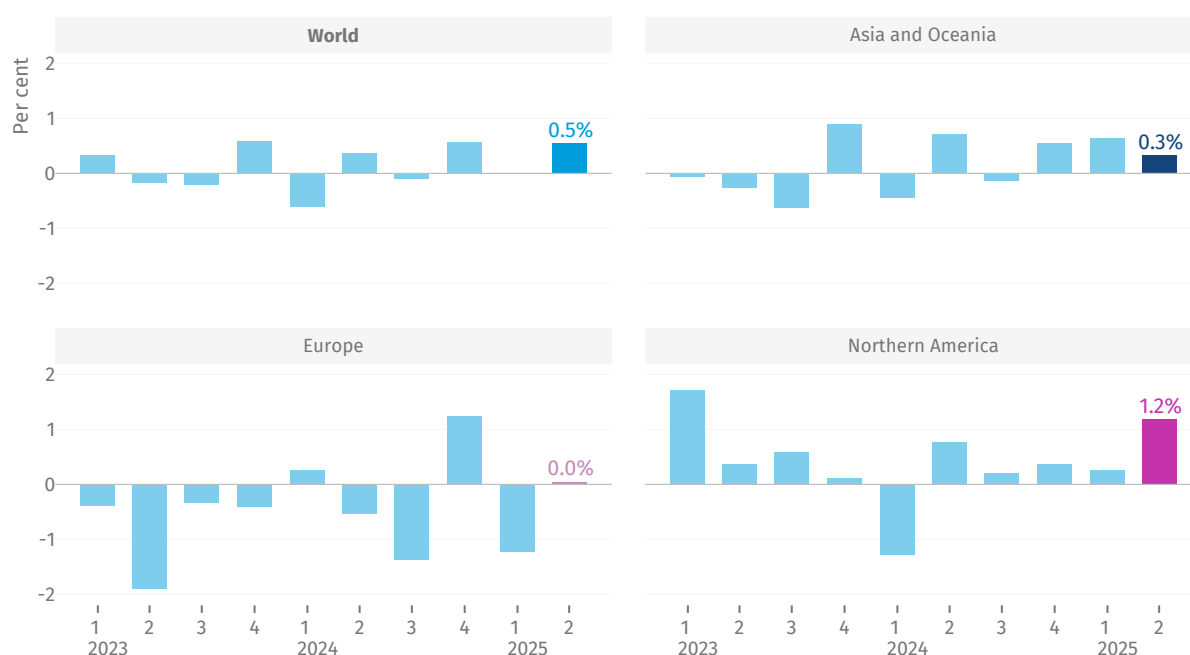
Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. The vertical axis scale differs across plots.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figures 4.4 and 4.5 present the latest regional trends in mining and electricity production. In 2024, Asia and Oceania accounted for more than half of the combined mining and utilities output, while Northern America and Europe contributed around 20 and 15 per cent, respectively. The recent growth in global mining production has been driven primarily by Asia and Oceania, alongside Northern America, which recorded a strong increase of 1.2 per cent in the second quarter of 2025. In contrast, global electricity production followed a more stable trajectory over time, with fewer fluctuations. The latest quarterly decline originates in reduced electricity generation in Europe and Northern America, where output fell by 1.3 and 2.4 per cent, respectively, offsetting the modest growth observed in Asia and Oceania.



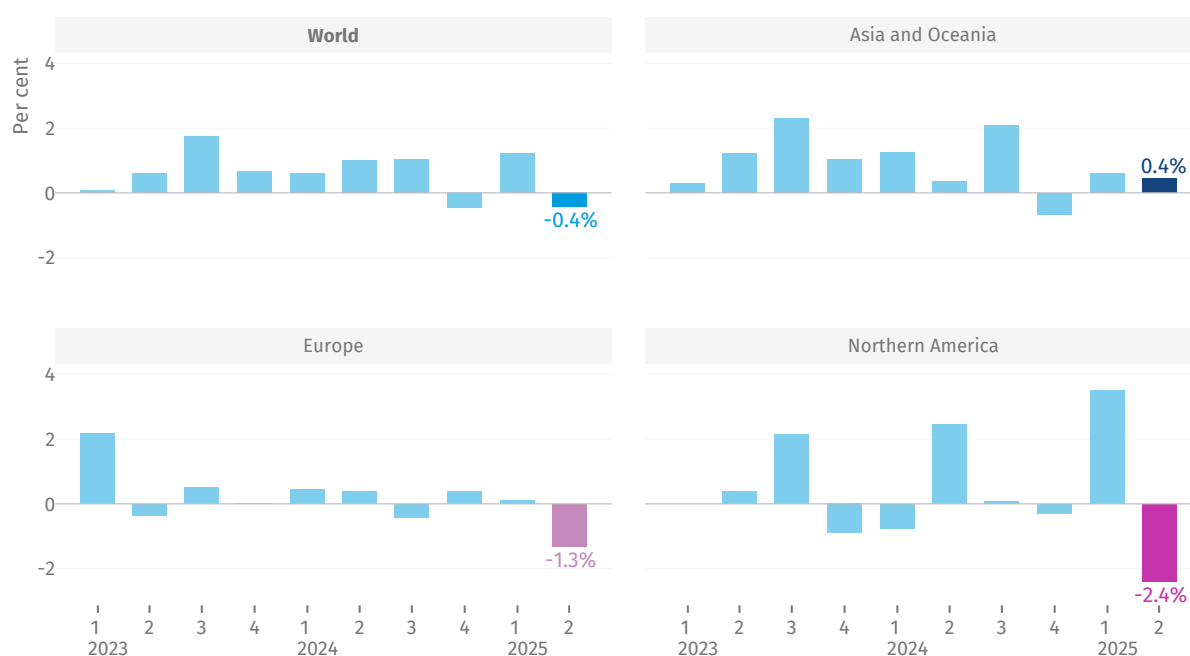
**Northern America
and Europe**
experienced a significant
reduction in **electricity**
production in Q2 2025

Figure 4.4 | Quarter-over-quarter growth rate of mining production by region

Source: [19]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Only regions with sufficient data coverage are included.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

Figure 4.5 | Quarter-over-quarter growth rate of electricity production by region

Source: [19]

Note: The labels indicate the most recent quarter-over-quarter growth rate, corresponding to the second quarter of 2025. Only regions with sufficient data coverage are included.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

4.2 Structure and transformation of mining and utilities industries

In previous editions of this Yearbook [37; 56; 15], the transformation of mining and utilities sectors has been analysed through the lens of structural business statistics (SBS), focusing on value added, employment, and investment by activity classification according to ISIC (e.g. ISIC Rev. 4 sections B, D, and E). While this approach provides essential insights into the economic structure, performance and transformation of these sectors, it often cannot reveal the underlying dynamics of specific mineral commodities that are increasingly shaping global industrial development.

In response to the growing importance of critical raw materials, strategic resource dependencies, and increasing energy demand, this section introduces a product-level perspective. Drawing on information published by World Mining Data (WMD) [54], the analysis shifts focus from aggregated industry indicators to monetary values and physical production volumes of individual mineral commodities. This approach enables a more granular understanding of global mining trends and a better alignment with resource security and sustainability topics. The emphasis on product-level data complements previous analysis based on industrial activities, offering a multidimensional view of the transformation underway in the combined mining and utilities sector.

Global production of mineral raw materials nearly doubled over the last four decades, underscoring the intensification of industrialization and the broadening of resource demand across sectors. According to WMD 2025 data [57], total world mining production (excluding bauxite) ⁱ surged from 9.6 billion metric tons in 1985 to 19.2 billion metric tons in 2023, a 99.0 per cent increase. This represents a compound average annual growth rate of approximately 1.8 per cent, underscoring the resilience of extractive industries. Sustained expansion led to the highest recorded level of production in 2023, reaffirming the pivotal role of mineral commodities in the global economy.

Global production of mineral raw materials: regional shifts and emerging leaders (1985–2023)

In the past forty years, global minerals production has experienced significant regional shifts, as illustrated in Figure 4.6. Asia has emerged as the leading contributor, tripling its output to surpass 12.0 billion metric tons in 2023, which accounts for more than 60 per cent of global production. Concurrently, Northern America, Latin America and the Caribbean, and Oceania have overtaken Europe, indicating a notable reconfiguration of the global mining landscape. Indeed, Northern

World Mining Data (WMD)

World Mining Data [54] is an internationally recognised annual statistical publication issued by the Austrian Federal Ministry of Finance. It provides comprehensive mineral production data from more than 160 countries, dating back to 1984, presented in both physical quantities and monetary terms. The data serves as a key resource for raw material policy analysis and commodity forecasting. The World Mining Congress (WMC) [55], founded in 1958 under UN auspices, plays a pivotal role in supporting data collection through dedicated questionnaires. National Committees of WMC member countries are among the primary contributors to the dataset, alongside other entities such as embassies and foreign trade representatives. Publicly accessible mining statistics from reputable sources, including the United States Geological Survey, are utilised for validation and cross-verification.

ⁱ Total production figures cannot be shown consistently including or excluding bauxite, due to the presentation of these figures in the source data.

Global minerals production
almost doubled
from **1985** to **2023**

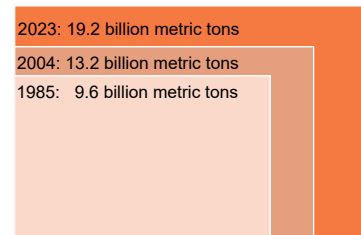
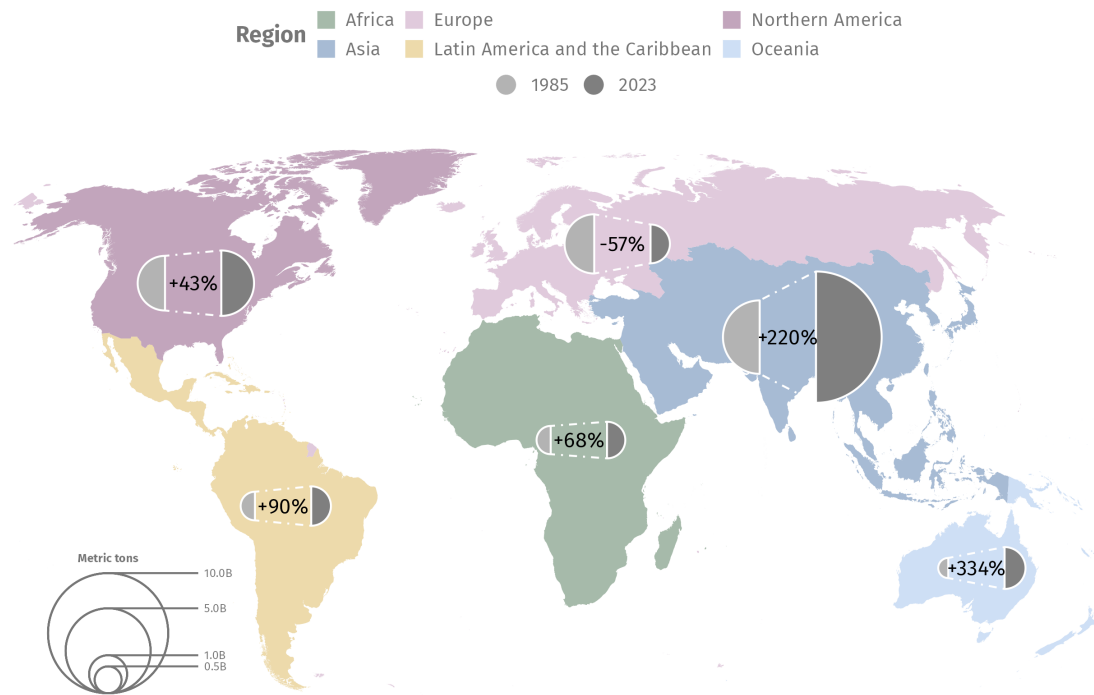


Figure 4.6 | Regional development of mineral raw materials production, 1985 and 2023

Source: [57]

Note: The pie charts embedded in the regional map depict mineral raw materials production in 1985 and 2023, measured in metric tons. The percentage labels indicate the regional growth rate between these two years.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

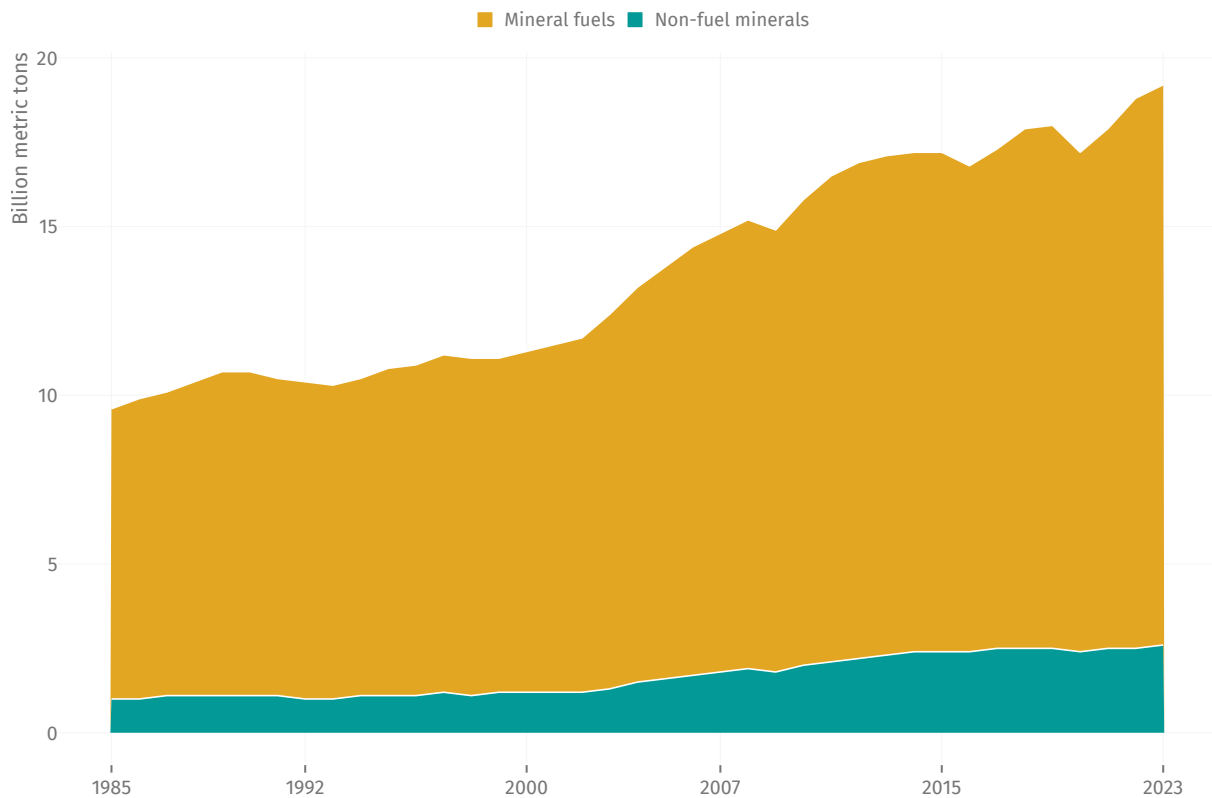
America saw a 43.3 per cent increase in production from 1985 to 2023. Latin America and the Caribbean nearly doubled their output during the same period, while Oceania experienced a remarkable 334.3 per cent increase, underscoring its expanding role in the supply of minerals. By contrast, Europe's production declined by 57.2 per cent, reflecting a regional shift towards reduced extraction, influenced by environmental and sustainability regulations in parts of Europe alongside cheaper and more accessible alternatives in other parts of the world. Africa showed significant growth, with an extraction of 0.93 billion metric tons in 2023, marking a 67.6 per cent increase compared to 1985.

The dynamic regional upswing and redistribution of mining activity, shaped by industrial expansion, evolving resource demand, and geopolitical transformations, is reflected at the national level, with China emerging as the world's leading producer, accounting for an impressive 26.9 per cent of global output in 2023. This figure alone underscores Asia's predominance, as China's share is more than twice that of the United States of America (12.3 per cent), the second-largest producer. Alongside the Russian Federation (8.0 per cent) and India (7.0 per cent), the top four countries collectively contribute with 54.2 per cent more than half of the world's total minerals output. This concentration high-

Leading producers in 2023 and their share in global production of mineral raw materials

- ▶ China (26.9%)
- ▶ United States (12.3%)
- ▶ Russian Federation (8.0%)
- ▶ India (7.0%)
- ▶ Australia (6.6%)
- ▶ Indonesia (4.5%)
- ▶ Saudi Arabia (3.3%)
- ▶ Canada (2.8%)
- ▶ Brazil (2.7%)
- ▶ Iran (2.6%)

In 2023, the
4 largest producers
accounted for
54.2%
of global minerals
production (incl. bauxite)

Figure 4.7 | Development of mineral raw materials production by broad category

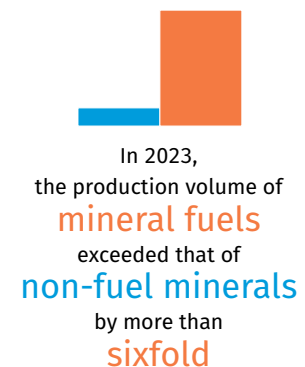
Source: [57]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

lights the strategic significance of these nations within global resource supply chains. Other notable contributors include Australia (6.6 per cent), Indonesia (4.5 per cent), and Saudi Arabia (3.3 per cent), reflecting increased investment and development across the Asia-Pacific and Middle East regions. Canada, Brazil, and the Islamic Republic of Iran complete the top ten, each accounting for between 2.6 and 2.8 per cent of global production.

Mineral composition: fuels dominate, yet non-fuel minerals continue to rise steadily

The composition of global minerals production can be broadly categorized into two groups: first, *mineral fuels*, which include coal, petroleum (incl. natural gas liquids), and natural gas, and second, *non-fuel minerals*. Although the term ‘non-fuel minerals’ is not used in the current version of WMD, it is nonetheless useful to employ such a category in further analysis to streamline findings. It encompasses iron and ferro-alloy metals like cobalt, nickel and titanium, non-ferrous metals such as aluminium, lithium, and rare earths, industrial minerals including phosphate rock and kaolin, and precious metals like gold and silver.



In terms of volume, production is overwhelmingly dominated by mineral fuels, as shown in figure 4.7. In 2023, these amounted to approximately 16.6 billion metric tons, exceeding the volume of non-fuel minerals, which were estimated at 2.6 billion tons, by more than sixfold. Over the past four decades, mineral fuels production has generally followed an upward trajectory, with occasional declines in specific years such as 1991–1993, 1999, 2009, 2016, and 2020. These fluctuations are likely attributable to geopolitical developments, global economic downturns, or shifts in energy policy. By contrast, non-fuel minerals have shown more gradual growth, yet their production has more than doubled since the mid-1980s, reflecting sustained demand for raw materials essential to industrial production, infrastructure, and technological advancement. Although petroleum, among other products, is classified as fuel, it also serves as a feedstock for a wide array of manufactured products. However, the available data do not differentiate between its use as a fuel for energy generation and its use as a material input. A further methodological consideration of significance pertains to the standardization of units within the data illustrated. To present all production figures in metric tons, data compilers, such as the contributors to WMD, have applied conversion factors to harmonise disparate mineral raw materials [57, pp. 10–11]. This practice enhances comparability across commodities, yet it also introduces interpretative nuances, particularly when analysing energy-related materials.

Mineral fuels from 1985 to 2023

A comparison of mineral fuels over time makes it essential to distinguish between their physical volume—measured in metric tons after harmonisation—and their energy content, expressed in tons of coal equivalent (TCE). This dual metric reveals important shifts in the global energy landscape between 1985 and 2023, as shown in Figure 4.8. In 2023, steam coal was the dominant fuel by mass, with a production volume of approximately 7.0 billion metric tons, contributing 41.7 per cent to the global mineral fuels production. However, its energy content was significantly lower, amounting to 5.2 billion TCE or a share of 27.7 per cent. This discrepancy underscores steam coal's relatively low energy density. Nonetheless, its share in both mass and energy terms increased compared to 1985, indicating a strong expansion in absolute output and a growing reliance on steam coal in certain regions. By contrast, petroleum, with a lower production volume in 2023 of 4.5 billion metric tons, yielded a much greater energy output of 6.5 billion TCE, thereby overtaking steam coal as the leading energy source when measured by calorific value. Interestingly, petroleum's share in the fuel mix declined between 1985 and 2023 from 32.3 per cent to 27.0 per cent by mass and from 40.6 per cent to 34.9 per cent



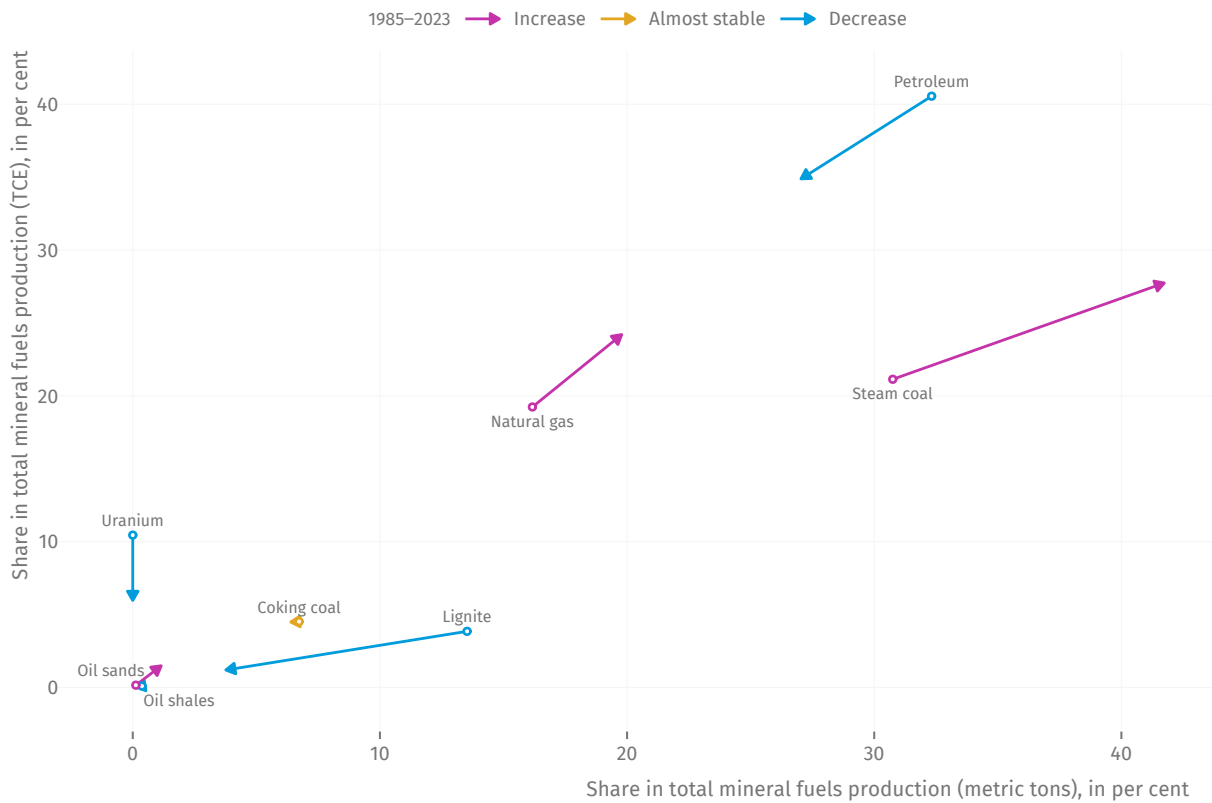
Mineral fuels
exhibited a steady
upward trajectory
with intermittent declines in
specific crisis years

Non-fuel minerals
have demonstrated
more gradual growth,
yet have more than doubled since
the mid-1980s



In 2023, **petroleum**
remained the **top energy**
source in the mineral fuels
production mix, though its share
has declined since 1985

Natural gas
grew in volume and energy
contribution due to its
efficiency and
strategic role

Figure 4.8 | Comparative overview of the change in mineral fuels production by individual fuel

Source: [57]

Note: Colours are used to categorize trends as increasing, decreasing, or almost stable.

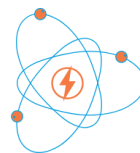
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by TCE, suggesting that its growth has not kept pace with other fuels, particularly steam coal and natural gas.

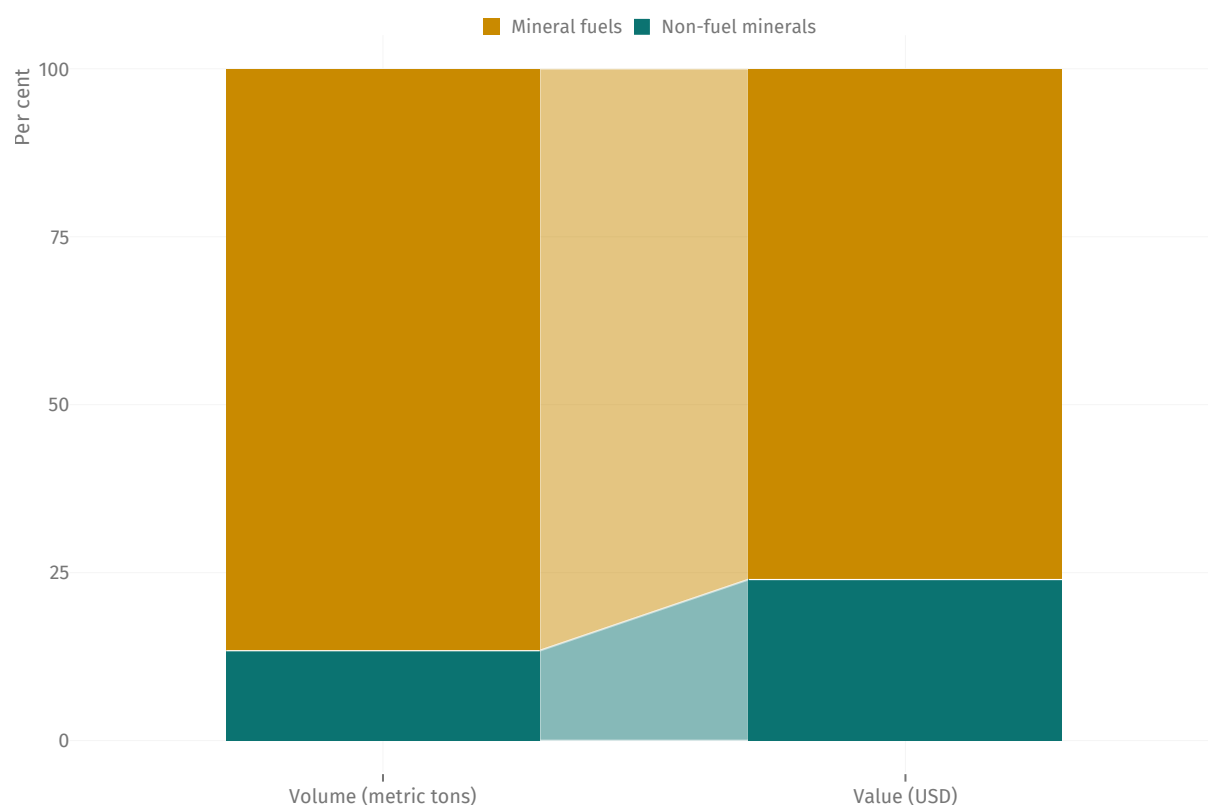
Natural gas demonstrated a notable discrepancy between mass and energy content. In 1985, its production stood at 1.4 billion metric tons (share of 16.2 per cent), yet its energy equivalent reached 1.9 billion TCE (share of 19.2 per cent). By 2023, both the volumes and shares increased to 3.3 billion metric tons and 4.5 billion TCE, representing 19.8 per cent and 24.2 per cent, respectively. This reflects natural gas's high energy efficiency and its increasing role in diversified energy portfolios, particularly in economies transitioning away from coal. Perhaps most striking is the case of uranium, which, despite a negligible production volume of just 62,297 metric tons (share almost 0.0 per cent), converted into an extraordinary 1.0 billion TCE (share of 10.4 per cent) in 1985. This highlights the exceptional energy yield of nuclear fuels and their disproportionate contribution to total energy supply. However, by 2023, uranium's energy share declined to 6.0 per cent, despite a slight increase in mass, pointing to a relative stagnation or decline in nuclear energy's role within the broader mix. Other fuels such as lignite, coking coal, and oil shales showed mixed trends. Lignite, for instance, saw a sharp decline in both volume and energy



Steam coal was the most extracted fuel by mass, yet its **energy contribution** was relatively lower



Uranium's small physical volume contrasts sharply with its substantial energy contribution, highlighting **nuclear energy's potency**

Figure 4.9 | Mineral raw materials production by volume and value, 2023

Source: [57]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

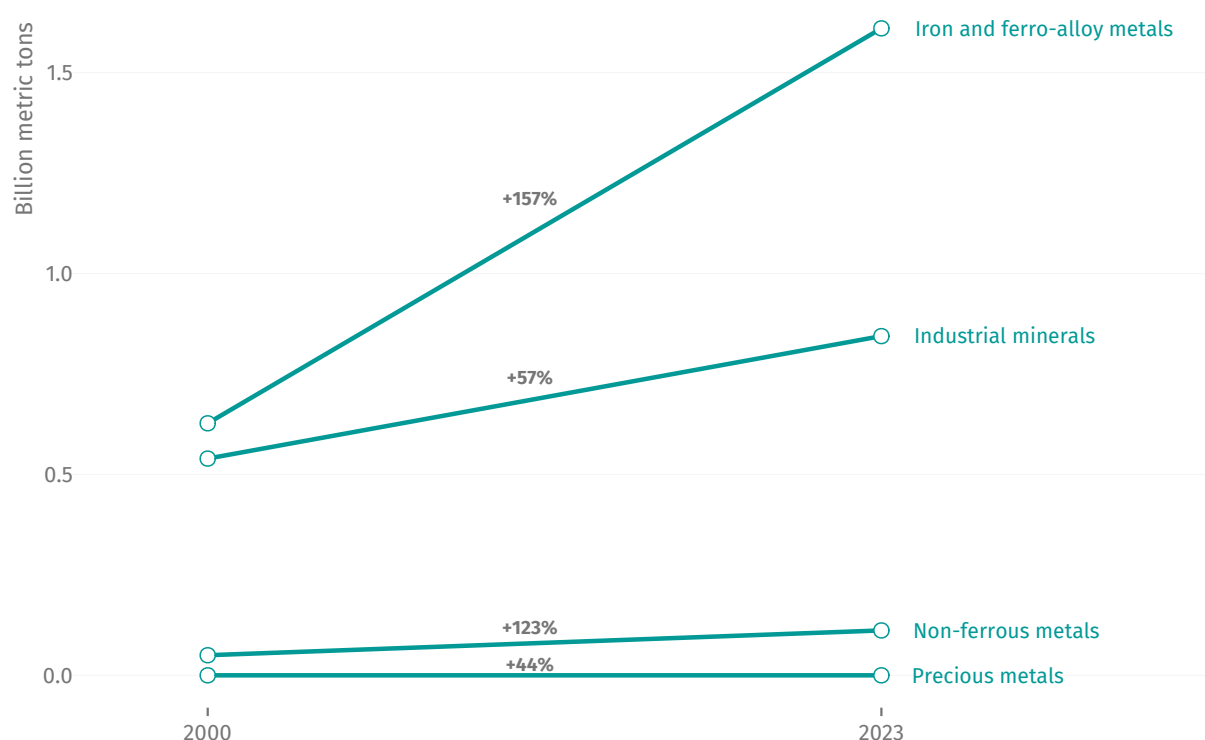
share, reflecting its comparatively low energy density and environmental concerns. Coking coal maintained a relatively stable share, while oil shales and oil sands remained marginal contributors despite some growth in absolute terms. This dual approach—tracking both volume in metric tons and energy content in TCE—offers a more nuanced understanding of global extraction patterns. It reveals not only the scale of resource use but also the shifting dynamics of energy efficiency, technological preferences, and strategic resource prioritization. The data suggest a gradual transition toward fuels with higher energy density and efficiency, even as traditional fuels like coal continue to expand in volume.

Monetary values of non-fuel minerals underscore their economic significance

Mineral fuels dominate global mineral raw materials production by volume. In 2023, mineral fuels accounted for 86.6 per cent of total production (excl. bauxite) by weight, while non-fuel minerals comprised only 13.4 per cent. However, when assessed in terms of value rather than volume, the landscape shifts, as also shown in Figure 4.9: non-fuel minerals represented 24.0 per cent of the total production value



With only
13% by weight,
non-fuel minerals
accounted for nearly
24% of total production value,
reflecting their strategic
**economic
importance**

Figure 4.10 | Change of non-fuel minerals production by mineral group

Source: [57]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

in USD (excluding diamonds due to difficulty in valuation), underscoring their strategic and economic significance. This disparity reflects to some extent the high market value of materials such as precious metals, non-ferrous metals and industrial minerals, which are indispensable for technological advancement, infrastructure development, and the transition to green energy. Nevertheless, for analytical consistency, production volume remained so far the preferred metric, as value-based assessments are subject to market volatility, particularly in the pricing of precious metals and energy commodities.

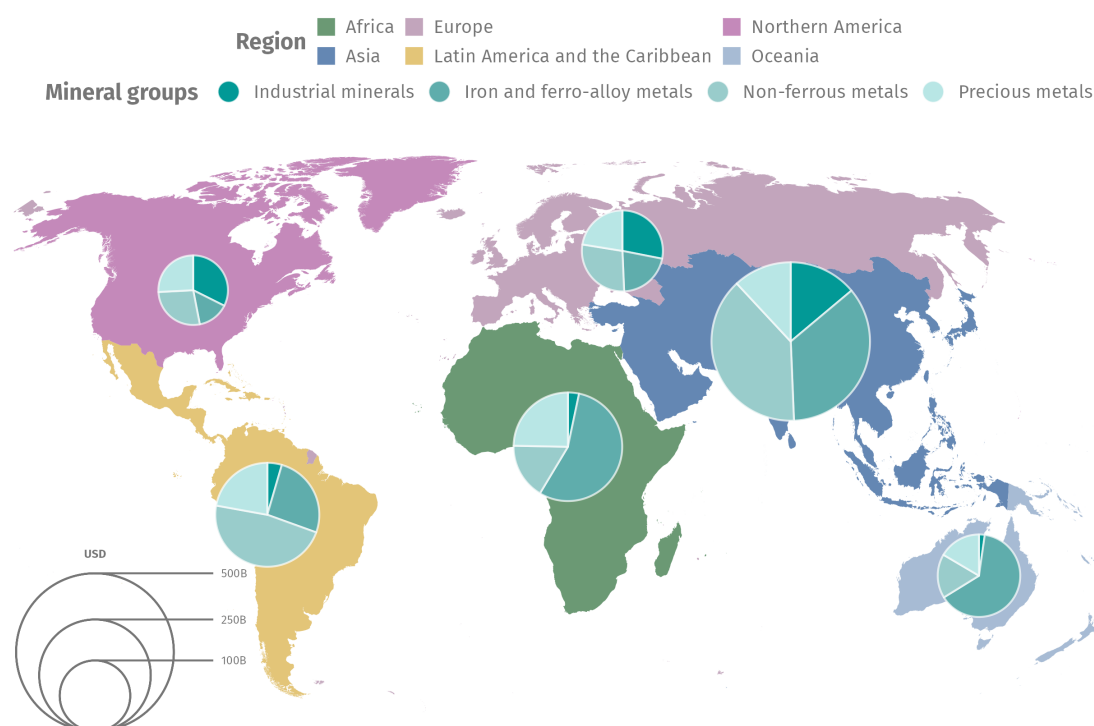
Iron and ferro-alloy metals lead the non-fuel minerals segment

Although representing a relatively small share by volume, precious metals, including gold, silver, palladium, and platinum constitute an economically significant category of non-fuel minerals. Between 2000 and 2023, this group recorded a notable growth rate of 44.1 per cent in volume. Iron and ferro-alloy metals emerged as the largest category by volume, reaching over 1.6 billion metric tons in 2023, almost twice the volume of industrial minerals and substantially surpassing both non-ferrous and precious metals. This expansion, amounting to nearly 157 per cent since 2000, highlights the sustained global demand for iron, manganese, chromium, and other materials. Other subcategories,

Iron and ferro-alloy metals far surpassed other categories by production volume in 2023



Growth in **non-ferrous metals, industrial minerals, and precious metals** reflects their **emerging roles**

Figure 4.11 | Regional composition of non-fuels minerals production, 2023

Source: [57]

Note: The pie charts illustrate the contribution of mineral groups to the total regional production of non-fuel minerals. The distribution is calculated using figures in USD.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

including non-ferrous metals and industrial minerals, have also experienced considerable growth. These trends underscore the multifaceted roles that non-fuel minerals play across various sectors, ranging from infrastructure development to manufacturing.

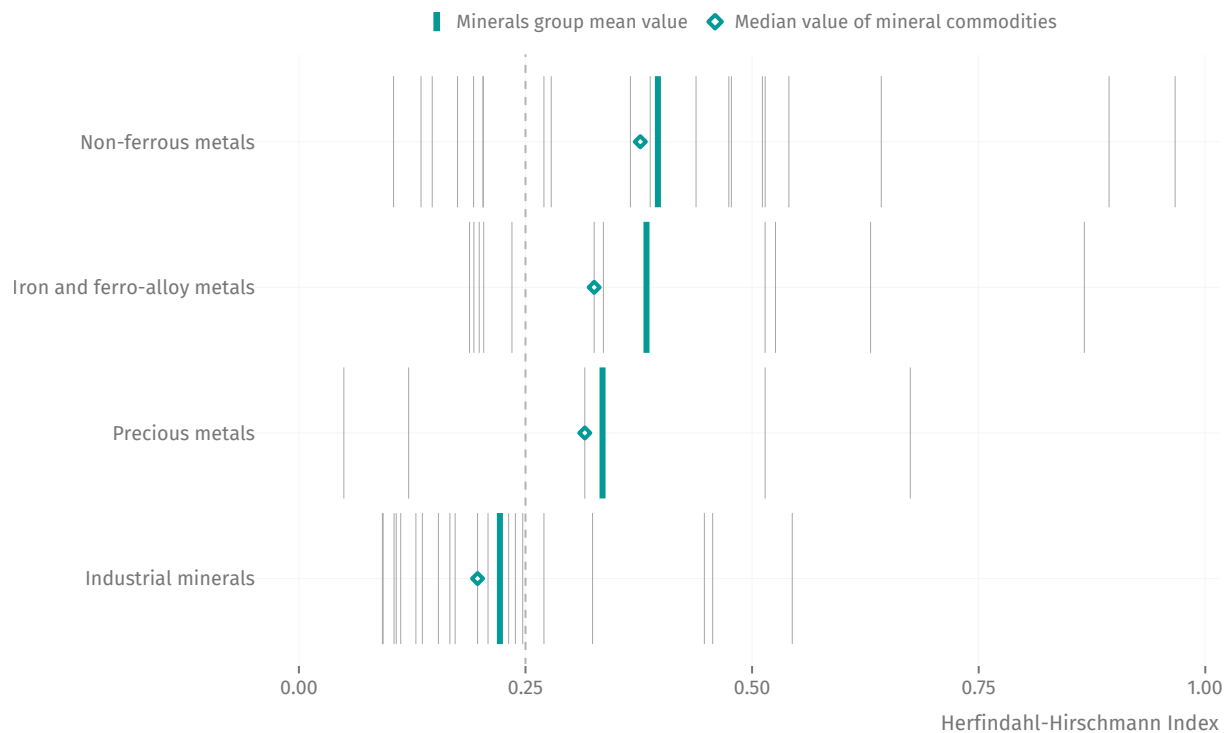
Regional value creation through non-fuel minerals

In 2023, the regional distribution of non-fuel minerals production, shown in Figure 4.11, highlights the considerable economic value generated by industrial minerals, non-ferrous metals, and precious metals, despite their relatively modest volumes compared to iron and ferro-alloy metals. Asia leads the global landscape, with a total non-fuel mineral output value of 503.9 billion USD. Notably, non-ferrous metals alone account for nearly 39 per cent of this total, underscoring their strategic importance in electronics and manufacturing.

Africa ranks second in overall output, showing significant contributions from iron and ferro-alloy metals as well as precious metals. Latin America and the Caribbean, although slightly behind in total value, stands out for its strong specialization in non-ferrous metals, which constitute nearly half of its non-fuel minerals production measured in USD. This reflects the region's vital role in supplying copper, aluminium,



Regions with
lower production volumes still
contribute significantly
to global value chains through
high-value minerals

Figure 4.12 | Global concentration of non-fuel minerals production by mineral group, 2023

Source: [57]

Note: The coloured lines represent the level of concentration, measured by the Herfindahl-Hirschman Index (HHI), for each mineral group. The point indicates the median HHI value across commodities, while the grey lines show individual commodities belonging to each group. The dashed vertical line marks the HHI threshold of 0.25, often used to indicate a high level of market concentration in economic analysis.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

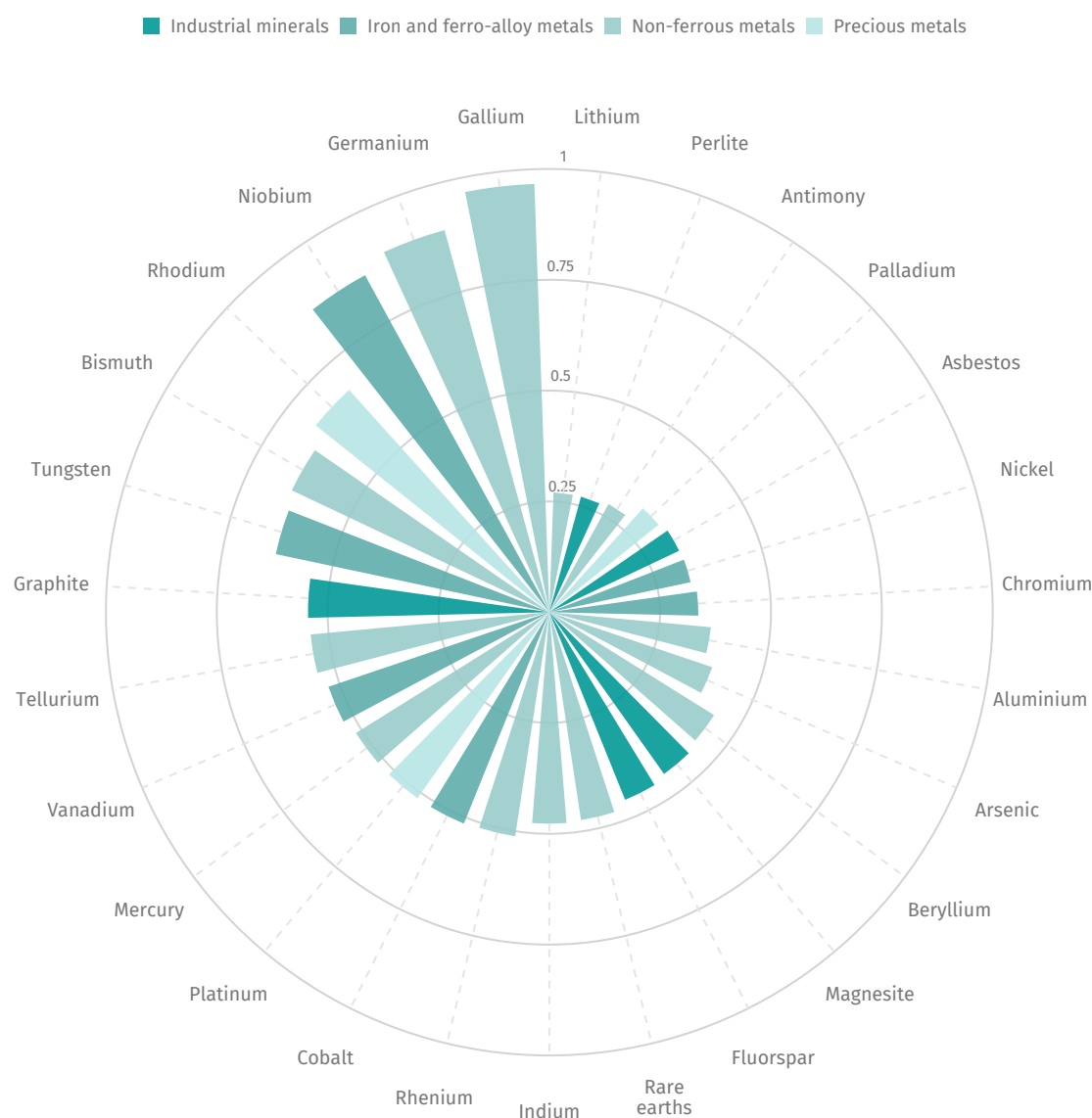
lithium, and other key raw materials. Oceania and Northern America, while lower in aggregated output value, also demonstrate notable concentrations in high-value categories such as industrial minerals and non-ferrous metals. Europe maintains a diverse mineral portfolio, with meaningful shares of all depicted mineral groups. These regional patterns illustrate that even regions with comparatively limited production volumes in minerals (see Figure 4.6 showing volume figures in metric tons for 1985 and 2023) can play a pivotal role in global value chains, owing to the economic significance and wide-ranging industrial applications of these mineral groups.

Strategic concentration in mineral supply chains

Although the regional distribution of non-fuel minerals appears relatively balanced when assessed by overall economic value, a more granular analysis of individual minerals reveals pronounced concentration in production volumes, linked to a limited number of producing countries. Figure 4.12 quantifies this phenomenon using the Herfindahl-Hirschman Index (HHI), where values above 0.25 are often



Production of **minerals**
essential for **advanced**
technologies like
germanium and rare earths is
highly concentrated

Figure 4.13 | Non-fuel minerals with a high producer concentration, 2023

Source: [57]

Note: The radar plot displays the Herfindahl-Hirschman Index (HHI) values for individual mineral commodities considered highly concentrated (exceeding a threshold of 0.25). Colours are used to categorize commodities according to mineral groups. Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

seen as indicating high concentration in economic analysis [58, p. 17]. In 2023, several minerals surpassed this threshold, indicating strategic dependencies within global supply chains. Among the various mineral groups, non-ferrous metals exhibit the highest number of highly concentrated resources. This includes gallium, germanium, bismuth, rhenium, tellurium, and rare earths, all of which are indispensable for electronics, renewable energy technologies, and advanced manufacturing processes. Within the iron and ferro-alloy metals category, minerals such as niobium, tungsten, and cobalt show elevated concentration levels, as further shown in Figure 4.13. These are essential for



China leads production of many **key minerals**, placing it at the **center** of global mineral **supply chains**

applications in aerospace, defence, and energy technologies. In the domain of precious metals, rhodium and platinum stand out due to their limited geographic availability and high economic value, further underscoring their strategic importance. Some industrial minerals also demonstrate notable concentration, particularly graphite, fluorspar, and magnesite. These materials are critical inputs for battery production, steelmaking, and the chemical industry. These figures showing concentration highlight the geopolitical and economic vulnerabilities inherent in mineral supply chains, especially with regard to materials that are vital for the transition to green technologies and the expansion of digital infrastructure.

Lithium, a non-ferrous metal among the highly concentrated commodities and known for its decisive role in energy storage solutions, has recorded the most pronounced production dynamics in most recent years with a growth rate of 160.2 per cent between 2019 and 2023. Australia led global production in 2023 with a share of 40.8 per cent, followed by Chile, which also anchors the strategically significant Lithium Triangle, a region comprising Chile, Argentina, and the Plurinational State of Bolivia. Collectively, the triangle contributes over 25 per cent of global lithium output, underscoring its importance in the global energy transition. China and Zimbabwe, with a global share of 21.3 and 6.8 per cent, are other major producers. Rare earths likewise witnessed substantial growth of 89.2 per cent during the same period, driven by increasing demand for components used in electronics, renewable energy technologies, and defence applications. China dominates mining of rare earths, accounting for more than 66 per cent of global production. Also, China emerged as the leading producer of 26 distinct mineral commodities, reaffirming its central role in global supply chains for numerous mineral raw materials.

This dominance also raises broader questions about strategic dependencies and development opportunities—issues explored in greater depth by the United Nations Trade and Development (UNCTAD)’s recent analysis of critical minerals and their role in sustainable industrial transformation [59]. The global reliance on a few key producers poses risks for countries seeking secure access to essential inputs for clean energy and digital technologies. UNCTAD emphasizes the need for resource-rich developing economies, particularly in Africa, to move beyond raw material extraction by investing in domestic refining and processing capabilities. Such investments can drive industrial development and enable these countries to capture greater value from their mineral wealth. The current boom in critical minerals presents a unique opportunity for sustainable transformation, provided it is supported by inclusive governance, transparent contracting, and equitable benefit-sharing. Also, strengthened international cooperation and adequate financing mechanisms are essential to bridge existing investment gaps.

Leading producers of non-fuel minerals in 2023 and the number of commodities led

- ▶ China (26)
- ▶ South Africa (5)
- ▶ Russian Federation (4)
- ▶ Australia (3)
- ▶ India (3)
- ▶ Türkiye (3)
- ▶ United States (3)
- ▶ Chile (2)
- ▶ Congo, Dem. Rep. of (2)
- ▶ Brazil (1)
- ▶ Canada (1)
- ▶ Guinea (1)
- ▶ Indonesia (1)
- ▶ Mexico (1)
- ▶ Peru (1)

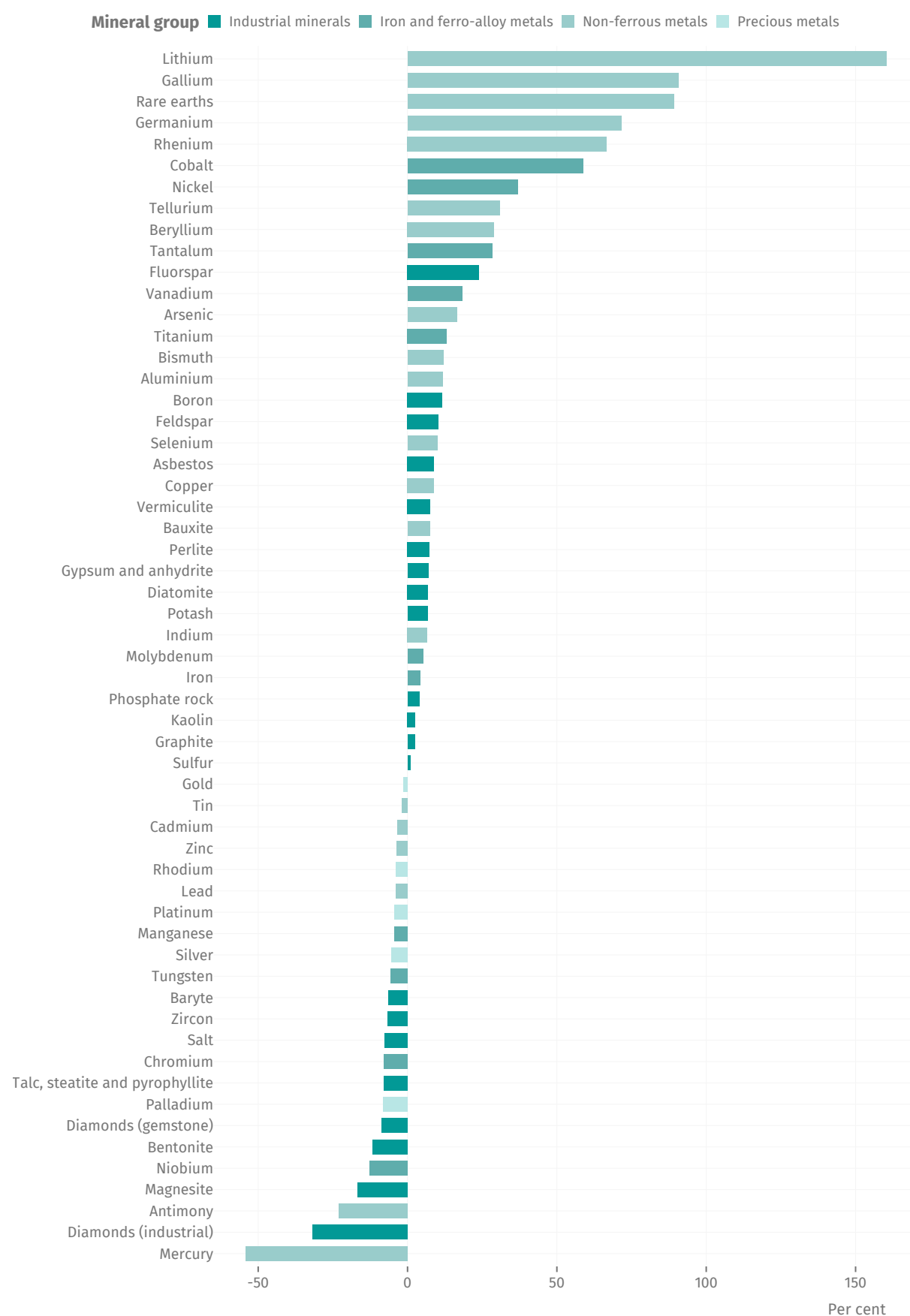


Lithium production
grew by **160%**
between **2019 and 2023**,
with Australia and the Lithium
Triangle leading global output



Critical minerals
offer a **chance** for economies
to **boost**
industrialization by
investing in local refining and
processing

Figure 4.14 | Recent growth dynamics for mineral commodities, 2019–2023



Source: [57]

Note: Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

4.3 Environmental impact of mining and utilities

Mining and utilities generate significant environmental pressures, including land degradation, water and air pollution, and GHG emissions. As a result, environmental considerations have become increasingly central to policy discussions in these sectors. Similar to the environmental analysis of the manufacturing sector in Section 3.6, this section offers a non-exhaustive overview of the main environmental challenges linked to mining and utilities. The use of examples and isolated figures throughout reflects the limited availability of consistent, comprehensive data on these topics.

Mining activities are a major driver of land degradation, contributing to deforestation, soil erosion, and water depletion. Gold and coal extraction together account for most mining-related forest loss globally, responsible for 36 per cent and 34 per cent, respectively. While the impacts of gold mining are distributed across several countries, coal-related deforestation is heavily concentrated in Indonesia, which accounted for 57 per cent of global tree-cover loss linked to coal extraction between 2000 and 2019 [60]. In Mongolia, an estimated 72 per cent of land is degraded due to desertification and mining activity, among other factors, with coal and gold extraction playing a significant role [61]. Moreover, mining frequently takes place in ecologically sensitive, biodiversity-rich areas, contributing to habitat fragmentation and species loss [62].

Mining operations also pose substantial risks of water and air pollution. Toxic by-products, waste rock, and tailings may contaminate water bodies for decades after mine closure. One study estimates that 23 million people globally live on floodplains contaminated with potentially harmful levels of toxic waste from current and historical metal-mining activities [63]. Air quality is also affected, as blasting, excavation, and ore transport release dust and fine particles containing heavy metals, which could increase the risk of respiratory and other health risks.

As expected, electricity generation accounts for a significant share of global CO₂ emissions, as it produces the energy required to power households and all other economic activities [64; 65]. But mining and water provision are also highly energy-intensive sectors. For example, examinations of drinking-water utilities have shown consistently high and rising energy intensity over time [66]. Mining also requires substantial energy, particularly during comminution and through the fuel used for mining and transport equipment. As easily accessible, high-quality mineral deposits become depleted and global demand for mining products grows, the sector's overall energy consumption is expected to rise further [67; 68].

Mining and utilities have significant **environmental impacts** though comprehensive data remain limited

Mining operations could exacerbate **deforestation** and **land degradation** including in **ecologically sensitive** areas

Toxic waste contaminates air and water and poses **health risks** over **decades**

Mining comprises **energy-intensive** activities and drives **CO₂ emissions**

Despite this high energy use, mining accounts for a relatively small share of global GHG emissions, between 4 and 7 per cent, because of its limited weight in the global economy.ⁱ Moreover, only a small portion of these emissions results from electricity use; the majority stems from fugitive methane released during the extraction and handling of coal, oil, and natural gas [69].

These environmental pressures are likely to be intensified by climate change, including through greater water scarcity and more frequent extreme weather events. Addressing them will require improved land rehabilitation, decarbonization of operations, stronger water-management systems, and enhanced engagement with affected communities.

Global policy frameworks increasingly target environmental sustainability in mining and utilities. Common instruments include environmental impact assessments, environmental taxes, green-finance incentives, and international cooperation to strengthen responsible practices. These measures seek to reduce habitat destruction, pollution, and emissions while advancing circular-economy approaches, technical capacity, and the transition to low-carbon energy sources.

Countries are increasingly adopting policies that aim to balance economic priorities with environmental sustainability by setting carbon-neutrality targets, promoting cleaner technologies, and reinforcing regulatory frameworks. However, gaps in implementation and enforcement remain widespread, including limited monitoring capacity. Expanded global data collection and strengthened statistical systems would further support these efforts, by enabling more robust assessment and monitoring of the ecological and health impacts associated with mining and utilities.

ⁱ These estimates include only emissions released directly during mining operations or indirectly from the use of electricity, heat, or steam; fugitive methane emissions are also included. However, to avoid double counting, indirect emissions from downstream processing and the end-use of mined products are excluded.

Global
policy frameworks
promote environmental
sustainability

Implementation gaps
persist and better
monitoring systems
are required



5 Industrial statistics for a sustainable industrial future

5.1 Transformations shaping the industrial landscape	112
5.2 Aligning industrial statistics with future needs	119
5.3 Cross-cutting challenges in the production of industrial statistics	125
5.4 Conclusion and the forthcoming revision of the international standards for industrial statistics	129

Industry remains central to development strategies around the world, but the context in which it operates is changing rapidly. As documented in the earlier chapters of the *Yearbook*, global manufacturing continues to expand, but trends in output, trade, productivity, and emissions have become increasingly uneven across regions and country groups. Progress towards the industry-related SDG 9 targets is generally off-track and unlikely to be achieved without a significant acceleration in the years remaining until 2030. At the same time, the sector is being reshaped by decarbonization pressures, digital technologies, shifting trade patterns, and changing labour markets.

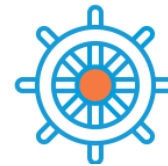
The thematic chapter of the *Yearbook 2024* [37] reviewed the last three decades of industrial development, summarizing major trends observed globally and across regions and industries, and documenting the scale of the sector's transformation.

In contrast, this thematic chapter of the *Yearbook 2025* takes a forward-looking perspective instead. It focuses on the main changes expected to shape industrial development in the coming years, including the shift towards production that is greener, more digital, more globalized, but also increasingly exposed to shocks. On this basis, the chapter considers strategic responses aimed at ensuring that industrial statistics continue to support evidence-based policymaking for a sustainable industrial future. Furthermore, it critically assesses existing gaps and challenges in this area of official statistics, outlining how standards and frameworks may need to evolve to meet the broader and more complex data needs in the years ahead.

Advancing the modernization of this field has become increasingly urgent. First, the rapid transformation of industrial activity risks rendering existing statistical standards obsolete or insufficient for future policy needs. Second, major international statistical frameworks have recently been revised, already incorporating many of the changes needed to better capture current and future economic structures. Together, these developments create a window of opportunity for developing the next generation of industrial statistics.

The objective of the chapter is therefore to identify the key emerging policy themes and structural transformations that will shape future demand for industrial statistics, outlining the response required to ensure they remain relevant, consistent, and fit for purpose. This analysis is framed within the broader context of recent changes to international statistical frameworks and the expected areas of methodological work in the coming years.

The remainder of this chapter is organized as follows. The next section presents the major transformations reshaping industrial development and industrial policy, and examines how these shifts alter the demand for industrial statistics. Section 5.2 then discusses the statistical implications of these changes and discusses how existing and emerging



Industrial statistics are at a
turning point,
requiring
modernization
to remain fit for purpose

Increasing demand
for information that explains
ongoing changes
and not just
historical performance



Industrial statistics are an
essential tool
for guiding reliable,
evidence-informed
policies

international standards can help address them. Section 5.3 turns to cross-cutting challenges in the compilation and production of industrial statistics, including response burden, data integration, data quality, and timeliness, and explores how future frameworks may mitigate these issues. The final section concludes by outlining the rationale and main directions for the forthcoming updates to the international industrial statistics standards, highlighting how these revisions can support the development of future-ready industrial statistics.

5.1 Transformations shaping the industrial landscape

A set of structural changes, operating on both the short- and long-term, are reshaping the trajectory of industrial development worldwide. Recent UNIDO publications have highlighted several cross-cutting, inter-related megatrends: the green and energy transition, digitalization and AI, reconfigured global value chains, demographic and labour market shifts, the growing importance of innovation and intangible assets, and an increasing emphasis on resilience to shocks [10; 70; 71].

Each of these transformations changes what needs to be measured and therefore has significant implications for industrial statistics. Three features are especially relevant from a statistical perspective. First, these changes are multidimensional and require information from domains that extend well beyond the traditional scope of industrial statistics. Second, they are interdependent, which calls for an integrated statistical system capable of linking information from multiple sources and domains. Third, they evolve over long periods but can shift abruptly in response to shocks. Monitoring them effectively therefore requires a combination of high-frequency indicators and long-term structural measures. The main megatrends are discussed below, with an emphasis on their implications for industrial statistics.

The green and energy transition: decarbonization, circularity, and resource efficiency

As discussed in Sections 2.2 and 3.6, manufacturing is a major source of global GHG emissions and resource use, accounting for 17.5 per cent of global CO₂ emissions from fuel combustion. If emissions from industrial processes are added, this share increases further. Energy-intensive industries, such as steel, cement, and chemicals, make particularly large contributions. The left panel of Figure 5.1 shows that the sector's energy consumption has continued to rise, although at a

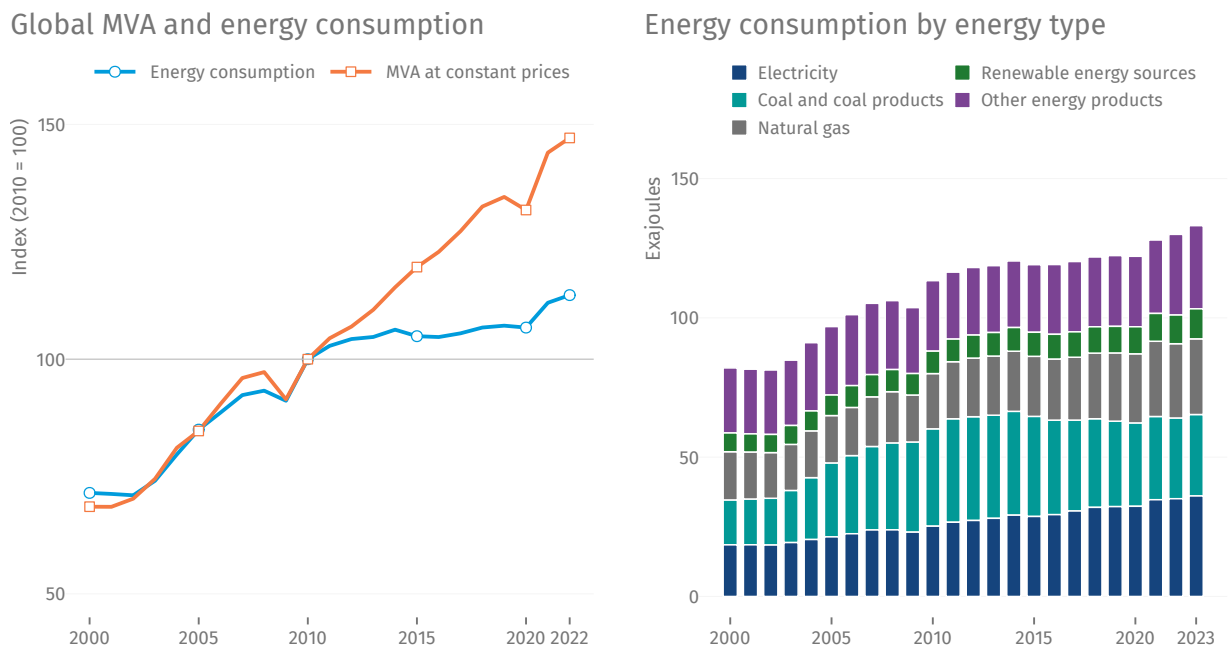


Major transformations are reshaping global industry, changing what **policymakers** need from industrial statistics



Megatrends are **multidimensional, interdependent,** and **long term**, yet sensitive to sudden shocks

Figure 5.1 | Global MVA and energy consumption of the manufacturing sector (left) and energy consumption of global manufacturing by energy type (right)



Source: [9; 72]

Note: In the left-side chart, both series have been converted to an index with base year 2010 to account for their differing measurement units. Only countries with available data in all series are included.

Data used to create this figure can be downloaded at the [UNIDO Statistics Portal](#).

slower pace than MVA, indicating improvements in energy efficiency. Moreover, the right-side panel illustrates a gradual, yet insufficient, shift towards cleaner energy sources, including a growing share of renewables and a declining reliance on coal and coal products.

At the same time, there is evidence of an emerging, but incomplete, decoupling between manufacturing activity and emissions. In a growing number of high- and middle-income economies, industrial emissions have started to decline even as output has continued to expand, driven by energy efficiency improvements, the increased reliance on low-carbon power generation, and structural shifts towards less carbon-intensive industries. However, current trajectories remain inconsistent with the steep reductions in industrial emissions required to meet the goals of the Paris Agreement [73].

The mining and utilities sectors have a key role to play, as they supply many of the critical resources required for the energy transition and provide essential activities such as energy generation, waste management, and recycling, all of which are fundamental to environmental sustainability. Industrial policy is increasingly expected to accelerate the diffusion of low-carbon technologies and promote energy and resource efficiency, while managing the distributional consequences of the transition, ensuring just outcomes for workers and regions that depend on emission- or material-intensive activities. At the same time, industrial strategies designed to support the emergence and



Manufacturing shows partial
decoupling
of emissions, but progress
remains misaligned with
climate goals



Mining and utilities
are critical
enablers
of the energy transition

scaling of green industries (e.g. electric vehicles, renewable energy technologies, or recycling machinery) are gaining prominence in many countries.

These policies significantly increase the demand for detailed environmental information on industrial activities, including disaggregated data at the level of specific industries or products, rather than aggregate measures for broad sectors. Beyond CO₂ emissions and energy use, a more comprehensive measurement framework is needed to capture other GHG emissions, water use and pollution, material extraction and resource depletion, land degradation and land use, and other environmental pressures.

A consistent identification of “green” manufacturing activities and products in industrial statistics has also become essential, including activities related to material recycling and product repurposing. Existing work under the system of environmental-economic accounting (SEEA) and the environmental goods and services sector provides useful building blocks for defining these activities and products in a way that is consistent with national accounts [74], as it will be described later.

Finally, the green transition requires statistical systems capable of linking industrial performance to broader environmental and social outcomes. This requires greater integration of industrial statistics with other sources such as energy balances, emission inventories, labour and household surveys, agricultural statistics, and land registers.

Digitalization and AI

Advanced digital production technologies, many of them powered by AI, are enabling new forms of smart, connected production, often referred to as Industry 4.0. These technologies are transforming how industrial firms produce, organize, and compete [25]. Evidence from recent research indicates that the adoption of digital technologies is associated with substantial productivity gains, and that investments in these technologies can help lagging firms catch up with the productivity frontier. However, the extent of those benefits depends heavily on the presence of key enabling conditions, particularly access to the required skills [75; 76; 77].

Digitalization and AI are not limited to production processes. They also change how industrial value is created and captured. A growing share of manufacturing companies derive income from the provision of digital services related to their products.

At the same time, digitalization raises new challenges. The diffusion of AI and automation is reshaping the organization of work, skill requirements, and the distribution of employment within and across



Green industrial policies

require
detailed data on
emissions, energy and material
use, and environmental impacts



Consistent classification of
green manufacturing
could help track emerging
clean industries
and markets



Advanced digital
and AI technologies
could offer substantial
productivity and
competitiveness gains

economies. Uneven adoption of digital technologies may exacerbate existing gaps in productivity and competitiveness between leading and lagging firms, sectors, and regions. Small and medium-sized enterprises (SMEs) face particular constraints: many continue to lack adequate digital connectivity, relevant skills, or access to advanced digital tools, reinforcing a widening digital divide [78]. Policies aiming at supporting the adoption of advanced digital production technologies should also consider critical enabling factors, such as digital infrastructure, governance systems, data security, access to finance, and skill development.

From the point of view of statistics, these transformations have several implications. First, there is a growing demand for systematic information on the adoption and use of digital technologies in manufacturing and related industries. This includes granular information disaggregated by industry, firm size, ownership, and region. Meeting this demand requires integrating questions on digital technologies into business and industrial surveys, and linking them with other sources of information on digital readiness, such as indicators on connectivity, cybersecurity, and digital skills.

Second, digitalization increasingly blurs traditional boundaries in industrial statistics, particularly between manufacturing and services and between tangible and intangible assets. As manufacturers expand their service-based activities, it becomes more difficult to capture their full value creation using concepts and classifications focused primarily on physical goods. This raises questions about how industrial statistics should treat service activities carried out within (digitally-enabled) manufacturing firms, as well as how to account for non-manufacturing companies that provide production-related services to the manufacturing sector.

And third, understanding the impact of digital technologies on productivity, workforce dynamics, and inclusiveness requires closer integration between industrial statistics and other domains, including labour, education, and skills statistics.

Globalization, GVCs, and supply-chain reconfiguration

The expansion of global value chains (GVCs) has been one of the defining features of industrialization and trade over the past decades. Production processes have become increasingly fragmented across borders, with different stages of design, component manufacturing, assembly, logistics, and support services located where firms perceive comparative advantages. GVCs have supported export growth, productivity gains, and structural transformation in many economies, while also creating new forms of dependency and vulnerability [79].



Many **SMEs** lag in digital adoption, risking **competitiveness** and **innovation gaps**

Industrial statistics need to measure **novel concepts** related to digital technology adoption and impacts



GVCs remain engines of export growth, but also channels of **vulnerability**

Recent crises, including the COVID-19 pandemic, regional conflicts, rising trade tariffs and climate-related disruptions, have exposed these interdependencies. They have revealed the extent to which industrial production in one country can be affected by events taking place in another. Disruptions can propagate rapidly through supply chains and severely impact globalized enterprises that rely on just-in-time delivery of key components, leading to production delays [23; 80; 81].

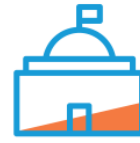
Partly in response to these developments, industrial policy has resurged worldwide, as countries seek to strengthen diversification, resilience, and economic security [10; 82]. The general policy challenge is therefore to deepen and upgrade GVCs participation as a channel for industrialization, technology diffusion, and job creation, while managing risks related to trade protectionism, logistical bottlenecks, climate shocks, and the concentration of critical inputs in a limited number of producers [83].

To support these policy objectives, timely and granular statistics are essential. Beyond measuring domestic production and trade flows separately, it is increasingly important to understand how firms and industries are integrated into cross-border production networks, what determines their participation, and where value added is generated along supply chains. Given the speed at which shocks circulate across production networks, developing high-frequency indicators of production and trade is also crucial.

Comprehensive information on the activities of multinational enterprises, the extent of foreign ownership, and the degree to which domestic firms participate in exporting, importing, and supplying inputs to GVCs is equally important. This underscores the need to integrate statistical business registers (SBRs), SBS, and trade microdata, and to strengthen statistics on foreign affiliates and enterprise groups.

Demographic change, jobs, informality, and inclusion

Demographic trends are reshaping the global supply of labour and the composition of the industrial workforce. UN population projections indicate that the world will continue to grow in the coming decades, but with marked regional differences: populations are ageing rapidly in many countries, while a large and growing youth population is concentrated in others [84]. Urbanization and international migration further alter the geographical distribution and skills profile of workers. These contrasting dynamics create very different policy challenges. In ageing economies, concerns focus on labour shortages, the sustainability of social protection systems, and the need to retain workers in productive employment. In younger economies, the priority is to



Industrial policy

is re-emerging as countries seek to balance benefits and risks from GVC participation



Detailed information

on MNEs, foreign ownership, and trade integration is crucial

Manufacturing's share in
global employment
declined to
14.1%

in 2023, down from 14.6% in 2012

generate sufficient decent jobs for a growing working-age population and prevent underemployment and informality [85].

The manufacturing sector remains an important source of decent work, particularly in low- and middle-income economies. However, as shown in Sections 2.2 and 3.4, its sustained capacity to provide employment appears to be at a turning point: over the past decade, manufacturing's share to global GDP has kept increasing, but its share in total employment has declined. Underutilization, labour market insecurity, limited access to social protection, and high rates of informality remain important concerns across regions [86].

In addition, industrial restructuring, globalization, digitalization, automation, and other factors affect different groups of workers unevenly, depending on their skills, gender, age, and location. Women, in particular, remain underrepresented in many industrial occupations, especially in higher-skilled, higher-paid jobs and in leadership roles, even as their overall participation in the manufacturing workforce has increased. Persistent wage gaps underscore these inequalities [39; 87]. Similar challenges apply to other groups, including youth, migrants, and persons with disabilities, who may face higher risks of exclusion or vulnerability during periods of rapid change.

Industrial statistics must therefore go beyond aggregates of production and value added to capture who participates in industry, under what conditions, and with what outcomes. To design policies that promote quality job creation and address inequalities, policymakers require granular data, at minimum disaggregated by sex, age, disability status, type of employment, occupation, and other relevant characteristics. Moreover, the focus on inclusion requires analysing the entire distribution of employment outcomes, rather than relying only on averages that may obscure important disparities. Meeting these needs calls for stronger integration between establishment-based industrial surveys, labour force and household surveys, and administrative records on employment and wages, supported by the use of common identifiers to facilitate data linkage.

Where informality is widespread, SBRs and standard establishment surveys capture only part of industrial activity. Measuring informal industrial employment and output requires special approaches, including dedicated modules in labour force or household surveys, mixed survey designs, and methods to estimate non-observed activities.

Innovation, R&D, and the growing importance of intangibles

In the digital age, value creation and productivity growth in manufacturing increasingly depend on knowledge, design, software, data, and other intangible assets, rather than on physical capital alone [88; 10;



Labour market insecurity, informality, and limited social protection are **crucial challenges**



Women remain underrepresented in manufacturing, especially in **higher-skilled jobs** and **MHT industries**

A **greater integration** of **industrial and labour statistics** is essential for addressing manufacturing's employment challenges

89; 90]. As documented in Section 3.2, this shift is reflected in the rising importance of technology and knowledge-intensive manufacturing industries. In many sectors, competitive advantage now depends on the ability to develop new products and processes, protect and leverage intellectual property, and use digital technologies to organize production and supply chains more efficiently. These changes are mirrored in companies' balance sheets and in national accounts, where investments in intangibles, such as R&D and innovation-related activities, constitute a rapidly growing share of GDP and are becoming stronger predictors of manufacturing performance than investment in physical assets [91; 92].

The growing prominence of R&D and innovation is closely linked to the megatrends discussed above. Digitalization expands opportunities to embed intangibles in products and production systems. Participation in GVCs often relies on product design, process know-how, quality control, and relational assets. In this context, R&D and innovation form part of a broader set of industrial capabilities that encompass technological and non-technological elements.

These trends have important implications for industrial statistics. First, they increase the need for robust and detailed statistics on R&D and innovation in industry, which remain limited or absent in many economies. Such information is necessary to understand which industries and firms invest in R&D, design, data, and other innovation-related activities; how this investment is financed; and what outcomes it generates in terms of new products, processes, and intellectual property.

Second, the growing importance of intangibles calls for a broader scope of investment and capital stock. Many traditional surveys remain rooted in outdated statistical frameworks and focus primarily on tangible assets such as machinery, equipment, and buildings, while treating intangibles as intermediate inputs, if they are captured at all. Aligning industrial statistics with the revised macroeconomic standards offers a pathway toward more comprehensive measurement of intangible assets.

Third, intellectual property indicators, such as patents, trademarks, and industrial designs, offer valuable complementary information. When compatible classifications and identifiers are used, these data can be linked to traditional industrial statistics, enabling analysis of innovation dynamics. Finally, industrial statistics must be complemented by broader innovation surveys and by both qualitative and quantitative data on management practices and organizational change. This is especially crucial for SMEs, which often innovate without formal R&D expenditure or rely on non-technological forms of innovation. Capturing these activities is essential for understanding the full breadth of innovation across industrial systems.



Intangible assets
are gaining predominance as
drivers
of industrial competitiveness
and productivity

Not enough is known about
firm-level **R&D** and
business innovation,
especially in SMEs



In modern statistical
accounting frameworks,
R&D expenditure,
data and **AI-related**
assets
are counted as
capital formation

Industrial resilience and shocks

The past decade has underscored the vulnerability of industrial sectors when facing large and overlapping shocks. Since the COVID-19 pandemic, different regions have faced climate-related events, geopolitical tensions, and conflicts, with worldwide impacts. The re-emergence of trade protectionism adds another layer of uncertainty.

The issue of industrial resilience was already discussed in the megatrend of digitalization and GVCs, but this is not purely a supply-chain issue. It also highlights issues about industrial structure, diversification, technological capabilities, and social protection.

From a statistical perspective, the key consequence is the need for flexible and responsive statistical systems that can react and adapt to emerging issues while at the same time preserving consistency. They should include a battery of industrial indicators that can capture both the short-term impact of shocks and the structural factors that shape resilience or vulnerability. This includes more timely indicators of production, employment, trade, and inventories, and information on exposures to specific risks. It also calls for greater use of high-frequency data, including from non-traditional sources.



Agile, responsive statistics,
capable of rapidly measuring
impacts
while maintaining
consistency,
will be increasingly needed

5.2 Aligning industrial statistics with future needs

The megatrends described in Section 5.1 directly affect what industrial statistics need to measure and how they are produced. Many of the conceptual tools required to respond to these new demands already exist in the international statistical frameworks, including the updated system of national accounts (SNA) 2025 [93], the system of environmental-economic accounting (SEEA) [74], the fifth revision of the ISIC [94], the third version of the Central Product Classification (CPC) [95], and the forthcoming 2028 update of the Harmonized Commodity Description and Coding System (HS). Other adjustments will require new methodological guidance and integration across statistical domains. This section outlines how these frameworks can be leveraged to keep industrial statistics relevant and fit for purpose amid ongoing and anticipated transformations in industrial activity.

Rethinking the scope and boundaries of industrial statistics

As summarized in Section 1.2, current international standards define industrial statistics as covering mining, manufacturing, and utilities. This core scope has long provided a solid conceptual foundation and remains valid. However, the transformations discussed above



New and revised
**international
standards**
provide a
toolbox
for modernizing
industrial statistics

increasingly challenge the traditional boundaries of what should be considered *industrial* for statistical and policy purposes.

GVCs spread industrial activities across borders and blur the separation between sectors. They involve the participation of domestic, foreign, and multinational enterprises (MNEs) connected through complex production networks. Although international manuals provide definitions and guidance on global production arrangements, such as contract manufacturing, outsourcing, and factoryless goods production (FGP), applying them in practice is often difficult and may lead to counterintuitive outcomes.

The green transition depends not only on what happens within manufacturing establishments, but also on upstream and downstream functions such as engineering, design, industrial R&D, maintenance and repair, recycling, and end-of-life management. Similarly, the digital transformation erodes the traditional boundary between manufacturing and services, as production processes become deeply integrated with software development, data processing, and platform-based business models. These developments suggest the need for information covering broader industrial ecosystems or production systems, rather than narrowly defined industrial sectors.

A review of the scope of industrial statistics could therefore consider how to define and classify industrial services and activities that support or are closely linked to industry, while maintaining coherence with ISIC, CPC, and SNA. Flexible thematic groups, such as green industries or digital-intensive industries, could be developed to answer specific policy questions. This would enable industrial statistics to better capture emerging realities while preserving the core definitional structure that ensures cross-country comparability.

Classifications for a changing industrial structure

Classifications of activities and products are at the core of official statistics. They determine how establishments are grouped into industries and how goods and services are categorized, ensuring that national data can be compared internationally. The ISIC classification of economic activities and the CPC classification of products therefore play a central role in ensuring that industrial statistics remain aligned with ongoing transformations.

ISIC Rev. 5, endorsed by the UN Statistical Commission in 2023–2024, preserves the broad structure of its predecessor while introducing some refinements to reflect new forms of production and the growing importance of services, digital activities, and environmental management. For industrial statistics, some of the improvements include:



The boundary between
manufacturing and
related services
is blurring, challenging traditional
sector-based measurement
approaches



**Classification
systems**
are at the core of
official statistics
and are essential
**modernization
building blocks**

- ▶ A finer breakdown of repair, waste management, and environmental services, supporting better identification and monitoring of “green” industries;
- ▶ Enhanced classification of information and communication, data processing, and platform-based activities, improving the statistical visibility of digital services that are increasingly integrates with industrial production;
- ▶ Clearer guidance on contract manufacturing, outsourcing, and FGP, helping ensure more consistent and transparent measurement of GVCs.

CPC Ver. 3, adopted in 2024–2025, provides a revised and more detailed product classification aligned with ISIC Rev. 5 and other macroeconomic standards. For industrial statistics, key implications include:

- ▶ Identification of green and low-carbon goods, circular economy products, and environmental services produced by industrial establishments;
- ▶ More explicit classification of digital products and services arising from industrial activities;
- ▶ Improved consistency in classifying industrial output by type of product, including intangible or knowledge-based outputs, in line with SNA concepts.

In addition, the HS classification remains the primary international standard for traded goods. The forthcoming 2028 revision, currently under development, is expected to introduce new subheadings for environmental and health-related goods, advanced “green” technologies, waste and scrap categories, and products relevant to circular economy and hazard management. These updates will allow trade statistics to better capture the evolving composition of manufactured exports and imports.

Looking ahead, it will be important to continue identifying research needs and potential future revisions to ensure that international classifications remain responsive to the transformations taking place in industrial sectors.

Aligning industrial statistics with SNA 2025

The adoption of SNA 2025 as the new international standard for national accounts marks a major milestone for economic statistics in general, and industrial statistics in particular. Table 5.1 shows the main international statistical frameworks relevant to industrial statistics and highlights the central role of the SNA within this architecture. While the update retains the core structure of SNA 2008, it introduces new or expanded treatment for digitalization, globalization, emerging

Main international classifications relevant for industrial statistics

- ▶ International Standard Industrial Classification of All Economic Activities (ISIC) [5]
- ▶ Central Product Classification (CPC) [95]
- ▶ Harmonized Commodity Description and Coding System (HS) [96]
- ▶ List of Industrial Products (LoIP) [97]
- ▶ Standard International Energy Product Classification (SIEC) [98]



Data collected at the appropriate activity and product detail levels lead to policy-relevant and analytically useful statistics

financial issues, the informal economy, and aspects of well-being and sustainability. Many of these revisions have direct implications for industrial statistics.

A key innovation in SNA 2025 is the updated guidance on digital products and technologies. Digital platforms, cloud computing, data, AI systems, and related digital services receive an enhanced and more comprehensive treatment. In particular, data and AI-related assets are now recognized as intellectual property products, broadening the existing category of software and databases within capital formation.

The **SNA 2025**
strengthens the treatment of
digitalization,
globalization,
well-being, and
sustainability

Table 5.1 | Selected international statistical standards relevant for industrial statistics

Framework	Main focus	Relevance for industrial statistics
International Recommendations for Industrial Statistics (IRIS) [4]	Concepts, definitions, and compilation main guidance for industrial statistics	Provides the core international standard for structural industrial statistics, covering scope, statistical units, variables, data sources, and key indicators
International Recommendations for the Index of Industrial Production (IRIIP) [99]	Concepts and methods for short-term indicators of industrial output	Sets standards for compiling IIPs, including index formulas, weighting schemes, and treatment of quality change and new products
System of National Accounts (SNA) [93]	Comprehensive framework for measuring production and income in the economy	Provides the accounting structure and core concepts (output, value added, assets) for compiling economic aggregates; clarifies treatment of digitalization, globalization, the informal economy, and sustainability
System of Environmental-Economic Accounts (SEEA) [74]	Integration of environmental and economic information in physical and monetary terms	Links industrial activities to energy, water, emissions, waste, and natural resources; provides structure for environmental goods and services
Balance of Payments and International Investment Position Manual (BPM) [100]	Framework for cross-border transactions and positions in goods, services, income, and financial assets and liabilities	Shapes the recording of trade in industrial services, charges for intellectual property, and other cross-border flows related to industrial activities
International Merchandise Trade Statistics (IMTS) [101]	Concepts and methods for compiling statistics on trade in goods	Provides standards for measuring imports and exports of manufactured products by partner and product
Manual on Statistics of International Trade in Services (MSITS) [102]	Concepts and methods for statistics on trade in services	Covers services supplied by industrial firms (e.g. maintenance, engineering, R&D, digital services), including modes of supply
Frascati Manual [103]	Concepts and methods for measuring research and experimental development	Defines R&D activities, expenditures, and personnel, providing a foundation for measuring industrial R&D and linking it to industrial performance
Oslo Manual [104]	Guidelines for collecting and using data on innovation in the business sector	Provides concepts and survey approaches for product, process, organizational, and marketing innovation in industrial firms

Source: UNIDO elaboration

For industrial statistics, this implies that surveys and administrative data used to compile industrial output and investment need to collect information on the expenditures on these assets; the bundling of physical goods with digital services or embedded software; and the integration of digital technologies into industrial production processes. These improvements will help industrial statistics to reflect more accurately the growing importance of digitalization, AI adoption, and the role of intangibles in industrial activity.

The SNA update devotes substantial attention to globalization and global production arrangements, providing refined guidance on MNEs, special purpose entities, GVCs, FGP, merchanting, and outsourcing arrangements. This has several implications for industrial statistics:

- ▶ The delineation between domestic and foreign production, and between resident and non-resident units, needs to be reflected in business statistics and in the profiling of MNE groups;
- ▶ Guidance on accounting for outsourcing, contract manufacturing, and ancillary activities, which affect how industrial output is attributed to establishments in different economies, thereby improving consistency between industrial statistics, trade statistics, and balance of payments data;
- ▶ The focus on thematic or satellite accounts (e.g. on the digital economy) increases the demand for detailed data from industrial establishments.

The SNA 2025 also broadens the national accounts framework to better account for well-being and sustainability, including more explicit treatment of natural resource depletion as a cost of production, distributional measures, and links to environmental-economic accounts. For industrial statistics, this will result in a more comprehensive reflection of the inclusiveness and sustainability dimensions of industrial production.

Integrating environmental-economic accounting into industrial statistics

Environmental pressures and the green transition feature prominently among the transformations shaping industry, but traditional industrial statistics remain largely focused on economic variables such as output, value added, and trade. The SEEA provides a conceptual and statistical framework to bridge this gap by integrating environmental information into the same accounting structure as the SNA.

The SEEA Central Framework, adopted in 2012, consists of several complementary elements. First, physical flow accounts, organized as physical supply and use tables, record the flows of natural inputs, products, and residuals between the environment and the economy,



The recent
**national accounts
update**
better reflects the
**global production
arrangements**
shaping manufacturing today



The **SEEA framework**
connects industrial activity
with energy, water, emissions,
waste, and natural resource use

by industry and product. Second, environmental asset accounts record stocks and changes in stocks of individual natural resources in volume and monetary terms. Third, accounts for environmental protection expenditure, environmental taxes and subsidies, and environmental goods and services link environmental policy instruments and markets to industries and institutional sectors. The SEEA Ecosystem Accounting, adopted in 2021, extends this approach further by quantifying ecosystem extent, condition, and services in a way that can be linked to economic activity [105].

This framework directly addresses several of the measurement challenges that industrial statistics must overcome to remain future-ready. First, it provides a coherent mechanism to link industrial activities to their resource use and environmental pressures, including energy and water use, GHG and air pollutant emissions, waste generation, and material flows. This information is essential for tracking resource efficiency, environmental performance, and the decoupling between emissions and production. Second, it provides a way to identify industries producing environmental goods and services, which can be useful for monitoring activity in “green” industries and the transition to sustainable production. And third, it supports the analysis of how industrial activities affect ecosystems and the services they provide, including land-use change, degradation, and restoration.

Despite these advantages, SEEA implementation remains uneven across countries and often partial in coverage, in part because of its significant data requirements. Moreover, environmental statistics are still often compiled separately from economic statistics, leading to inconsistencies and limited integration across domains. A more comprehensive and harmonized implementation of the SEEA offers significant potential benefits for industrial statistics. It would strengthen the capacity of NSOs to measure the green transformation, assess environmental impacts with greater level of detail, and provide policymakers with integrated information to design and monitor sustainable industrial strategies.

Linking industrial statistics with other domains

As emphasized throughout this chapter, many emerging policy questions concerning industry cannot be addressed using industrial statistics alone. Understanding the implications of industrial transformations for decent jobs, wages, innovation capacity, external competitiveness, inclusiveness, and other dimensions requires integrating information across multiple statistical domains. This integration could draw on several established international standards, including but not limited to:

Two main manuals
in the SEEA family:
Central Framework
and **Ecosystem
Accounting**



The **adoption**
of environmental accounting
remains low worldwide,
limiting its
contribution to
green industrial
policymaking

- Labour statistics, guided by International Conference of Labour Statisticians (ICLS) resolutions on statistics related to labour force, employment, working time, wages, and occupational classifications. Linking business statistics with labour force surveys and other demographic and social data could provide information to analyse the distributional and inclusiveness dimensions of industrialization, including gender and youth employment, informality, skills, and job quality.
- R&D and innovation statistics, guided by the Frascati Manual [103] and the Oslo Manual [104], respectively. These frameworks define concepts, classifications, and survey methodologies for measuring R&D expenditure and personnel, as well as innovation activities in the business sector. Integrating these statistics with structural and performance indicators from industrial statistics would provide a more complete understanding of the role of innovation and intangible assets in industrial development.
- Trade and balance-of-payments statistics, including the 2025 update of the Balance of Payments Manual [100] and forthcoming revisions to the International Merchandise Trade Statistics and International Trade in Services Statistics manuals. These standards offer guidance for measuring cross-border flows of goods, services, and capital, including digital trade and services supplied by industrial firms.
- Energy statistics, which can be linked with industrial statistics to obtain detailed information on energy production and consumption by industrial establishments and on their progress along the energy transition [106].

Taken together, these options suggest that the future of industrial statistics may lie not in expanding as a standalone statistical domain, but in deeper synergies within an integrated system of economic, social, and environmental statistics.

5.3 Cross-cutting challenges in the production of industrial statistics

The adaptations described in the previous section would allow industrial statistics to better respond to the data needs arising from the transformations reshaping industry. However, these conceptual advances also interact with multiple, often simultaneous operational pressures faced by NSOs. This section highlights some of the most important cross-cutting challenges in the compilation and dissemination of industrial statistics that must also be addressed to maintain data quality, increase efficiency, and ensure industrial statistics remain fit for purpose.



Complex policy questions
require **industrial data**
to be integrated with
labour, trade, and
innovation statistics



The future role of
industrial statistics
lies within a broader
integrated system,
not in isolated efforts

Response burden, non-response, and declining survey participation

Business surveys are typically demanding for establishments. They request detailed information on output, inputs, employment, investment, products, and technologies, often at high frequency and for multiple reference periods, using concepts that do not always align with business accounting practices. At the same time, firms face increasing and overlapping reporting obligations, including tax, social security, regulatory, and other statistical requirements. These pressures are particularly acute for SMEs, which have fewer administrative resources. International frameworks highlight the importance of considering response burden when collecting statistical information [107; 108].

A related challenge is declining response rates. Many NSOs report increasing difficulties in recruiting and retaining survey respondents, particularly among small firms and in sectors with high turnover. The United Kingdom's labour force survey, although a household survey, illustrates this trend: full response rates declined from 48.2 per cent in 2010 to 17.4 per cent in 2024 [109; 110]. Similar patterns are observed in many other countries. If not adequately addressed, these trends can introduce bias and threaten the representativeness of industrial statistics.

To mitigate these challenges, many NSOs are adopting an administrative data-first approach, surveying establishments only when comparable data cannot be obtained from existing sources, such as tax filings, financial statements, or employment reports. Non-traditional data sources, including big data, could also serve as complements. Implementing such approach requires well-established data flows, robust data governance frameworks, and clear methodologies for integrating data sources. When surveys remain necessary, several improvements could help reduce burden and improve participation: careful survey design and sampling strategies that avoid repeatedly burdening the same units; the use of web-based questionnaires and pre-filled forms; responsive collection strategies targeted at high-impact non-respondents; and clear communication about the purpose and benefits of industrial statistics.

Survey frames and complex enterprise structures

Producing high-quality industrial statistics depends critically on the quality of the underlying survey frames. SBRs are widely recognized as the backbone of business statistics, providing coherent frames for sampling and integrating data across surveys and administrative



Response burden management

is an increasingly important **priority** for NSOs



Survey **response rates** are declining all over the world, undermining the **representativeness** and **data quality** of official statistics

sources [111; 112]. However, in many countries SBRs remain incomplete or outdated. Gaps are common with respect to SMEs, industries with high firm turnover, and establishments operating in the informal sector.

Globalization and new business models introduce additional complexity. Enterprise groups are becoming increasingly multifaceted, with operations spread out in multiple industries and jurisdictions. FGP and outsourcing blur the link between the location where goods are physically produced and the units that own the products, technologies, and intellectual property.

These issues challenge the ability of SBRs and survey frames to correctly identify statistical units and to attribute industrial activity to the appropriate resident entities. Meeting these challenges requires prioritizing the coverage, quality, and maintenance of the SBR; establishing a dedicated large case unit (LCU); expanding the use of administrative data for business profiling; and aligning national practices with international guidance on enterprise groups and global production [112].

Data sources and data integration

While surveys remain indispensable in many cases, the pressure to reduce response burden and increase timeliness and granularity is accelerating the shift towards multi-source statistical systems. Administrative data, including tax records, social security registers, customs records, business licensing, and many others, can play a central role in this transition, especially for structural variables like turnover, trade, employment, and firm characteristics [113; 114].

At the same time, new types of data are becoming increasingly relevant for industrial statistics. Satellite imagery, scanner data, web-scraped data, and other digital traces hold significant potential for measuring specific aspects of economic and environmental activity (see, for instance, the resources available in [115]).

A major challenge lies in linking survey and non-survey data sources. The adoption of unique identifiers that allow consistent linkage across datasets would significantly facilitate integration. When they are not available, probabilistic matching can achieve reliable results when based on robust methodologies. Additional challenges commonly arise, including data access restrictions, limitations in legal frameworks, and the presence of intrinsic biases or quality issues in non-survey data.



High-quality
business registers
are essential for
sampling,
data integration,
and analysing
business
demography



Industrial statistics
increasingly combine
surveys,
administrative
records,
big data, and
other non-traditional sources

Timeliness and quality trade-offs

Industrial statistics are used both for structural analysis and short-term monitoring of economic activity. Data users increasingly expect timely, high-frequency indicators of industrial activity with minimal publication lags. Meeting these expectations requires balancing difficult trade-offs between timeliness, coverage, granularity, and accuracy.

For industrial statistics, improving timeliness without jeopardizing quality often requires a combination of measures. These include, among others, redesigning survey instruments and collection modes; expanding the use of administrative and high-frequency data sources; and applying model-based estimation and nowcasting techniques to produce preliminary estimates. New methodologies, including machine learning tools, can greatly accelerate data production while maintaining or even improving quality. These include, for example, automated classification, anomaly detection, or imputation techniques [116].

Methodological modernization and process integration

In addition to new data sources and methods, addressing the challenges presented requires a modernization of statistical production processes and infrastructure. Applying a statistical business process model to industrial statistics can help identify opportunities for standardization, automation, and reuse of tools across surveys and domains, ultimately leading to improved efficiency and consistency. It can also contribute to a more robust data and metadata management, supported by stronger integration and clear governance structures [117; 118].

For industrial data in particular, given its complex classifications, multiple disaggregations, and frequent revisions, well-designed metadata systems are essential to ensure that users can correctly interpret statistical outputs and that data can be consistently linked across domains and over time. Strengthening these systems is therefore essential for making industrial statistics more agile, interoperable, and responsive to the evolving needs of policymakers and other users.

Dissemination, communication, and AI-readiness

There has never been a greater demand for reliable, accessible, and timely data. As evidence-based policymaking becomes more deeply embedded and as analytical tools grow more sophisticated, the need for disaggregated, high-frequency indicators continues to rise. At the

Machine learning methods

applied across the data production cycle can improve

timeliness while maintaining or improving other dimensions of **data quality**



Statistical business process models

can streamline industrial data production and support

end-to-end modernization

same time, the way industrial statistics are disseminated and communicated is undergoing rapid transformation. Some users look for concise summaries and policy-ready dashboards, while others expect open, machine-readable datasets with clear licences, rich metadata, and tools that enable automated processing and reproducible analysis. Ensuring that dissemination systems serve this increasingly diverse spectrum of users is now a central challenge for producers of industrial statistics.

Confidentiality and data protection remain essential considerations. As dissemination becomes more granular (by industry, region, firm size, or other attributes), the risk of disclosing sensitive information grows. Statistical offices must continue to apply robust confidentiality safeguards, including SDC techniques, when releasing detailed data to the public.

A new and rapidly growing dimension is the AI-readiness of official statistics. While dissemination platforms were traditionally optimized for direct human use, industrial data are now increasingly accessed by AI systems that retrieve, read, and synthesize information on behalf of users. This shift rises the importance of standardized formats, semantic consistency, consistent identifiers, and machine-readable metadata, enabling AI tools to interpret and combine statistical content correctly. Dissemination systems must therefore be optimized simultaneously for human interpretation and automated consumption, ensuring clarity, transparency, and interoperability [119].

5.4 Conclusion and the forthcoming revision of the international standards for industrial statistics

This chapter has shown that industrial statistics are at a turning point. Structural transformations in industry, including the green and energy transition, digitalization, the reconfiguration of GVCs, labour market shifts, the rising importance of innovation, and growing concerns about resilience, are reshaping what policy makers need to know about industry and how industrial development should be measured. At the same time, statistical producers face mounting operational pressures related to response burden, non-response, integration of multiple data sources, timeliness, data quality, confidentiality, and dissemination.

Section 5.2 described how many of the conceptual tools needed to respond to these demands already exist in the form of revised international frameworks and classifications. However, the current industrial statistics standards themselves still reflect an earlier generation of industrial structures, policy priorities, and statistical practices. They



Dissemination

should cater both human and AI users through **open, interoperable, and machine-readable** data and metadata standards

Both the megatrends reshaping industry and the mounting cross-cutting challenges call for a **renewed approach** to industrial statistics

no longer fully match the information needed to design, monitor, and evaluate industrial policy in a rapidly changing world.

The existing manuals have provided a robust foundation for industrial statistics for many years. They standardized key concepts, classifications, and indicators; supported the development of national industrial survey programmes; and guided international reporting, including for SDG 9. However, they pre-date many of the conceptual, policy, and statistical developments discussed in this chapter, leaving important gaps where updated guidance is needed.

In response, UNIDO and the UN Statistics Division have initiated exploratory work on the future of international standards for industrial statistics. The directions under consideration include:

- ▶ Aligning definitions and measurement concepts in industrial statistics with the SNA 2025 and the SEEA, including for digitalization, globalization, and environmental and distributional aspects;
- ▶ Clarifying the scope of industrial statistics, including the treatment of industrial services and industry-related activities, drawing on updated SNA guidance;
- ▶ Fully leverage updated classifications to identify green, digital, and high-technology industries and products;
- ▶ Providing guidance on the measurement of R&D, innovation, intangibles, MNE activity, the informal economy, and other concepts within an industrial context;
- ▶ Strengthening methodological guidance on the use of administrative and other data sources, the role of the SBR, and integration of industrial statistics with labour, innovation, trade, and environmental data;
- ▶ Updating the list of industrial performance indicators to better reflect contemporary policy priorities and data needs.

This would result in a new generation of international standards for industrial statistics, replacing the existing manuals. Importantly, this should not be seen as an isolated technical exercise, but as part of a broader global effort to adapt official statistics so that they remain relevant, coherent, and fit for purpose amid accelerating economic, technological, and environmental change.

The transformations described in this chapter will continue to evolve, and statistical systems will need to remain flexible and forward-looking while at the same time maintaining consistency over time. By building on the strengths of current standards while addressing emerging demands in a pragmatic manner, the forthcoming revision can help ensure that industrial statistics remain a solid analytical foundation for evidence-based policies in support of an inclusive, resilient, and sustainable industrial future.

Current industrial statistical standards

- ▶ International Recommendations for Industrial Statistics (IRIS) [4]
- ▶ International Recommendations for the Index of Industrial Production (IRIIP) [99]
- ▶ List of Industrial Products (LoIP) [97]



The forthcoming revision of the **industrial statistics manuals**

aims to align them with recently updated frameworks and make

them more

fit for purpose

ANNEXES



A Key industrial indicators

Table A.1 | Industrial indicators by country/area, latest available year

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Industry value added share in GDP 2024 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2024 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)	CIP Index 2023 (rank)
Afghanistan	28	-0.7	7.1	1.4	11.70	10.1	18.2	-70.8	6.4	...	148
Albania	343	4.8	6.2	4.8	12.21	71.3	18.6	-34.3	11.2	1.03	108
Algeria	477	3.6	10.1	2.1	15.61	29.5	6.8	-24.1	10.7	0.69	88
Andorra	1,381	2.4	3.4	...	4.59	92.2	64.6	-68.2	...	...	...
Angola	213	5.1	7.0	1.7	21.75	12.0	34.0	-15.2	2.5	0.19	126
Anguilla	626	4.0	2.6	...	9.08	96.7	51.3	-70.6	2.4	...	...
Antigua and Barbuda	431	3.0	2.3	...	6.98	89.2	66.8	-78.0	4.1	...	...
Argentina	1,843	-2.7	13.4	26.4	18.28	26.1	38.2	-22.4	11.8	0.24	55
Armenia	590	5.5	10.7	4.9	17.22	44.0	21.8	-18.8	8.2	0.34	89
Aruba	1,092	12.3	3.3	...	7.22	84.6	28.4	-62.3	4.0	...	...
Australia	2,874	-1.6	5.1	26.4	13.02	46.2	14.2	-17.3	6.6	0.35	34
Austria	8,596	-1.8	18.7	50.0	21.59	86.8	63.5	1.6	16.6	0.12	13
Azerbaijan	445	2.4	7.7	11.4	27.37	10.2	42.6	-25.8	5.6	0.54	103
Bahamas	129	-0.3	0.4	0.0	2.33	82.5	52.0	-67.5	2.8	...	138
Bahrain	3,810	2.6	16.2	22.3	29.00	71.1	20.1	9.0	11.9	0.38	65
Bangladesh	577	5.0	24.9	17.3	28.16	97.3	1.8	-7.0	11.3	0.20	64
Barbados	1,170	1.9	5.7	4.7	8.80	85.1	31.2	-50.3	2.5	...	116
Belarus	1,692	6.5	24.7	38.5	28.92	84.7	45.9	6.0	18.8	0.31	56
Belgium	5,065	0.1	11.2	60.3	12.79	86.6	59.0	4.6	11.7	0.23	14
Belize	402	1.1	6.4	...	9.70	63.8	7.8	-51.3	8.0	...	130
Benin	124	3.1	9.4	...	10.22	15.7	19.3	-33.3	14.2	0.34	...
Bermuda	399	-2.1	0.4	18.9	2.09	91.3	90.8	-83.0	1.7	...	144
Bhutan	200	3.2	5.5	...	19.52	51.9	55.1	-36.6	6.7	...	...
Bolivia (Plurinational State of)	325	1.0	10.2	9.6	19.42	42.7	7.2	-27.4	11.6	0.52	102
Bosnia and Herzegovina	780	3.8	11.6	21.0	17.53	86.8	33.7	-23.2	17.6	0.61	77
Botswana	410	-2.0	5.9	7.1	21.44	95.5	4.4	-19.3	6.5	0.07	92
Brazil	875	2.3	9.1	33.4	13.45	50.4	32.5	-12.1	11.5	0.46	42
British Virgin Islands	546	1.2	1.3	...	2.65	48.5	32.0	-80.6	2.5	...	...
Brunei Darussalam	5,238	4.5	17.7	94.5	54.05	60.5	8.9	19.3	5.9	0.23	62
Bulgaria	1,451	3.4	15.1	34.9	19.80	69.8	50.2	-9.5	18.1	0.49	47
Burkina Faso	79	4.4	10.3	20.6	19.26	7.9	18.6	-44.3	14.6	0.03	...
Burundi	28	2.5	10.7	1.9	11.81	30.7	6.9	-71.8	1.2	...	147

Table A.1 | Industrial indicators by country/area, latest available year (*continued*)

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Industry value added share in GDP 2024 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2024 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)	CIP Index 2023 (rank)
Cabo Verde	218	4.4	4.8	4.2	7.92	88.7	22.4	-77.2	9.6	...	140
Cambodia	697	10.2	32.1	4.6	35.84	82.0	14.9	-5.7	16.5	0.27	69
Cameroon	215	4.1	14.5	4.5	18.08	19.9	17.6	-39.2	9.6	0.05	125
Canada	4,108	0.1	9.0	40.4	15.55	58.2	56.0	-12.8	8.8	0.32	20
Cayman Islands	685	3.0	0.8	...	3.47	82.6	44.4	-83.7	1.1	...	...
Central African Republic	73	3.5	18.6	...	19.40	42.9	21.4	-65.6	...	...	145
Chad	76	5.9	9.1	0.2	42.92	0.3	54.2	-29.1	7.0	0.18	...
Chile	1,360	1.4	9.4	19.6	18.76	59.3	6.5	-5.0	9.4	0.28	49
China	3,563	4.9	28.0	56.7	32.90	93.8	63.3	25.6	29.1	0.57	2
China, Hong Kong SAR	471	-3.4	1.0	20.5	2.22	87.9	88.6	-2.5	2.4	0.11	91
China, Macao SAR	401	-1.6	0.7	0.0	1.39	93.4	43.4	-67.0	1.6	...	141
China, Taiwan Prov.	11,099	7.8	35.7	76.3	37.82	97.6	82.1	16.5	26.4	0.13	6
Colombia	809	0.0	11.8	20.4	18.63	35.3	37.9	-33.7	10.7	0.26	71
Comoros	113	3.2	6.8	...	7.43	53.2	46.0	-72.5	5.2	...	...
Congo	158	2.1	9.4	5.4	40.63	13.8	24.6	-26.6	10.8	0.12	119
Cook Islands	307	2.4	1.1	...	0.07	52.4	75.0	-80.4	3.8	...	...
Costa Rica	2,073	3.3	13.9	14.7	16.97	75.0	58.5	-15.0	11.1	0.12	59
Croatia	2,115	2.9	12.0	29.9	14.86	81.3	48.8	-21.2	16.8	0.23	54
Cuba	641	-0.6	8.5	12.7	10.12	75.1	15.0	-46.2	9.3	0.59	100
Curaçao	408	4.1	2.7	...	7.58	85.9	38.4	-54.7	6.7	6.90	...
Cyprus	910	0.0	4.0	20.4	5.53	85.5	43.0	-41.3	6.2	0.42	93
Czechia	5,196	2.2	25.1	56.4	27.07	95.1	73.0	8.6	25.8	0.21	15
Côte d'Ivoire	307	5.5	12.9	9.6	17.13	25.9	21.1	-17.1	9.6	0.19	97
Korea, Dem. People's Rep. of	103	-0.6	17.6	...	32.35	89.7	21.4	-67.3	...	...	...
Congo, Dem. Rep. of	73	3.1	13.2	...	52.53	30.0	0.6	-24.7	5.8	0.00	...
Denmark	10,795	6.0	17.1	68.1	19.65	82.0	55.4	2.9	11.2	0.06	17
Djibouti	182	6.6	5.6	...	10.67	84.9	41.3	-5.3	0.2	...	...
Dominica	253	-0.7	2.9	...	8.35	76.2	49.7	-69.8	7.8	...	...
Dominican Republic	1,236	4.1	13.4	25.9	15.38	74.8	47.3	-36.0	9.9	0.35	...
Ecuador	837	-3.6	13.8	11.5	21.37	21.9	9.6	-30.8	9.4	0.27	85
Egypt	548	1.8	13.7	32.5	25.86	70.4	32.4	-28.2	12.5	0.55	68
El Salvador	592	2.0	12.9	8.6	18.09	92.0	19.4	-33.9	13.7	0.57	82
Equatorial Guinea	1,292	6.6	25.7	...	40.79	13.8	43.8	-19.9	...	0.46	...

Table A.1 | Industrial indicators by country/area, latest available year (*continued*)

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Industry value added share in GDP 2024 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2024 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)	CIP Index 2023 (rank)
Eritrea	44	2.6	6.1	5.2	15.92	51.5	1.3	-15.0	...	0.16	149
Estonia	2,627	0.1	13.0	26.2	15.09	79.4	52.9	-7.2	17.2	0.06	50
Eswatini	1,277	2.3	32.9	2.0	34.12	91.8	18.5	10.7	11.6	0.09	84
Ethiopia	50	9.7	5.6	16.1	6.84	17.8	33.4	-72.2	3.2	1.06	146
Fiji	516	2.2	8.7	6.5	11.34	63.6	11.3	-46.9	5.6	...	122
Finland	6,572	0.0	14.4	45.5	17.76	84.4	52.1	1.9	12.8	0.14	27
France	4,055	0.8	10.0	47.7	11.27	88.8	61.9	-3.8	11.1	0.14	12
French Polynesia	1,044	1.9	4.8	...	6.88	78.6	13.3	-75.0	6.2	...	...
Gabon	561	6.4	8.5	...	41.88	37.2	23.7	8.3	...	0.34	90
Gambia	12	0.1	1.6	...	4.66	58.7	12.8	-63.3	8.1	...	150
Georgia	573	11.3	8.7	12.1	12.41	88.8	61.6	-39.7	5.9	0.78	99
Germany	8,579	-3.6	19.7	61.8	21.25	88.9	73.1	11.7	18.2	0.11	1
Ghana	236	4.2	10.9	11.4	23.97	10.9	13.1	-30.1	11.4	0.26	117
Greece	2,237	1.8	10.0	25.4	13.84	78.4	30.3	-11.7	9.9	0.19	43
Greenland	1,364	5.1	2.6	...	6.20	32.2	36.3	-39.2	...	0.30	...
Grenada	441	4.3	4.4	...	8.01	70.8	60.9	-79.9	4.0	...	...
Guatemala	640	2.8	14.0	19.8	16.24	66.3	21.9	-39.6	13.8	0.33	76
Guinea	116	4.8	11.0	4.8	32.62	39.6	2.1	-7.1	2.8	...	...
Guinea-Bissau	91	3.3	12.6	...	13.51	15.8	11.1	-47.3	5.1	...	...
Guyana	394	5.7	1.2	...	77.58	8.0	85.4	-18.3	10.3	0.59	...
Haiti	194	-3.3	17.6	12.6	19.43	96.2	3.8	-44.5	1.4	0.24	134
Honduras	408	3.4	15.9	7.0	19.59	63.2	22.7	-31.2	12.8	0.29	94
Hungary	2,855	1.1	17.5	63.5	18.87	90.7	78.0	7.1	20.8	0.18	26
Iceland	5,222	0.0	8.9	11.8	13.95	31.2	33.0	-40.9	9.2	0.01	75
India	356	6.8	14.8	44.5	18.63	86.9	42.5	-3.1	12.0	1.36	37
Indonesia	868	5.1	19.9	35.2	27.39	75.9	32.9	2.9	13.7	0.71	36
Iran	936	6.4	15.8	46.0	27.95	63.6	39.8	10.5	17.8	1.77	52
Iraq	198	-0.2	4.8	17.7	38.14	9.7	3.7	-32.1	6.2	1.78	150
Ireland	32,270	1.4	34.0	75.6	33.99	96.1	62.5	26.3	9.9	0.02	3
Israel	5,592	-0.5	12.4	60.3	14.10	85.1	71.3	-19.4	9.1	0.07	32
Italy	4,842	0.5	14.2	44.9	16.12	90.3	56.1	11.0	18.3	0.13	11
Jamaica	442	-1.7	8.3	...	12.52	86.1	3.8	-46.3	3.5	0.28	110
Japan	7,981	-0.8	21.4	58.4	24.54	86.0	80.9	6.3	15.4	0.17	8

Table A.1 | Industrial indicators by country/area, latest available year (*continued*)

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Industry value added share in GDP 2024 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2024 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)	CIP Index 2023 (rank)
Jordan	727	2.4	18.2	34.6	22.34	80.9	39.6	-21.2	8.4	0.24	72
Kazakhstan	1,352	4.4	11.2	11.2	24.55	22.6	44.9	-26.7	6.8	0.87	63
Kenya	156	3.3	8.4	13.1	11.81	46.2	25.0	-50.6	6.8	0.39	112
Kiribati	95	3.4	4.6	...	6.08	14.3	74.6	-74.5	4.0	...	...
Kosovo	788	5.6	14.8	...	20.18	...	...	...	9.1	0.55	...
Kuwait	2,166	-6.5	9.4	29.3	47.75	66.8	39.1	17.5	5.6	1.06	51
Kyrgyzstan	216	11.4	15.0	2.1	18.09	25.0	29.2	-64.3	11.5	0.35	115
Lao People's Dem. Rep	253	6.5	8.7	4.6	25.57	51.4	30.6	-11.9	3.4	0.78	105
Latvia	1,839	-2.8	11.0	28.9	13.14	76.2	40.4	-7.4	13.2	0.15	61
Lebanon	240	-11.9	4.2	23.2	7.08	75.9	33.3	-47.5	10.8	0.36	101
Lesotho	156	3.6	15.8	1.4	28.97	90.8	6.0	-21.0	22.6	...	...
Liberia	34	-0.9	5.2	...	25.18	38.4	48.8	-28.8	1.7	...	...
Libya	118	0.5	2.3	10.8	39.43	7.2	12.6	-30.3	...	3.08	...
Liechtenstein	63,505	1.8	35.5	52.2	36.79	...	...	...	...	...	...
Lithuania	3,480	3.4	18.0	30.3	20.02	85.0	43.2	3.4	15.9	0.11	38
Luxembourg	4,622	-2.2	4.4	25.2	5.58	87.3	44.9	-19.6	2.9	0.24	46
Madagascar	41	4.9	9.1	10.2	11.85	43.1	4.1	-42.3	5.4	0.00	135
Malawi	69	4.2	12.5	11.3	16.20	32.1	66.7	-62.3	3.8	...	142
Malaysia	2,782	5.3	23.5	47.9	31.97	88.0	66.8	7.0	16.8	0.39	23
Maldives	227	4.9	1.9	2.6	4.32	60.0	29.6	-70.7	9.7	...	143
Mali	143	6.4	15.7	14.8	16.54	6.3	46.8	-47.8	4.8	0.07	...
Malta	2,039	3.7	5.7	17.1	6.38	84.4	61.4	-40.5	10.6	0.04	74
Marshall Islands	79	14.3	1.4	...	5.88	93.4	94.9	-22.7	4.3	...	...
Mauritania	105	3.7	6.2	4.4	13.57	41.3	2.7	-26.9	6.5	0.62	...
Mauritius	1,304	3.4	11.1	10.6	13.25	84.5	20.3	-40.0	11.2	0.21	98
Mexico	2,096	1.1	20.3	43.5	23.78	83.8	80.1	0.0	16.6	0.16	18
Micronesia, Fed. States of	18	0.3	0.6	...	2.78	2.8	58.2	-49.8	2.4	...	...
Monaco	6,417	0.4	2.8	47.7	2.80	...	...	...	...	...	...
Mongolia	373	3.7	8.0	9.1	23.12	51.4	1.1	-10.1	8.3	1.20	107
Montenegro	289	1.1	3.4	10.6	9.62	64.3	38.4	-65.0	6.2	0.81	127
Montserrat	717	5.1	3.5	...	7.56	74.8	71.7	-65.7	2.5	...	...
Morocco	615	3.2	17.3	32.0	22.79	79.3	62.5	-23.7	11.0	0.26	57
Mozambique	40	0.5	6.6	14.4	18.26	26.8	7.7	-33.3	2.8	0.13	133

Table A.1 | Industrial indicators by country/area, latest available year (*continued*)

	MVA per capita	MVA growth rate	MVA share in GDP	MHT share in MVA	Industry value added share in GDP	Manuf. share in exports	MHT share in manuf. exports	Norm. manuf. trade balance	Manuf. share in employment	Manuf. CO ₂ intensity	CIP Index
	2024	2024	2024	2023	2024	2024	2024	2024	2024	2022	2023
	(2015 USD)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(%)	(kg/USD)	(rank)
Myanmar	314	-5.2	21.9	5.5	27.98	48.5	11.0	-17.0	12.2	0.18	86
Namibia	414	3.1	10.3	13.6	25.90	60.7	22.1	-26.9	7.2	0.10	106
Nauru	2,178	5.4	20.6	...	33.11	7.8	59.4	-51.6	1.4	...	...
Nepal	55	2.1	4.8	9.7	8.19	83.2	6.8	-64.6	3.1	3.60	131
Netherlands	5,783	1.1	11.4	57.9	12.42	78.5	56.5	4.9	8.2	0.18	10
New Caledonia	4,474	8.4	14.8	...	20.15	97.9	54.0	-19.0	3.4	...	...
New Zealand	3,584	-4.7	8.5	19.2	11.55	50.3	22.2	-24.2	8.6	0.25	48
Nicaragua	348	4.4	15.1	13.0	21.71	56.0	25.3	-30.3	11.5	0.22	95
Niger	41	7.0	6.8	0.2	14.77	35.1	37.7	-36.4	7.6	0.35	139
Nigeria	209	1.7	8.4	3.5	11.91	11.6	20.1	-32.0	12.9	0.23	109
North Macedonia	712	3.2	10.8	45.5	14.27	90.3	66.4	-10.0	19.6	0.59	73
Norway	4,854	0.9	6.1	48.0	24.86	21.9	52.8	-19.3	7.0	0.20	40
Oman	1,639	3.0	9.5	18.1	42.36	52.1	46.2	-1.2	13.2	1.50	60
Pakistan	194	2.5	12.0	20.9	16.45	74.6	10.9	-17.3	14.9	1.04	81
Palau	153	8.7	1.1	...	2.82	74.8	79.5	-76.5	1.5	...	...
Panama	754	-0.9	4.4	6.3	10.63	61.1	43.8	-27.7	7.7	0.51	96
Papua New Guinea	44	2.7	1.7	...	24.41	30.4	2.6	-2.8	1.2	...	129
Paraguay	1,270	2.4	19.2	26.6	27.54	23.0	33.9	-44.4	10.3	0.03	87
Peru	806	2.3	12.0	16.1	22.08	54.5	3.9	-4.2	8.4	0.31	66
Philippines	727	4.9	18.5	44.6	22.49	89.6	77.7	-21.9	7.9	0.13	45
Poland	2,893	4.2	17.0	33.9	20.78	89.0	54.7	2.5	18.7	0.25	21
Portugal	1,961	-1.1	8.5	25.4	11.49	89.7	45.8	-8.1	16.5	0.23	39
Puerto Rico	13,999	3.1	45.5	86.9	47.79	...	...	...	9.0	...	...
Qatar	5,158	2.1	8.9	56.0	42.46	43.0	18.6	7.4	5.1	1.41	44
Korea, Rep. of	10,115	1.6	27.3	71.7	29.83	97.1	75.7	15.9	15.5	0.13	4
Moldova, Rep. of	299	-1.2	9.8	11.8	12.19	69.0	37.2	-41.2	7.6	0.71	111
Romania	1,875	-0.1	15.0	52.5	17.47	88.8	64.1	-11.9	19.5	0.29	35
Russian Federation	1,456	4.8	13.1	42.2	23.30	49.8	32.6	-5.7	14.0	1.27	33
Rwanda	89	9.6	8.3	16.1	11.11	39.2	12.7	-39.9	4.7	0.35	136
Saint Kitts and Nevis	898	-0.5	4.1	...	5.57	91.1	73.8	-72.2	1.8	...	...
Saint Lucia	394	2.1	3.3	...	6.88	80.4	23.2	-50.0	4.5	...	137
St. Vincent and the Grenadines	343	0.9	3.6	...	7.50	78.1	59.8	-68.1	...	...	...
Samoa	154	1.7	3.7	...	6.32	71.6	60.4	-72.7	6.0	...	...

Table A.1 | Industrial indicators by country/area, latest available year (continued)

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Industry value added share in GDP 2024 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2024 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)	CIP Index 2023 (rank)
San Marino	17,414	0.4	33.5	...	33.47	...	...	...	25.4	...	...
Sao Tome and Principe	107	1.7	6.5	...	9.39	45.2	60.2	-68.4	6.4	...	...
Saudi Arabia	2,845	4.4	12.2	38.3	32.17	40.8	50.8	-12.5	8.9	1.03	41
Senegal	238	4.2	15.7	17.9	19.82	54.3	11.4	-31.3	15.3	0.41	104
Serbia	1,094	3.4	13.9	30.4	20.22	83.2	47.7	-2.9	18.2	0.39	58
Seychelles	774	4.0	4.9	...	7.10	67.8	11.4	-41.9	5.3	...	...
Sierra Leone	60	2.4	5.3	...	11.79	85.2	10.7	-9.4	4.1	...	...
Singapore	14,322	3.7	20.7	82.4	21.83	93.4	69.9	8.4	9.0	0.16	9
Sint Maarten (Dutch part)	740	5.1	2.5	...	6.19	13.4	47.3	-72.7	...	...	...
Slovakia	3,936	4.9	20.3	51.2	22.78	94.0	73.2	7.8	16.7	0.46	25
Slovenia	5,395	1.9	20.6	50.5	24.16	92.2	76.4	2.2	22.6	0.14	31
Solomon Islands	144	0.1	7.4	...	10.33	85.7	1.6	-17.1	5.5	...	...
Somalia	12	5.1	2.2	...	2.81	11.4	42.5	-64.2	10.7	...	...
South Africa	640	-0.9	11.2	25.8	17.47	63.1	46.2	-4.3	...	0.96	53
South Sudan	16	-19.3	3.8	...	50.38	10.7	4.1	-44.9	1.0	0.18	...
Spain	3,271	2.9	11.0	40.8	13.60	77.7	56.6	-2.8	8.3	0.16	22
Sri Lanka	637	5.2	16.0	7.4	19.16	78.2	11.8	-19.1	17.3	0.06	79
State of Palestine	195	-28.3	9.8	7.6	11.74	82.8	14.6	-48.2	12.4	0.04	114
Sudan	83	-20.2	7.0	...	13.04	3.9	21.0	-50.5	2.4	0.18	...
Suriname	951	2.3	13.0	...	16.87	39.9	7.8	-31.1	12.7	0.10	120
Sweden	7,624	-0.9	14.0	58.6	16.39	88.0	61.1	5.6	9.8	0.07	16
Switzerland	19,015	1.9	20.9	71.0	22.47	69.8	67.9	9.2	12.2	0.03	7
Syrian Arab Republic	15	-4.2	1.9	10.0	10.81	58.5	23.2	-28.7	14.2	5.60	132
Tajikistan	292	8.1	19.5	2.2	27.48	55.4	8.3	-55.5	5.4	0.52	121
Thailand	1,638	2.5	25.0	40.2	30.12	86.0	63.2	6.3	15.7	0.46	28
Timor-Leste	20	2.4	1.6	...	9.07	15.7	39.0	-75.2	1.5	...	...
Togo	131	8.1	14.0	1.0	19.02	54.4	23.1	-42.2	14.3	0.16	...
Tonga	184	0.0	4.1	...	7.90	43.0	30.2	-79.4	12.9	...	150
Trinidad and Tobago	2,099	3.9	14.2	2.8	27.18	63.7	48.3	-12.1	9.0	0.58	67
Tunisia	553	1.3	13.1	28.3	17.08	89.3	48.5	-8.9	19.2	0.72	70
Turkmenistan	2,185	-0.2	29.2	7.8	30.92	21.9	28.8	-4.9	...	0.97	...
Turks and Caicos Islands	116	-1.0	0.6	...	4.74	65.7	37.2	-64.3	1.4	...	...
Tuvalu	5	1.8	0.1	...	0.95	15.1	62.7	-87.9	10.4	...	...

Table A.1 | Industrial indicators by country/area, latest available year (*continued*)

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Industry value added share in GDP 2024 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2024 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)	CIP Index 2023 (rank)
Türkiye	2,510	3.3	16.9	35.5	19.42	87.0	45.0	-4.3	19.8	0.35	24
Uganda	143	6.8	15.6	12.2	22.22	28.5	45.4	-42.6	4.3	0.23	118
Ukraine	206	3.3	10.0	28.6	15.14	59.2	25.7	-30.0	13.2	1.60	80
United Arab Emirates	5,211	6.9	12.4	31.2	31.20	51.9	41.1	-8.2	8.6	1.35	29
United Kingdom	4,143	1.3	8.8	45.3	11.06	70.4	71.7	-17.0	7.8	0.10	19
Tanzania	97	6.2	8.2	6.1	14.95	25.9	14.0	-51.9	4.4	0.47	128
United States	6,870	2.7	10.5	48.9	13.65	71.9	66.1	-25.0	9.7	0.17	5
Uruguay	1,899	1.3	10.0	18.2	12.81	39.5	31.5	-27.0	9.9	0.14	78
Uzbekistan	527	5.3	13.7	26.6	17.71	42.5	36.2	-40.5	8.9	0.82	83
Vanuatu	75	-2.3	2.9	...	5.83	59.4	9.5	-74.9	4.0	...	...
Venezuela (Bolivarian Rep. of)	56	3.4	1.4	18.6	13.75	35.8	32.2	-23.9	10.8	5.88	113
Viet Nam	1,038	10.7	25.9	35.0	30.45	89.8	57.7	4.9	21.7	0.93	30
Yemen	42	-0.6	9.0	2.1	15.66	17.4	35.2	-68.6	5.5	0.62	150
Zambia	107	3.6	8.1	13.5	22.03	19.6	30.3	-34.9	4.9	0.83	124
Zimbabwe	110	-1.8	7.8	21.8	17.67	29.2	24.0	-33.9	4.9	0.89	123

Source: [9; 19; 20; 34–36; 40]

Note: MVA per capita figures are in constant 2015 US dollars. Figures based on national accounts variables for 2024 are UNIDO estimates. CO₂ intensity is calculated as CO₂ emissions in kilograms per unit of MVA in constant 2015 US dollars. With the objective of maximizing data availability, the latest observed value for manufacturing share in employment is used: 2024 for 5.6 per cent of the cases, 2023 for 41.9 per cent, 2022 for 14.6 per cent, and before 2022 for the remaining cases. Normalized manufacturing trade balance expresses the manufacturing trade balance as a share of total trade (exports plus imports). Manuf. = manufacturing. Norm. = Normalized.

Data used to create this table can be downloaded at the [UNIDO Statistics Portal](#).

Table A.2 | Industrial indicators by country/area group, latest available year

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2023 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)
Geographical regions									
World	1,937	2.9	16.5	51.2	78.7	62.0	...	14.1	0.40
Africa	212	2.2	10.6	17.5	41.0	35.3	-25.0	8.5	0.47
<i>Northern Africa</i>	447	1.4	12.5	29.1	53.4	41.9	-24.8	12.1	0.53
<i>Sub-Saharan Africa</i>	161	2.8	9.6	11.9	35.5	31.0	-25.0	8.0	0.43
Asia and Oceania	1,862	4.0	22.4	53.8	82.8	62.7	7.4	16.1	0.53
<i>Central Asia</i>	827	3.8	14.5	14.5	26.4	39.5	-32.0	9.3	0.85
<i>Eastern Asia</i>	4,126	3.9	26.6	58.6	92.8	70.6	17.6	20.8	0.46
<i>South-eastern Asia</i>	1,106	5.1	21.7	41.6	87.2	60.3	4.3	14.6	0.51
<i>Southern Asia</i>	373	6.2	15.4	37.9	82.7	36.6	-3.9	12.6	1.21
<i>Western Asia</i>	1,613	3.0	13.0	37.3	54.2	43.3	-9.6	11.5	0.67
<i>Oceania</i>	2,145	-2.1	5.5	24.9	46.4	15.2	-18.4	5.9	0.33
Northern America and Europe	5,003	1.3	12.1	50.8	78.8	62.7	-4.7	12.8	0.19
<i>Europe</i>	4,186	0.4	14.1	52.8	81.8	62.3	2.3	14.5	0.20
<i>Northern America</i>	6,584	2.5	10.4	48.3	69.0	64.3	-23.0	9.9	0.18
Latin America and the Caribbean	1,191	1.1	13.2	35.1	63.2	56.8	-10.8	12.0	0.30
<i>Caribbean</i>	1,703	2.6	19.5	61.6	73.1	40.1	-40.5	8.0	0.45
<i>Central America</i>	1,695	1.3	18.7	40.2	82.1	76.2	-4.1	15.8	0.18
<i>South America</i>	928	0.7	10.2	26.8	45.0	24.5	-15.6	10.9	0.38
Groups by income and stage of industrial development									
High income	5,591	1.3	13.4	53.6	78.2	64.6	-2.8	12.9	0.21
<i>High-income industrial economies</i>	5,755	1.2	13.6	54.2	80.2	64.7	-2.4	13.2	0.18
<i>High-income industrializing economies</i>	2,699	3.9	9.2	35.8	62.5	64.5	-6.0	7.2	1.13
Middle income	1,310	4.5	21.5	48.9	80.3	57.3	5.0	15.3	0.60
<i>Middle-income industrial economies</i>	2,846	4.6	24.9	53.6	88.3	62.5	13.9	19.9	0.54
<i>Middle-income industrializing economies</i>	442	4.2	14.3	29.8	57.6	34.9	-15.8	11.9	0.80
Low income	64	2.0	9.0	11.7	26.9	15.1	-42.1	5.7	0.34
Selected regional groups									
Common Market for Eastern and Southern Africa (COMESA)	174	1.8	11.0	25.1	45.3	30.8	-32.1	6.2	0.51
ECCAS	128	4.6	10.5	3.6	21.1	16.1	-21.3	6.0	0.14
Economic Community of West African States (ECOWAS)	204	2.8	9.3	5.8	20.7	15.7	-28.0	12.3	0.23
East African Community (EAC)	95	4.6	10.0	10.9	31.3	13.6	-39.7	5.5	0.27

Table A.2 | Industrial indicators by country/area group, latest available year (*continued*)

	MVA per capita 2024 (2015 USD)	MVA growth rate 2024 (%)	MVA share in GDP 2024 (%)	MHT share in MVA 2023 (%)	Manuf. share in exports 2024 (%)	MHT share in manuf. exports 2024 (%)	Norm. manuf. trade balance 2024 (%)	Manuf. share in employment 2023 (%)	Manuf. CO ₂ intensity 2022 (kg/USD)
Southern African Development Community (SADC)	187	1.2	10.1	18.0	45.4	35.2	-17.4	5.5	0.65
Caribbean Community (CARICOM)	438	0.7	7.6	7.2	32.6	40.6	-30.9	7.0	0.39
Southern Common Market (MERCOSUR)	941	0.8	9.9	30.2	44.7	32.6	-15.9	11.3	0.42
United States Mexico Canada Agreement (USMCA)	5,447	2.4	10.9	47.8	71.8	67.8	-19.3	11.6	0.18
Association of Southeast Asian Nations (ASEAN)	1,108	5.1	21.7	41.6	87.2	60.3	4.3	14.6	0.51
Gulf Cooperation Council (GCC)	3,262	4.1	11.7	36.0	49.7	41.1	-6.2	7.9	1.15
Eurasian Economic Union (EAEU)	1,394	4.9	13.2	38.9	48.0	34.9	-9.0	13.2	1.16
European Union	5,289	-0.3	15.2	54.0	86.9	62.9	5.0	15.6	0.13
Other groups									
Emerging industrial economies	1,641	5.1	25.3	53.5	91.2	59.2	16.7	16.4	0.64
Least developed countries (LDCs)	175	3.6	14.8	12.0	45.9	11.5	-26.6	7.7	0.28
Landlocked developing countries (LLDCs)	216	3.9	11.7	14.5	31.7	31.4	-32.9	6.2	0.71
Small island developing States (SIDS)	2,270	3.2	18.3	71.7	87.8	67.7	1.4	7.3	0.23
BRICS	1,648	4.9	22.8	53.3	81.2	55.7	12.0	16.2	0.68
G20	2,178	2.7	16.5	52.0	80.6	62.2	0.4	14.4	0.39
Org. for Economic Co-operation and Development (OECD)	5,411	0.9	13.4	52.3	80.2	64.8	-3.2	13.4	0.16
Org. of the Petroleum Exporting Countries (OPEC)	658	4.3	10.9	32.1	44.0	40.7	-10.6	13.1	1.17

Source: [9; 19; 20; 35; 36; 40]

Note: MVA per capita figures are in constant 2015 US dollars. Figures based on national accounts variables for 2024 are UNIDO estimates. CO₂ intensity is calculated as CO₂ emissions in kilograms per unit of MVA in constant 2015 US dollars. With the objective of maximizing data availability, the latest observed values for manufacturing share in employment as well as proportion of medium and high-tech MVA in total value added are used to calculate the group aggregates. Normalized manufacturing trade balance expresses the manufacturing trade balance as a share of total trade (exports plus imports). Manuf. = manufacturing. Norm. = Normalized.

Data used to create this table can be downloaded at the [UNIDO Statistics Portal](#).

B Explanatory notes

Unless otherwise indicated, *manufacturing* refers to ISIC Rev. 4 section C, while the combined *mining and utilities* sector includes sections B, D, and E of ISIC Rev. 4 [5].ⁱ

ISIC code numbers are accompanied by a descriptive title. For example, the descriptive title of ISIC division 16 is “Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials”. However, the description is sometimes shortened for space considerations. In this example, ISIC division 16 may be described simply as “Wood products, excluding furniture”. A list of ISIC codes and their corresponding descriptive titles is given in Annex E.1.

In the presentation of statistical tables and visualizations, individual countries and areas are referred to as economies. They are listed alphabetically, unless another ranking is considered more appropriate. Economies are classified according to their stage of industrial development and their geographical regions. Additional country groupings may be used whenever necessary. For additional information on UNIDO’s classification system, see Annex C. The lists of economies according to the main country groups used in the *Yearbook* are provided in Annexes E.3, E.4, and E.5.

In some cases, data on economic activity and traded commodities are classified according to technological intensity. For more details on the sources of these classifications, as well as the lists of higher-technology industries and traded commodities, see Annex E.2.

Unless otherwise specified, data for China do not include Hong Kong SAR of China, Macao SAR of China, and Taiwan Province of China, which are reported separately. References to Kosovo should be understood within the framework of United Nations Security Council Resolution 1244 (1999). References to Kosovo and to Taiwan Province of China are included exclusively for statistical completeness and do not imply any opinion whatsoever by UNIDO regarding their legal status.

ⁱ ISIC provides a comprehensive framework for collecting and reporting economic data for purposes of economic analysis, decision-taking and policy-making.

International Standard Industrial Classification of All Economic Activities (ISIC)



References to dollars (\$) are to dollars of the United States of America. National currencies have been converted into dollar equivalents by using period average exchange rates as published in [120] and other sources. Data converted into dollars by using current exchange rates may be influenced by exchange rate fluctuations. Annual variations in data converted in that manner may not reflect movements in national currency data.

Growth rates and proportions are expressed as percentages. Periods indicated with a dash (e.g. 2010–2022) include both the initial and final years. Apparent arithmetic discrepancies, such as disaggregated figures that do not sum exactly to reported totals, may arise from the rounding of underlying data or from figures available at different levels of precision.

Throughout this publication, three dots (...) indicate that data are not available or are not separately reported.



C Overview of methodological aspects

One of the most important objectives of UNIDO's statistical databases and regular publications, such as this *Yearbook*, is providing statistical information that facilitate international comparisons related to the manufacturing sector and the combined mining and utilities sector. Bearing in mind the requirements of international comparability when maintaining a common platform for global industrial statistics, the data presented were compiled using the international recommendations and the standards endorsed by the UN. Concepts and definitions are drawn from [4; 5].

ISIC is the recommended international reference classification of productive activities. Since the adoption of the first version in 1948, the majority of countries around the world have used ISIC as their national activity classification or have developed national classifications derived from it. Besides, it has become an essential tool for comparing statistical data on economic activities at the regional and international levels.

In this publication, information on industrial sectors and their groups are presented according to ISIC Rev. 4 [5], since this revision remains the most used classification of economic activities worldwide. Data reported in accordance with ISIC Rev. 3 [121] or other national/regional classifications of economic activities were converted to ISIC Rev. 4 using appropriate correspondence tables. A new version of the classification, ISIC Rev. 5, was endorsed by the UN Statistical Commission at its 54th session in 2023 [94]. All economies are encouraged to reflect this new classification in the coming years, as far as national requirements permit, in their industrial classification schemes.

In some cases, UNIDO applies imputation methods when data are missing. The primary objective is extending the length of time series and enabling the calculation of country group aggregates. Minor discrepancies may occur due to data revisions or during the conversion and estimation processes. As a result, users may observe differences when

ISIC

ISIC has provided guidance for developing national activity classifications since its adoption in 1948. It is an essential tool for international comparison of statistical data on economic activities.

UNIDO databases
provide statistical indicators to
facilitate **long-term**
analysis of economic
performance across
economies

comparing data for the same reference year across various editions of the *Yearbook* or when contrasting earlier and later data releases.

The *Yearbook* presents information sourced from a wide variety of data, including national accounts, structural business statistics, industrial performance indicators, and short-term statistics. These official data compiled by UNIDO were originally collected by individual national authorities and NSOs through surveys and censuses conducted periodically, as well as from business registers and other administrative sources. The data collected were generally complemented with estimations made by national experts.

UNIDO relies on appropriate statistical benchmarking techniques to combine statistical information from multiple data sources in order to produce estimates with the highest degree of accuracy, reliability, coherence, and detail possible [4].

Statistical indicators in the *Yearbook* are presented using percentage distributions, cross-country averages, ratios and shares, real growth rates, and per capita measures. These indicators are designed to facilitate regional, international, and time-based comparisons.

National accounts and population

Industrial statistics based on the SNA provide a continuing flow of key information that is indispensable for the monitoring, analysis, and evaluation of an economy's performance over time. The main economic indicators presented in the *Yearbook*, including value added of the manufacturing sector (ISIC Rev. 4 section C) and those for the mining and utilities sectors (ISIC Rev. 4 sections B, D, and E), are obtained based on the SNA. They represent the net contribution of the respective sectors to GDP. Information on MVA and MUVA serve as significant benchmarks for measuring performance in those industrial sectors. In this publication, both MVA per capita at constant prices and the share of MVA in GDP are used as markers of a country's level of industrialization and its progress towards structural transformation.

UNIDO obtains data on GDP, MVA, and other national accounts aggregates mostly from the National Accounts Main Aggregates Database [122]. The database is compiled by the United Nations Statistics Division (UNSD) from official national accounts submitted by NSOs, supplemented with estimates generated by UNSD experts, whenever appropriate. It includes information for more than 200 economies and is updated annually in December. Complete information on the estimation methodology is available in [123]. The base year of these data is currently 2015.

Quality dimensions of official statistics:

Relevance,
accuracy and reliability,
timeliness and punctuality,
accessibility and clarity,
coherence and comparability

National Accounts

Statistics based on the SNA provide a continuous flow of key information indispensable for monitoring, analysing, and evaluating an economy's performance over time.

UNIDO Statistics complements this database with other national and international sources, as needed. In addition, UNIDO generates estimates of the main aggregates with the purpose of improving timeliness and facilitating comparability over time and across economies.

Population figures are based on data compiled by the Population Division of the UN Department of Economic and Social Affairs (DESA) [84].

Index of industrial production (IIP)

Another important economic benchmark featured in the *Yearbook* is the IIP, which tracks economic trends by measuring growth of industrial production in volume terms, independent of price fluctuations. For many economies, the index is available with a breakdown of manufacturing industries at the 2-digit level of ISIC Rev. 4. While the focus is primarily on the manufacturing sectors (ISIC Rev. 4 section C), it often includes data on the mining and utilities sectors (ISIC Rev. 4 sections B, D, and E) as well. The time series extend back to 2005. Figure C.1 provides an overview of the economies for which the IIP are available.

Coverage of global value added by available quarterly IIPs

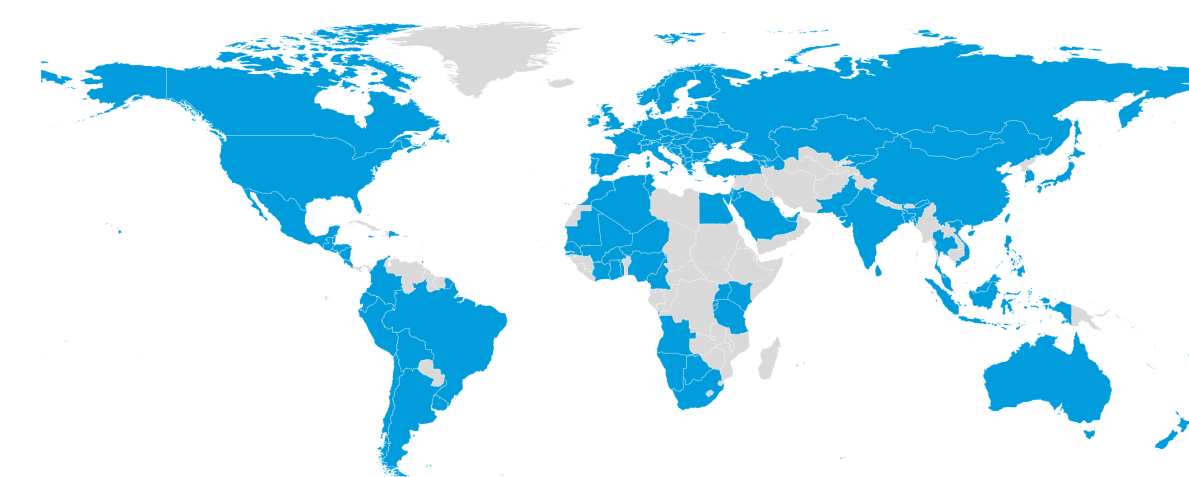
- ▶ Mining and quarrying (ISIC B): 83.2%
- ▶ Manufacturing (ISIC C): 97.5%
- ▶ Electricity (ISIC D): 76.3%
- ▶ Water supply (ISIC E): 41.5%

Features of the IIP and UNIDO's IIP databases

- ▶ The IIP measures the volume of industrial output in real terms, free from price fluctuations.
- ▶ It is one of the most important short-term statistics (STS), used to keep track of economic activity in industrial sectors, including manufacturing.
- ▶ It is usually available in quarterly and monthly frequencies, although with different coverage and level of detail.
- ▶ UNIDO regularly collects and harmonizes IIP data from across the world.
- ▶ It is available as the UNIDO IIP database [19], with a coverage of around 80 economies on a monthly basis and about 120 economies with quarterly data.
- ▶ The database includes original and seasonally-adjusted data.

The information is compiled by UNIDO from NSOs' publications and other resources. For the majority of European countries, information on IIP was obtained from the Statistical Office of the European Union (Eurostat). Most national indices are calculated using the Laspeyres formula and a combination of volume extrapolation and deflation methods. The methodologies most commonly followed are described in [99].

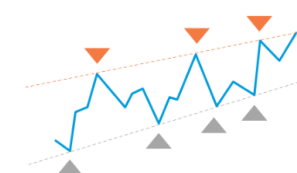
UNIDO
databases
include **81** economies
(95% of global MVA)
with both **monthly and
quarterly IIP**
plus around **40**
additional economies
with **quarterly IIP only**

Figure C.1 | Data availability of quarterly IIP indices, second quarter of 2025

Source: [19]

Short-term estimates are often influenced by seasonal and calendar-related effects, which can complicate the analysis of economic trends. To address this, UNIDO conducts seasonal adjustment of the data when necessary, using the TRAMO/SEATS method in JDemetra+ software [124].

Currently, the base year of indices in the IIP database is 2015. If a country uses a different base year or a different classification system other than ISIC Rev. 4, the national indices are converted as appropriate. Further information on the methodology and the estimation procedure of these indices is available in [125; 126].



UNIDO IIP time series are **seasonally adjusted** to minimise the influence of seasonal- and calendar-related factors

Manufacturing trade

The *Yearbook* presents annual and monthly/quarterly manufacturing trade data to assess the international performance of manufacturing across the world. Trade statistics capture the physical movement of products across borders and reflect a key component of global economic integration and competitiveness. In principle:

- Imports comprise all foreign goods entering a country, including purchases of goods manufactured abroad and re-imports (i.e. domestically produced goods that were previously exported and later returned).

Both **re-exports** and **re-imports** are included in trade statistics

- Exports include domestically produced goods and re-exports (i.e. foreign goods that were previously imported and later exported without significant domestic transformation).

Two main data sources underpin the *Yearbook's* trade statistics. First, yearly data on exports and imports come from the UNIDO Annual Manufacturing Trade Database [40], which contains country, region, and group-level aggregates of trade in manufactured goods, expressed in current USD. Time series generally extend back to 1995 and have full country coverage. The primary source is UNCTAD, complemented by additional official data where appropriate.

Second, sub-annual manufacturing trade data, used to capture short-term dynamics, are drawn from the UNIDO Monthly Manufacturing Trade Database [41]. Figure C.2 shows the economies covered in this database. Although the data are collected on a monthly basis, the *Yearbook* reports them at a quarterly frequency to smooth short-term volatility.

The UNIDO database includes 60+ economies (around 90% of global exports and imports) with monthly trade available

Features of the UNIDO Monthly Manufacturing Trade Database

- Trade data capture exports and imports in current USD.
- These short-term indicators reflect the linkages between producers and consumers across borders within the global trading system.
- Data are typically available at quarterly and monthly frequencies, though country coverage and detail vary.
- UNIDO regularly collects and harmonizes trade data released by national authorities.
- The database currently provides monthly information for approximately 60 economies.
- The database includes original as well as seasonally-adjusted time series.

UNIDO compiles this database from official releases by national authorities; for most European economies, the source is Eurostat. Values are expressed in current USD, converted from national currency when needed using monthly exchange rates published by the [120].

Quarterly and monthly trade data can be affected by calendar and seasonal patterns, which may obscure underlying economic trends. When necessary, UNIDO applies seasonal adjustment using the TRAMO/SEATS method in the JDemetra+ software [125; 124]. Further methodological information on this database is available in [127].

Both UNIDO trade databases rely on the Revision 3 of the SITC, which is consistent with a modified version of the Lall classification for identifying manufactured goods [42]. According to this classification, man-

Coverage of global manufacturing trade by available quarterly trade data

- Exports: 89.9%
- Imports: 90.8%

annually to NSOs, data compiled jointly by regional and international organizations such as Eurostat and the OECD, and relevant publications issued by individual national authorities.

It is well understood that individual national data compilation schemes vary, which could affect comparability across economies. Hence, national authorities are encouraged to adhere to UN's concepts and definitions on industrial statistics, whenever possible.

Production (output) is an essential economic indicators in industrial statistics. While some data are still reported using census concept covering only activities of industrial nature, current recommendations call for approximating the national accounts measurement by including revenue from activities of a non-industrial nature.

Output cannot be directly observed from the accounting records of enterprises/establishments [4]. The value of output in the case of estimates compiled on a production basis comprises:

- a) The value of the sale of all products of the establishment;
- b) The net change between the beginning and end of the reference period in the value of work in progress and stocks of goods to be shipped in the same condition as received;
- c) The value of industrial work performed or industrial services rendered to others;
- d) The value of goods shipped in the same condition as received less the amount paid for these goods; and
- e) The value of fixed assets produced during the period by the unit for its own use.

In the case of estimates compiled on a shipment basis, the net change in the value of stocks of finished goods between the beginning and the end of the reference period is also included. Gross output is equivalent to census output plus the revenue from activities of a non-industrial nature.

Valuation methods of output vary across countries/areas. The main difference is the inclusion or exclusion of taxes and subsidies on products and production (see Figure C.3).

- *Values at factor costs* exclude taxes on production (taxes that the producing units are liable to pay as a result of engaging in production) and includes subsidies on production (subsidies that resident enterprises may receive as a result of engaging in production, with the exception of direct subsidies on products).
- *Values at basic prices* equal values at factor costs including taxes on production and excluding subsidies on production.
- Finally, *values at producers' prices* equal values at basic prices plus taxes on products (taxes payable per unit of a product, excluding value added tax (VAT)) minus subsidies on products

Manufacturing output refers to the more refined or higher value merchandise for use or sale. It is the **end-product** of industrial processes that use labour, machines, tools, and chemical and biological processing or formulation

Census and gross output

The difference between census output and gross output is the exclusion or inclusion of the output from the activities that are non-industrial in nature.

Value at factor costs

excludes taxes on production and includes subsidies on production

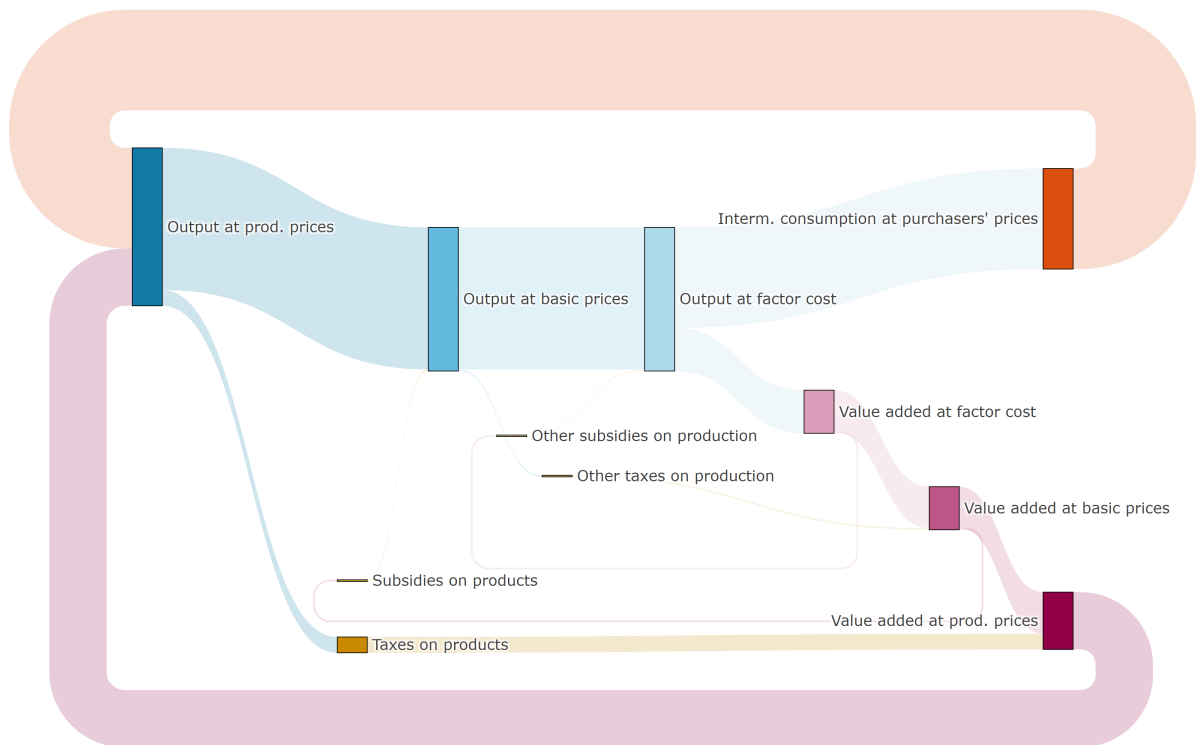
Value at basic prices =

Value at factor costs + taxes on production - subsidies on production

Value at producers' prices =

Value at basic prices + taxes on products (excl. VAT) - subsidies on products

Figure C.3 | Valuation methods in SBS



Source: UNIDO elaboration

Note: The proportions were chosen individually and do not have generally valid implications.

(subsidies receivable per unit of a product). For more details, see [128, p. 5].

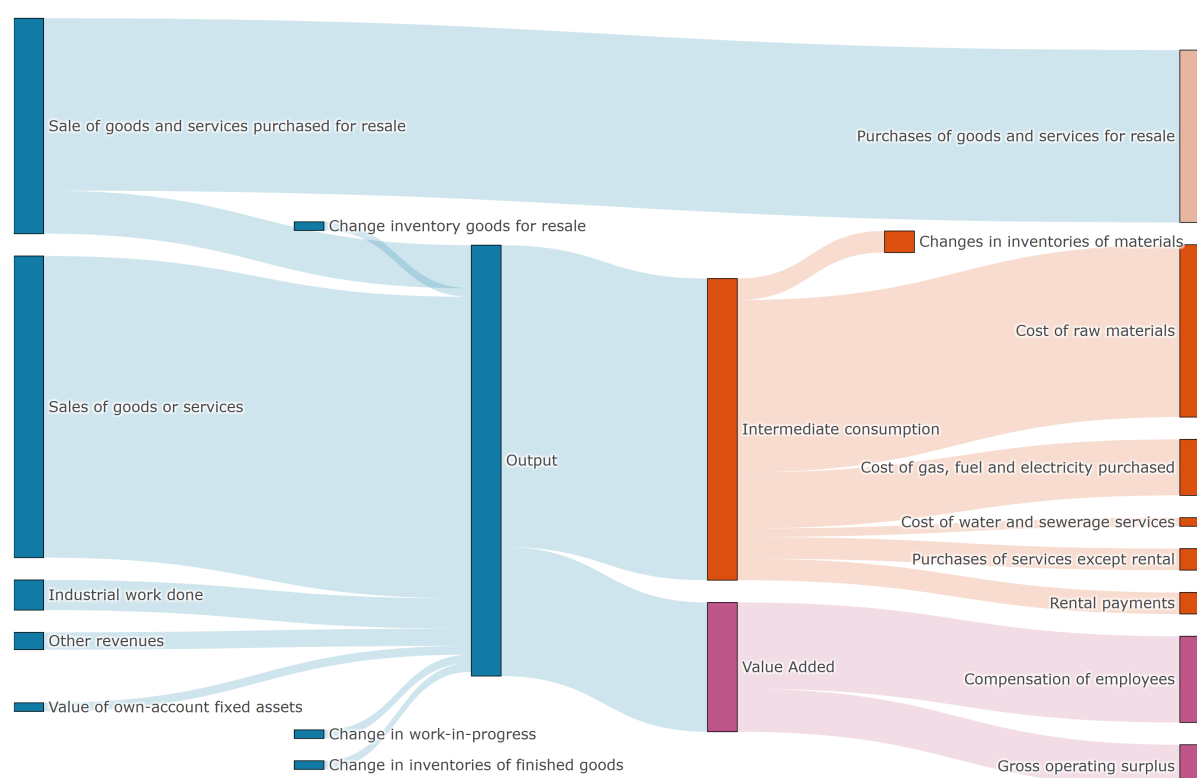
Another important economic measurement is value added. In SBS, the value added of manufacturing industries is a survey concept that refers to the given industries' net output derived from the difference of gross output and intermediate consumption/input. Items covered in the latter include:

- a) value of materials and supplies for production, including the cost of all fuels and electricity purchased; and
- b) the cost of services received (mainly payments for contract and commission work and repair and maintenance work).

If input estimates are compiled on a "received" rather than on a "consumed" basis, the result is adjusted for the net change between the beginning and the end of the period in the value of stocks of materials, fuel, and other supplies. The estimates of value added are gross of depreciation and other provisions for capital consumption, unless otherwise stated. Similar to output, the valuation of value added may be at factor costs, at basic prices, or at producers' prices, depending on the treatment of indirect taxes and subsidies as described above and summarized in Figure C.3.

Beyond output,
value added and
employment
are essential economic
measurements

Figure C.4 | Relationship between output, intermediate consumption, and value added



Source: UNIDO elaboration

Note: The proportions of the variables were chosen individually and do not reflect any generally valid pattern. The categories needed to be abbreviated. The interested reader can find the full description and definition of the variables in [4]. Additionally, the terms primarily reflect the situation of a producer in manufacturing.

Figure C.4 provides a visual overview of the key components of output, intermediate consumption, and value added, as well as their linkages.

One more important statistical concept presented in UNIDO's databases and in this publication is employment. This is one of the central elements in the 2030 Agenda for Sustainable Development. Within the framework adopted at the 19th International Conference of Labour Statisticians in 2013, employment is defined as work performed in return for pay or profit [129]. It has long been recognized as a pathway to economic development, social inclusion, and well-being. Estimates on the number of employees include both full-time and part-time workers, excluding working proprietors, active business partners, unpaid family workers, and home workers. The figures reported usually refer to the average number of workers during the reference year, obtained as the sum of the "average number of employees" during the year or the total number of employees measured for a single period of the year.

National authorities may use the data on the number of persons engaged when those for the employees are not separately available. However, both terms are not interchangeable. Figures for persons en-

Employment

Industrial statistics on employment are essential tools for measuring progress towards national and international policy goals.

gaged cover employees, working proprietors, active business partners, and unpaid family workers.

It is worth noting that this publication also uses employment estimates obtained through household or labour force surveys; although these estimates are more timely and may provide a more complete coverage than employment estimates from SBS, they are based on different statistical units and not available with the same granularity by industry of economic activity.

Finally, it is important to highlight that in this publication, national accounts data, SBS, and other sources may be combined with the objective of providing comprehensive statistics for analysing and comparing industrial performance across country groups at a relatively detailed level of economic activity. For instance, it is possible to construct indicators with the combined information from the IIP and value added estimates obtained from SBS.

For each economy and industrial division, value added estimates were generated by applying IIP figures to the 2015 value added base weights. These weights were estimated by UNIDO using various national and international sources. For more country-specific SBS, readers can refer to the databases available at the [UNIDO Statistics Portal](#) [3].

SBS vs. IIP

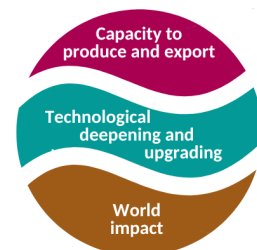
SBS refer to low-frequency data (annual or less frequent) that are comprehensive but not very timely. IIP relates to high-frequency data (quarterly and monthly) that are timely but have less detail and reduced coverage.

Competitive industrial performance (CIP) Index

The CIP Index is a composite indicator that measures the capacity of countries/areas to increase their presence in international and domestic markets, while developing industrial sectors and activities with higher value added and higher technological level. It is constructed by combining eight variables across three dimensions:

- a) *Capacity to produce and export manufactured goods.* This dimension captures a country's ability to increase the presence of its manufactured goods in international and domestic markets. It covers comparable measures of the country's manufacturing production and exports, namely MVA and manufacturing exports in per capita terms. These indicators allow for cross-country comparisons, independent of the differences in population or economy size.
- b) *Technological deepening and upgrading.* The second dimension measures the technological complexity of manufacturing processes in a country by using two composite sub-indices. First, the degree of industrialization intensity that estimates the complexity of production processes. This indicator consists of the share of MVA originating from MHT industries and the share of MVA in GDP. Second, export quality, another composite indicator

Three dimensions of the CIP Index



that measures the quality of the integration process of the country's manufacturing sector. This indicator is estimated based on the share of MHT manufacturing exports in total manufacturing exports and the share of manufacturing exports in total exports.

- c) *World impact*. The third dimension serves as a proxy of economies of agglomeration, scope, and scale present in a country's manufacturing sector. It captures the world impact based on two indicators: the country's share in world MVA and in world trade of manufactured goods.

All variables are first standardized to a $[0, 1]$ interval. Scores for each dimension are obtained by calculating the geometric mean of the underlying variables. Finally, the overall CIP score is calculated as the geometric mean of the scores for the three dimensions. The higher the score in any of the variables, the higher the country's industrial competitiveness and its rank in the overall CIP Index. The detailed conceptual framework and methodology of the Index are described in [130]. The Index's scores, ranks, and underlying indicators are available in the [UNIDO Statistics Portal](#) [34].

Sustainable Development Goal (SDG) 9 indicators

Following the adoption of the 2030 Agenda for Sustainable Development, the global indicator framework for SDGs was developed by the UN Statistical Commission through its Inter-agency and Expert Group on SDG Indicators (IAEG-SDGs). This was adopted by the UN General Assembly in July 2017 and is contained in the Resolution adopted by the General Assembly on Work of the Statistical Commission pertaining to the 2030 Agenda for Sustainable Development [131]. The latest reference metadata information for all SDG indicators is available in [132]. The current global SDG database can be consulted in [133].

SDG 9, one of the 17 global Goals, is composed of eight targets and 12 indicators aiming at building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation.

UNIDO is a custodian agency of six industry-related SDG indicators under Goal 9:

- ▶ 9.2.1 Manufacturing value added as a proportion of GDP and per capita
- ▶ 9.2.2 Manufacturing employment as a proportion of total employment
- ▶ 9.3.1 Proportion of small-scale industries in total industry value added based on national and international classifications
- ▶ 9.3.2 Proportion of small-scale industries with a loan or line of credit

The current edition of the
CIP Index
evaluates the industrial
performance of
153 economies



Shared blueprint for **peace,**
prosperity for **people**
and the **planet**



SDG 9 aims to
build resilient infrastructure,
promote inclusive
and sustainable
industrialization,
and foster innovation

- ▶ 9.4.1 CO₂ emission per unit of value added
- ▶ 9.b.1 Proportion of medium and high-tech industry value added in total value added.

UNIDO has developed advanced tools for monitoring the performance of countries/areas towards achieving the industry-related targets of the SDG 9. These include an overall composite index, the SDG 9 Industry Index described below, as well as a progress metric and an outlook assessment. They have the objective of identifying lagging cases and thus support economies in their successful implementation of ISID. The data and relevant analytical tools are available in the [UNIDO Statistics Portal](#) [20]. Detailed methodological notes about these tools are available at [28].



What drives SDG 9 industry performance?
Explore the latest progress on the
UNIDO Statistics Portal

SDG 9 Industry Index

The SDG 9 Industry Index is a composite indicator that summarizes countries' performance on the industry-related targets of SDG 9. It combines the information of the indicators for which UNIDO is the custodian agency and for which sufficient data are available (i.e. the two series of indicator 9.2.1, plus 9.2.2, 9.4.1, and 9.b.1).

A short explanation of the Index's calculation is presented below. A more complete description on the methodology can be found in [28; 134].

All indicators are first normalized through the min-max method to standardize the variables for further data aggregation, as they have different measurement units. The normalization is applied with three caveats: (i) the minimum and maximum values for normalizing each indicator are calculated over the entire sample of observations, which consists of all countries with available information over the period from 2000 to the latest year; (ii) to avoid the excessive influence of outliers in the transformation, the observed minimum and maximum values are capped at three standard deviations from the mean; (iii) an inverse normalization is applied to SDG indicator 9.4.1 (*CO₂ emission intensity of manufacturing*), so that all normalized indicators follow the same direction.

After these steps, normalized indicators have a range of [0, 1], with zero indicating the case with the lowest performance and one indicating the case with the best performance. All normalized indicators have the same direction: a higher value indicates a better performance. Finally, since the normalization is calculated using common parameters, the resulting indicators can be used for comparisons across countries and time.

SDG indicators on
small manufacturing firms
lack sufficient data coverage
to be included in the
SDG 9 Industry Index

The Index score is then calculated as the geometric mean of the five normalized indicators. Geometric aggregation allows to limit compensability among indicators. However, it has the disadvantage that a country with a value of zero in one normalized indicator (i.e. when it coincides with the minimum value observed in the sample) also has a composite Index equal to zero in that period. It is worth noting that all indicators are aggregated under an equal weighting scheme. This choice makes the Index transparent and facilitates interpretation [135]. Ranks are also calculated to facilitate country comparison.

As described in the methodology summary above, the construction of the Index requires available information for all indicators. The Index currently covers 138 countries for the period from 2000 to 2022.

Country/area classifications

The main purpose of the UNIDO country classification is defining a relatively homogeneous set of country groups that have common features in terms of their stage of industrial development, determined by objective criteria derived from official indicators of industrial performance. It adheres to the basic principles of statistical classifications, as defined in [136]. The country groups are revised annually with the objective of reflecting countries' current economic situation, while the classification methodology remains stable. More information on the classification methodology can be found in [38].

The main country classification proposed by UNIDO divides countries/areas according to a combination of their income level and their stage of industrial development. It therefore relies on two parameters: (i) per capita income levels, sourced from the World Bank's 2025 income groups [137]; and (ii) structural transformation metrics (MVA per capita in constant USD, historical maximum of the share of manufacturing employment in total employment, and exports of MHT products per capita), calculated from UNIDO databases. This classification system allocates countries/areas among five groups:

- *High-income industrial economies*: Countries/areas that have achieved a high national income through a development path that resulted in highly industrialized economies.
- *High-income industrializing economies*: Countries/areas with a high income level but with relatively low levels of industrialization.
- *Medium-income industrial economies*: Countries/areas classified as medium-income economies that have already achieved positive outcomes in terms of structural transformation indicators.

SDG 9 Industry Index

is a summary measure of
inclusive and
sustainable
industrial
development
as declared in the 2030 Agenda
for Sustainable Development

UNIDO country classification

is based on a combination of
income groups
and
industrialization
indicators

Five UNIDO country groups:
two groups of
industrial economies
and three groups of
industrializing
economies

- *Medium-income industrializing economies*: Countries/areas classified as medium-income but that still remain at comparatively low levels of industrialization; these are some of the economies that could benefit the most from prioritizing industrial development in their policy strategies.
- *Low-income economies*: Countries/areas that remain at low income levels and also underperform in industrial development indicators; these economies would greatly gain from an accelerated structural change process that could help them escape the development trap.

This classification has the objective of summarizing large amounts of country-level data into relevant insights and common trends observed in countries facing similar development challenges. The list of economies according to these five groups is included in Annex E.4.

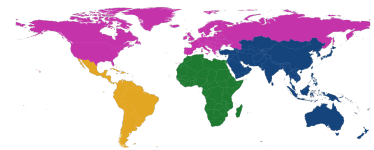
A special economic group proposed by UNIDO is that of emerging industrial economies (EIEs). These are the most dynamic low- and middle-income economies in terms of their industrial performance. It includes economies that have continuously registered positive spells of growth in MVA. This group includes 12 economies in 2025: three middle-income industrial economies, six middle-income industrializing economies, and three low-income economies. The list is given in Annex Table E.5.1.

Countries/areas are also classified according to geographical region. UNIDO follows the UN Standard Country or Area Codes for Statistical Use (M49) [138]. According to this standard, all world economies are assigned to one main regions, which are then further divided into subregions. These regional groupings are, inter alia, the main classification system used for SDG indicators. The complete classification is given in Annex E.3.

Finally, the report also considers the three country groups with specific development challenges, according to official UN lists. These are LDCs, LLDCs, and SIDS, presented in Annex E.5.

Emerging industrial economies

includes the most dynamic low- and middle-income economies in terms of industrial performance



Geographical regions

follow the official UN classification of countries and areas for statistical use: M49 Standard

D UNIDO Statistics and the International Yearbook of Industrial Statistics

D.1 UNIDO Statistics

Following the recommendations of the United Nations Statistical Commission, UNIDO has the mandate to compile and disseminate international statistics on the manufacturing, mining, and utilities sectors. This involves the regular collection, screening, and validation of industrial data from official sources and presenting them according to harmonized definitions and classifications. UNIDO generates analytical indicators based on official data, which can be used to monitor a country's prevailing performance and progress in terms of ISID.

The statistics are disseminated through various databases available at the [UNIDO Statistics Portal](#) [3], as well as publications such as this *Yearbook*. These products offer statistical information to a wide range of users, including national governments, international development partners, academic and research institutions, as well as the business and statistical communities. Priority is given to quality assurance of statistics, especially in terms of international comparability and alignment with UN recommended concepts and classifications.

As a custodian agency of six industry-related indicators under SDG 9, UNIDO has been actively involved in the process of developing and monitoring the global indicator framework for the 2030 Agenda for Sustainable Development. UNIDO contributes to the *UN SDG Global Database* [133], reports to the IAEG-SDGs, and collaborates with countries on improving data availability and quality for better tracking their progress towards the 2030 Agenda.

The organization cooperates closely with other international agencies, namely the ILO on employment; the International Energy Agency (IEA) on CO₂ emissions; and the World Bank on access to finance, to produce data and compile the corresponding SDG indicators. Furthermore, UNIDO publishes the *SDG 9 Industry Index*, benchmarking countries' performance towards achieving the industry-related targets of SDG 9 [134]. Countries can monitor their progress and prospects towards



Open access to
statistical products
at the [UNIDO Statistics Portal](#)

UNIDO industrial statistics databases

- ▶ INDSTAT [35; 36]
- ▶ IDSB [139]
- ▶ IIP [19]
- ▶ SDG 9 [20]
- ▶ National accounts [9]
- ▶ Manufacturing trade [40; 41]
- ▶ CIP Index [34]



SDG 9 monitoring

achieving industry-related SDG targets by using the tools available at the [UNIDO Statistics Portal](#) [3]. These insights are summarized in a biennial UNIDO report, presenting progress achieved towards the industry-related SDG 9 targets [39].

Industrial statistics are essential for formulating industrial development strategies and policies, analysing structural change and monitoring economic growth. UNIDO shares its expertise in industrial statistics with national and regional statistical offices by providing technical assistance and trainings on a wide range of topics. By implementing customized programmes to target countries' specific needs, UNIDO assists in strengthening the institutional capacity of NSOs and line ministries to collect industrial data, maintain business registers, set up short-term statistical indicators and information systems, carry out data analyses of industrial performance, and track progress on SDGs. UNIDO furthermore provides capacity development activities on the usage of empirical data for economic analyses, which are relevant for promoting industrial development through evidence-based policymaking.

D.2 The International Yearbook of Industrial Statistics

This *Yearbook* presents a summary of the most recent data in industrial sectors. It outlines the current performance and latest developments, allowing the reader to analyse patterns of growth, business cycle fluctuations, and longer-term trends, including progress towards structural change, ISID and SDG 9. While the *Yearbook* focuses on aggregate global trends observed in industrial sectors, it also highlights the main regional trends and sector-specific developments. More detailed information for specific regions and country groups are presented in Factsheets accompanying the *Yearbook*.

This publication relies on visualizations and short analytical texts to summarize the prevailing situation in global industry and to highlight key insights extracted from the data. It is essentially sourced from information that is available on [UNIDO Statistics Portal](#) [3]. Interested readers are invited to visit this resource to access the full datasets and related statistical products.

All data presented in this *Yearbook* were compiled bearing in mind the coherence of industrial statistics derived from various sources, the requirements of international comparability, and the standards of this work endorsed by the UN Statistical Commission. Concepts and definitions were drawn from international recommendations applicable to industrial statistics [4] and the classification of economic activities according to ISIC [5].

SDG 9 Progress Report 2025



Technical assistance
and trainings



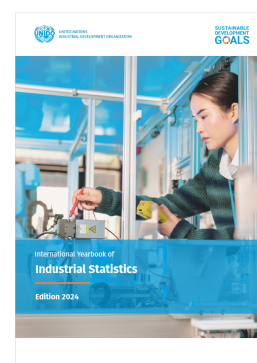
All data featured in this *Yearbook*
can be accessed at the
**UNIDO Statistics
Portal**

This is the thirty-first edition of the *International Yearbook of Industrial Statistics*, published by UNIDO since 1995. For historical reference, this publication replaced two reports: the *Handbook of Industrial Statistics* [140], which was published biennially by UNIDO until 1992, and the *United Nations Industrial Statistics Yearbook Volume I (General Industrial Statistics)* [141], which was discontinued after its 1991 edition was published in 1993 by UNSD. These changes were introduced in line with the recommendations of the UN Statistical Commission at its 27th session.

This four most recent editions of the *Yearbook* feature several novelties compared to earlier editions. The current format represents an effort to modernize this long-standing publication, expanding its coverage to include all aspects of industrial statistics. It merges the contents of earlier editions of the *Yearbook* (see [142] for the last edition under the previous format) and the biennial report *World Statistics on Mining and Utilities* (see [143] for the last edition), at the same time expands its coverage to include the CIP Index, SDG 9 indicators, and other industrial performance indicators.

The *Yearbook* will continue upgrading and expanding its coverage of ISID indicators in coming years, reaffirming UNIDO's role as the international reference point on industrial statistics.

International Yearbook of Industrial Statistics 2024



E Classifications

E.1 International Standard Industrial Classification of All Economic Activities

Table E.1.1 | ISIC Rev. 4 B – Mining and quarrying

Division	Group	Description
05		Mining of coal and lignite
	051	Mining of hard coal
	052	Mining of lignite
06		Extraction of crude petroleum & natural gas
	061	Extraction of crude petroleum
	062	Extraction of natural gas
07		Mining of metal ores
	071	Mining of iron ores
	072	Mining of non-ferrous metal ores
08		Other mining and quarrying
	081	Quarrying of stone, sand and clay
	089	Mining and quarrying n.e.c.
09		Mining support service activities
	091	Support activities for petroleum
	099	Support activities for other mining, quarrying

Source: [5]

Table E.1.2 | ISIC Rev. 4 C – Manufacturing

Division	Group	Description
10		Food products
	101	Processing/preserving of meat
	102	Processing/preserving of fish, etc.
	103	Processing/preserving of fruit,vegetables
	104	Vegetable and animal oils and fats
	105	Dairy products
	106	Grain mill products,starches and starch products
	107	Other food products
	108	Prepared animal feeds
11		Beverages
	110	Beverages
12		Tobacco products
	120	Tobacco products
13		Textiles
	131	Spinning, weaving and finishing of textiles
	139	Other textiles
14		Wearing apparel
	141	Wearing apparel, except fur apparel
	142	Articles of fur
	143	Knitted and crocheted apparel
15		Leather and related products
	151	Leather;luggage,handbags,saddlery,harness;fur
	152	Footwear
16		Wood products, excluding furniture
	161	Sawmilling and planing of wood
	162	Wood products, cork, straw, plaiting materials
17		Paper and paper products
	170	Paper and paper products
18		Printing and reproduction of recorded media
	181	Printing and service activities related to printing
	182	Reproduction of recorded media
19		Coke and refined petroleum products
	191	Coke oven products
	192	Refined petroleum products
20		Chemicals and chemical products
	201	Basic chemicals,fertilizers, etc.
	202	Other chemical products
	203	Man-made fibres
21		Pharmaceuticals,medicinal chemicals, etc.
	210	Pharmaceuticals,medicinal chemicals, etc.
22		Rubber and plastics products
	221	Rubber products
	222	Plastics products
23		Other non-metallic mineral products
	231	Glass and glass products
	239	Non-metallic mineral products n.e.c.
24		Basic metals
	241	Basic iron and steel
	242	Basic precious and other non-ferrous metals
	243	Casting of metals

Table E.1.2 | ISIC Rev. 4 C – Manufacturing (continued)

Division	Group	Description
25		Fabricated metal products, except machinery
	251	Struct.metal products, tanks, reservoirs
	252	Weapons and ammunition
	259	Other metal products;metal working services
26		Computer, electronic and optical products
	261	Electronic components and boards
	262	Computers and peripheral equipment
	263	Communication equipment
	264	Consumer electronics
	265	Measuring,testing equipment; watches, etc.
	266	Irradiation/electromedical equipment,etc.
	267	Optical instruments and photographic equipment
	268	Magnetic and optical media
27		Electrical equipment
	271	Electric motors,generators,transformers,etc.
	272	Batteries and accumulators
	273	Wiring and wiring devices
	274	Electric lighting equipment
	275	Domestic appliances
	279	Other electrical equipment
28		Machinery and equipment n.e.c.
	281	General-purpose machinery
	282	Special-purpose machinery
29		Motor vehicles, trailers and semi-trailers
	291	Motor vehicles
	292	Automobile bodies, trailers and semi-trailers
	293	Parts and accessories for motor vehicles
30		Other transport equipment
	301	Building of ships and boats
	302	Railway locomotives and rolling stock
	303	Air and spacecraft and related machinery
	304	Military fighting vehicles
	309	Transport equipment n.e.c.
31		Furniture
	310	Furniture
32		Other manufacturing
	321	Jewellery, bijouterie and related articles
	322	Musical instruments
	323	Sports goods
	324	Games and toys
	325	Medical and dental instruments and supplies
	329	Other manufacturing n.e.c.
33		Repair and installation of machinery/equipment
	331	Repair of fabricated metal products/machinery
	332	Installation of industrial machinery/equipment

Source: [5]

Table E.1.3 | ISIC Rev. 4 D – Electricity, gas, steam and air conditioning supply

Division	Group	Description
35		Electricity, gas, steam & air conditioning
	351	Electric power generation, transmission
	352	Manufacture of gas
	353	Steam and air conditioning supply

Source: [5]

Table E.1.4 | ISIC Rev. 4 E – Water supply; sewerage, waste management and remediation activities

Division	Group	Description
36		Water collection, treatment and supply
	360	Water collection, treatment and supply
37		Sewerage
	370	Sewerage
38		Waste collection,treatment,disposal activities
	381	Waste collection
	382	Waste treatment and disposal
	383	Materials recovery
39		Remediation activities
	390	Remediation activities

Source: [5]

E.2 List of higher technology industries and traded commodities

Data on economic activity and traded commodities can be classified according to technological intensity. In the case of industries, the classification is based on R&D expenditure relative to value added, also known as R&D intensity, as calculated by the OECD [26; 27]. In this publication, “higher-technology” industries refer to the group of medium-high and high technology industries (MHT) given in Table E.2.1*. In the case of traded commodities, the classification is based on technological content as calculated in [42; 144]. In this publication, “higher-technology” commodities refer to the group of medium- and high-technology commodities, provided in Table E.2.2

Table E.2.1 | Medium-high and high technology manufacturing industries by ISIC Rev. 3 and 4

Description	
ISIC Rev. 3	
24	Manufacture of chemicals and chemical products
29	Manufacture of machinery and equipment n.e.c.
30	Manufacture of office, accounting and computing machinery
31	Manufacture of electrical machinery and apparatus n.e.c.
32	Manufacture of radio, television and communication equipment and apparatus
33	Manufacture of medical, precision and optical instruments, watches and clocks
34	Manufacture of motor vehicles, trailers and semi-trailers
35	Manufacture of other transport equipment, excluding
	351 = Building and repairing of ships and boats
ISIC Rev. 4	
20	Manufacture of chemicals and chemical products
21	Manufacture of basic pharmaceutical products and pharmaceutical preparations
252	Manufacture of weapons and ammunition
26	Manufacture of computer, electronic and optical products
27	Manufacture of electrical equipment
28	Manufacture of machinery and equipment n.e.c.
29	Manufacture of motor vehicles, trailers and semi-trailers
30	Manufacture of other transport equipment excluding
	301 = Building of ships and boats
325	Manufacture of medical and dental instruments and supplies

Source: [26; 27; 145]

* Although this classification includes both the 2-digit and 3-digit levels of ISIC Rev. 4, the majority of the analysis presented in the *Yearbook* is based on the 2-digit level of ISIC Rev. 4 because of data availability constraints. Nevertheless, whenever data at 3-digit level of ISIC Rev. 4 were available, they were also considered in UNIDO estimates. For detail country-specific SBS, readers can refer to the databases available at UNIDO [3].

Table E.2.2 | Medium and high technology commodities by SITC Rev. 3

Description	
SITC Rev. 3	
266	Synthetic fibres suitable for spinning
267	Other man-made fibres suitable for spinning; waste of man-made fibres
512	Alcohols, phenols, phenol-alcohols, and their halogenated, sulphonated, nitrated or nitrosated derivatives
513	Carboxylic acids and their anhydrides, halides, peroxides and peroxyacids; their halogenated, sulphonated, nitrated or nitrosated derivatives
525	Radioactive and associated materials
533	Pigments, paints, varnishes and related materials
541	Medicinal and pharmaceutical products, other than medicaments of group 542
542	Medicaments (including veterinary medicaments)
553	Perfumery, cosmetic or toilet preparations (excluding soaps)
554	Soap, cleansing and polishing preparations
562	Fertilizers (other than those of group 272)
571	Polymers of ethylene, in primary forms
572	Polymers of styrene, in primary forms
573	Polymers of vinyl chloride or of other halogenated olefins, in primary forms
574	Polyacetals, other polyethers and epoxide resins, in primary forms; polycarbonates, alkyd resins, polyallyl esters and other polyesters, in primary forms
575	Other plastics, in primary forms
579	Waste, parings and scrap, of plastics
581	Tubes, pipes and hoses, and fittings therefor, of plastics
582	Plates, sheets, film, foil and strip, of plastics
583	Monofilament of which any cross-sectional dimension exceeds 1 mm, rods, sticks and profile shapes, whether or not surface-worked but not otherwise worked, of plastics
591	Insecticides, rodenticides, fungicides, herbicides, anti-sprouting products and plant-growth regulators, disinfectants and similar products, put up in forms or packings for retail sale or as preparations or articles (e.g., sulphur-treated bands, wicks and
593	Explosives and pyrotechnic products
597	Prepared additives for mineral oils and the like; prepared liquids for hydraulic transmission; anti-freezing preparations and prepared de-icing fluids; lubricating preparations
598	Miscellaneous chemical products, n.e.s.
653	Fabrics, woven, of man-made textile materials (not including narrow or special fabrics)
671	Pig-iron, spiegeleisen, sponge iron, iron or steel granules and powders and ferro-alloys
672	Ingots and other primary forms, of iron or steel; semi-finished products of iron or steel
678	Wire of iron or steel
711	Steam or other vapour-generating boilers, superheated water boilers, and auxiliary plant for use therewith; parts thereof
712	Steam turbines and other vapour turbines, and parts thereof, n.e.s.
713	Internal combustion piston engines, and parts thereof, n.e.s.
714	Engines and motors, non-electric (other than those of groups 712, 713 and 718); parts, n.e.s., of these engines and motors
716	Rotating electric plant, and parts thereof, n.e.s.
718	Power-generating machinery, and parts thereof, n.e.s.
721	Agricultural machinery (excluding tractors), and parts thereof
722	Tractors (other than those of headings 744.14 and 744.15)

- 723 Civil engineering and contractors' plant and equipment; parts thereof
- 724 Textile and leather machinery, and parts thereof, n.e.s.
- 725 Paper mill and pulp mill machinery, paper-cutting machines and other machinery for the manufacture of paper articles; parts thereof
- 726 Printing and bookbinding machinery, and parts thereof
- 727 Food-processing machines (excluding domestic); parts thereof
- 728 Other machinery and equipment specialized for particular industries; parts thereof, n.e.s.
- 731 Machine tools working by removing metal or other material
- 733 Machine tools for working metal, sintered metal carbides or cermets, without removing material
- 735 Parts, n.e.s., and accessories suitable for use solely or principally with the machines falling within groups 731 and 733 (including work or tool holders, self-opening die-heads, dividing heads and other special attachments for machine tools); tool holder
- 737 Metalworking machinery (other than machine tools), and parts thereof, n.e.s.
- 741 Heating and cooling equipment, and parts thereof, n.e.s.
- 742 Pumps for liquids, whether or not fitted with a measuring device; liquid elevators; parts for such pumps and liquid elevators
- 743 Pumps (other than pumps for liquids), air or other gas compressors and fans; ventilating or recycling hoods incorporating a fan, whether or not fitted with filters; centrifuges; filtering or purifying apparatus; parts thereof
- 744 Mechanical handling equipment, and parts thereof, n.e.s.
- 745 Non-electrical machinery, tools and mechanical apparatus, and parts thereof, n.e.s.
- 746 Ball- or roller bearings
- 747 Taps, cocks, valves and similar appliances for pipes, boiler shells, tanks, vats or the like, including pressure-reducing valves and thermostatically controlled valves
- 748 Transmission shafts (including camshafts and crankshafts) and cranks; bearing housings and plain shaft bearings; gears and gearing; ball screws; gearboxes and other speed changers (including torque converters); flywheels and pulleys (including pulley bloc
- 749 Non-electric parts and accessories of machinery, n.e.s.
- 751 Office machines
- 752 Automatic data-processing machines and units thereof; magnetic or optical readers, machines for transcribing data onto data media in coded form and machines for processing such data, n.e.s.
- 759 Parts and accessories (other than covers, carrying cases and the like) suitable for use solely or principally with machines falling within groups 751 and 752
- 761 Television receivers (including video monitors and video projectors), whether or not incorporating radio-broadcast receivers or sound- or video-recording or reproducing apparatus
- 762 Radio-broadcast receivers, whether or not incorporating sound-recording or reproducing apparatus or a clock
- 763 Sound recorders or reproducers; television image and sound recorders or reproducers; prepared unrecorded media
- 764 Telecommunications equipment, n.e.s., and parts, n.e.s., and accessories of apparatus falling within division 76
- 771 Electric power machinery (other than rotating electric plant of group 716), and parts thereof
- 772 Electrical apparatus for switching or protecting electrical circuits or for making connections to or in electrical circuits (e.g., switches, relays, fuses, lightning arresters, voltage limiters, surge suppressors, plugs and sockets, lamp-holders and junct

- 773 Equipment for distributing electricity, n.e.s.
- 774 Electrodiagnostic apparatus for medical, surgical, dental or veterinary purposes, and radiological apparatus
- 775 Household-type electrical and non-electrical equipment, n.e.s.
- 776 Thermionic, cold cathode or photo-cathode valves and tubes (e.g., vacuum or vapour or gas-filled valves and tubes, mercury arc rectifying valves and tubes, cathode-ray tubes, television camera tubes); diodes, transistors and similar semiconductor devices;
- 778 Electrical machinery and apparatus, n.e.s.
- 781 Motor cars and other motor vehicles principally designed for the transport of persons (other than motor vehicles for the transport of ten or more persons, including the driver), including station-wagons and racing cars
- 782 Motor vehicles for the transport of goods and special-purpose motor vehicles
- 783 Road motor vehicles, n.e.s.
- 784 Parts and accessories of the motor vehicles of groups 722, 781, 782 and 783
- 785 Motor cycles (including mopeds) and cycles, motorized and non-motorized; invalid carriages
- 786 Trailers and semi-trailers; other vehicles, not mechanically-propelled; specially designed and equipped transport containers
- 791 Railway vehicles (including hovertrains) and associated equipment
- 792 Aircraft and associated equipment; spacecraft (including satellites) and spacecraft launch vehicles; parts thereof
- 793 Ships, boats (including hovercraft) and floating structures
- 811 Prefabricated buildings
- 812 Sanitary, plumbing and heating fixtures and fittings, n.e.s.
- 813 Lighting fixtures and fittings, n.e.s.
- 871 Optical instruments and apparatus, n.e.s.
- 872 Instruments and appliances, n.e.s., for medical, surgical, dental or veterinary purposes
- 873 Meters and counters, n.e.s.
- 874 Measuring, checking, analysing and controlling instruments and apparatus, n.e.s.
- 881 Photographic apparatus and equipment, n.e.s.
- 882 Photographic and cinematographic supplies
- 884 Optical goods, n.e.s.
- 885 Watches and clocks
- 891 Arms and ammunition

Source: [42; 144]

E.3 Country/area classification by geographical region

Table E.3.1 | Country/area classification by geographical region

Africa	Americas	China, Hong Kong SAR	Northern Europe
Central Africa	Caribbean	China, Macao SAR	Denmark
Angola	Anguilla	China, Taiwan Province	Estonia
Cameroon	Antigua and Barbuda	Democratic People's Republic of Korea	Finland
Central African Republic	Aruba	Japan	Iceland
Chad	Bahamas	Mongolia	Ireland
Congo	Barbados	Republic of Korea	Latvia
Democratic Republic of the Congo	British Virgin Islands	South-eastern Asia	Lithuania
Equatorial Guinea	Cayman Islands	Brunei Darussalam	Norway
Gabon	Cuba	Cambodia	Sweden
Sao Tome and Principe	Curaçao	Indonesia	United Kingdom
Eastern Africa	Dominica	Lao People's Democratic Republic	Southern Europe
Burundi	Dominican Republic	Malaysia	Albania
Comoros	Grenada	Myanmar	Andorra
Djibouti	Haiti	Philippines	Bosnia and Herzegovina
Eritrea	Jamaica	Singapore	Croatia
Ethiopia	Montserrat	Thailand	Greece
Kenya	Puerto Rico	Timor-Leste	Italy
Madagascar	Saint Kitts and Nevis	Viet Nam	Kosovo
Malawi	Saint Lucia	Southern Asia	Malta
Mauritius	Saint Vincent and the Grenadines	Afghanistan	Montenegro
Mozambique	Sint Maarten (Dutch part)	Bangladesh	North Macedonia
Rwanda	Trinidad and Tobago	Bhutan	Portugal
Seychelles	Turks and Caicos Islands	India	San Marino
Somalia	Central America	Iran (Islamic Republic of)	Serbia
South Sudan	Belize	Maldives	Slovenia
Uganda	Costa Rica	Nepal	Spain
United Republic of Tanzania	El Salvador	Pakistan	Western Europe
Zambia	Guatemala	Sri Lanka	Austria
Zimbabwe	Honduras	Western Asia	Belgium
Northern Africa	Mexico	Armenia	France
Algeria	Nicaragua	Azerbaijan	Germany
Egypt	Panama	Bahrain	Liechtenstein
Libya	Northern America	Cyprus	Luxembourg
Morocco	Bermuda	Georgia	Monaco
Sudan	Canada	Iraq	Netherlands (Kingdom of the)
Tunisia	Greenland	Israel	Switzerland
Southern Africa	United States of America	Jordan	Oceania
Botswana	South America	Kuwait	Australia and New Zealand
Eswatini	Argentina	Lebanon	Australia
Lesotho	Bolivia (Plurinational State of)	Oman	New Zealand
Namibia	Brazil	Qatar	Melanesia
South Africa	Chile	Saudi Arabia	Fiji
Western Africa	Colombia	State of Palestine	New Caledonia
Benin	Ecuador	Syrian Arab Republic	Papua New Guinea
Burkina Faso	Guyana	Türkiye	Solomon Islands
Cabo Verde	Paraguay	United Arab Emirates	Vanuatu
Côte d'Ivoire	Peru	Yemen	Micronesia
Gambia	Suriname	Europe	Kiribati
Ghana	Uruguay	Eastern Europe	Marshall Islands
Guinea	Venezuela (Bolivarian Republic of)	Belarus	Micronesia (Federated States of)
Guinea-Bissau	Asia	Bulgaria	Nauru
Liberia	Central Asia	Czechia	Palau
Mali	Kazakhstan	Hungary	Polynesia
Mauritania	Kyrgyzstan	Poland	Cook Islands
Niger	Tajikistan	Republic of Moldova	French Polynesia
Nigeria	Turkmenistan	Romania	Samoa
Senegal	Uzbekistan	Russian Federation	Tonga
Sierra Leone	Eastern Asia	Slovakia	Tuvalu
Togo	China	Ukraine	

Source: [138]

E.4 Country/area classification by stage of industrial development

Table E.4.1 | Country/area classification by stage of industrial development

High-income industrial economies			
Australia	Estonia	Liechtenstein	Romania
Austria	Finland	Lithuania	Russian Federation
Bahrain	France	Luxembourg	San Marino
Belgium	Germany	Malta	Singapore
Bulgaria	Greece	Nauru	Slovakia
Canada	Hungary	Netherlands (Kingdom of the)	Slovenia
Chile	Iceland	New Zealand	Spain
China, Taiwan Province	Ireland	Norway	Sweden
Croatia	Israel	Poland	Switzerland
Cyprus	Italy	Portugal	United Kingdom
Czechia	Japan	Puerto Rico	United States of America
Denmark	Latvia	Republic of Korea	Uruguay
High-income industrializing economies			
Andorra	Brunei Darussalam	Guyana	Qatar
Anguilla	Cayman Islands	Kuwait	Saint Kitts and Nevis
Antigua and Barbuda	China, Hong Kong SAR	Monaco	Saudi Arabia
Aruba	China, Macao SAR	Montserrat	Seychelles
Bahamas	Cook Islands	New Caledonia	Sint Maarten (Dutch part)
Barbados	Curaçao	Oman	Trinidad and Tobago
Bermuda	French Polynesia	Palau	Turks and Caicos Islands
British Virgin Islands	Greenland	Panama	United Arab Emirates
Middle-income industrial economies			
Belarus	Costa Rica	Mauritius	South Africa
Bosnia and Herzegovina	Dominican Republic	Mexico	Thailand
Brazil	Eswatini	North Macedonia	Türkiye
China	Malaysia	Serbia	Viet Nam
Middle-income industrializing economies			
Albania	Egypt	Lao People's Democratic Republic	Saint Vincent and the Grenadines
Algeria	El Salvador	Lebanon	Samoa
Angola	Equatorial Guinea	Lesotho	Sao Tome and Principe
Argentina	Fiji	Libya	Senegal
Armenia	Gabon	Maldives	Solomon Islands
Azerbaijan	Georgia	Marshall Islands	Sri Lanka
Bangladesh	Ghana	Mauritania	State of Palestine
Belize	Grenada	Micronesia (Federated States of)	Suriname
Benin	Guatemala	Mongolia	Tajikistan
Bhutan	Guinea	Montenegro	Timor-Leste
Bolivia (Plurinational State of)	Haiti	Morocco	Tonga
Botswana	Honduras	Myanmar	Tunisia
Cabo Verde	India	Namibia	Turkmenistan
Cambodia	Indonesia	Nepal	Tuvalu
Cameroon	Iran (Islamic Republic of)	Nicaragua	Ukraine
Colombia	Iraq	Nigeria	United Republic of Tanzania
Comoros	Jamaica	Pakistan	Uzbekistan
Congo	Jordan	Papua New Guinea	Vanuatu
Cuba	Kazakhstan	Paraguay	Venezuela (Bolivarian Republic of)
Côte d'Ivoire	Kenya	Peru	Zambia
Djibouti	Kiribati	Philippines	Zimbabwe
Dominica	Kosovo	Republic of Moldova	
Ecuador	Kyrgyzstan	Saint Lucia	
Low-income economies			
Afghanistan	Eritrea	Mali	Sudan
Burkina Faso	Ethiopia	Mozambique	Syrian Arab Republic
Burundi	Gambia	Niger	Togo
Central African Republic	Guinea-Bissau	Rwanda	Uganda
Chad	Liberia	Sierra Leone	Yemen
Democratic People's Republic of Korea	Madagascar	Somalia	
Democratic Republic of the Congo	Malawi	South Sudan	

Source: [38]

E.5 Other common country/area groups

Table E.5.1 | List of countries/areas: Emerging industrial economies

Emerging industrial economies		
Bangladesh	India	Myanmar
Cambodia	Indonesia	Rwanda
China	Lao People's Dem. Rep	Tanzania
Ethiopia	Malaysia	Viet Nam

Source: [38]

Table E.5.2 | List of countries/areas: Least developed countries (LDCs)

Least developed countries (LDCs)		
Afghanistan	Guinea	Rwanda
Angola	Guinea-Bissau	Senegal
Bangladesh	Haiti	Sierra Leone
Benin	Kiribati	Solomon Islands
Burkina Faso	Lao People's Dem. Rep	Somalia
Burundi	Lesotho	South Sudan
Cambodia	Liberia	Sudan
Central African Republic	Madagascar	Tanzania
Chad	Malawi	Timor-Leste
Comoros	Mali	Togo
Congo, Dem. Rep. of	Mauritania	Tuvalu
Djibouti	Mozambique	Uganda
Eritrea	Myanmar	Yemen
Ethiopia	Nepal	Zambia
Gambia	Niger	

Source: [146]

Table E.5.3 | List of countries/areas: Landlocked developing countries (LLDCs)

Landlocked developing countries (LLDCs)		
Afghanistan	Ethiopia	North Macedonia
Armenia	Kazakhstan	Paraguay
Azerbaijan	Kyrgyzstan	Rwanda
Bhutan	Lao People's Dem. Rep	South Sudan
Bolivia (Plurinational State of)	Lesotho	Tajikistan
Botswana	Malawi	Turkmenistan
Burkina Faso	Mali	Uganda
Burundi	Moldova, Rep. of	Uzbekistan
Central African Republic	Mongolia	Zambia
Chad	Nepal	Zimbabwe
Eswatini	Niger	

Source: [147]

Table E.5.4 | List of countries/areas: Small island developing States (SIDS)

Small island developing States (SIDS)		
Anguilla	Grenada	Saint Kitts and Nevis
Antigua and Barbuda	Guinea-Bissau	Saint Lucia
Aruba	Guyana	Samoa
Bahamas	Haiti	Sao Tome and Principe
Barbados	Jamaica	Seychelles
Belize	Kiribati	Singapore
British Virgin Islands	Maldives	Sint Maarten (Dutch part)
Cabo Verde	Marshall Islands	Solomon Islands
Comoros	Mauritius	St. Vincent and the Grenadines
Cook Islands	Micronesia, Fed. States of	Suriname
Cuba	Montserrat	Timor-Leste
Curaçao	Nauru	Tonga
Dominica	New Caledonia	Trinidad and Tobago
Dominican Republic	Palau	Tuvalu
Fiji	Papua New Guinea	Vanuatu
French Polynesia	Puerto Rico	

Source: [148]

Table E.5.5 | List of countries/areas: BRICS

BRICS		
Brazil	India	South Africa
China	Indonesia	United Arab Emirates
Egypt	Iran	
Ethiopia	Russian Federation	

Source: [149]

Table E.5.6 | List of countries/areas: United States Mexico Canada Agreement (USMCA)

United States Mexico Canada Agreement (USMCA)		
Canada	Mexico	United States

Source: [38]

Table E.5.7 | List of countries/areas: Southern Common Market (MERCOSUR)

Southern Common Market (MERCOSUR)		
Argentina	Brazil	Uruguay
Bolivia (Plurinational State of)	Paraguay	Venezuela (Bolivarian Rep. of)

Source: [38]

Table E.5.8 | List of countries/areas: European Union

European Union		
Austria	France	Malta
Belgium	Germany	Netherlands
Bulgaria	Greece	Poland
Croatia	Hungary	Portugal
Cyprus	Ireland	Romania
Czechia	Italy	Slovakia
Denmark	Latvia	Slovenia
Estonia	Lithuania	Spain
Finland	Luxembourg	Sweden

Source: [38]

Table E.5.9 | List of countries/areas: African Union

African Union		
Algeria	Eswatini	Namibia
Angola	Ethiopia	Niger
Benin	Gabon	Nigeria
Botswana	Gambia	Rwanda
Burkina Faso	Ghana	Sao Tome and Principe
Burundi	Guinea	Senegal
Cabo Verde	Guinea-Bissau	Seychelles
Cameroon	Kenya	Sierra Leone
Central African Republic	Lesotho	Somalia
Chad	Liberia	South Africa
Comoros	Libya	South Sudan
Congo	Madagascar	Sudan
Congo, Dem. Rep. of	Malawi	Tanzania
Côte d'Ivoire	Mali	Togo
Djibouti	Mauritania	Tunisia
Egypt	Mauritius	Uganda
Equatorial Guinea	Morocco	Zambia
Eritrea	Mozambique	Zimbabwe

Source: [38]

Table E.5.10 | List of countries/areas: Gulf Cooperation Council (GCC)

Gulf Cooperation Council (GCC)		
Bahrain	Oman	Saudi Arabia
Kuwait	Qatar	United Arab Emirates

Source: [38]

Table E.5.11 | List of countries/areas: Eurasian Economic Union (EAEU)

Eurasian Economic Union (EAEU)		
Armenia	Kazakhstan	Russian Federation
Belarus	Kyrgyzstan	

Source: [38]

Table E.5.12 | List of countries/areas: South Asian Association for Regional Cooperation (SAARC)

South Asian Association for Regional Cooperation (SAARC)		
Afghanistan	India	Pakistan
Bangladesh	Maldives	Sri Lanka
Bhutan	Nepal	

Source: [38]

Table E.5.13 | List of countries/areas: Regional Comprehensive Economic Partnership (RCEP)

Regional Comprehensive Economic Partnership (RCEP)		
Australia	Japan	New Zealand
Brunei Darussalam	Korea, Rep. of	Philippines
Cambodia	Lao People's Dem. Rep	Singapore
China	Malaysia	Thailand
Indonesia	Myanmar	Viet Nam

Source: [150]

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Concepts and definitions

C | E | F | G | H | I | L | M | S | U | W

C

CO₂ emission intensity Indicator computed as the ratio between CO₂ emissions from fuel combustion and the value added of associated economic activities. It can be computed for the entire economy (total CO₂ emissions per unit of GDP) or for specific sectors, notably the manufacturing sector (CO₂ emissions from manufacturing per unit of MVA). 25

E

Economies of scale In economics, this term refers to the reduced cost per unit that arises from a higher production level. 5

Electricity, gas, steam and air-conditioning supply This sector refers to ISIC Rev. 4 section D. It includes the provision of electric power, natural gas, steam, hot water, air-conditioning supply and the like through a permanent infrastructure of lines, mains and pipes. This section also includes the operation of electric and gas utilities, which generate, control and distribute electric power or gas. 4, 18, 186

F

Factoryless goods production (FGP) Factoryless goods production represents a distinctive organizational model within the broader landscape of global manufacturing arrangements. A principal controls the production of a good by undertaking the entrepreneurial steps and providing the technical specifications required to produce the good, but outsources all or most of the material transformation process required to produce the output [151]. x, 120

G

Gross domestic product (GDP) GDP basically derives from the concept of gross value added. GDP is the sum of gross value added of all resident producer units plus the part (possibly the total) of taxes on products that is not included in the valuation of output, less subsidies on products [152, p. 34]. x, 5

H

Human Development Index (HDI) Long-standing indicator measuring human development, defined as the process of expanding people's freedoms and opportunities and improving their well-being. The HDI is a composite index of three dimensions: 1) the ability to lead a long and healthy life, measured by life expectancy at birth; 2) the ability to acquire knowledge, measured by mean years of schooling and expected years of schooling; and 3) the ability to achieve a decent standard of living, measured by gross national income per capita. A higher HDI value indicates a higher human development level. This indicator is calculated and maintained by UNDP [16]. x, 7

I

Index of industrial production (IIP) Indicator that describes changes in the volume of goods produced in industrial sectors over time. Its main purpose is to provide a measure of short-term changes in value added over a given reference period. However, since it is difficult to collect high-frequency data to accurately measure value added, gross output measures such as the value of production or turnover data are more commonly used. The IIP, being a volume index, is not influenced by price fluctuations [99, p. 11]. x, 18, 146

Industrial statistics Field of statistics related to the characteristics and economic activities of all resident units in the reporting country that are primarily active in the following productive activities:

- a. Mining and quarrying (ISIC Rev. 4 section B);
- b. Manufacturing (ISIC Rev. 4 section C);
- c. Electricity, gas, steam and air-conditioning supply (ISIC Rev. 4 section D);
- d. Water supply; sewerage, waste management and remediation activities (ISIC Rev. 4 section E).

Industrial statistics form part of the broader domain of structural and short-term business statistics [4, pp. 2 and 12]. 4, 5, 159, 160

Industry This term refers to the set of all production units engaged primarily in the same or similar kinds of productive activity [5, p. 9] The ISIC provides the international guidelines for cataloguing economic activity into specific industries, such as agriculture, mining, manufacturing or services. 4

Industry value added (IVA) This is a national accounts aggregate measuring the exclusive and exhaustive contribution of industrial sectors to GDP. x, 13

Inter-agency and Expert Group on SDG Indicators (IAEG-SDGs) Working group tasked by the United Nations Statistical Commission to develop and implement the global indicator framework for the SDGs and the Targets of the 2030 Agenda for Sustainable Development. It is composed of representatives from Member States from all regions, with custodian agencies participating as observers and experts. 23

ISIC The International Standard Industrial Classification of All Economic Activities, abbreviated as ISIC, is a standard classification of economic activities arranged so that entities can be classified according to the activity they carry out. Currently, the fourth revision of ISIC [5] is the most commonly used, and it is the reference used throughout this Yearbook. The hierarchically structured ISIC Rev. 4 classification contains sections, divisions, groups and classes. The ISIC classification is maintained by UNSD. x, 4, 121

L

Large case unit (LCU) A large case unit is a specialized unit within an NSO that focuses on ensuring the quality and consistency of data for large establishments, especially MNEs. x, 127

Least developed country (LDC) The LDC category was established by the United Nations (UN) General Assembly in 1971. The UN defines LDCs as countries that have low-income levels and face severe structural impediments to sustainable development. x

M

Manufacturing This industry refers to ISIC Rev. 4 section C. It includes activities related to the physical or chemical transformation of materials, substances or components into new products, although such a definition cannot be used as the single universal criterion for determining what constitutes manufacturing (see [5] for more details on activities included and not included in manufacturing). The materials, substances or components transformed are raw materials that are products of agriculture, forestry, fishing, mining or products of other manufacturing activities. Substantial alteration, renovation or reconstruction of goods is generally considered to be manufacturing. The output of a manufacturing process may be finished in the sense that it is ready for utilization or consumption, or it may be semi-finished in the sense that it is to become an input for further manufacturing. 4, 18, 93, 146, 186

Manufacturing value added (MVA) This is a national accounts aggregate measuring the exclusive and exhaustive contribution of manufacturing to GDP [153, p. 5]. xi, 6

Mining and quarrying This industry corresponds to ISIC Rev. 4 section B. It includes activities related to the extraction of minerals that occur naturally as solids (coal and ores), liquids (petroleum) or gases

(natural gas). Supplementary activities aimed at preparing the crude materials for marketing are also included, if they are undertaken in conjunction with mining, for example, crushing, grinding, cleaning, drying, sorting, concentrating ores, liquefaction of natural gas and agglomeration of solid fuels. 4, 18, 93, 146, 186

Mining and utilities value added (MUVA) This is a national accounts aggregate measuring the exclusive and exhaustive contribution of mining and utilities sectors to GDP. xi, 89

Multinational enterprise (MNE) A multinational enterprise, sometimes also called multinational corporation (MNC), just multinational, or international corporation, is an enterprise producing goods or delivering services in more than one country. It has its management headquarters in one (or rarely more than one) country, the home country, while also operating in other countries, the host countries [154]. xi, 120

S

Short-term statistics (STS) Infra-annual production-related statistics collected to monitor the business cycle. They are suitable for the short-term evaluation of supply, demand and production factors [4, p. 3]. Although available more frequently and in a timelier manner, they usually cover only some variables of interest and are published with a limited level of detail. 146

Statistical confidentiality Principle 6 of the *Fundamental Principles of Official Statistics* [155] stresses that data collected by national statistical agencies are to be strictly confidential and used exclusively for statistical purposes. Many NSOs therefore implement national rules on statistical confidentiality. The two main reasons for declaring data to be of primary confidentiality are:

- a. too few units in a cell;
- b. dominance of one or two units in a cell.

The limits of what constitutes "too few" or "dominance" vary between statistical domains. Statistical confidentiality is ensured through appropriate methods:

Physical protection: data is securely stored and not accessible to anyone without explicit authorization.

Statistical disclosure control (SDC): methods for reducing the risk that statistical units are identified when the statistical data is being published, including:

Tabular data protection: for aggregate information on respondents presented in tables (using suppression, rounding, combinations and interval publication).

Microdata protection: for information on statistical units (using local suppression, sampling, recoding, top and bottom coding, rounding, rank swapping and microaggregation).

xi, 89, 187

Structural business statistics (SBS) These are production-related statistics that are collected and compiled to determine the structure, activity, competitiveness and performance of enterprises at national, regional and international levels. They generally provide annual information with respect to a reference year [4, p. 3]. Although available only annually and published with some delay, they usually cover a large number of variables at a highly granular level. Other common abbreviations used by NSOs are ABS (annual business statistics/survey) or AIS (annual industrial survey). xi, 96, 149

Sub-annual Manufacturing Trade Data Manufacturing trade is a key component of global trade, accounting for the majority of traded goods and offering valuable, timely insights into a country's productive structure and the competitiveness of its manufacturing sector. xi

Sustainable Development Goals (SDGs) The 2030 Agenda for Sustainable Development [17], adopted by all United Nations Member States in 2015, provides a shared blueprint for peace and prosperity

for people and the planet, now and into the future. At its core are the 17 SDGs, which are an urgent call for action by all countries in a global partnership. They recognize that ending poverty and other deprivations must go hand-in-hand with strategies that improve health and education, reduce inequality and spur economic growth—all while tackling climate change and working to preserve our oceans and forests. xi, 3, 154 **Goal 9** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation. 3, 9, 13, 20, 29–31, 44, 54, 82, 130, 154, 155, 158, 159.

System of environmental-economic accounting (SEEA) An international statistical framework that organizes data to show the interrelationships between the economy and the environment, as well as the stocks and changes in stocks of environmental assets as they bring benefits to humanity. It provides a standardized approach with consistent definitions and classifications, similar to the system of national accounts (SNA), to measure environmental assets and their contributions to the economy [74]. xi, 114, 119

System of national accounts (SNA) A coherent, consistent and integrated set of macroeconomic accounts, balance sheets and tables based on a set of internationally agreed concepts, definitions, classifications and accounting rules [156]. xi, 119, 188

U

UNIDO industrial statistics databases UNIDO Statistics currently maintains a variety of databases, as described below. These can be accessed at the [UNIDO Statistics Portal](#). 158 The **Annual Manufacturing Trade Database** [40] contains aggregate data at the country, region and group level for variables related to the trade of manufactured goods. The database provides historical time series in current United States dollar (USD). The data are collected primarily from the United Nations Trade and Development (UNCTAD); other sources of official data may be used when appropriate. 61, 158. The **Competitive Industrial Performance (CIP) index** benchmarks the ability of countries to produce and export manufactured goods competitively. 158. The **Industrial Demand-Supply Balance Databases (IDSB)** [139] contain disaggregated data on the manufacturing sector, including a correspondence to international trade data. 158. The **IIP Database** [19] contains monthly, quarterly and annual data on the index of industrial production for the manufacturing as well as mining and utilities sectors. 146, 158. The **Industrial Statistics Databases (INDSTAT)** [35; 36] contain disaggregated data for selected variables on the manufacturing, mining and utilities sectors. 52, 158. The **Monthly Manufacturing Trade Database** [41] contains monthly data on manufacturing trade with special emphasis on higher-technology goods. 61, 158. The **National Accounts Database** [9] contains aggregate data at the country, region and group level for several national accounts indicators related to industrial economic activity. 158. The **SDG 9 Database** [20] contains the industry-related indicators of SDG 9. 158.

W

Water supply; sewerage, waste management and remediation activities This industry, corresponding to ISIC Rev. 4 section E, includes activities related to the management (including collection, treatment and disposal) of various forms of waste, such as solid or non-solid industrial or household waste, as well as contaminated sites. Activities of water supply are also grouped in this section, since they are often carried out in connection with, or by units also engaged in, the treatment of sewage. 4, 18, 186



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